

The Energy Balance Project

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Part I: Physical Intuition and the EBU Idea

**From life-supporting function to an exact
account of physical action**

Revised Gaussian explanatory edition

This book begins with the reason for the project and develops its mathematics patiently. Physical stories, notation, diagrams, worked calculations, guided problems and practice studios form one continuous course. Curiosity and ordinary arithmetic are enough to begin.

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AUTHOR-REVIEW EDITION

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Author, rights, and scientific scope

Konrad Grzyb is the author and originator of the Energy Balance Project. This revised manuscript was prepared with AI assistance under his direction. It is supplied for author review; completion of the typesetting is not a claim of author acceptance, independent peer review or publication.

The software repository has a separate MIT licence. That software licence does not automatically grant rights in the books. This revision introduces no new rights grant. The original unified explanatory edition is preserved, as are Parts II and III. Their historical findings remain findings about their declared models.

Scientific boundary

Definitions, conditional mathematical consequences, implementation checks and observed scientific results are different kinds of statement. This book explains a declared Gaussian potential and derives exact identities. It does not report an experiment establishing recovery, fairness, planetary homeostasis or economic adoption.

The original Part I is the editorial baseline. Its original editable manuscript was unavailable in the inspected project files, so this is a reconstructed, editable revision. The companion revision ledger identifies retained subjects, rewritten examples and removed model-specific material. The shorter introductory candidate remains a separate preserved draft; it is not this book's scale or layout template.

All numbered tank calculations are illustrative mathematics on explicitly supplied states. They are not measurements of hospitals or reservoirs, recommended operating quantities, registered experimental parameters, or model trajectories. A reference or scale in such an example is a teaching choice. A deployed system would require measurement, calibration, institutions and separate evidence.

Preface: the question behind the project

A clinic can have an orderly financial account while its water system is failing. The reverse is also possible: its equipment works, but a patient cannot afford access. These are distinct failures, and neither disappears because the other is measured well. The motivation of this book is to bring physical function into sharper view without losing that distinction.

Money coordinates claims, obligations and exchanges. It can support investment, maintenance, cooperation and public provision. A monetary record does not, by itself, reconstruct the water withdrawn, electricity transformed, equipment worn, resources displaced or service delivered in an action. An additional physical account might help people ask better questions about these effects. Whether a particular account actually improves decisions is a question for investigation.

The Energy Balance Project proposes a way to value a specified physical change relative to a declared representation of functional balance. It begins with a state: what is present, where it is and in which units. It declares a reference, a scale and a potential measuring departure from that reference. Then it compares the same represented world before and after a finite action. The comparison has a sign. Reducing the declared burden produces positive field value; increasing it produces negative field value.

There is no moral score for the person in this equation. A medicine delivery may be essential and physically burdensome. A perfectly calculated tank redistribution may fail to represent water quality. A useful repair may create a benefit the current state description does not yet contain. Throughout the book, mathematical exactness is paired with a question about representation: exact relative to what?

This revision retains the learning method of the original Part I. We draw a physical scene before writing its matrix. We read subscripts and transposes aloud before using them. We calculate the same action by more than one route. We include zero and negative cases. The studios at the end repeat the entire chain through longer worked problems rather than assuming that recognizing an equation is the same as understanding it.

The central change is the introductory potential. Earlier editions taught one threshold-based model in detail. Here the main example uses a fixed reference and a positive scale to build a quadratic, Gaussian-shaped deviation geometry. Biological threshold laws, lower and upper homeostatic bands, and a particular preservation controller are no longer prerequisites for understanding EBU.

Their original theorems remain scoped to the historical models in which they were established.

This change does not make all living systems Gaussian. It gives us a transparent, differentiable example whose finite differences can be calculated exactly. It also makes an important limitation visible: a mathematical landscape does not contain its own actor. Someone or something must choose and perform an action. A field that measures change is not automatically a controller that restores balance.

The book can therefore be read before new scientific tests. Definitions and conditional proofs can be explained now. Recovery, persistence, useful incentives and real institutional effects require their own evidence. The final chapter returns to those questions after the reader has the tools to state them precisely.

The route through this volume

The opening seven chapters develop the reason for the project: monetary signals, scarcity, unequal exposure, planetary pressures, homeostasis and equilibrium. They distinguish a motivating concern from a conclusion about EBU. None asks the reader to accept a replacement economy in advance.

Chapters 8–16 build the physical and mathematical objects. We follow a pump, map cells and edges, check conservation, declare references and scales, construct the potential, interpret its slope, and calculate the full effect of a finite action. Permission is explained before the exact quote is used, because a favourable value cannot make an impossible movement possible.

Chapters 17–23 explain records and composition. They separate a prediction from verified occurrence, a local calculation from a global audit, an exact group total from an individual attribution, and a route from an instantaneous remote effect. These chapters preserve the original volume's concern with real needs, repairs, privacy, accountability and institutional choice.

Chapters 24–31 are practice studios. Keep a notebook open. Write the units next to every value, calculate before consulting an answer and explain each sign in words. A recurring Ridge–Vale example provides continuity; variations show which conclusions depend on its special numbers. Chapter 32 closes the course with the mathematical statements, their assumptions and the research questions still open.

The combined Part II continues directly at Chapter 33, joining mathematical derivation, implementation and evidence into one course. It develops group value before attribution, attribution before capacity accounts, and physical permission before execution. Detailed historical proofs and experiments sit within the relevant chapters, with their assumptions clearly marked. References to *historical Part II* or *historical Part III* still identify the preserved original editions, not the new numbering. Their threshold models must not be substituted into Gaussian examples without changing the declaration.

The forward series now has eight parts: I introduces the idea; II joins mathematics, implementation and evidence; III develops measurement and falsification; IV studies homeostasis and stochastic dynamics; V develops multiple-action structure; VI addresses distance; VII addresses coordination and memory; VIII addresses institutions. The later six are a research and teaching plan, not established findings.

How to read an equation without fear

Begin with the nouns. In a statement such as $x_i = 18X$, x_i is the quantity at one named location, 18 is its numerical value and X is its stock unit. The subscript identifies a coordinate. It is not a multiplication by i . A reference x_i^* is a declared comparison value; the star is a label, not an instruction to multiply.

Read a subtraction as a comparison. If before-burden is 16 and after-burden is 4, then $16 - 4 = 12$ means the represented burden decreased by twelve units of the declared account. The arithmetic alone does not identify a payer, establish a wage, or certify that all human needs were satisfied.

A vector is a list whose order matters. A matrix is a rule for combining entries in lists. A derivative describes how a function changes nearby. An integral adds those changing nearby contributions along a specified path. These descriptions will be made precise through small examples before the general forms are needed.

When an equation has a condition beside it, read the condition as part of the equation. A claim for fixed references and scales is not automatically true if either is revised halfway through an account. A proof for complete state records cannot detect damage that the state never represented. A proof for a common action path does not identify a physical path that was never observed.

Four questions beside every calculation

What physical object does each symbol name? Which quantities are held fixed? What operation is being performed? What conclusion follows, and which conclusion would need additional evidence?

Use the glossary when a familiar letter changes role. In particular, stock, scale, potential, marginal, action quantity, attribution and spending capacity are different objects. Similar typography is never permission to exchange their meanings. The same discipline that helps a first-time reader also protects a scientific calculation.

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Why the physical question matters

1.1 Begin with the clinic before it opens

The first patient has not arrived, but the clinic is already depending on an extraordinary arrangement. Water has reached a clean tank. Electricity is available to a refrigerator. A road is passable. A trained person has checked the medicine. A drain carries waste away from the room in which another person will receive care. None of these conditions is produced by the word “healthcare”. Each depends on particular things being present, in a particular condition, at the right place and time.

Imagine that this clinic stands in Vale, below a settlement called Ridge. Ridge has a water store and a workshop. Vale has its own store, the clinic and a collection of homes. People travel between them, repair equipment, make agreements and pay bills. We shall return to these places throughout the book. They are invented teaching places: their simplicity helps us examine one question at a time. They are not observations from a successful EBU economy.

On Monday, the clinic pays an invoice for water. Its accounts show the payment correctly. On Tuesday, the inlet valve fails and the clinic’s tank stays nearly empty. Nothing about the correctness of Monday’s payment changes this fact. A nurse can hold a valid receipt while standing beside a dry tap. Equally, water might arrive because a neighbour helps, even though no sale occurs. The physical event and the monetary event can be connected without being the same event.

This is the point from which the Energy Balance Project begins. We want to understand how an economic arrangement could remain answerable to the physical conditions on which its participants depend. That ambition is larger than the mathematics developed here. The mathematics supplies a carefully bounded account of physical changes; whether an institution built around such an account would be useful, fair or reliable is a further question.

The first separation

An invoice records a recognised transaction. A physical account records a declared change in the world. A service record asks whether the intended function was available. All three may concern the same morning at the clinic, but none automatically answers the other two questions.

1.2 What it means to support life

The expression “support life” can become so broad that it explains nothing. Here it begins with identifiable functions. A household needs water suitable for its intended use. A medicine store needs conditions that preserve the medicine. A person needs access to food, shelter, care and the relationships and institutions through which these are obtained. A landscape may sustain processes on which a settlement depends without being owned by any single participant.

A useful description therefore specifies the function before selecting a number. Litres of water measure quantity. They do not, by themselves, measure quality, accessibility or timeliness. Kilograms of food do not identify its nutritional suitability for a particular person. A count of hospital beds does not say whether trained staff and essential supplies are present. The temptation to find one impressive number is strong precisely because the full question is difficult.

Our response will be to keep the representation explicit. A model includes some variables and omits others. When it includes water stock but omits water quality, its conclusions concern stock under the stated quality assumption. The omitted variable does not become unimportant. It becomes a limit of the conclusion. Declaring that limit is part of building a useful theory, rather than a reason to abandon theory altogether.

The word “functional” will be important. A system is functional relative to something it is meant or understood to do. A full tank disconnected from every tap may be physically intact and functionally useless for washing. A bridge can retain its mass after losing its ability to carry a vehicle. Conversely, a repair can restore a vital function while adding little material. Physical description must sometimes represent configuration and condition as well as quantity.

The first mathematical model in this edition deliberately chooses a narrower object: one homogeneous, conserved scalar distributed among places. Within that model, units are interchangeable in the declared physical sense. The choice is a teaching and research boundary. It does not claim that water,

medicine, biodiversity and electricity are interchangeable, or that they should be added as though their units were identical.

1.3 Why monetary success can leave a physical question open

Suppose the clinic's accountant reports that expenditure matched the budget. This is valuable information. It helps establish that purchases were authorised and that obligations can be met. Yet the sentence leaves open whether the tank filled, whether the pump will work tomorrow and whether every patient can reach the building. Those questions require additional observations and decisions.

The same distinction applies at a larger scale. A profitable enterprise may preserve a resource well, or may use it in a way whose consequences fall outside its accounts. A public programme may provide an essential service while operating at a monetary cost. These are logical possibilities, not a claim that profit is always harmful or public spending always beneficial. The point is that one outcome cannot be inferred solely from the sign of another account.

It would consequently be too strong to say that money has no connection to life or the environment. People buy food, finance water treatment, pay carers and fund restoration with money. Contracts, taxes and regulations can connect monetary consequences to physical behaviour. The precise concern is that a monetary balance, by itself, contains no general theorem guaranteeing preservation of every life-supporting function. Such preservation requires additional institutions, information and physical action.

EBU asks whether an explicit account of represented physical change could contribute something useful to that arrangement. The proposal is not justified simply by naming shortcomings elsewhere. It must make its own definitions clear, expose its own failures and eventually face comparisons that could show it unnecessary. A theory gains credibility by making a difficult question examinable, not by assuming that dissatisfaction with the present proves the alternative.

1.4 Three objects that will accompany us

The first object is the *physical state*. In the small teaching world this can be a list of stocks, one for each place. If Ridge has eighteen units and Vale has two, we write the pair as (18, 2). This list reports where the represented resource is. It does not state a price, an entitlement or a moral ranking.

The second object is the *represented burden*. A declared function assigns a number to the state relative to a declared reference and scale. The reference

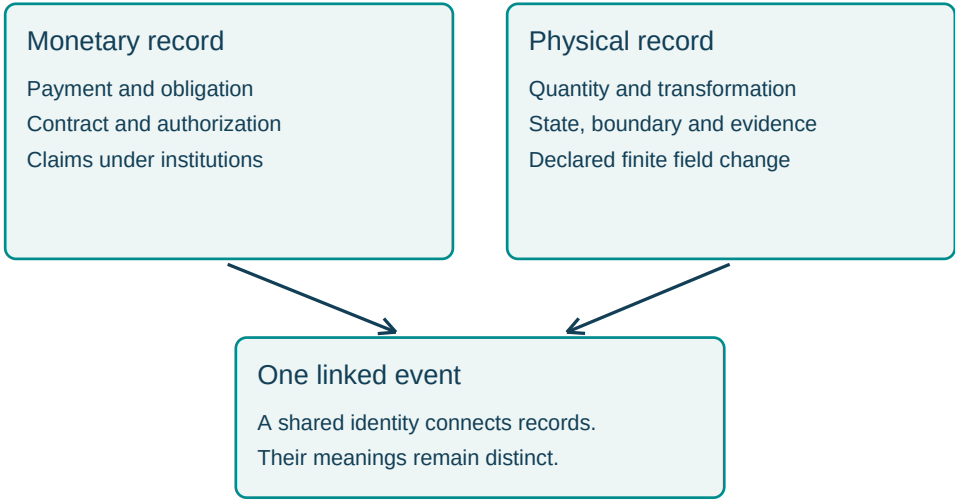


Figure 1.1: A financial record and a physical record describe different aspects of the same provision. Neither automatically contains the other.

says which configuration the account uses for comparison. The scale says how differences are measured. Later we shall develop a smooth quadratic construction related to a Gaussian reference. Neither the reference nor its scale can be discovered merely by writing a formula on a page.

The third object is the *signed action result*. Once a physical action is specified, its result compares represented burden before and after the action and subtracts any separately justified process burden. The sign reports the direction of that declared comparison. It does not announce whether an actor is virtuous, whether a patient deserves care or whether a wallet should receive a payment.

These objects answer successive questions: what is present, how the represented state is evaluated, and what difference a particular action makes to that evaluation. Confusing them produces surprisingly persuasive mistakes. A larger stock is not automatically a better state. A lower burden is not automatically a complete social improvement. A positive result is not automatically a right to compensation.

1.5 A first account without a valuation

Let the two stores contain twenty units in total. Ridge holds eighteen and Vale holds two. A declared lossless transfer moves four units from Ridge to

Vale. The physical account is

$$(18, 2) \longrightarrow (14, 6), \quad 18 + 2 = 14 + 6 = 20. \quad (1.1)$$

This is an account of a hypothetical change. It is not a simulation, and the numbers are not registered study parameters. The equality simply checks that the same twenty units are represented on both sides of the transfer.

Several useful questions remain unanswered. Were four units legally available for transfer? Could the pipe carry them? Did the clinic ask for them? What quantity should remain at Ridge? Does six at Vale serve the intended function? What payment, if any, was agreed? The conservation equality cannot answer these questions because their information is absent from the equality.

This restraint is a strength. If the physical equality also purported to settle ownership and need, it would conceal institutional choices inside arithmetic. Instead, we can attach additional records to the same event. A permission record describes what could be accepted. A demand record describes the request. A valuation record applies the declared function. A monetary record describes payment. Common event identity allows the records to be compared while their meanings remain distinct.

Read only what the record contains

The transfer from $(18, 2)$ to $(14, 6)$ establishes a four-unit decrease at Ridge and a four-unit increase at Vale under the lossless declaration. It establishes no EBU value until the reference, scales and burden function have been declared. It establishes no payment until an institutional rule or agreement supplies one.

1.6 What changes when the example becomes real

A real installation would need observations, not just written numbers. The source and destination meters must refer to compatible quantities and time intervals. Water held in the pipe might need representation. A leak would require an explicit account. Quality assumptions would need checking. An event that was requested but never performed must not be recorded as a completed physical contribution.

Measurement uncertainty remains real even though this edition removes numerical-error controllers from its instructional core. The removal simplifies the theory being taught; it does not create perfect instruments. A future measurement system would have to declare calibration, uncertainty, missing-data handling and correction procedures. Exact algebra tells us what follows from declared inputs. It cannot certify those inputs by itself.

Boundaries matter as much as instruments. If the account includes only the two tanks, water temporarily stored in the pipe appears to have disappeared unless transit is declared. If it includes an external source, inflow across the boundary must be distinguished from an actor's internal redistribution. If it includes electricity consumption, the relevant carrier and conversion must be specified. The reader should get into the habit of asking, "Where did the thing go?" before asking, "What number did the action earn?"

A full causal chain may therefore contain several linked actions: producing electricity, delivering it, using it in a pump and transferring water. These actions should not be compressed into one unexplained event merely because one customer receives one invoice. Nor should an extra burden be invented at every link simply because a chain exists. Each represented physical effect needs one declared accounting place.

1.7 An event needs an identity

Suppose two people both report that they delivered the same four units. One opened the valve and the other read the meter. Both may have contributed useful work, but there was one physical transfer. Recording two complete four-unit transfers would describe a world that did not occur. Recording one transfer does not imply that only one person deserves recognition; it means that the physical event and the attribution of human contributions need distinct records.

The event record should therefore identify the source, destination, interval and measured quantity before any attribution is attempted. It can then identify the associated participants and their declared roles. A correction should point back to the same event rather than silently insert a second event. These are ordinary documentary requirements with important mathematical consequences: an exact field change must not be counted repeatedly under several descriptions.

The converse mistake is also possible. Two genuinely separate transfers may have equal quantities and use the same route at different times. They should not be merged merely because their descriptions look alike. Their baselines can differ, so their EBU results can differ as well. A useful physical account preserves enough identity to distinguish repetition of a record from repetition of an action. This is one reason that verification remains part of the subject even when the arithmetic fits on a single line.

1.8 Why a simple world is worth studying

The small Ridge–Vale world is not offered as a miniature society with all the important details retained. It is a controlled language for learning. Two stocks make it possible to see the source decrease and destination increase on the same line. One resource makes conservation unambiguous. A fixed reference makes it possible to distinguish a physical change from a change in evaluation.

The simplifications also create obligations. We must not call the model regenerative when no regeneration appears in it. We must not infer continuing service from a single transfer. We must not claim that a conserved total prevents local deprivation. We must not use mathematical smoothness as evidence that institutions will be smooth, responsive or fair. Each extension requires its own stated mechanism.

There is nevertheless substantial work to do within this narrow world. A finite transfer differs from an infinitesimal suggestion. Several actions can share a nonlinear field. A path can be used for an accounting decomposition without being an observed physical trajectory. Local records can close against a global potential under specific assumptions. These are meaningful questions before any claim about a national economy is attempted.

1.9 The human purpose does not disappear into the model

An account can help reveal that maintaining a clinic is physically expensive in a particular location. It cannot decide that people in that location should lose care. The appropriate response might be better infrastructure, shared support, a different route, a public guarantee or a reconsideration of the representation. The burden number provides information for a decision; the decision remains accountable to people.

Likewise, a person who cannot perform a measured action still has needs. Children, disabled people, ill people and carers must not become invisible because a convenient model records certain transfers more easily than other contributions. The project's motivation includes a commitment to essential support. That commitment is ethical and institutional; it is not a theorem produced by the sign of an equation.

This book consequently asks the reader to hold two ideas together. Physical consequences should remain visible, including inconvenient consequences of necessary actions. Human rights and access should not be reduced to the ability to produce a favourable physical score. Honest accounting and humane institutions have different jobs, and a responsible design must explain how they interact.

1.10 Guided problems and answers

1.10.1 The missing two units

The clinic pays for ten units. Its inlet meter records eight. What can be concluded, and what must be investigated?

The payment record establishes the monetary transaction or contractual claim according to its terms. The inlet reading supplies evidence about arrival at that meter. Their difference does not identify a cause. Two units could remain in transit, have leaked, have been measured differently or have been incorrectly invoiced. The next useful step is to reconcile boundaries and timestamps, then inspect the relevant observations. Assigning an EBU result before identifying the physical event would conceal the uncertainty.

1.10.2 The helpful action with no sale

A neighbour transfers four units to the clinic without requesting money. Does the absence of a sale prevent a physical account?

No. The source, destination, quantity and resulting states can be recorded independently of payment. A signed EBU result would additionally require the declared valuation. Whether the neighbour should receive recognition, reimbursement or credit is a separate institutional question. Recording the event does not require inventing a customer or treating the clinic's gain as a monetary loss to someone else.

1.10.3 The balanced total

Two arrangements contain twenty units: $(18, 2)$ and $(10, 10)$. Are they equally good because their totals agree?

Conservation establishes that the totals agree. It does not establish equivalent function. If the places have different needs or capabilities, even equal stocks need not be an appropriate reference. A later valuation must declare what it compares and why. The equal pair will be useful in our teaching fixture because we explicitly choose equal references, not because equality of allocation is derived from conservation.

What money records and what it leaves open

2.1 The pump has two kinds of balance

The clinic manager places two sheets on the table. The first records a payment to the pump workshop. The second records water entering and leaving the clinic's tank. On the first sheet, a debit has a corresponding credit. On the second, an increase at one represented place must be explained by a physical source or a boundary flow. Both sheets speak of balance, but they balance different things.

Imagine that the workshop charges one hundred monetary units for a repair. The clinic pays, the workshop records revenue, and the transaction is complete. Suppose the repaired pump then delivers four units of water. The four water units cannot be obtained by subtracting the two monetary balances. They require a physical record. A change in the negotiated repair price would not, by itself, change the measured water delivery.

This distinction is elementary, yet much confusion arises from ignoring it. The word “value” may mean a price, a preference, a benefit, a replacement cost or a declared physical evaluation. The same word does not make the underlying objects identical. Before comparing two values, we should ask which question each number answers and which unit each carries.

2.2 Money as a social coordination instrument

Money allows people to make payments without matching every exchange through direct barter. Prices express exchange terms in a common unit. Saving and borrowing connect claims across time. Accounting makes obligations visible, and contracts specify what parties have agreed to provide. These functions can support exceptionally complicated cooperation among people who never meet.

Consider the clinic's refrigerator. Its components may come from many workshops; its installer needs food and housing; the electricity supplier needs maintenance; the clinic receives income from several sources. A monetary arrange-

ment can coordinate these relationships without making each participant exchange directly with every other participant. Recognising that achievement does not require believing that every resulting allocation is fair or physically sustainable.

The useful question is therefore not whether money is “real” in the same sense as water. A contract is real as an institutional relationship, even though it is not a liquid in a tank. The question is how institutional claims are connected to material capabilities. A claim to receive water matters because people accept and enforce it; actual delivery also requires water, equipment and work.

Claims can change without an immediate matching transfer of a particular physical stock. The Bank of England’s 2014 account of the modern UK monetary system explains that commercial bank lending can create a corresponding deposit; it rejects the idea that banks merely lend out a fixed pool of prior deposits. Its account also describes constraints on lending and the role of monetary policy. This is an institutional explanation, not a claim that lending creates physical resources without limit.[1]

For our purposes, the consequence is modest but important. The creation or transfer of a monetary claim does not satisfy the conservation equation for water, fuel or medicine. Such an equation must refer to the represented physical resource. Conversely, conserving water does not settle who is entitled to claim payment. Each account needs its own rules.

2.3 A price carries information, but not every kind

Suppose a replacement valve becomes difficult to obtain. A higher price may encourage repair, substitution or additional supply. It may also prevent a clinic with little purchasing power from obtaining the valve. The physical scarcity and the distribution of purchasing power both affect the situation, but they are different facts.

A price need not be meaningless to be incomplete. It can communicate something about current exchange conditions without being a direct measure of biological need, ecological effect or engineering condition. If two buyers offer different amounts for the last valve, the bids reveal willingness and ability to pay under the arrangement. They do not independently measure the consequences of each buyer going without it.

Nor should we imagine that one can read a price backwards into a unique physical history. The same quoted price might reflect transport expense, bargaining, taxation, quality, scarcity, contractual risk or a temporary discount. One number can result from many combinations. An engineer who needs to

know how much electricity a route used must measure or otherwise establish that quantity.

This is why a proposed EBU account should be evaluated as an additional declared description. It cannot acquire accuracy merely by disagreeing with a price. If its state variables are poor, its reference unjustified or its observations wrong, it can also mislead. The comparison should be between explicit information structures, not between an idealised new metric and every defect of actual markets.

2.4 Effects outside the immediate exchange

Imagine two workshops in our invented valley. Both sell a repaired component for the same amount. One contains a cleaning fluid and pays for appropriate treatment; the other allows it to enter a shared stream. If the buyer's payment record is the only account consulted, the two transactions may appear equivalent. The stream record distinguishes them.

This hypothetical example illustrates an external effect: the transaction changes conditions experienced by people or places outside the immediate agreement. It does not prove how common such behaviour is, or specify the best remedy. A physical measurement, a liability rule, a standard, a tax, a collective agreement and a change in production practice might play different roles. The remedy needs evidence about the actual setting.

Positive effects can also be incompletely captured. A person who repairs a shared path may improve access for many people who do not pay that person directly. A family member may perform care whose importance is not represented by a sale. The absence of a market payment is therefore not equivalent to an absence of work or social significance.

An EBU model does not automatically see these effects either. If the stream or the path is outside its declared state, a calculated result may miss them. Extending the representation can improve coverage, but increases measurement and modelling demands. It is preferable to state a limited account honestly than to call it comprehensive because its symbol has an ambitious name.

2.5 Existing institutions already connect money and physics

It would be inaccurate to present the contemporary economy as having no controls, no environmental rules or no public purpose. Institutions already attempt to constrain harmful actions, support essential services and attach consequences to measured physical behaviour. Their effectiveness and coverage vary, but their existence matters to any fair comparison.

For example, the European Commission describes the EU Emissions Trading System as a cap-and-trade arrangement. Covered emissions are monitored and reported; operators surrender allowances corresponding to them. The system therefore connects a declared physical quantity to an institutional obligation and a tradable allowance. This brief description concerns the mechanism, not a claim about its total effectiveness or the status of every proposed reform.[2]

Notice what this example teaches. The connection requires a defined scope, a unit, a measurement practice, a registry and enforcement rules. It is not generated by the general existence of money. A different physical concern requires its own carefully specified connection. Water availability, access to care and habitat condition are not interchangeable merely because they can appear in financial discussions.

EBU must be situated among such existing efforts. It should explain what information it adds, which decisions it could support and which burdens its use would impose. A useful comparison might eventually show that an existing physical account or a simpler rule performs the relevant task as well. That outcome must remain possible if the programme is to be research.

2.6 Four accounts for one repair

Return to the clinic's pump. We can describe the repair through four linked accounts. The monetary account records the agreed charge and payment. The physical account records the changed equipment condition and any declared resource flows. The service account records whether water became available when needed. The institutional account records authorisation, responsibility and access.

These accounts overlap without collapsing. If the repair is paid for but fails, the monetary event may have occurred while the expected service did not. If a volunteer fixes the pump, physical and service changes can occur without a sale. If the pump works but the clinic excludes a patient, engineering success coexists with an access failure. If the tank fills after rain rather than repair, the physical record must identify the different cause.

The fourth case is especially important for later EBU reasoning. An actor's account should not silently claim a change produced by an independently occurring event. That is why we will compare actions at a declared common baseline and record external changes separately. This distinction is an accounting requirement before it becomes a calculus exercise.

A payment is neither proof nor disproof of a physical contribution

A workshop receives one hundred monetary units, and the clinic's store rises by four physical units. To connect these observations, identify the performed action and the measurement interval. The payment alone does not prove the increase; the increase alone does not prove that this workshop caused it. A shared event record makes the proposed connection examinable.

2.7 Why financial solvency and physical preservation can diverge

Take a hypothetical store that begins with ten replacement parts. A manager sells eight and receives more money than the parts originally cost. The monetary result may be favourable. Whether the arrangement remains physically capable depends on replenishment, future needs, alternatives and timing. A positive financial balance does not identify those conditions by itself.

The reverse divergence is equally possible. The manager might keep all ten parts available for emergencies and make no sales. Availability is preserved over that interval, but the financial arrangement may fail to cover wages or rent. A physical account does not pay a bill unless institutions explicitly arrange that relationship. The existence of one kind of reserve cannot be substituted for the other without explaining the conversion.

These examples do not imply that enterprises cannot plan or that monetary analysis ignores the future. They show why planning requires information beyond a single current balance. The EBU research question concerns whether a specific physical valuation and action account can make selected consequences more explicit and useful. It does not begin with a proof that every existing form of planning fails.

Time makes the distinction sharper. Deferring maintenance may improve a current financial result while increasing later risk; performing maintenance may consume resources now to preserve future function. A present-state model must be careful here. It may represent equipment condition directly, but it cannot claim all future benefits merely because the story sounds plausible. Forecasts require an additional model and evidence.

2.8 What an EBU result could and could not do

The signed EBU quantity introduced later describes a change in a declared potential after a declared process burden. It can therefore support a statement such as: "Relative to this representation and baseline, this accepted action

reduced the evaluated burden by this amount.” The statement is precise enough to check, but narrower than “this action benefited everyone”.

An institution might choose to display that result, use it in planning, require an explanation for certain patterns or connect it to an incentive. Each choice changes the social arrangement and requires justification. The mathematical account alone does not establish legal tender, ownership, settlement rights, taxation or a conversion rate between EBU and money.

This separation also protects the account. If an essential action has an unfavourable physical result, the institution should be able to support the action without falsifying the record. An emergency delivery over a difficult route may remain necessary. Its consequences can be acknowledged while collective resources cover them. A truthful account is most useful when it survives uncomfortable decisions.

Do not infer an institution from an identity

An accounting equality can establish that defined quantities close. It cannot establish that an actor owns the resulting number, that a bank must accept it, or that people without such numbers should lose access to essential services. Those are additional rules with additional responsibilities.

2.9 A change of price without a change of water

The clinic and workshop renegotiate their agreement. Yesterday’s four-unit delivery cost twenty monetary units; today’s identical delivery costs fifteen. Assume, for this example only, identical physical baselines, routes, quantities and declared valuation. The monetary comparison changes. The EBU calculation does not change merely because the negotiated price did.

Now reverse the situation. The charge remains twenty, but the physical baseline differs because Vale already has more water. The same delivered quantity may then have a different represented field effect. The monetary record can remain unchanged while the physical evaluation changes. These paired examples show why a fixed conversion between price and EBU cannot be assumed from their common association with a transaction.

A future institution might nevertheless choose an explicit conversion or reward rule. That would be a policy linking two accounts, with consequences for incentives, access and risk. It would need to explain how revisions are made and what happens when the physical result is negative or uncertain. The link could be studied, but it should not be hidden inside the definition of the physical unit. Keeping the accounts distinct allows readers to inspect

whether the proposed institutional link is sensible rather than treating it as an arithmetic necessity.

2.10 Common misunderstandings

“Money measures nothing real” is too crude. Monetary records describe real institutional relationships and can organise real provision. Their physical interpretation depends on what they are connected to.

“A high price proves high physical burden” also goes too far. The price and the burden may be associated in a setting, but one cannot be deduced from the other without a model of that setting. A donated resource can carry a substantial physical burden; an expensive service can use little material.

“A physical account can replace every monetary function” is unsupported. Measuring an action does not automatically supply credit, insurance, dispute resolution or a workable payment system. A proposed institutional extension must specify how those functions would operate.

“If the physical account is exact, institutions become objective” confuses arithmetic with choice. The representation, reference and permitted uses still require reasons. Exact calculation makes the consequences of those declarations clearer; it does not remove the declarations.

2.11 Guided problems and answers

2.11.1 Same price, different event

Two couriers charge the same amount to deliver the same quantity. One uses a direct route and the other a detour. Can the payment record distinguish their physical effects?

The common price does not establish identical effects. A physical comparison requires the declared routes, carriers, quantities and boundaries. It also requires evidence about what happened. If the teaching model represents only the source and destination stock change, both routes may have the same represented endpoint value; the unrepresented route difference must remain a limitation or be represented through an additional justified account.

2.11.2 No income, valuable maintenance

A community repairs a path through unpaid work. Does zero monetary revenue imply zero EBU?

No inference follows in either direction. The absence of revenue describes the payment arrangement. An EBU calculation would require an appropriate

state, reference, physical change and process-burden declaration. Calling the repair socially valuable does not supply those data. The right response is to distinguish the ethical judgement, the physical observations and any mathematical valuation.

2.11.3 A claim and a tank

A bank account increases while the town's water stocks remain unchanged. Has conservation been violated?

Not merely for that reason. The monetary and water accounts track different objects. A violation would occur if the water account reported an unexplained physical creation or disappearance under a closed boundary. Whether the monetary change is valid depends on its own institutional rules. The exercise is a reminder to identify the conserved quantity before applying conservation language.

2.12 What to carry forward

Money is a powerful instrument for organising claims, exchange and time. Its records need explicit connections to particular physical and social objectives. EBU investigates one such physical account and must justify its usefulness on its own terms. The next chapter turns to a consequence that a total or a price can easily hide: resources may exist while people still lack effective access to them.

Scarcity, access and unequal exposure

3.1 A full store and an empty cup

At Ridge, the storekeeper can see water through the inspection window. At Vale, a resident opens a tap and receives none. Before deciding that the valley has “run out of water”, we need to identify what has failed. The stock may be insufficient, the pipe may be broken, the water may be unsuitable, or the resident may be unable to obtain permission or afford delivery. Each situation can produce the same immediate experience: an empty cup.

The distinction changes the response. Repairing a pipe does not solve the absence of water at its source. Increasing the source stock does not fix a locked gate. Reducing a delivery charge does not remove contamination. A diagnosis expressed only as scarcity is incomplete until the relevant resource, location, time, quality and access arrangement are specified.

FAO’s water-scarcity overview explicitly distinguishes physical scarcity, access scarcity and infrastructure scarcity. Its categories separate insufficient suitable water from water that exists but is inaccessible because of institutional arrangements or missing delivery infrastructure. The distinction gives a useful real-world reason to avoid treating a regional total as a complete account of provision.[3]

Our invented valley will let us practise that distinction without pretending that a two-place model represents every cause of deprivation. In particular, a stock variable does not by itself encode a person’s income, legal status, disability, available time or ability to travel. Those matters may determine access even when the physical stock is measured correctly.

3.2 Scarcity relative to a task

The statement “there are six units” is a quantity statement. The statement “six units are enough” relates that quantity to a purpose. Six may be enough for one declared task and insufficient for another. It may be enough now but insufficient over a longer interval. It may be available in aggregate but divided among places in a way that makes the intended service impossible.

This is why need and demand require careful language. A *need* identifies something required for a declared human or functional purpose. A *request* expresses a desired service to a provider. An *effective market demand* additionally involves willingness and ability to transact under particular terms. These concepts may coincide in an example, but they are not definitions of one another.

Suppose a clinic requests four units and a decorative fountain requests four units. Equal quantities do not establish equal urgency. Nor does a larger offer of payment prove a larger health consequence if the request is refused. A priority rule might account for purpose and urgency, but that rule must be declared. It cannot be extracted from the arithmetic fact that four equals four.

The physical teaching model can represent quantities and transfers faithfully while leaving such priorities open. This may feel unsatisfying at first. Yet silently embedding a priority rule would be less satisfactory, because it would prevent readers from seeing where a decision entered. A transparent research programme should state both the represented physical comparison and the additional institutional choices needed to use it.

3.3 Stocks, routes and access form different bottlenecks

Consider three hypothetical mornings. On the first, Ridge holds too little suitable water to meet all declared uses. On the second, Ridge holds enough, but the only pipe is closed for repair. On the third, the pipe works and the stock is sufficient, but a household cannot obtain delivery under the payment rule. In each case, adding the same extra quantity at Ridge may have a different effect.

On the first morning, added suitable supply could directly relieve the physical shortfall, subject to distribution. On the second, it may merely enlarge an already adequate inaccessible store. On the third, it may leave the household excluded. These are logical consequences of the declared situations; they are not measured estimates of how often each bottleneck occurs.

A useful account can therefore include a chain of questions. Is the resource present? Is it suitable? Is a route available? Can the route carry the required quantity in time? Is someone authorised and able to perform the action? Can the intended person use the resulting service? The first negative answer may explain the immediate failure, but repairing it can reveal another constraint farther along the chain.

This chain helps explain why a scalar score is never a substitute for inspection. If the score improves because more resource is stored upstream while the de-

livery route remains broken, the relevant function may not improve. A model intended to evaluate that function would need to represent the relationship adequately. A limited stock model should describe its conclusion as limited stock redistribution.

3.4 Unequal exposure means more than unequal totals

Two households may face the same service interruption and experience different consequences. One may have stored supplies, flexible work and transport. Another may have none of these and include a person whose care depends on reliable access. Equality of the external interruption does not imply equality of exposure, coping capacity or loss.

The World Health Organization's 2025 report on social determinants of health equity identifies living conditions and access to power, money and resources as important influences on avoidable health gaps. Its overview describes variation associated with social position and circumstances; it does not reduce every health outcome to income alone.[4]

For the present book, the lesson is that socioeconomic conditions should not disappear behind a regional average. A hypothetical average delivery of ten units can arise when everyone receives ten or when one group receives twenty and another receives zero. Those are the same average and different arrangements. Whether they are acceptable depends on purpose, needs and rights as well as arithmetic.

The phrase "socioeconomic inequality" also covers several dimensions. Income concerns a flow over time; wealth concerns accumulated claims or assets; access concerns practical ability to obtain a service; power concerns influence over rules and decisions. Combining them into one vague disadvantage measure may obscure how a particular failure can be repaired.

EBU's physical account cannot settle all these dimensions. Indeed, an institution that rewarded only easily measured actions could create new exclusions. Its design would need to ask who can participate, whose work is observable, who bears measurement costs and who can challenge an incorrect record. These questions arise from the proposal's intended use; they are not answered by a favourable conservation residual.

3.5 A total can conceal a distribution

Return to our fixed twenty-unit example. Ridge and Vale can hold (18, 2), (14, 6) or (10, 10). The total remains twenty. We can write

$$18 + 2 = 14 + 6 = 10 + 10 = 20. \quad (3.1)$$

The equality says that the represented quantity is conserved. It does not say that the arrangements provide the same service. A particular reference may prefer one distribution, but the reference must be stated rather than inferred from conservation.

Suppose, for illustration, that ten units at each place are the declared comparison configuration. Then $(10, 10)$ matches that declaration and $(18, 2)$ does not. Now suppose Ridge houses a much larger function than Vale and the declared compatible reference is instead $(14, 6)$. The same physical arrangements receive a different interpretation. We have changed the evaluation, not the total mass.

Neither reference is justified by being easy to write. An applied reference would require reasons connected to the represented function. It might need evidence about service requirements, physical capacity or agreed provision. The introductory arithmetic does not supply such evidence. Its purpose is to make the point of choice visible before the later potential assigns a smooth numerical burden.

Distribution is visible only when places remain visible

An aggregate can be conserved while one place is deprived. Retaining separate state coordinates allows us to examine distribution. Choosing how to evaluate those coordinates requires a declared reference and does not follow from the conservation law.

3.6 The cost of reaching a service

Access is not always a yes-or-no gate at a shop. Consider a hypothetical patient who can obtain medicine only by making a long journey. The medicine is present, the price is affordable, and the transaction is legal. Yet the journey may require time away from paid work, assistance from another person and reliable transport. A simple inventory reports availability while the patient faces a substantial practical obstacle.

This example shows why the boundary of an account matters. If the account stops at the warehouse, it can correctly record medicine in stock while missing the final access problem. If it extends to the clinic, it can record arrival there while still missing the patient's journey. Every extension must be justified by the question being studied and supported by relevant data.

There is no honest shortcut in which the word “local” makes all these consequences disappear. Local observation can be sufficient for a specified mathematical field difference when all affected terms are known. It is not a guaran-

tee that the representation includes every socially important effect. The local theorem and the adequacy of the model are different questions.

Nor should the patient be charged an invented physical burden simply because several people helped. A route has real transformations and real resource uses, each requiring its own declared account. Counting the same electricity once at generation, again at delivery and again as an unexplained surcharge would exaggerate burden. Linked actions should reveal a chain, not multiply one effect by the number of records.

3.7 Essential support and the danger of circular qualification

Imagine an institution that says a person may receive water only after generating enough positive EBU. A person who is ill and unable to act could then lose the support needed to recover. The qualification rule would create a circle: access requires contribution, while contribution requires access. No physical accounting identity resolves that institutional choice.

The ethical position adopted in this introduction is that essential support deserves explicit protection. The precise mechanism remains a matter for institutional design and public justification. The important mathematical boundary is that an action's signed field result is not a measure of a person's worth or a complete entitlement rule.

Similarly, a settlement may need to make a physically costly action available because the alternative is unacceptable. Delivering medicine over a difficult route may increase some represented burdens while meeting an urgent need. The account should preserve that information. Hiding the burden to make the action look favourable would obstruct learning; denying the need because the burden is unfavourable would confuse evaluation with permission.

The better question is what the account helps the community learn. Perhaps a local store would reduce repeated emergency journeys. Perhaps a more reliable route would help. Perhaps the chosen potential omits a crucial function. These possibilities require investigation. They do not follow automatically from the sign of a single action.

3.8 What a fair comparison would have to show

If EBU is eventually tested as guidance for a resource arrangement, an apparent improvement in a total cannot be the whole result. A comparison needs to report who received service, which constraints were respected and what happened under difficult conditions. It also needs to examine cases in which EBU guidance performs poorly.

For example, a rule that never transfers anything can preserve every source stock over a short action-only interval. That does not establish satisfactory service. A rule that serves every immediate request may deplete a source needed elsewhere. A rule that improves an aggregate potential may leave a particular essential request unmet. Each behaviour reveals a different dimension.

The book does not select a final controller or freeze study metrics here. Instead, it develops the account needed to state such questions precisely. A research design should declare its service, physical and distributional criteria before observing outcomes. Otherwise, the evaluator can choose whichever interpretation makes its favoured rule look successful.

3.9 The deadline changes the access question

Two deliveries bring the same quantity to Vale. One arrives before the clinic opens; the other arrives after the day's procedures have been cancelled. An end-of-day stock report may show the same quantity in both cases. The service histories are different. Access therefore includes timing whenever the function has a deadline.

To represent that difference, an account needs an appropriate time resolution and a declared service record. It would be misleading to repair the problem simply by assigning a more attractive final reference after the late delivery. The reference evaluates a state; a deadline concerns whether a service occurred during a specified interval. Both may matter, but neither substitutes for the other.

Timing can also interact with unequal circumstances. In an invented example, one resident can return tomorrow, while another has only one available journey. A uniform postponement can therefore impose different practical burdens. Establishing those burdens in a real setting requires evidence about the affected people, rather than a presumption based on their group label.

This is a useful design lesson for future studies. If sustained demand is the intended subject, the study must preserve the time structure of demand and delivery. A total delivered eventually may be an inadequate outcome. The criterion should be declared before results are inspected, so that delayed service is neither excused nor condemned according to which controller produced it.

3.10 Common misunderstandings

"Scarcity is always just a shortage of total material" overlooks location, timing, suitability and access. Conversely, "all scarcity is artificial" overlooks genuine

physical limits. A careful diagnosis can recognise both without treating either as the explanation of every case.

“Equal distribution is automatically fair” mistakes one possible numerical pattern for a complete ethical argument. Equal quantities may support unequal functions or needs. A reference must be justified in its intended setting, and a scalar allocation cannot settle every dimension of fairness.

“A positive physical score proves that inequality has improved” makes a claim the score may not represent. If access or distributional outcomes matter, the account must measure them separately or explicitly include a justified representation. A label cannot add missing information.

“The model is useless unless it includes everything” sets an impossible standard. A small model can expose an accounting error or prove a conditional identity. Its usefulness depends on maintaining the connection between the narrow result and the question it actually answers.

3.11 Guided problems and answers

3.11.1 Three identical complaints

Three residents say, “There is no water.” In one case the source is empty, in another the pipe is broken, and in the third delivery is refused for nonpayment. Would the same intervention solve all three?

Not necessarily. The first requires suitable supply or a change in uses; the second requires a route or repair; the third requires an access decision under the relevant institution. Each intervention can also encounter additional constraints. A useful record identifies the failure rather than replacing all three with a common headline.

3.11.2 The average service report

Two places each require attention. A report states that average delivery was ten units. Can we infer that each received ten?

No. Many pairs have that average, including $(20, 0)$ and $(10, 10)$. The distribution must be reported if it matters to the claim. We must also establish whether equal delivery is the relevant goal. Averages summarise information; they do not preserve every fact from which they were calculated.

3.11.3 The unfavourable essential action

An accepted emergency delivery has a negative signed EBU result in the declared model. Must the delivery be refused?

The sign alone supplies no such rule. Permission, essential support and institutional financing need their own declarations. The result should remain visible so that future arrangements can be improved and the model's adequacy questioned. A truthful account can coexist with an explicit decision to perform a costly necessary action.

3.12 What to carry forward

Scarcity can concern physical quantity, infrastructure or access, and unequal circumstances can turn the same interruption into very different experiences. Conservation is essential, but totals alone cannot establish provision. These distinctions prepare us to consider planetary pressures without losing sight of the people and functions through which those pressures are experienced.

Planetary pressures and conditional futures

4.1 The valley has a boundary, but the world does not stop there

Ridge and Vale are useful because we can draw them on one sheet. A water transfer appears as a decrease at one place and an increase at another. The real clinic, however, depends on conditions outside that drawing. Its electricity, equipment, medicines and food have histories elsewhere. Its water source depends on a larger landscape. Its waste crosses boundaries that the first diagram may not include.

Expanding the boundary does not mean that every local account must contain a complete model of Earth. It means that a claim must respect its boundary. A local transfer can be accurately described while remaining silent about upstream extraction or downstream pollution. The silence becomes misleading only when the local result is presented as a complete planetary judgement.

This chapter provides motivation for keeping those connections visible. It uses a small selection of dated scientific assessments, followed by original explanatory examples. The assessments describe pressures and possible futures. They do not validate EBU, identify its parameters or show that its proposed account can resolve the problems described.

4.2 Read an environmental number with its date and unit

A statement about planetary change should carry at least four pieces of information: what was measured or estimated, the period concerned, the comparison baseline and the relevant uncertainty or conditional assumptions. Without these, a number can travel into a claim it was never intended to support.

The IPCC's 2023 Synthesis Report assessed global surface temperature in 2011–2020 as 1.09°C higher than in 1850–1900, with a very likely range of 0.95 to 1.20°C. This is a global decadal comparison from that assessment. It is not the temperature of every location, a single-year value or a claim about the latest calendar year.[5]

The distinction between a global average and a local experience is worth pausing over. A hypothetical average increase of one degree across two places could accompany different changes in each. The average is meaningful for its

purpose, but cannot answer every question about local exposure. In the same way, an average resource stock does not reveal every empty tank.

The distinction between a decade and a year also matters. A long-period statistic smooths some short-term variation and answers a different question from a single observation. Choosing the appropriate timescale depends on the claim. A clinic manager planning tomorrow's delivery and a planner assessing infrastructure over decades need different kinds of information, even when both concern water.

4.3 A projection is a conditional statement

The same IPCC assessment gives different late-century warming estimates under different greenhouse-gas scenarios. For 2081–2100 relative to 1850–1900, its best estimates include 1.4 °C for SSP1-1.9, 2.7 °C for SSP2-4.5 and 4.4 °C for SSP5-8.5. The corresponding very likely ranges are 1.0–1.8, 2.1–3.5 and 3.3–5.7 °C. These are conditional scenario results, not probabilities that societies will choose those pathways.[5]

The word “if” is essential to this kind of reasoning. A conditional model says what follows under a specified collection of assumptions. It does not establish that those assumptions will hold. If a town asks how long a store would last under a declared withdrawal schedule, the answer changes when the schedule changes. That dependence does not make the calculation useless; it makes the assumptions part of the answer.

Nor is an uncertainty interval a permission to substitute any preferred conclusion. A range records a stated kind of uncertainty within an assessment. The uncertainty about a physical response and the uncertainty about future human decisions are not the same object. Combining them casually can make a conditional projection sound like a definite forecast or make a serious risk sound like an arbitrary guess.

For this book, the methodological lesson is more important than memorising a headline. Any future EBU study must likewise identify its forcing, initial condition, action rules and interpretation. A favourable result in one declared world would not become a forecast of every institution that uses the same acronym.

Keep the condition attached

“Under this scenario, the model projects this outcome” is a different statement from “this outcome will occur”. Removing the scenario converts a conditional result into an unsupported forecast. The same discipline applies to climate assessments and to the much smaller models developed in this project.

4.4 Living systems cannot be reduced to one temperature number

Climate is one dimension of planetary condition. The 2019 IPBES Global Assessment estimated that around one million animal and plant species were threatened with extinction. This was an assessment estimate of threatened species, not a census of species already extinct or a statement that every threatened species would disappear on a fixed date.[6]

We do not need to turn this estimate into an EBU coordinate to recognise the modelling challenge it poses. A single homogeneous resource cannot represent the identities and relationships of different species. Exchanging one unit of an arbitrary scalar for another cannot by itself describe the loss of a particular ecological function. A future ecological application would require a much richer state and justification for any aggregation.

Consider an invented garden with two functions: food production and water infiltration. A change could improve one while worsening the other. Reporting only the first would give an incomplete picture; adding them requires a declared scale or decision rule. The example does not claim to model a real ecosystem. It shows why an apparently comprehensive score can hide choices about what counts and how.

The lesson for our introductory Gaussian model is therefore a boundary, not a defect to conceal. We use a conserved scalar to establish exact accounting relations. We do not announce that biodiversity is that scalar. The conceptual desire to preserve life-supporting conditions should make us more careful about representation, because those conditions include differences that simple addition can erase.

4.5 Resources, services and material throughput

The UNEP and International Resource Panel’s *Global Resources Outlook 2024* describes rising resource extraction and a conditional continuation in which extraction could increase by about 60% between 2020 and 2060 without urgent concerted action. The headline is a global material-extraction comparison

with a specified baseline and horizon. It is not a forecast of every country's consumption, nor an EBU study result.[7]

An important conceptual question follows: what service is obtained from a material flow? Two hypothetical arrangements might provide the same number of refrigerated medicine-days while using different equipment lifetimes or maintenance practices. Conversely, equal material use could support different amounts or qualities of service. Throughput and function are connected but cannot be treated as synonyms.

A repair example makes the distinction concrete. Replacing an entire pump and replacing one failed seal may both restore flow under specified conditions. The material accounts differ. Whether the seal repair is actually adequate depends on inspection and performance, not merely on its smaller mass. A resource-efficient story is not evidence until the relevant function and duration are established.

This is also why "use less" is incomplete as a universal instruction. A place lacking safe infrastructure may require additional material investment to obtain essential service. Another arrangement may be able to reduce waste while preserving service. A responsible account needs to distinguish deprivation from avoidable throughput. The physical numbers inform that distinction; the purpose and institutional judgement must remain explicit.

4.6 Three clocks in one decision

The clinic operates on several timescales. Water must be available during today's procedures. A pump must remain serviceable through a maintenance interval. A watershed or energy infrastructure may be affected over much longer periods. A decision that looks favourable on one clock can impose a burden on another.

Suppose an invented temporary delivery solves today's shortage but requires repeated emergency journeys. The first journey may be necessary. The pattern may nevertheless suggest that a different arrangement deserves investigation. Calling the first action necessary does not make the long-term arrangement efficient; calling the arrangement inefficient does not make today's patient dispensable.

A present-state potential provides one declared view of the current arrangement. It does not automatically predict every delayed effect. To represent equipment wear, accumulated pollution or future availability, the state or an additional model must contain those quantities with suitable evidence. Merely enlarging the time interval in a sentence does not add them to an equation.

This distinction will matter when the book later proves a telescoping identity. A sequence of exact potential differences can be added exactly under stated conditions. The identity concerns the represented account. It is not a theorem that long-run services continue, that disturbances remain manageable or that every delayed ecological effect has been included.

4.7 The importance of boundaries and displacement

Imagine a town that reports a decrease in its own workshop waste after purchasing more finished components from outside. Local waste may indeed have fallen. Whether total waste associated with the town's use fell requires an account that includes the external production. Both statements can be investigated, but they have different boundaries.

This is an original teaching example of displacement, not a quantified claim about a particular trade pattern. It warns against evaluating a change through a boundary selected after its attractive result is seen. If the research question concerns a whole route, the route must be declared before the comparison. If it concerns one facility, the conclusion should remain at facility scope.

The same discipline applies to physical losses. A transfer cannot be called conservative merely because a source and destination are both listed. The sum must close within the declared coordinates or through an explicit boundary account. In the lossless teaching model, this is deliberately simple. In an extended model, a loss ledger would record where the represented quantity goes; it would not automatically become an additional term in the potential.

Nor does including a loss ledger justify counting the same effect twice. Conservation concerns the physical balance. Valuation concerns how the declared state is evaluated. Process burden concerns a residual declared effect not already represented in that evaluation. Keeping these roles separate will allow later chapters to make exact statements without pretending that every physical consequence is measured by one number.

4.8 Why the future is neither guaranteed nor unknowable

Environmental discussion can fall into two opposite errors. One treats a severe scenario as an inevitable destiny. The other treats uncertainty as a reason to avoid any conclusion. Conditional reasoning offers a more useful position: identify the plausible consequences of declared assumptions, examine uncertainty and ask which decisions remain sensible across relevant cases.

Our small teaching world offers an analogy. If Ridge can send water only through one pipe, a closed pipe prevents that transfer. We do not need to

know the next year's exact demand to recognise the constraint. Yet knowing the constraint does not determine the best investment, which depends on alternatives, costs, priorities and evidence. Some facts are decisive at one level while leaving other questions open.

The EBU programme should be assessed with the same discipline. Exact accounting may be established before performance is known. Performance in a registered world may be observed before institutional usefulness is established. Neither uncertainty about the last question nor confidence about the first should be used to collapse the sequence.

The practical motivation is to improve the connection between decisions and the conditions they affect. That aim is compatible with uncertainty, provided uncertainty is recorded rather than hidden. It also demands that a proposal be open to revision if the representation or guidance proves unhelpful.

4.9 A stock and a rate answer different environmental questions

Imagine a hypothetical waste container with twelve units already inside it. A new process adds two units per day. The twelve is a stock; the two per day is a rate. Reducing the rate from two to one does not, by itself, reduce the existing stock. It slows the increase if nothing else changes. Removing three units per day would produce another account, which must also identify where the removed material goes.

The arithmetic is elementary, but the language can easily obscure it. "Less added" is not always "less present". "Removed from this place" is not automatically "destroyed". A complete account needs the initial stock, the flows and the boundary. This hypothetical example is not a model of a particular pollutant; different materials and processes require their own physical descriptions.

The same care will be needed in the EBU equations. A rate multiplied by a duration gives a quantity under the declared convention. Substituting a rate where a finite quantity is required changes the units and the result. Setting the duration to one can conceal the mistake numerically, which is why symbols and units should remain visible even in simple examples.

These distinctions also improve interpretation. A change that reduces an adverse inflow may be useful while leaving a large accumulated burden. A temporary improvement in a rate does not establish a completed recovery of the state. The account should preserve both facts instead of compressing them into one optimistic verb.

4.10 Reading a headline as a careful student

Suppose a headline states, “Resource use will rise by sixty percent.” A careful reader asks: which resources, relative to which year, by when, under what assumptions, and according to which source? The question is not an attempt to dismiss the finding. It restores the content needed to understand it.

Suppose another headline says, “The model proves balance.” The same habit applies. Which model? Which conserved quantity? Which definition of balance? Does the statement concern an algebraic identity, an equilibrium, a controller’s performance or an empirical observation? The word “proves” should direct us to assumptions and an argument, not end the inquiry.

These habits help protect readers from both exaggerated criticism and exaggerated promises. A serious account can state severe risks without announcing that every detail is known. A serious research proposal can state exact mathematics without claiming to have solved those risks.

4.11 Guided problems and answers

4.11.1 A temperature without a period

A speaker quotes 1.09°C and calls it “today’s warming”. What is missing?

The figure used here concerns the IPCC’s assessed 2011–2020 global average relative to 1850–1900. The period and baseline are essential. It should not be relabelled as a present-day annual observation. The answer illustrates why a correct number can still be used incorrectly.

4.11.2 A conditional future

A projection follows a declared high-emissions scenario. Does that establish that the scenario will occur?

No. The projection describes a response conditional on the scenario and the assessment’s methods. It does not assign a probability to the human pathway merely by reporting a temperature range. Decisions can change the assumptions; physical uncertainty remains a separate issue.

4.11.3 The clean local workshop

A workshop reduces its local waste by buying a previously homemade component elsewhere. Has the whole route improved?

The local record alone cannot answer. The external component’s relevant production and delivery must be considered under a declared boundary. The

local improvement may be real while the wider claim remains unestablished. Expanding the claim requires expanding the evidence.

From economic feedback to a research question

5.1 A gauge is useful only in an arrangement

Vale installs a clear gauge on the clinic's tank. The gauge makes the level visible. For several days, everyone admires the improvement. Then the level falls and nobody opens the valve, because nobody has been assigned responsibility. The gauge has worked perfectly. The arrangement has not provided water.

Now imagine the opposite case. A caretaker opens the valve whenever a neighbour asks, but cannot see the tank level and has no reliable record of what arrived. The arrangement contains action and concern, yet lacks essential feedback. A third case is also possible: the level is visible and responsibility is assigned, but the pump has no power. Information, decision and physical capability are distinct requirements.

These simple cases help clarify the claim that an economy needs a connection to physical functioning. The claim should not be expressed as though economies contain no feedback. Prices, inventories, contracts, public decisions and engineering controls can all participate in feedback arrangements. The research question is whether selected physical consequences are observed, evaluated and acted upon adequately for a specified purpose.

EBU initially addresses one part of this arrangement: an explicit signed account of a represented physical action. It does not become a controller merely because its result can inform a controller. The distinction will prevent us from attributing behaviour to an equation that only evaluates a change.

5.2 Five questions behind a feedback loop

The first question is what is being observed. A gauge may show stock, pressure or flow. Those are different quantities. A meter showing yesterday's delivery cannot directly tell the caretaker how much is in the tank now unless withdrawals and other changes are also known. Observations need a declared meaning and time.

The second question is what reference is used. A caretaker cannot decide whether a reading calls for action without some criterion. The criterion might concern an intended stock, a service schedule or a safety condition. In our later Gaussian account, a declared reference and positive scale determine the represented evaluation. They do not establish the complete service policy.

The third question is what action is available. A valve can transfer water only if an admissible route and sufficient source stock exist. A displayed shortage does not create a pipe. The set of physically possible actions therefore constrains every decision rule, including any rule that uses EBU.

The fourth question is how an action is selected. A person, protocol or controller may choose among available actions. Selection might prioritise service, reduce a declared potential or follow another justified rule. Different selectors can use the same physical account and produce different behaviour. Their differences must be tested rather than erased by giving them a common label.

The fifth question is how the result is verified. The requested action may differ from the accepted action, and the accepted action may differ from what was actually performed. A reliable record distinguishes these stages. If a request for four units produces a measured transfer of three, the final physical account should not retain four simply because that was the plan.

The account and the action rule

A potential evaluates a represented state. An action account compares represented states. A selector chooses an action from an allowed set. Observations establish what happened. These components can work together, but specifying one does not silently specify the others.

5.3 Turning an aspiration into a question

“Make the economy support life” expresses a purpose, but is too broad to be a single scientific hypothesis. To investigate it, we must identify a smaller claim. For example, one could ask whether a declared action rule maintains specified services in a particular resource world under declared disturbances. Such a question needs explicit variables, criteria, alternatives and conditions of failure.

The work in this book comes before that complete study. It asks whether the proposed physical account can be defined consistently and what identities follow from the definition. Can a finite action be evaluated without replacing it with a tangent? Can a local computation recover the affected global dif-

ference? Can a sequence of records close when external changes are recorded separately? These are bounded mathematical questions.

The fact that an identity can be proved does not establish that a decision rule using it will achieve the broader purpose. A speedometer can measure speed accurately while a driver takes the wrong road. Likewise, an exact EBU quote can be attached to an action selected by a poor policy. Accuracy of the account and adequacy of the action rule must each be examined.

Conversely, a successful-looking trajectory does not repair an inconsistent account. If a method double-counts resource creation or attributes an external change to an actor, a desirable endpoint does not make the accounting correct. The programme needs both a trustworthy representation and a fair test of behaviour.

5.4 The four kinds of statement

A *definition* declares what a symbol or model object means. Defining a potential does not demonstrate that it measures every important burden. It makes the subsequent mathematical question precise. The definition can be useful, inadequate or inappropriate for a proposed application.

A *theorem* establishes a conclusion from specified assumptions. If a fixed potential is evaluated before and after a sequence of actions, certain sums may telescope exactly. The theorem remains conditional on the stated sequence, representation and accounting rules. Removing an assumption requires a new argument.

An *implementation check* establishes something about a program at the checked boundary. A test that verifies an arithmetic identity on a fixture is evidence about the implementation of that identity for those conditions. It is not an experiment establishing long-run homeostasis, and the number of passing checks does not change the category.

An *empirical finding* records what an executed, adequately documented investigation observed. Its interpretation depends on the design, measurements, comparisons and declared limitations. A result in one world is not automatically a result in another. The distinction matters even when the same code and the same mathematical notation appear in both.

A candidate design belongs to none of these categories by wish alone. It may contain definitions and proposed procedures, but it remains prospective until the required adoption, implementation and execution occur. This vocabulary will appear throughout the book because the status of a claim is part of its meaning.

5.5 The temptation to build a favourable world

Suppose we wish to compare two hypothetical rules. One transfers toward a declared stock reference. Another serves requests in their arrival order. It would be easy to choose examples in which every request happens to move stock toward the reference, then announce agreement between EBU guidance and service. It would be equally easy to choose only examples in which the request rule performs badly.

A useful investigation would include cases that separate the rules. A necessary service may move stock away from a reference. A reference-restoring action may provide no requested service. A source-preserving rule may refuse all transfers. These cases expose what the rules actually prioritise.

The reference itself must also be declared before inspecting the outcomes that are meant to test it. Otherwise, an evaluator can move the target after seeing where the favoured rule went. An action that previously increased burden can be made to appear helpful by redefining the reference around its endpoint. That is a change in evaluation, not a discovery about the original action.

Examples in a textbook serve a different purpose from registered trials: they illustrate a distinction openly. The Ridge–Vale quantities are chosen to make arithmetic legible. We may change them to teach a different point, provided the change is stated. We may not present the resulting teaching convenience as evidence that an untested controller succeeds generally.

5.6 What the first physical model includes

The core mathematical world uses a finite collection of places and one homogeneous resource. Stocks are nonnegative, and internal lossless transfers preserve a fixed total. The declared reference is compatible with that total. Positive fixed scales determine how deviations from the reference are measured. These assumptions are strong enough to support exact calculations and narrow enough to remain visible.

The world does not acquire regeneration because Ridge is called a source in a story. A source of an individual transfer is simply the place from which that transfer withdraws stock. If natural production, consumption or disturbance is introduced, its physical effect needs its own declaration and record. It must not be treated as an actor's contribution merely because it improves the state.

Nor does the Gaussian potential create movement. A state away from the reference can remain there if no action occurs. A graph can make some desired redistributions unavailable. A permission rule can limit an otherwise attractive transfer. These possibilities are not edge cases added to spoil an

elegant theory; they are consequences of distinguishing state evaluation from dynamics.

The model's simplicity also helps us see a basic problem in some proposed demonstrations. If a world starts exactly at its reference and contains no disturbance or drive, a rule that acts only on a restoring field may have no reason to move. That does not test recovery from disturbance or sustained service. The scientific question must fit the declared world.

5.7 A worked comparison with no winner declared

Consider two proposals from the same stock state, (18, 2). Proposal A transfers four units, producing (14, 6). Proposal B transfers eight units, producing (10, 10). Both preserve the total twenty. Suppose the reference is (10, 10) and the scales will later be declared equally. Proposal B reaches that reference; proposal A moves toward it.

This information still does not establish a universal winner. A request may have been for four units. The route may allow only four. A separate requirement may prevent sending eight. The physical action set and service purpose must be included before choosing. Even when both proposals are permitted, a controller's objective must be explicit.

Now add Proposal C: do nothing. It preserves the current total and avoids the immediate work of transfer. If success is defined only as "no stock becomes negative", C may appear perfectly successful. If a declared service must be supplied, C may fail. This is why success criteria must include the intended task rather than only the easiest invariant.

The worked comparison teaches an evaluation habit. Record physical validity, represented field change and service outcome separately. Then apply the declared interpretation rule. Do not hide all three inside a single word such as "good", and do not change the interpretation after discovering which proposal your preferred method selected.

5.8 What would count as progress

Progress can occur at several levels. Clarifying a unit prevents an invalid calculation. Proving an identity establishes a conditional relation. Correcting code makes implementation more faithful. Designing a fair comparison makes a future result interpretable. A negative experimental result can reveal that an elaborate method adds no useful benefit over a simpler one.

None of these achievements needs to masquerade as another. A mathematical book can be valuable before a long-run experiment is performed. Its value

lies in explaining definitions and consequences accurately, including what remains unproved. An experimental programme can then target the remaining questions rather than pretending that they were settled by notation.

The central long-run question is demanding: can physically permitted actions guided by a declared account support the intended functions under specified changing conditions? Answering it would require a plant with those changes, a controller, comparison rules, observations and failure criteria. This edition provides the conceptual and mathematical foundation for asking the question. It does not report that the answer is yes.

5.9 A question needs a possible disappointing answer

Suppose a proposed comparison is written so that EBU is declared successful whenever it improves the potential, but a comparator is declared successful only when it improves the potential and serves every request. The asymmetric criteria have already favoured one method before anything runs. The remedy is to define the relevant claims and apply their criteria consistently to every eligible arm.

A comparison can also become uninformative when every possible result is given a favourable interpretation. If EBU acts, the evaluator praises responsiveness; if it does nothing, the evaluator praises preservation; if service fails, the evaluator calls the world unrealistic after the fact. A useful design states in advance which observations would count against its central claim and why.

That requirement does not mean that every failure disproves every part of the project. A controller may fail while the accounting identity remains correct. An unsuitable world may reveal a design error rather than a universal impossibility. The response should locate the failed proposition precisely. This lets the programme learn without turning every setback into total rejection or every success into total vindication.

The explanatory book has a related obligation. Its worked examples should include limits, negative signs and competing purposes, so that readers can see what the equations actually say. A sequence of exclusively favourable stories would teach advocacy more effectively than it taught the model.

5.10 The role of institutions in a physical programme

Even a successful resource controller would not settle ownership, consent, privacy or access. Measurements might concern people or facilities whose participation requires agreed rules. An incorrect receipt might need correction. A reference could disadvantage a community. A decision could impose burdens on people outside the measured boundary.

These are not peripheral issues to be solved by adding a positive coefficient. They concern who may decide, who can contest a decision and which rights constrain the system. A mathematically explicit account can help expose choices, but cannot legitimate them merely by calculating consistently.

One useful design principle follows: keep the route from observation to consequence inspectable. A person should be able to distinguish a meter reading, a model assumption, an action-selection rule and an institutional sanction. When all four are compressed into one unexplained score, neither scientific review nor public accountability can identify where an error entered.

5.11 Guided problems and answers

5.11.1 The perfect gauge

The tank gauge is accurate, but the clinic repeatedly runs out of water. Has the gauge failed?

The observations alone do not show a measurement failure. Responsibility, available actions, decision rules or physical supply may have failed. A feedback arrangement requires more than a sensor. The analogous lesson for EBU is that an exact account does not prove effective control.

5.11.2 The passing arithmetic check

A program reproduces an exact potential difference on one hundred synthetic examples. Has it established long-run homeostasis?

No. It has checked a computational property on the declared examples, assuming the test itself is adequate. Long-run behaviour requires a specified evolution and an authorised investigation. Increasing the fixture count does not turn a static arithmetic question into a dynamic service result.

5.11.3 The target moved after the run

A proposed rule performs poorly relative to its original reference. The evaluator chooses a new reference equal to its final state and reports zero final burden. What changed?

The evaluation changed. The original physical history did not. The new account may be mathematically calculable, but it cannot be substituted for the original registered comparison. The change must be disclosed and any new study treated according to its own prospective rules.

Homeostasis and functional balance

6.1 The tank level stays steady while water moves

Stand beside the clinic's tank for a few minutes. Water enters through one pipe and leaves through another. The level can remain nearly unchanged even though the water present is continually being replaced. A photograph would show stability. A flow meter would show activity. Both observations can be correct.

Now close the inlet while leaving the outlet open. The level begins to fall. A caretaker notices and restores the inlet. If the available inflow can match the withdrawal, the level may recover. If the source is empty or the pump cannot provide enough, noticing the problem will not be sufficient. A useful account of regulation must include both information and physical capability.

This simple story introduces homeostasis: the maintenance of selected conditions compatible with continuing function despite ongoing processes and disturbances. The word does not require complete immobility, perfect constancy or a universal target shared by every system. It directs attention to what is maintained, how deviations are detected and which processes respond.

In physiology, McFarland and colleagues developed and validated a teaching framework for homeostasis through negative feedback at the organism level. Their framework concerns regulated variables and the mechanisms that influence them. It is a conceptual educational source, not a quantitative model of society or proof that every ecosystem follows one feedback design.[8]

The analogy with a resource network is useful only if we preserve that difference. A body and a town are not the same kind of system. A town contains people with rights, conflicting purposes and institutions. We borrow a question about continuing function; we do not infer a social controller from biological vocabulary.

6.2 What is being maintained?

Before calling something homeostatic, identify the variable and the function. In the tank example, the stock supports a declared service. In a different example, the relevant condition might be temperature, pressure or an equipment

state. These quantities have different units and mechanisms. They cannot be merged by describing all of them as “balance”.

The variable may also be only a proxy. A measured tank level can indicate available water quantity, but not necessarily water quality. A measured room temperature may not capture every condition relevant to storage. If a proxy is used, the relationship to function needs evidence. Good regulation of the proxy can coexist with poor regulation of what was actually intended.

For a model, we therefore distinguish the represented condition from the wider purpose. The state variable is what the mathematics contains. The functional interpretation explains why that variable was chosen. The evidence for the interpretation belongs to the application. Keeping those layers visible allows a model to be corrected when its proxy proves inadequate.

The same care applies to a reference. A declared reference stock may be convenient for a teaching account, but an applied reference needs justification. It might depend on the intended service, physical design or an agreed planning criterion. The reference is not automatically a natural constant, and different places may appropriately have different values.

6.3 Negative feedback in ordinary language

In a negative-feedback arrangement, a detected departure elicits an influence that tends to oppose that departure under the stated conditions. If the tank falls, the arrangement increases inflow or reduces outflow. If it rises beyond the intended level, the arrangement decreases inflow or permits additional outflow. The word “negative” concerns opposition to a departure; it does not mean morally bad.

This description contains several assumptions. The departure must be observed with sufficient timeliness. The relevant action must be available. The sign of the response must be appropriate. The response must not be so delayed or strong that it creates a different problem. A diagram showing an arrow back to a comparator does not prove that all these conditions hold.

Consider a hypothetical caretaker who responds to yesterday’s low reading by opening the valve fully today. If the tank has already filled through another route, the delayed response may overshoot. A different caretaker may react immediately but lack enough source stock. The first arrangement has an information problem; the second has an action-capacity problem. Both can fail despite an intention to restore the same condition.

This book does not derive a controller that resolves such dynamic questions. It develops a potential and exact finite action account. The account may help describe whether a particular accepted action moves the represented

state toward or away from its reference. A theorem about that description is different from a theorem that repeated decisions produce stable regulation.

A restoring description is not a restoring mechanism

Drawing a landscape with its lowest point at a reference does not make the physical system travel downhill. Movement requires an action law, available routes, permission and resources. A proof about the landscape alone establishes no recovery rate or long-run homeostatic performance.

6.4 Functional balance is not equality everywhere

Ridge and Vale need not hold identical stocks in every useful arrangement. A storage facility, a clinic and a transit point may perform different functions. Even in a single-resource model, the declared reference may assign different stocks to different nodes. Equal numbers can be a convenient teaching choice without being a universal law.

Suppose a hypothetical two-place system contains twenty units and uses a reference (14,6). That reference is compatible with the total. A state (10,10) has equal stocks but does not match the declared reference. The physical equality of the two numbers therefore cannot settle functional balance. We must ask why the reference was chosen and whether it suits the intended application.

The scales matter as well. A difference of two units may have different significance at two places. The later Gaussian construction divides each deviation by a declared positive scale before squaring it. This creates a dimensionless comparison within the model. It does not prove that the scale is objectively correct or that unlike services have become interchangeable.

The phrase “functional balance” will therefore refer to an arrangement’s relation to its declared functional description, rather than to an insistence that every coordinate be equal. Where the description is incomplete, the conclusion must remain incomplete. The mathematical clarity lies in making the declaration inspectable.

6.5 A stable stock is not necessarily a healthy arrangement

Imagine a tank sealed shut with ten units inside. Its level is constant. If the intended function is to supply the clinic, the absence of delivery may be a failure. Now imagine a tank whose level stays at ten because inflow matches useful withdrawal. The same stock observation accompanies a different process.

This distinction is central to any future study of sustained service. A rule that does nothing can leave an action-only world unchanged. If no disturbance or consumption exists, the world may remain within its initial conditions indefinitely. That is not evidence that the rule can meet changing needs. The test must contain the processes relevant to the claim.

Conversely, a temporarily changing stock is not automatically a failure. A deliberate transfer can reduce a source stock while improving the represented arrangement. A service can require withdrawal. Maintenance can temporarily interrupt operation. The judgement depends on the declared purpose and constraints, rather than on a general preference for every variable to remain constant.

Homeostasis is therefore better understood through the maintenance of function over an appropriate interval than through the absence of visible change. The interval and the function must both be stated. A snapshot can support a state description; it cannot by itself establish recovery after disturbance or continuing provision.

6.6 A physical balance before a valuation

For the tank, an elementary stock account is

$$\text{new stock} = \text{old stock} + \text{inflow} - \text{outflow}, \quad (6.1)$$

where every term uses the same physical unit and the flows refer to the declared interval. This is a balance statement. It does not yet say whether the new stock is desirable.

If the old stock is ten units, inflow is four and outflow is four, the new stock is ten. If inflow is two and outflow is four, the new stock is eight. The arithmetic is simple, but its interpretation depends on the boundary. If some water remains in a connecting pipe, the account needs to say whether that water is included in inflow, transit or neither yet.

The fixed-total teaching model later in the book considers internal transfers among represented places. A lossless transfer subtracts a quantity from one place and adds the same quantity to another. External inflow and withdrawal are not silently present. When discussed, they are declared separately, because they change the physical accounting conditions.

This ordering is deliberate. We first establish what changed. We then evaluate that change through a declared potential. If we reverse the order, a desirable interpretation can tempt us to overlook a missing physical term. No reduction in represented burden excuses an unexplained creation of resource.

6.7 The role of reserves and limits without importing a controller

Ordinary life contains limits. A tank cannot contain a negative quantity, a pipe has physical capabilities, and essential functions can require availability. Recognising these facts does not require importing every historical threshold model into this introduction. The present edition develops a smooth reference-based potential and keeps physical permission as a separate declaration.

A future application may impose a source reserve or a storage-capacity limit. Such a restriction would constrain the allowed action set. It would not automatically become a term in the potential, and its numerical value could not be inferred merely from a desire for safety. A reserve protects a specified purpose under specified assumptions; its meaning must be stated.

This distinction helps prevent one of the most common modelling confusions. A permission rule that limits export does not necessarily direct resource toward a place that needs it. Refusing to empty a source and restoring a destination are different actions. The former may be necessary for safety while remaining insufficient for service.

The Gaussian landscape likewise should not be read as a complete permission rule. An action can move toward the reference while violating a physical restriction. Another can be physically permitted while moving away from the reference. The account becomes clearer when permission and valuation are checked separately.

6.8 Why a smooth potential is a useful teaching choice

The later potential assigns burden continuously as a state moves away from its declared reference. Its quadratic form allows exact finite arithmetic and makes curvature visible. A derivative describes the local slope, while a finite difference describes a completed finite change. The relation between these objects can be derived rather than guessed.

This is a mathematical advantage, not a claim that every biological or social response is quadratic. A convenient model can be useful for establishing concepts even when a future application needs a different representation. The correct question is what the construction assumes and what follows from those assumptions.

It also gives us a clean way to discuss overshooting. Moving toward a reference can reduce burden up to a point; continuing past it can eventually undo the improvement. A positive initial slope along an action direction therefore does not justify multiplying that slope by an arbitrary finite quantity. Later chapters calculate the exact correction.

At this stage, the essential insight is qualitative. “More action” is not the same as “more restoration”. The state changes while the action proceeds, so the value of an additional increment changes as well. A finite account must evaluate the actual change rather than freeze the first impression indefinitely.

6.9 Several functions can make competing demands

Suppose the clinic and the workshop draw from the same store. The clinic needs water for current service; the workshop needs water to repair the equipment that will support later service. Sending everything to one destination may impair the other function. Calling both needs homeostatic does not remove the competition.

A physical model can expose the shared constraint: the same source quantity cannot be sent twice. A potential can express a declared evaluation of the resulting states. A permission rule can exclude physically invalid proposals. Yet a choice among valid competing uses still depends on the declared policy and purpose. No general biological metaphor supplies that choice automatically.

The example also shows why independent local proposals require a common acceptance step when they contend for the same stock. Each proposer may see an available quantity and request it. Accepting all requests separately could exceed the source. The final account must evaluate the vector actually accepted, because the physical world receives the combined action rather than each proposal’s private imagined world.

Later chapters will examine this shared-field issue mathematically. Here it is enough to see its functional meaning. Maintaining one variable cannot be treated as success if the action quietly disables another declared essential function. Whether the representation captures both functions is an application question; whether the resource is double-spent is an accounting question. Both deserve a clear answer.

6.10 The caretaker’s questions

Before using a homeostatic description, our imaginary caretaker can ask a sequence of practical questions. Which function are we trying to keep available? Which state variable represents a condition relevant to it? What reference is being used? Which observations support the current state? Which actions are permitted? What will be measured after the action?

These questions do not demand that the caretaker solve every philosophical issue before opening a valve. They demand that a research model state the assumptions on which its conclusion depends. A routine operational rule can

be simple because much of that design work has already been done. A new theory must expose that work rather than borrow the appearance of simplicity. The caretaker should also ask what happens when the assumptions fail. If a sensor is unavailable, the account needs a declared response. If the resource changes quality, the homogeneous model may no longer apply. If a new service is added, the old reference may need review. Such review changes the model prospectively; it should not rewrite past actions under an undisclosed new evaluation.

6.11 Guided problems and answers

6.11.1 The unchanging tank

Two tanks remain at ten units for an hour. One is sealed. The other supplies four units while receiving four. Do the equal final stocks establish equal service?

No. The state snapshots agree, but the flow histories differ. A service claim requires a record of the relevant delivery. The sealed tank illustrates why preservation of a stock alone is an insufficient success criterion for a system intended to provide something.

6.11.2 The well-drawn landscape

A model has a potential with a unique minimum at its declared reference. Does a state away from the reference necessarily move toward it?

No. The potential is an evaluation. A state changes only through the declared physical dynamics or actions. Available routes, restrictions and a selection rule determine whether and how movement occurs. A unique minimum does not imply a particular trajectory.

6.11.3 The export limit

A rule prevents Ridge from sending more than a declared quantity. Does that guarantee that Vale receives enough?

No. The rule limits an action at the source. It does not by itself select a destination, create a route or require delivery. Service requires additional conditions. This is why the language of protection should always identify what is protected and by which mechanism.

Equilibrium, reference and continuing life

7.1 Three different meanings of “balanced”

The clinic manager says the accounts are balanced. The caretaker says the tank is balanced because inflow equals outflow. A mathematician says a model is at equilibrium. The sentences sound similar, but their meanings depend on different objects. Confusion begins when a conclusion established for one is carried into another without explanation.

A monetary balance concerns claims under accounting rules. A stock-flow balance concerns quantities within a boundary and interval. An equilibrium concerns a specified law of change. A reference concerns a chosen comparison configuration. A system may possess one of these properties while lacking another.

The chapter’s purpose is to make those distinctions comfortable before the book introduces a Gaussian landscape. We want the reader to see precisely why a smooth minimum is useful, and equally precisely why it is not a proof that a living or economic system will settle there. Words such as equilibrium, balance and stability should earn their meaning from the declared model.

7.2 A reference is a comparison, not a prediction

Suppose we declare a two-place reference of ten units at Ridge and ten at Vale. We write it as $x^* = (10, 10)$. The star marks the reference; it does not mark an observed event, a historical average or a forecast. The actual state could be $(18, 2)$, $(14, 6)$ or something else.

We can compare each actual state with the reference. At $(18, 2)$, the deviations are eight and minus eight. At $(14, 6)$, they are four and minus four. At $(10, 10)$, both are zero. These are straightforward differences. A potential will later turn the vector of deviations into a declared scalar burden.

Nothing in the reference declaration says that a valve will open. Nothing says which participant may use a pipe, or whether a pipe exists. The declaration also does not prove that ten units is the appropriate stock for either place. An applied reference needs reasons and evidence linked to the function it is meant to represent.

References can serve different purposes in different disciplines. A calibration standard provides a comparison for an instrument. A design value specifies intended operating conditions. A statistical mean summarises a distribution. Our x^* is a declared model reference. If an application wants to identify it with one of these other objects, it must supply the connection explicitly.

7.3 An equilibrium belongs to a law of change

To call a state an equilibrium, we must state the dynamics. In a simple discrete description, suppose a rule maps a current state x to a next state $T(x)$. A state $\bar{x}$ is a fixed point of that rule when

$$T(\bar{x}) = \bar{x}. \quad (7.1)$$

This equation defines a relation between a state and a particular rule. It does not say whether the state is desirable, fair or reachable from another state.

Consider the rule “do nothing”. Every state is then a fixed point in an action-only world. The state (18,2) is as stationary as (10,10). If the clinic needs a transfer, this stationarity can coexist with failed service. A fixed point therefore cannot be treated as a synonym for satisfactory function.

Now consider a different hypothetical rule that always transfers a prescribed quantity when permitted. Its fixed points, if any, depend on the permission conditions and update law. Changing the rule can change the equilibria while leaving the potential unchanged. Conversely, changing the potential can change the evaluation of a fixed point without changing the physical rule.

The lesson is that equilibrium is not a free-standing property of a list of numbers. It is a property of numbers under specified evolution. Whenever a claim says that an economy or resource system “returns to equilibrium”, ask which variables, which dynamics, which disturbances and which sense of return are meant.

7.4 A steady state can contain continuing flow

The tank example adds another distinction. If equal quantities enter and leave during each interval, the stock can remain constant. The state is steady in that coordinate, but activity continues. A record of the stock alone cannot reveal whether the water is still or moving through.

This matters for living systems and for infrastructure. Continuing function often depends on throughput, replacement, maintenance and exchange with surroundings. Describing a stable condition does not imply the absence of

these processes. A model that represents them must specify both the state and the flows or transformations that sustain it.

For our teaching model, the fixed-total internal-transfer world is deliberately narrower. It isolates redistribution of one conserved quantity. When we discuss ongoing withdrawal, regeneration or external supply conceptually, those processes are not thereby inserted into the equations. An extended world would need explicit terms and an appropriate physical boundary.

The distinction prevents a misleading inference: a potential minimum in a closed redistribution model is not automatically the operating state of an open system providing continuous service. Withdrawal can move the open system away from the reference; replenishment can change the total; maintenance can affect available actions. Each process requires its own account.

7.5 Conservation restricts the possible reference

In the fixed-total model, the sum of stocks is M . A compatible reference must have the same total if exact matching is to be possible through internal lossless transfers. For two places with $M = 20$, the reference (10, 10) is compatible. The reference (12, 12) requires a total of twenty-four and cannot be reached by merely rearranging twenty.

This is not a subtle optimisation issue. It is a physical compatibility check. No controller, field or incentive can create the missing four units while the model remains closed and conservative. A calculation that appears to do so has changed a boundary, a state definition or an assumption, whether or not that change was admitted.

Even compatible totals do not guarantee reachability. If Ridge and Vale have no connecting route, the difference cannot be corrected by the allowed actions. If a route points only one way, some states may permit correction and others may not. If physical limits restrict quantities, the path to the reference may require additional actions or may be unavailable.

Thus three questions must be kept separate: is the reference arithmetically compatible, is it reachable through the allowed physical actions, and will a declared selector actually choose those actions? The first is a constraint check. The second concerns the action structure. The third concerns dynamics or policy.

A reference that asks for missing resource

The actual state $(18, 2)$ contains twenty units. The proposed reference $(12, 12)$ contains twenty-four. Internal lossless transfer preserves the total, so it cannot reach that reference. Either the reference must be reconsidered for the closed model or an external addition must be explicitly declared. A potential can still be written around the incompatible reference, but zero burden would be physically unattainable within the stated world.

7.6 Stability is stronger than being an equilibrium

A state can be a fixed point and still behave differently when perturbed. To discuss stability, we must describe what counts as a small disturbance and what the subsequent dynamics do. Does the state remain near the fixed point? Does it return? Does the departure grow? These are different mathematical questions.

An elementary analogy is a ball at rest on different surfaces, but even that analogy needs its own physical laws. A bowl suggests restoring motion under gravity and dissipation; a flat shelf suggests a different response; a crest suggests another. The shape alone is not a complete dynamics. Friction, constraints and applied forces determine the actual motion.

The Gaussian potential in this book is a declared landscape for accounting. We shall use its derivatives to calculate marginal directions and its finite differences to evaluate completed changes. We do not silently assume a ball rolling under that potential, nor identify it with physical energy. A restoring controller would require an independently declared relation between the field and accepted action.

This distinction is particularly important when the word “homeostasis” appears. A state with low represented burden has a property of the evaluation. A controller that repeatedly restores relevant function under disturbance has a property of behaviour. Establishing the first does not establish the second, although the first may be useful in designing or analysing the second.

7.7 What the Gaussian name contributes

The term Gaussian will enter through a mathematical shape. A positive scale and a reference define a normal-shaped reference expression; taking an appropriate relative negative logarithm produces a quadratic deviation measure.

The later chapter derives that relation explicitly. The resulting potential is convenient because its gradient and finite changes can be computed exactly.

The name does not claim that observed tank stocks follow a normal distribution. It does not claim that future disturbances are Gaussian, that people's needs are normally distributed or that a network has reached statistical equilibrium. Each of those would be an additional empirical or model assumption.

Nor does the scale automatically represent measurement uncertainty. It is a declared physical scale for the deviation in the chosen coordinate. A sensor's uncertainty might influence an applied design, but identifying the two requires justification. Treating every sigma as a standard error simply because the symbol is familiar would change the model's meaning.

The distinction is useful for readers approaching the mathematics without statistics. They can understand the potential as a smooth squared distance from a declared reference, with different scales changing the importance of a given physical deviation. The Gaussian relation explains the chosen form; it does not ask the reader to believe an unobserved probability law.

7.8 A reference can be revised, but not invisibly

Suppose Vale's clinic expands and the intended resource arrangement changes. A revised reference may be justified. The physical state at the instant of revision might remain exactly the same while the evaluated burden changes. That difference is caused by changing the evaluation, not by a resource transfer.

If the account uses a fixed potential to value a sequence of actions, such a revision must be recorded as a separate change. Otherwise, an actor could appear to gain value merely because the target moved closer to the current state. Conversely, an actor could appear to lose value because a new requirement was introduced after the action.

This is not an argument against learning or democratic revision. References should be reviewable when their purpose or evidence changes. It is an argument for preserving the identity of the account under which each claim was made. A new version can govern future evaluation while the old version remains available for interpreting earlier records.

The same reasoning applies to changing scales. A larger scale can make the same deviation appear smaller in the potential. That is a change in evaluation, not proof that the physical condition improved. A record that separates physical change from model revision allows both to be discussed honestly.

7.9 The value of an action depends on its starting state

Imagine moving four units from Ridge to Vale. Starting from (18, 2), the endpoint is (14, 6). Starting from (10, 10), the endpoint is (6, 14). The action direction and quantity are the same; the starting states differ. Relative to the symmetric reference (10, 10), the first moves toward the reference and the second moves away.

A name such as “water delivery” therefore cannot carry a universal sign. The physical context matters. The same route can relieve one imbalance and deepen another. Later the exact finite formula will make this dependence explicit, including the change of field during the transfer.

This contextual character should not be confused with arbitrary judgement. Once the baseline, action, potential and process burden are fixed, the calculation is determined. What remains open is whether those declarations are appropriate for the intended application and how the institution uses the result. Determinacy within a model and justification of a model are separate achievements.

The example also explains why a reference point is not an instruction to stop every useful activity. A service may require moving resource away from a reference at one moment and restoring it later through a separate process. The account should show those changes. A service policy must decide what actions are necessary and how continuing supply is arranged.

7.10 Reaching a point and staying there are different tasks

Suppose an allowed transfer takes the two-place world exactly to its declared reference. That calculation establishes an endpoint. To say that the system stays there, we need to know what happens next. A further request, an external withdrawal or a different actor’s action can change the state. The first successful arrival supplies no general protection against those events.

In the closed action-only teaching world, choosing no further actions does leave the state unchanged. This is a conditional fact about that simple world. In a world with continuing service, choosing no action may allow withdrawals to change the state. A claim about sustained balance must therefore include the process that creates the need for continuing response.

There is also a distinction between one reachable point and a robust operating arrangement. A route might permit reaching the reference once while leaving no capacity for a later disturbance. To investigate robustness, a study must declare the disturbances and the available responses. The potential itself does not specify their frequency, magnitude or distribution.

Keeping these tasks distinct helps organise the programme. Exact finite accounting explains a single accepted change. Sequential identities explain how declared changes add. Dynamic analysis or experiments examine behaviour under a specified evolution. Institutional investigation asks how those mechanisms operate among people. Each layer can build on earlier work while retaining its own unanswered questions.

7.11 The questions to ask before accepting an equilibrium claim

A claim that “the field restores balance” needs several questions. Is the field a potential derivative, an observed physical influence or a metaphor? What action law moves a quantity, under which constraints? Is the reference compatible and reachable?

Identify the timescale and external processes: one action, an undisturbed sequence, or continuing demand and supply? Then identify the evidence: a conditional proof, software check, illustration or registered observation?

A failed service demonstration may reflect an absent driving process, not a general impossibility. A static calculation is not a dynamic theorem. Precise vocabulary locates the remaining question.

7.12 Guided problems and answers

7.12.1 The stationary unequal state

In a closed two-place world, a rule always chooses no action. Is $(18, 2)$ an equilibrium under that rule? Does this establish satisfactory provision?

It is a fixed point: the no-action update leaves the state unchanged. This establishes no service result. Service depends on the declared function and its criteria, neither of which the update rule supplies. Equilibrium and desirability are distinct questions.

7.12.2 The unattainable target

A closed system contains twenty units and declares reference $(12, 12)$. Can a sequence of conservative transfers achieve zero deviation from this reference?

No. The reference total differs from the physical total. Every internal lossless transfer preserves twenty, while the reference requires twenty-four. The mismatch must be resolved through a revised reference or an explicitly different open-system model; changing the action selector cannot remove it.

7.12.3 The revised scale

The physical state remains fixed, but the evaluator doubles a coordinate's deviation scale. Has a physical action improved the state?

No physical change has been described. The numerical evaluation changes because the model changed. A proper record identifies that revision separately from actor-caused resource changes. Exact arithmetic cannot turn a change in definition into a performed action.

7.12.4 The Gaussian inference

A book uses a Gaussian-derived potential. May a reader conclude that observed stock fluctuations are normally distributed?

No. A chosen reference shape does not establish a distribution of observations. That claim would require a statistical model and evidence appropriate to it. The potential can be used as a declared mathematical evaluation without making that empirical assertion.

7.13 What to carry forward

A reference states a comparison; equilibrium belongs to a law of change. Steady stock can coexist with continuing flow, and stability requires a dynamic argument. We now use these distinctions to construct physical states, actions and their exact signed account.

The physical question beneath an economic action

8.1 Begin with something familiar: a water pump

Imagine a small town whose clinic receives water from a storage tank on a nearby ridge. A pump moves water through a pipe. The clinic pays an electricity bill, the operator receives wages, and the supplier sends an invoice. These monetary records are useful. They identify obligations, payments, and claims under the institutions in which the participants live.

An engineer responsible for the same event needs another record. How much water left the source? How much reached the clinic? Which tanks changed? Was the water of the declared usable quality? What powered the pump? Did a pipe leak? What happened outside the chosen measurement boundary? A balanced invoice does not supply these answers. Conversely, a complete engineering account does not determine the legal price of water or the operator's wage.

The introductory chapters have explained why this separation matters. We now build the mathematical account patiently, beginning with the physical event. The simple world used throughout the central chapters contains one homogeneous resource, measured in a common stock unit, written X . Its transfers are lossless unless an example explicitly announces another boundary. This is a deliberate teaching model. Real pumps and water networks use several carriers and can lose usable material. Their more complete representation must name those processes rather than hide them inside a convenient word such as “delivery.”

Two records of one event

A monetary account records institutional claims and payments. A physical account records stocks and transformations. EBU adds a declared valuation of the physical change: how much the change reduces or increases the represented deviation from a reference. Each record answers a different question.

8.2 Why the accounting object is an action

Ordinary language can compress a long chain into “the clinic bought medicine.” Physically, a precursor is produced, ingredients are transformed, a refrigerator maintains a temperature, a vehicle transports a package, and a nurse administers a dose. Several places, carriers, time intervals, and by-products can be involved. The invoice may combine them; a physical account cannot assume they have become one atomic event.

An action in this book is a declared physical transformation with an identified affected state. For a simple transfer, that means a source, destination, resource identity, quantity, and event boundary. More complex transformations require a correspondingly richer description. “Repair the pump” is a meaningful practical instruction, but it becomes a quotable physical action only when the model specifies what measurable coordinate the repair changes. A stock-only model cannot see an improvement in machinery reliability merely because the prose calls it useful.

This requirement can sound restrictive until we consider the alternative. If a model pays for any story about a better future, an actor can claim improvements that no represented variable records. If it pays only for today’s material displacement, it may miss a valuable repair. The honest response is to identify the missing state or to state that this benefit lies outside the present account. Neither refusal to invent a number nor a zero value in a narrow model establishes that the work has no human value.

8.3 Five objects on the desk

Before calculating, place five distinct records beside one another. The first is the physical state: how much resource is present at each place. The second is the reference: the declared configuration against which deviation will be measured. The third is the action: the proposed change to the state. The fourth is permission: the rules that determine whether this action may occur. The fifth is the signed value: the finite change of a declared potential, with any separately justified process burden accounted once.

Confusion usually begins when one record silently performs another’s job. A large stock is treated as a high monetary balance. A positive value is treated as permission. A mathematical minimum is treated as proof of social justice. An action’s owner is treated as its uniquely identified causal contributor. These substitutions may make a story move quickly, but they make its equations unreliable.

Object	Question it answers
Physical state	What quantity is present, where, in which unit?
Reference	Against what declared configuration is deviation measured?
Action	Which physical coordinates change, and by how much?
Permission	Is that change admissible under the declared rules?
Signed EBU	What exact represented change of burden accompanies it?

8.4 Our running world: Ridge, Vale, and the clinic

Let Ridge be cell 1 and Vale be cell 2. Vale contains the clinic, but the scalar state records the resource in its tank, not the clinic's complete health or its patients' welfare. Initially Ridge has 18X and Vale has 2X. The total is 20X. We shall write this state as $z = (18, 2)^T$, introducing the vector notation fully in Chapter 9.

For the main example the declared reference is $x^* = (10, 10)^T$, and each cell has scale 2X. The reference says what the model treats as its centre; the scale says how stock deviations are standardized. The potential is

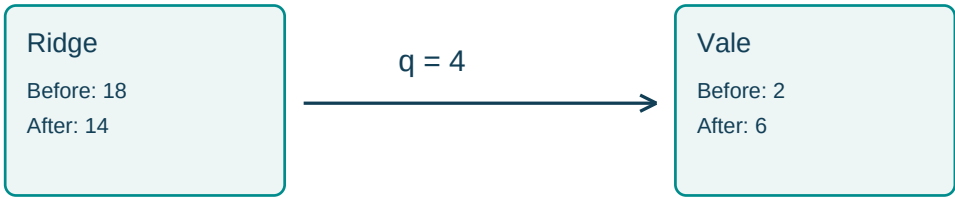
$$V(z) = \frac{1}{2} \left(\frac{18 - 10}{2} \right)^2 + \frac{1}{2} \left(\frac{2 - 10}{2} \right)^2 = 16. \quad (8.1)$$

This equation will be derived and interpreted carefully in Chapters 11 and 12. For now, read each fraction as a deviation measured in units of the declared scale. Squaring makes its contribution nonnegative. Adding the two contributions gives the represented burden of this particular state.

Consider a lossless transfer of 4X from Ridge to Vale. Ridge then holds 14 and Vale 6. Total stock remains 20. Potential becomes

$$V(14, 6) = \frac{1}{2}(2)^2 + \frac{1}{2}(-2)^2 = 4. \quad (8.2)$$

In the ideal example there is no separate process burden, so the signed EBU is $16 - 4 = 12\text{EBU}$. The stock ledger and the EBU account have different balancing terms. Four physical units leave Ridge and enter Vale. Twelve EBU describe the decrease of the chosen potential. They are not twelve extra physical units created by the pipe.



Total stock: 20 before and after

Potential: 16 before, 4 after; field value: +12

Figure 8.1: A four-unit lossless transfer preserves twenty physical units while reducing the declared potential from sixteen to four.

8.5 What the first calculation does and does not say

The result 12EBU is exact arithmetic under the declared model. It does not establish that the reference is empirically correct, that the transfer is permitted by every relevant institution, or that the clinic has received every service it needs. It also does not identify how the number should be assigned to the pump owner, engineer, electricity supplier, and courier. Those are further questions.

We can nevertheless learn something precise. Conserving total stock does not preserve its location. A lossless redistribution can reduce deviation without increasing mass. The change of potential accounts for positive EBU without requiring a matching negative receipt from another person. Whether an experimental capacity account or a future institution recognizes such a receipt is a separately declared rule.

The same formula can also give a negative result. Sending 17X is physically possible in this simple two-tank example because Ridge begins with 18. The endpoint is (1, 19). Both cells are nine units from their references, and potential is 20.25. The signed field change is therefore $16 - 20.25 = -4.25\text{EBU}$. A beneficial initial direction can become excessive when carried far enough. This is why a finite action, rather than only an initial tendency, must be valued.

8.6 Need does not disappear into a number

The choice to begin with a clinic keeps the purpose visible. It does not turn the potential into a complete theory of need. A clinic's water stock can be near its declared reference while an ambulance urgently needs fuel. A water transfer can lower this field's burden while leaving an unrelated medical shortage untouched. The model has one carrier; the world has many interdependent functions.

An essential action may also have negative EBU in a declared field. Moving a scarce resource from one already deficient place to another may increase the chosen total deviation, even if an institution decides that an emergency justifies the action. Honest accounting retains the negative number and states the policy that allowed the action. Changing the reference or the formula after seeing the emergency would obscure the physical and institutional decision rather than explain it.

In later research, a signed capacity rule may restrict which negative actions can be afforded. That proposal needs its own ownership map, account balances and acceptance rules. Subtracting two potentials does not supply those decisions. The mathematics here provides an exact valuation object with which the proposals can be stated and tested.

8.7 Every noun needs a measurement boundary

"Water" is already a modelling choice. Potable water, contaminated water, steam, and water locked in soil are not interchangeable for every function. A homogeneous scalar model assumes that one declared unit has the same relevant meaning wherever it appears. If that assumption fails, the state needs more coordinates or a different domain.

Similarly, a place must have a clear boundary. If the source meter measures withdrawal before a pump and the destination meter measures arrival after a leaking pipe, the two readings refer to different physical positions. A difference between them is information about the intervening process. Calling both readings "the quantity" erases that information.

Time matters as well. A starting gauge reading and an integrated flow reading must belong to compatible event windows. If a tank receives rain between two observations, the action cannot claim that contribution merely because the final tank is fuller. External changes must be identified before the actor's baseline is frozen, or accounted separately under a declared event model. Our first closed examples contain no such external change during the action.

8.8 The useful discipline of an incomplete model

A small model is valuable when its omissions are explicit. It lets the reader see a conservation identity, a sign convention, and a finite calculation without attributing to the model every feature of a town. The danger is not simplicity itself. The danger is retaining the small equations while allowing the prose to claim the whole town.

Suppose a technician repairs the clinic's insulation, lowering tomorrow's electricity need while leaving today's water stock unchanged. The present water field gives zero state-change value. That is an accurate report of its limited representation. To value the repair in a physical field, one might need a verified insulation or heat-loss coordinate, a declared potential for it, and a transformation connecting the repair to that coordinate. This book does not fabricate those definitions. It preserves the lesson: a physical consequence needs a represented carrier if the field is to see it.

Exact does not mean complete

An exact finite difference can be computed from an incomplete or poorly calibrated model. Exactness concerns the relation between the equation and its declared inputs. Empirical adequacy concerns whether those inputs and definitions represent the intended physical problem. The second question requires evidence beyond the algebra.

8.9 Guided problem: write the action before pricing it

A courier moves four sealed containers of the represented resource from Ridge to Vale. Each container contains one stock unit. The route is declared lossless in this exercise. Write the physical action and its limits before using the potential.

First identify the carrier. The stock unit is one container's verified contents, with a common quality definition at both places. Second identify the endpoints: source Ridge, destination Vale. Third state the finite quantity: $q = 4X$, not four units per hour. Fourth write the changes: source -4 , destination $+4$. Fifth check availability: Ridge has 18 and can supply four under the exercise's simple nonnegative-stock rule. Sixth name omissions: the transporter's fuel, labour, and vehicle wear are not physical coordinates in this two-stock illustration.

The last point does not authorize arbitrary deductions. The ideal example explicitly sets process burden to zero. A more complete model would declare

additional carriers or justify a separate physical process term with compatible units and no double counting. An invoice for fuel is not automatically such a term.

Now calculate. The before-state potential is 16; the after-state potential is 4; signed EBU is 12. Report the conclusions in separate sentences: four stock units arrived, total stock was conserved, and represented deviation fell by twelve potential units. No statement about fairness or full clinical success has been established.

8.10 A useful question for the clinic's committee

Imagine the committee receives two reports about the same delivery. One says that four stock units arrived. The other says that the declared field improved by twelve EBU. The committee should not ask which report is the “real” one. It should ask what each establishes and which relevant facts remain outside both.

The delivery record establishes quantity under its measurement boundary. The field account adds a comparison with the declared reference and scale. Neither report establishes whether the water arrived before a medical procedure, whether its quality was independently verified, or whether the remaining source stock is sufficient for functions absent from the model. These may be decisive practical facts.

A good EBU account therefore invites additional questions instead of swallowing them. What carrier does the state represent? How was the reference chosen? Did an external change occur between measurements? Is the process burden complete enough for the intended use? Which claims are mathematical consequences, and which would require a field study?

This posture is compatible with ambition. A research programme can aim to connect economic action more closely to the conditions supporting life while beginning with a model it can audit. The point of the narrow calculation is to establish dependable pieces of reasoning that can be extended carefully. Grand aims do not reduce the need for a clear four-unit transfer.

For the reader, an especially productive habit is to translate every strong sentence back into a record. “The clinic was helped” might refer to delivered quantity, a lower potential, a measured service outcome, or a judgment about urgent need. Ask which one is meant. Then the statement can be tested or qualified at the appropriate level rather than accepted because its language sounds benevolent.

8.11 Common questions answered carefully

Is EBU another price? It is a value defined from a declared physical field. A market price can be related to it by an institution, but the relationship requires its own rule. The equation does not supply one automatically.

Does negative EBU mean a bad person? The sign belongs to an action evaluated from a state under a fixed field. The same action at another state can have another sign. An actor's character is not a state coordinate.

Does positive EBU prove permission? No. A value can be calculated for a hypothetical transformation that has insufficient stock or violates a declared boundary. Permission must be checked independently, and an inadmissible point must not become an executable promise.

Does zero EBU mean nothing happened? No. A finite transfer can preserve potential while moving substantial stock. Our $q = 16$ transfer sends the main state from $(18, 2)$ to $(2, 18)$, with potential 16 at both ends. Equal potential does not mean identical state.

8.12 Practice and reflection

For a delivery you know, write a monetary description and a physical description. In the physical description identify the carrier, source, destination, finite quantity, timing, and any boundary crossings. Mark facts that a one-resource stock model would omit.

For the main Ridge–Vale example, calculate the after-state and potential for $q = 8$. The endpoint is $(10, 10)$, the potential is zero, and the field reduction is 16EBU. Explain why this does not mean that eight water units became sixteen water units. Then calculate $q = 0$: the state is unchanged and EBU is zero. These two answers establish the anchors of the finite schedule developed later.

Finally, explain in ordinary words why measuring an event and deciding who may perform it are separate tasks. A convincing answer should retain the clinic's practical need while refusing to make a valuation equation supply an undeclared social rule.

What to carry forward

Begin with a physical action and a declared boundary. Keep stock, reference, permission, potential, and signed EBU distinct. The running example is small enough to audit completely. Its exact calculations teach an accounting structure; they do not establish an economy's long-run behaviour.

Cells, nodes, and edges

9.1 Why the map comes before the actor

A model of economic activity can begin by giving an actor a goal. The actor seeks a resource, maximizes a reward, or chooses a trading partner. Physical accounting must first establish a world in which these actions have a definite meaning. Where is the resource stored? Which transformations are available? Which places can exchange it? Otherwise a clever decision rule can deliver stock that was never withdrawn or use a route that the represented world does not contain.

The simplest useful map is a directed graph. A graph is a collection of places together with connections between them. A directed connection has an orientation. Here a place is called a cell or node, and a directed connection is called an edge. The graph describes available physical transformations; it does not prescribe which one an actor selects.

A cell need not be a square on a geographical grid. It may be one well-mixed tank, a warehouse, or a region for which a particular stock can be measured. Good boundaries keep important exchanges visible. Combining a reservoir and treatment plant into one cell may conceal a transformation between them. Dividing one well-mixed tank into a thousand cells may add notation without adding physical information.

9.2 Vocabulary before notation

Word	Meaning in the present model
Cell or node	One represented place carrying a scalar stock.
Index	A stable label used to identify a cell or an edge.
Edge	One declared directed transfer possibility.
Source	The cell from which an action withdraws stock.
Destination	The cell that receives stock.
Path	An ordered sequence of adjoining edges.
Topology	The pattern of available connections.
Support	The coordinates changed by a specified action.

An index is a label, not a size or a ranking. Writing $i = 1$ for Ridge and $i = 2$ for Vale does not declare Ridge more important. It fixes the order in which their numbers appear. That order must remain stable throughout a calculation and its record.

We write x_i for the stock at cell i . Thus $x_1 = 18$ and $x_2 = 2$ are two scalar statements. A scalar is one number with its declared unit. A vector collects several numbers in a fixed order.

9.3 A vector is an ordered physical record

The complete state of N cells is written

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \in \mathbb{R}^N. \quad (9.1)$$

The notation $\mathbb{R}^N$ means the set of ordered lists of N real numbers. Our physical domain is narrower: every stock is nonnegative, and their sum is the fixed total M . Writing the larger mathematical space is convenient for derivatives; it does not permit negative water in an executable state.

For Ridge and Vale the state is $x = (18, 2)^T$. The superscript T means transpose. It turns a row into a column or a column into a row. It is neither a power nor a time label. A row multiplied by a matching column gives the dot product:

$$a^T b = \sum_{i=1}^N a_i b_i. \quad (9.2)$$

For $a = (2, 3)^T$ and $b = (5, 7)^T$, the result is $2 \cdot 5 + 3 \cdot 7 = 31$. Later, a marginal vector multiplied by a state change will add the local contributions to a change of potential. The operation remains this ordinary multiply-and-add procedure.

9.4 An edge is a directed possibility

Write an edge as $e = (i \rightarrow j)$. One unit sent on this lossless edge removes one stock unit from source i and adds one to destination j . The existence of $i \rightarrow j$ does not imply that $j \rightarrow i$ exists. A pipe may operate only in one direction. A permitted road route may differ from its reverse. If both directions are available, represent them as separate edges.

The graph also does not say how much will move. A road's existence does not fill a truck. The action contains a finite nonnegative quantity q_e , expressed in

stock units. Direction is carried by the edge; we do not silently use a negative quantity to create a reverse process that the graph has not declared.

Nor is an edge a universal relation of distance. Two distant tanks may have a direct pipe, while neighbouring tanks may have no connection. Distance can matter to delay, equipment, or an extended physical model. It does not enter this field through an assumed inverse-square law. The graph and the potential have explicitly defined jobs.

9.5 One column for one transfer

For edge $e = (i \rightarrow j)$, define the column S_e by

$$(S_e)_k = \begin{cases} -1, & k = i, \\ +1, & k = j, \\ 0, & \text{otherwise.} \end{cases} \quad (9.3)$$

Read the subscript outside the parentheses as “entry k of the column.” It does not multiply two symbols. The minus sign marks withdrawal, the plus sign marks delivery, and zero marks an unchanged coordinate.

The action’s state change is

$$\delta_e = S_e q_e. \quad (9.4)$$

For Ridge to Vale, $S_e = (-1, 1)^T$. Sending four units gives $\delta_e = (-4, 4)^T$, so the successor is $(18, 2)^T + (-4, 4)^T = (14, 6)^T$. This is the calculation already performed in words. The column has made its signs reusable.

Read the column aloud

One lossless edge column says: withdraw one unit here, deliver one unit there, change nothing else directly. Multiplying by a finite quantity gives the physical increment. The column records where the change occurs; it does not choose the quantity or value the action.

9.6 From columns to a matrix

Place all edge columns side by side to form the incidence matrix,

$$S = [S_1 \ S_2 \ \cdots \ S_m]. \quad (9.5)$$

There is one row per cell and one column per edge. The letter m here counts graph edges; the symbol E is reserved elsewhere for an EBU value.

If $q = (q_1, \dots, q_m)^T$ collects finite quantities, then Sq is the sum of their state increments.

Consider a separately declared three-cell illustration: Ridge is cell 1, a store is cell 2, and the clinic tank is cell 3. Edges run $1 \rightarrow 2$ and $2 \rightarrow 3$. Both are lossless. Then

$$S = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}. \quad (9.6)$$

For quantities $q = (4, 2)^T$, multiplication gives

$$Sq = \begin{bmatrix} -4 \\ 4 - 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ 2 \end{bmatrix}. \quad (9.7)$$

The store receives four and sends two, for a net gain of two. The net entries add to zero. This matrix product is physical bookkeeping, not a proof that the proposed actions are jointly feasible. If the store begins empty and the event rule forbids use of simultaneous incoming stock, its outgoing proposal still fails. Event timing is not determined by matrix notation.

9.7 A shared source has one stock

Now let Ridge supply three destinations: the clinic, a food store, and a fire tank. In this new four-cell example,

$$S = \begin{bmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (9.8)$$

With quantities $(2, 1, 0.5)^T$, the state increment is $(-3.5, 2, 1, 0.5)^T$. Every source entry appears in the same row. The total withdrawal is 3.5, not three independent uses of the same stock.

This geometry matters twice. Physical feasibility must consider aggregate use of the source. Valuation must consider the combined change of any shared potential coordinate. Independently asking whether each action fits can miss a joint conflict. Independently valuing each action from the full initial state can also count the same improvement more than once. Those are distinct problems, and later chapters address them separately.

9.8 A path is not a single edge

A route from Ridge through a depot to Vale is a sequence of edges. Its intermediate stock may matter. So may timing, physical availability, and

the order in which each action becomes possible. Combining the route into one arrow can be appropriate only if the model explicitly justifies the coarser transformation and retains every relevant boundary.

Suppose the depot begins with two units. A simultaneous event requests four units into the depot and three out. The endpoint has three units, so it is nonnegative. But the initial depot does not contain the three outgoing units. A model may allow synchronized passage through a real continuous pipe; another may require receipt before onward dispatch. Both can be specified. Neither follows from the positive endpoint alone.

The same caution applies more strongly to fuel, electricity, and delivered water. They are different carriers, and an energy conversion is not a lossless relocation of a homogeneous scalar. Their actions should form a linked physical chain with appropriate units and conversion accounts. The simple graph notation can teach how to record support, but it cannot erase the engineering of the chain.

9.9 Three maps that must stay distinct

The physical graph tells us which transformations exist. The potential's dependency structure tells us which coordinates each potential term reads. The action-interaction structure tells us which actions overlap in a way relevant to joint valuation or feasibility. These maps need not coincide.

In our separable potential, each local term reads one cell, so the potential has no cross-coordinate terms. Nevertheless two actions withdrawing from Ridge both change Ridge's local quadratic. They interact in the joint finite value. A diagonal potential therefore does not imply independent actions.

Conversely, two geographically close places may have no common action support. Their values can be independent within this model if no potential factor couples them. A richer model could add such a factor, but then the affected neighbourhood used in a local quote must include it. The general principle is simple: locate dependence from the actual equations, not from a picture or a metaphor.

9.10 Building a graph from a practical description

Begin with the carrier and its unit. Next choose cells whose stocks can be measured consistently. Name each transfer process and its direction. Assign stable row and column indices. For each lossless edge put -1 at its source, $+1$ at its destination, and zeros elsewhere. Keep a separate record of physical quantity limits, timing, and permitted information.

Do not place EBU values or reference stocks inside S . The incidence matrix does not depend on whether Ridge currently has excess or deficit relative to its reference. It describes the change per unit of the named transfer. Stocks belong in x , references in x^* , scales in σ , and finite quantities in q . Separating these objects makes later changes auditable.

An edge record might state its identifier, endpoints, carrier, quantity unit, and physical validity conditions. A version identifier can freeze that declaration for an event. This helps prevent a route or parameter from changing halfway through an account. Versioning does not validate a bad physical definition; it makes the definition traceable.

9.11 Notation laboratory

For a four-cell transfer from cell 2 to cell 4, the column is $S_e = (0, -1, 0, 1)^T$. Its second entry is -1 , its fourth is $+1$, and its dot product with the all-ones vector is zero. A quantity $q_e = 3$ gives $(0, -3, 0, 3)^T$.

If there are N cells and m edges, the matrix has shape $N \times m$. The quantity vector has shape $m \times 1$, so the result has shape $N \times 1$. The inner dimensions match. This simple size check catches many transcription mistakes before any arithmetic is attempted.

For any cell i , the component formula is

$$(Sq)_i = \sum_{e: \text{dest}(e)=i} q_e - \sum_{e: \text{src}(e)=i} q_e. \quad (9.9)$$

It says to add all incoming quantities and subtract all outgoing quantities. A reader who finds the full matrix abstract can always return to this one-cell account. Matrix form is especially useful for proofs; component form is especially useful for tracing a real event.

9.12 Guided problem: build a returning route

A depot sends resource to a clinic and to a water plant. The clinic can return unused resource to the depot. In this illustration all transfers involve the same homogeneous carrier and are lossless. Index depot 1, clinic 2, plant 3, and edges $1 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 1$. The matrix is

$$S = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}. \quad (9.10)$$

For $q = (2, 1, 0.5)^T$, the change is $(-2.5, 1.5, 1)^T$. Its entries add to zero. The reverse edge is its own event component. It is not a negative number concealed in the first quantity.

Now suppose the starting state is $(5, 1, 0)^T$. The endpoint is $(2.5, 2.5, 1)^T$, with total six units before and after. The depot's outgoing total is three, and the clinic's outgoing quantity is one-half, each available from its initial stock. Under a source-availability rule excluding unreceived inflow, those particular source checks pass. Other physical constraints would still need their own declared checks.

9.13 Disconnected regions and unreachable references

The graph can impose conservation statements stronger than the global total. Suppose there are two disconnected regions and every edge begins and ends inside one region. Summing the incidence entries over either region gives zero for every edge. Each region therefore retains its own stock under the declared internal actions.

For a concrete four-cell illustration, let cells 1 and 2 form one connected pair and cells 3 and 4 another, with no edge between the pairs. Starting stocks are $(8, 2, 3, 7)$. The two regional totals are both ten. A reference $(5, 5, 5, 5)$ is compatible with these regional totals. A reference $(6, 6, 4, 4)$ has the same global total 20 but demands twelve units in the first region and eight in the second. It cannot be reached using only the declared internal edges.

This is an exact obstruction supplied by topology. No improvement in quote arithmetic removes it. A new cross-region route would change the physical graph and could remove the obstruction, but its construction is another physical and institutional question. The field should expose the remaining deviation without pretending that the actor has an unavailable action.

More generally, a cut is a partition of the cells into two sets. The only internal transfers that can change the total on one side are edges crossing that cut. Writing the sum of the component ledger on that side cancels every wholly internal edge. This simple cancellation is useful when auditing a region, even when its individual cells are numerous.

9.14 Renaming cells should not change the physics

If Ridge and Vale exchange their numerical indices in a data file, every associated object must be reordered together: state, references, scales, and incidence rows. The physical result should be unchanged. A consistent permutation changes notation, not the model.

For example, the main state $(18, 2)$ and column $(-1, 1)$ become $(2, 18)$ and $(1, -1)$ when the two row labels are exchanged. The same four-unit transfer produces the reordered endpoint $(6, 14)$. This is the earlier endpoint $(14, 6)$

written in the new order. It must not be confused with physically reversing the action while keeping the original labels.

This distinction is elementary but important in large records. A matrix with correct numbers and the wrong identifier-to-row map can conserve total stock while moving it between the wrong places. Physical identity belongs in the record alongside arithmetic consistency. Stable identifiers make an account reproducible across software ordering choices.

9.15 Common questions and practice

Does S calculate the best route? No. It records effects of declared edge quantities. A route selector is another object.

Are all entries of a physical state allowed real numbers? The state is represented by real coordinates, but admissibility restricts them to the declared nonnegative fixed-mass domain and any further physical conditions.

Does a zero entry mean a place cannot be affected indirectly? It means this action column does not directly change that coordinate. Later actions or a richer potential can create other dependencies, which must be represented separately.

For practice, draw Ridge connected to two destinations and add a return edge from one destination. Write all columns and compute their sum for quantities $(3, 2, 1)$. Your source increment should be -4 , the returning destination's net increment $+2$, and the other destination's increment $+2$. Explain how the result depends on the declared ordering of coordinates but not on the order in which the three columns are added.

Next remove the return edge. A proposal to return one unit is now unavailable even if it would improve the potential. This illustrates a central boundary: an algebraically attractive direction is not automatically an edge in the physical world.

What to carry forward

A cell is a state location. An edge is a directed physical possibility. Its incidence column records source withdrawal and destination delivery. The product Sq adds finite increments. Feasibility, timing, value, and action selection require their own declarations.

With the map established, we can audit quantities, units, and boundaries. The next chapter follows one physical change without letting a mathematical shorthand create or destroy stock.

Quantities, boundaries, and one physical change

10.1 The ledger underneath every valuation

Before asking whether an action improves the represented world, ask whether its physical account closes. A transfer cannot deliver more of the homogeneous resource than the declared source withdraws in a lossless model. A missing quantity cannot disappear because a potential decreases. An extra quantity cannot enter because an account has enough capacity to pay for it.

The main model has a finite number of nonnegative stocks and a fixed total M . It represents internal relocation of one homogeneous scalar. The physical state space is

$$\mathcal{X}_M = \left\{ x \in \mathbb{R}^N : x_i \geq 0, \quad \sum_i x_i = M \right\}. \quad (10.1)$$

This set is called a simplex. The unfamiliar name adds no new assumption: every coordinate is nonnegative and all coordinates share one total. In two cells with total 20, specifying Ridge's stock determines Vale's stock as $20 - x_1$. The two numbers cannot vary independently while the total remains fixed.

10.2 Quantity is not rate

We use q for a finite source-side quantity. It has the same stock unit as x_i . If a flow rate j is constant over a duration Δt , the quantity is

$$q = \Delta t j. \quad (10.2)$$

The rate has units X/time; multiplying by duration gives X. Once the finite quantity has been formed, the action increment is $S_e q$. Multiplying it by Δt again would count time twice.

For example, a rate of eight units per hour acting for half an hour moves four units. The same quantity is moved by a rate of two units per hour for two hours, if the declared process is constant and no other event intervenes. The endpoint stock change is the same in this ideal example. Other physical

consequences, such as equipment heating or delay, may differ in a richer model and would need representation there.

If the rate varies, quantity is the integrated rate, $q = \int j(t) dt$. One cannot replace this integral by initial rate times duration without an assumption that makes that replacement valid. The finite account begins once the event's quantity has been established; it need not pretend every physical flow is constant.

Object	Unit	Example
Stock x_i	X	18 units in Ridge
Rate j_e	X/hour	8 units per hour
Duration Δt	hour	one-half hour
Quantity q_e	X	4 units sent
Increment $S_e q_e$	X per coordinate	$(-4, 4)^T$

10.3 The exact internal-transfer identity

Let $\mathbf{1}$ be the column vector whose entries are all one. Its transpose sums a state: $\mathbf{1}^T x = \sum_i x_i$. Every lossless incidence column has one -1 and one $+1$, so

$$\mathbf{1}^T S_e = 0. \quad (10.3)$$

For a group of additive transfer increments $\delta_G = \sum_{a \in G} S_a q_a$, linearity gives $\mathbf{1}^T \delta_G = 0$. Hence

$$\mathbf{1}^T (z + \delta_G) = \mathbf{1}^T z = M. \quad (10.4)$$

This is an exact algebraic conservation identity under the declared transfer model. It does not require that the action lowers potential or that an actor acts wisely. A damaging relocation conserves the same total as a helpful one.

The identity also does not by itself ensure nonnegative stocks. A proposal to send 25 units from Ridge's initial 18 has changes $(-25, 25)$, whose sum is zero, but its endpoint is $(-7, 27)$. Conservation and feasibility are separate conditions. The negative coordinate rejects the proposal under our physical domain.

10.4 Worked ledger for Ridge and Vale

The starting state is $z = (18, 2)^T$. For $q = 4$, the source loses four and the destination gains four. The physical residual is

$$r_M = (14 + 6) - (18 + 2) = 0. \quad (10.5)$$

At $q = 8$, the state is $(10, 10)^T$, with the same zero residual. At $q = 17$, it is $(1, 19)^T$, again with zero residual. Yet their finite field values are different: 12, 16, and -4.25 , respectively, under the main potential.

This comparison is worth reading slowly. The physical total cannot determine the distributional potential. The potential cannot replace the stock ledger. They answer complementary questions about the same action. A programme that reports only one of them has left the other question unanswered.

10.5 The event boundary determines what is credited

The symbol z denotes the shared pre-action baseline. In an undisturbed illustrative event, it is simply the current state. In a later forcing-first model, an external event changes an earlier state x to z ; the actor then acts from z . Both sides of the actor's finite comparison use that same baseline.

Suppose an external conservative relocation moves the system from $(10, 10)$ to $(18, 2)$. Its total stock is still 20, but it has increased potential from zero to 16. An actor then transfers four back toward the reference, producing $(14, 6)$ and potential 4. The actor's field reduction is 12. The external increase of 16 belongs in the external account, not as an actor debit or credit created by relabelling time.

These are three declared states used in an algebraic illustration, not a simulated trajectory or scientific observation. Their purpose is to make the baseline visible. If an account instead compared the final potential 4 directly with the earlier reference potential zero and called the result the actor's effect, it would combine the external event and the action. The resulting -4 would answer another question.

10.6 No hidden creation by clipping

Suppose a proposed update produces $(-1, 21)$. Replacing -1 by zero may seem like harmless housekeeping. It changes total stock from 20 to 21. If the positive coordinate is also adjusted, a new redistribution has occurred. Either way, the operation is a physical or mathematical correction requiring an explicit account.

For the main model, infeasible actions are excluded before execution. We do not repair their successors silently. A broader model may include spill, saturation, external replenishment, or emergency corrections, but those processes must carry named increments or boundary entries. A state projection can be a useful numerical procedure; it is not automatically a law of physical conservation.

This principle is more general than water. A inventory programme that sets a negative medicine count to zero has not produced medicine, but it has changed its reported record. An account should expose the disagreement so that its cause can be investigated. Concealing it creates an apparently consistent state while destroying the information needed to explain the event.

10.7 What changes when a route loses usable stock

The central study model is lossless. To understand the boundary discipline, consider a separate accounting illustration with efficiency $\eta = 0.8$. A source sends $q = 5$, a destination receives four, and one unit enters a declared loss compartment ℓ . The extended increment is

$$(\Delta x_i, \Delta x_j, \Delta \ell) = (-5, 4, 1). \quad (10.6)$$

Its entries sum to zero. The usable-stock total falls by one, while the extended carrier account remains conserved. Alternatively, a model may record the one unit crossing an explicit external boundary. It may not leave the unit unnamed.

The loss coordinate need not enter the potential used in the main model. A physical ledger and a valuation functional have different scopes. But its absence from V does not by itself authorize a second charge. Any proposed process burden must identify exactly which physical burden is unrepresented by the finite field difference and satisfy the applicable non-duplication rule. A stock-accounting coordinate alone neither establishes nor resolves that valuation question.

A zero residual is necessary but incomplete

The numbers $(-5, 3, 2)$ also sum to zero, but they do not satisfy the declared $\eta = 0.8$ transfer rule. Check the source withdrawal, destination delivery, and loss relation separately. Conservation of a total cannot certify every component of a physical transformation.

This illustrative extension does not adopt a loss-aware runtime or modify the lossless first domain. It shows why an eventual extension needs more than changing a coefficient in the EBU equation. The physical transformation and its boundary accounts must be specified first.

10.8 A homogeneous carrier is a strong simplification

A conserved scalar is not a claim that all useful things can be added in one physical unit. One litre of water and one kilowatt-hour of electricity cannot

become two units of a single conserved stock merely because both support a clinic. Their linked use requires separate carriers and a transformation relation with consistent dimensions.

The same applies to physical degradation. Energy can remain conserved while its usefulness changes. A stock of usable doses may fall when medicines expire although their material constituents remain. The carrier named by a coordinate must therefore be explicit: usable resource under a declared quality definition, elemental mass, energy, or another measured quantity. Its conservation statement applies to that declaration and boundary.

Our ideal transfers preserve the homogeneous scalar exactly. They are a clean setting for learning local finite accounting. Extending them to heterogeneous resource chains is scientific work, not a change of labels. It must preserve what is conserved, represent relevant changes of quality, and avoid an unlicensed conversion between different EBU accounts.

10.9 Net change and gross movement

Two opposite transfers can leave stocks unchanged while moving a substantial gross quantity. Suppose the two lossless directions between Ridge and Vale are both declared, and an event sends three units each way. The increments $(-3, 3)$ and $(3, -3)$ add to zero. The endpoint is the initial state.

The stock ledger correctly reports zero net change. A separate action record correctly reports six units of gross source-side movement, three on each route. Neither description should erase the other. If the complete physical model includes process energy or equipment wear, those can change even when the represented stock increment cancels. The ideal cost-free scalar example records zero field value for the net event because its endpoints coincide; it does not make a claim about an actual pump's energy use.

This distinction prevents another kind of hidden correction. An auditor should not delete both action lines merely because their increments cancel. The lines identify what was attempted or executed, while the endpoint describes the resulting stock distribution. Physical feasibility may depend on gross use of a route or shared machine, so it too can differ between an empty event and a nonempty cancelling group.

10.10 Conservation under a change of accounting boundary

Imagine observing only Vale. A four-unit delivery raises its stock by four. Within this smaller boundary the delivery is an inflow. Now enlarge the boundary to include Ridge. The same event is an internal transfer with zero total change. The physical event did not change; the boundary did.

Write this explicitly before comparing two ledgers. A local total can rise while the global total is conserved, because the local boundary has an incoming edge. A statement such as “stock increased” is incomplete until its region and carrier are named. EBU’s local potential difference can still be calculated, but a source contribution must not disappear just because the reader looks only at the destination’s tank.

For any region, add the stock changes of its cells. Every transfer with both endpoints inside contributes minus and plus the same quantity, cancelling internally. Edges entering the region contribute positively; edges leaving contribute negatively. This cancellation is the regional version of the all-ones-vector identity. It explains why enlarging a boundary changes which terms are called external without changing the complete physical account.

In a real study, changing the boundary can also change which consequences are represented and hence which potential terms are needed. It must therefore be recorded as a model revision, not as a way to make an inconvenient burden vanish from a report. A narrower account can be useful if its scope is stated honestly.

10.11 Measurement is not a new flow

Real gauges have finite resolution, calibration limitations, and timing uncertainty. Removing numerical error-controller material from the introductory model does not make these practical issues disappear. They belong in the interpretation of observed records and in later domain validation.

If a gauge reports 18 units, the underlying quantity may not be known with infinite precision. The algebraic example treats the input as exact so that its mathematical implications are visible. An empirical application must state how the input was obtained and what uncertainty accompanies it. It must not insert measurement uncertainty into the stock equation as though uncertainty were additional water.

A measured residual can reflect an omitted flow, incompatible timestamps, sensor bias, or computational rounding. A declaration of acceptable measurement agreement must precede judging a real record. Choosing a convenient allowance after observing an inconvenient mismatch would not establish conservation. The equations here specify the physical target against which such a measurement protocol would be assessed.

10.12 Guided problem: reconstruct a branching event

Consider a new three-cell state $(9, 3, 0)^T$, with lossless edges from cell 1 to cells 2 and 3. The finite quantities are two and four. The increments are $(-2, 2, 0)^T$ and $(-4, 0, 4)^T$. Their sum is $(-6, 2, 4)^T$, and the endpoint is $(3, 5, 4)^T$.

Check the ledger three ways. Cell by cell, source withdrawal is six and total delivery is six. By totals, the starting 12 equals the ending 12. By matrix structure, each column sums to zero, so every linear combination does too. The agreement is not three independent empirical tests; it is three views of one declared algebraic identity.

Now alter only the second quantity to eight. Aggregate withdrawal becomes ten, exceeding the source's nine. The formal endpoint is $(-1, 5, 8)^T$. The total remains 12, but the state is infeasible. Do not clip its first entry or report the eight-unit delivery as executed service. The example demonstrates why a joint source check belongs before an accepted event.

Finally convert rates correctly. If the valid quantities two and four were delivered over one-quarter hour at constant rates, the rates were eight and sixteen units per hour. If those same rates acted for half an hour, the quantities would double. A duration change can therefore alter physical feasibility even when the rates appear unchanged.

10.13 Practice with answers

A lossless edge carries a constant rate of three units per hour for two-thirds of an hour. The finite quantity is two units. Starting from $(7, 1)$, the endpoint is $(5, 3)$, and total stock remains eight. Explain why writing $x' = x + \Delta t S q$ with this already integrated $q = 2$ would be wrong.

In a separate extended loss ledger, a source sends ten units at $\eta = 0.9$. Destination delivery is nine and the loss-compartment increment is one. A record showing destination eight and loss two has zero total residual but violates the stated efficiency. Identify the missing component check.

For the main model, a no-action event has $q = 0$, so $\delta = 0$ and state is unchanged. If a real observer nevertheless sees a change, at least one premise of the no-action, no-external-change illustration does not describe the observation. The response is to locate the process or measurement issue, not to declare that zero caused a nonzero transition.

Declaring a reference and a scale

11.1 A comparison needs a declared centre

A tank reading of ten units is a physical fact within a measurement protocol. Whether it represents shortage, adequate provision, or excess cannot be read from the number alone. The same quantity can support a small clinic and be inadequate for a large hospital. A model that measures deviation must therefore declare what each stock is being compared with.

Write the reference for cell i as x_i^* . The star identifies a reference value; it is not multiplication and does not mean that the value has been proved optimal. The vector x^* collects all references in the same cell order as the state. In our main example it is $(10, 10)^T$.

The reference is a model declaration whose practical meaning requires domain justification. It might be informed by engineering function, an agreed service profile, or a research design. Algebra cannot decide which interpretation is appropriate. The important first step is to record that interpretation explicitly and keep it fixed during a comparison.

11.2 Reference, observation, and equilibrium

Three ideas are easily conflated. An observed mean summarizes data. A reference states the centre used by a model. A dynamic equilibrium is a state that remains unchanged under a particular evolution rule. These objects may coincide in a carefully justified application, but their symbols do not make them identical.

If a community's water distribution is persistently unequal, its historical mean distribution may reflect that inequality. Choosing a reference equal to that mean is a substantive decision. Conversely, choosing a more balanced reference does not prove that available routes or institutions can reach it. A reference makes the question precise; it does not answer every practical question associated with it.

For the Gaussian potential taught here, x^* is the unique zero of the full quadratic when all scales are positive. That mathematical minimum is a

statement about the chosen function. It does not prove that an actor process converges to the point, that a society should adopt it, or that all living functions are satisfied there.

Three sentences with different meanings

“The mean stock was ten” describes a dataset. “The reference is ten” defines a comparison. “Ten is an equilibrium” describes a specified dynamic law. Use the sentence supported by the available definition or evidence.

11.3 Compatibility with a conserved total

The first model preserves total stock M . We therefore choose references satisfying

$$x_i^* \geq 0, \quad \sum_i x_i^* = M. \quad (11.1)$$

This ensures that the reference lies in the same nonnegative fixed-mass domain as the physical states. In the main example, $10 + 10 = 20$. A reference of $(12, 12)$ would require 24 units, which a closed 20-unit world cannot supply.

The compatibility condition is simple but scientifically important. Without it, a positive minimum deviation may reflect an impossible reference rather than poor action choices. A model might intentionally study an incompatible target in another project, but it would need to explain that problem. We do not introduce it silently here.

Compatibility is weaker than reachability. If the only physical edge runs from Vale to Ridge, then starting from $(18, 2)$ the direct transfer toward $(10, 10)$ is unavailable. If the graph is disconnected, each component preserves its own total under internal transfers. The reference must then also be compatible with those component totals to be reachable by those actions. Declaring a global total alone does not construct a route.

11.4 The scale tells us how to compare deviations

Let $\sigma_i > 0$ be a declared physical scale in the same units as x_i . The standardized deviation is

$$y_i = \frac{x_i - x_i^*}{\sigma_i}. \quad (11.2)$$

The numerator says how many stock units the cell differs from its reference. Dividing by the scale expresses that difference in units of the declared width. The result is dimensionless.

With reference ten and scale two, stock fourteen has standardized deviation two. Stock six has standardized deviation minus two. The signs distinguish above from below. Both are the same standardized distance from the reference. A quadratic potential will give them equal local burden because that symmetry is part of its definition.

The scale is not automatically a measurement error, tolerance, acceptable range, or statistical standard deviation. The symbol σ is familiar from probability, but notation is not evidence of a probability model. Here it controls the geometry of a declared potential. A later calibration could connect it to data; that connection would need to be stated and justified.

11.5 A scale is not a boundary

At $x_i = x_i^* + \sigma_i$, the standardized deviation is one. Nothing discontinuous happens there. There is no switch from safe to unsafe and no automatic prohibition. The potential and its derivative change smoothly through this point.

This distinguishes the Gaussian construction from a model that uses a flat acceptable interval with penalties outside it. The present field has one declared centre and positive curvature everywhere. Every nonzero local deviation contributes to its potential. Physical feasibility still includes nonnegative stock and any separately declared domain constraints; those are not manufactured by treating σ_i as a threshold.

If an application requires a hard storage capacity or a protected ecological condition, it must declare that physical or institutional rule separately. Calling a scale a safety certificate would give it a job its definition does not perform. Conversely, the lack of an introductory threshold term does not claim that real systems lack critical boundaries.

11.6 Worked standardization table

For one cell with $x^* = 10$ and $\sigma = 2$, the relevant values are

Stock x	Deviation $x - 10$	Standardized y	$\frac{1}{2}y^2$
2	-8	-4	8
6	-4	-2	2
8	-2	-1	0.5
10	0	0	0
12	2	1	0.5
14	4	2	2
18	8	4	8

Read the third column before the fourth. Standardization is a physical-unit conversion within the model; squaring and halving is the declared valuation shape. Separating these operations helps a reader locate where an assumption enters.

A deviation of four stock units is two scales, and its quadratic contribution is two. Doubling that physical deviation to eight makes the contribution eight, four times as large. This is a transparent consequence of the square. It is not an empirical theorem that doubling every real shortage multiplies its harm by four.

11.7 Different cells may have different scales

Consider a new illustrative reference (10,10) with scales (2,4). The main stocks remain (18,2). Ridge is four scales above reference and Vale is two scales below it. Their contributions are eight and two, for total ten.

The field now assigns a smaller contribution to Vale's eight-unit deviation than before. No physical stock changed; the declared geometry changed. This example is a sensitivity calculation across two different models. It must not be inserted into one action record as though the actor caused the difference between potential 16 and potential 10.

Unequal scales can be useful when a stock deviation has different functional significance in different places. They can also conceal unjustified weights. A narrow scale steepens the field and makes changes at that cell count more strongly. Its provenance belongs in the model definition, not in an after-the-fact explanation of why a preferred action won.

11.8 Changing physical units must not change the answer

Suppose one stock unit represents one cubic metre, and we switch to litres. Every stock, reference, scale, and finite quantity is multiplied by 1000. The standardized deviation remains unchanged:

$$\frac{1000x_i - 1000x_i^*}{1000\sigma_i} = \frac{x_i - x_i^*}{\sigma_i}. \quad (11.3)$$

The potential and its finite difference are therefore unchanged. The marginal's numerical value changes because it is now per litre rather than per cubic metre. Multiplying it by a correspondingly converted quantity restores the same dimensionless product.

If an implementation changes stocks to litres but leaves σ numerically unchanged, it has changed the model, not just the unit. Its potential becomes

much steeper. This is a useful audit test because every piece of the declaration is visible. A physical unit conversion must transform all quantities with that unit consistently.

11.9 Why a reference must remain fixed during an action

An action comparison uses one potential before and after. If the reference or scale changes between the two evaluations, the result includes a model change. A zero physical increment can then appear to create value.

For example, take a cell with stock 14, reference 10, and scale 2. Its contribution is two. If an editor moves the reference to 14 without moving stock, the contribution becomes zero. The number fell by two, but no physical action supplied that reduction. The field definition changed.

References can be revised in a research programme or calibrated between declared epochs. Such revisions need an explicit version boundary and an account of the redefinition. They cannot be credited to an actor as though the same physical field had improved. The same principle applies to a scale revision or to adding and removing potential terms.

11.10 Changing cell granularity changes the declaration

Suppose one well-mixed tank has stock x , reference x^* , and scale σ . An editor divides it into two bookkeeping cells with equal half-stocks and half-references. It is tempting to copy the old scale into both cells and assume the sum of their potentials still means the same thing. It does not.

With copied scales, each deviation is half the original, and the two squared contributions sum to half the original potential. If each new scale is instead set to $\sigma/2$, each standardized deviation equals the old one, and their total potential doubles. Neither result is a physical consequence of moving water; it comes from changing the accounting representation.

If the halves are constrained always to remain equal and the aim is to preserve this particular aggregate quadratic exactly, choosing each scale as $\sigma/\sqrt{2}$ makes the two contributions sum to the original. That is a conditional algebraic mapping, not a universal rule for spatial refinement. Once the halves can vary independently, the finer model can represent an internal imbalance that the original one-cell state could not see.

The example shows why cell boundaries, references, and scales should be calibrated together. Merely increasing the number of cells is not a neutral way to gain apparent precision. A change of granularity may reveal useful

new physical structure, but it also changes the field's meaning and must be declared.

11.11 The reference is not a quota for each transaction

A ten-unit reference at Vale does not mean that every delivery should contain ten units. The reference is a state coordinate, whereas an action quantity is a change. If Vale already contains six, adding four reaches ten; if it contains nine, adding four reaches thirteen. The same delivery has different consequences at the two baselines.

Nor does a cell's reference assign ownership of that stock to a particular actor. The physical index identifies the represented place. Ownership and permission require their own maps and rules. Keeping those objects separate allows a potential to be studied without pretending that an equation has decided all institutional arrangements.

Finally, a reference at one time does not guarantee adequacy at every future time. A continuing-life application may have changing demands and carriers. Such extensions require a time-varying model or additional physical state. They cannot be simulated simply by shifting the reference after an outcome and reporting the resulting lower potential as earned improvement.

11.12 A reference document that a reader can audit

A useful declaration identifies the carrier, cell boundary, stock unit, reference value, scale, and provenance for each coordinate. It states the total-mass compatibility condition and whether the action graph can connect the relevant places. It also states the interpretation intended for the potential: a standardized model deviation rather than a complete measure of welfare.

For teaching, the provenance is openly illustrative. We choose (10,10) and (2,2) to make arithmetic readable. These are not estimated parameters or a registered scientific configuration. In an empirical application, the same fields would need evidence and review. The algebra does not change its standard of precision; the origin and adequacy of its inputs become additional questions.

The document should also distinguish fixed parameters from state variables. If carrying capacity, demand, or another functional characteristic is allowed to change physically, it may need its own represented coordinate. Quietly editing a reference whenever conditions change can make a poor process appear successful by moving its target after the fact.

11.13 Guided problem: asymmetric needs

For a separate two-cell exercise, declare reference (12, 8), scales (2, 2), total stock 20, and current state (18, 2). Both coordinates are six units from reference, or three scales. Potential is $\frac{1}{2}(3)^2 + \frac{1}{2}(-3)^2 = 9$.

A lossless six-unit transfer reaches (12, 8), the declared reference. An eight-unit transfer reaches (10, 10), leaving deviations (−2, 2), each one scale, and potential one. Equal raw stocks are therefore not the minimum of this changed model. The reference embodies different declared functional stocks at the two cells.

Do not conclude that either reference is ethically correct from the calculation. The example establishes the mathematical dependence of value on the declaration. An author who changes from equal to asymmetric references must explain the domain reason, rather than claim that the equations discovered it.

11.14 Guided problem: incompatible mass

Suppose total stock is 20 and someone proposes reference (12, 12). The reference lies outside the fixed-mass domain. With equal scales, the best balanced feasible point is (10, 10), but it still has a positive deviation from the proposed reference. In fact each coordinate differs by minus two and contributes one-half at scale two, for total one.

This is not a result showing that an actor failed to recover to a reachable reference. The reference itself demands more stock than the closed model contains. The correct first-model response is to require a compatible reference or explicitly change the physical boundary and problem. The example explains why compatibility is checked before controller performance is interpreted.

11.15 Misconceptions and practice

“Sigma is the noise level.” Only if a separate declaration connects it to measurement uncertainty. In this model it is a physical reference scale.

“One scale from the reference is a safe boundary.” No such boundary is implied. It is a location on a smooth quadratic.

“Gaussian means actual stocks follow a normal distribution.” The Gaussian shape motivates the potential geometry. An actual distribution requires a stochastic process and evidence or a mathematical derivation for that process.

“A compatible reference must be reachable.” Conservation compatibility removes one obstruction; graph structure and allowed actions may create others.

For practice, take reference 15, scale 3, and stocks 9, 15, and 21. Their standardized deviations are minus two, zero, and two; their local potentials are two, zero, and two. Now convert every physical quantity to a unit one-tenth as large. The numerical stocks and scale become ten times larger, while all three potentials remain unchanged.

For a three-cell reference (4, 6, 10) and total stock 20, check compatibility. Then imagine no edge connects the first two cells to the third, and the third begins with eight. The reference is globally compatible but unreachable by internal transfers within the two disconnected components. Explain the missing condition in words before attempting any controller design.

What to carry forward

The reference names the centre of comparison; the positive scale measures deviations in physical units. Neither is automatically an empirical mean, uncertainty bound, or equilibrium. Keep them fixed within an action account and compatible with the declared physical domain.

We can now assemble these declarations into a potential and see why a Gaussian landscape gives especially transparent finite mathematics.

Homeostasis as a Gaussian landscape

12.1 From comparison to a function

A function is a repeatable rule that maps an input to an output. Our input is a physical state. The output will be a nonnegative number measuring standardized deviation from the declared reference. Calling this number a potential gives it a mathematical role: its differences and derivatives can be evaluated consistently across actions.

For cell i , define

$$V_i(x_i) = \frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2, \quad \sigma_i > 0. \quad (12.1)$$

The global potential is

$$V(x) = \sum_{i=1}^N V_i(x_i). \quad (12.2)$$

The word global describes the argument of this mathematical function. It does not require every actor to read the whole state. A later locality identity shows exactly which terms a local action needs.

The factor one-half is a convenient normalization. It makes the derivative especially simple. Omitting it would create a consistently rescaled model if every dependent equation were changed, but mixing the two conventions would change numerical quotes. We retain the one-half throughout this edition.

12.2 Why the landscape is called Gaussian

Consider the unnormalized shape

$$g_i(x_i) = \exp \left[-\frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2 \right]. \quad (12.3)$$

It has the familiar Gaussian bell shape, with its highest value at the reference. Taking the negative logarithm gives $-\log g_i(x_i) = V_i(x_i)$. The potential is therefore the relative negative-log geometry of that shape, with its minimum set to zero.

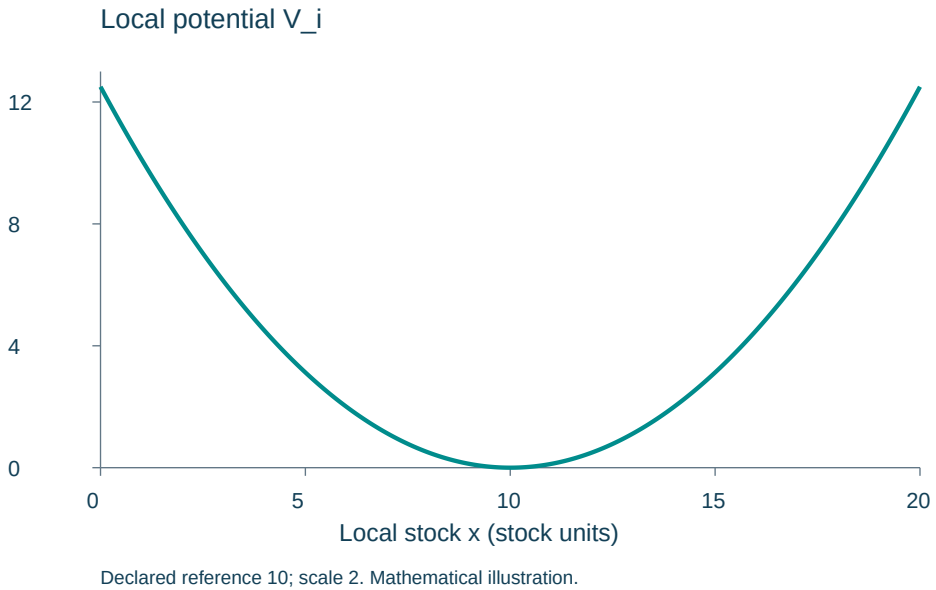


Figure 12.1: The one-coordinate quadratic penalty with reference ten and scale two. Its height is a declared deviation measure, not an amount of physical stock.

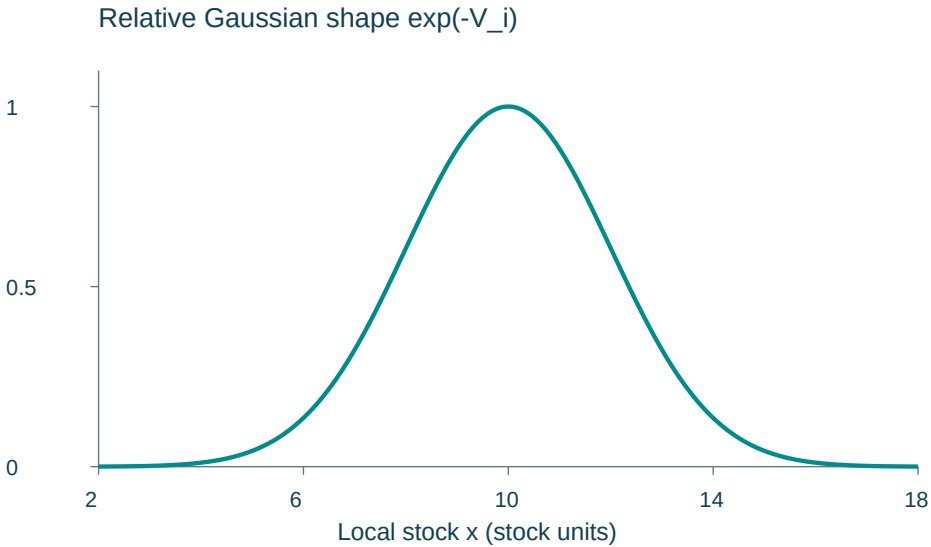
This construction is a definition, not a claim that nature universally follows a Gaussian law. No probability density has yet been fitted to stock observations. No random mechanism has been specified that makes the system's stationary distribution equal to g_i . The name explains the shape used for deviation; it does not supply missing dynamics or evidence.

On the fixed-mass nonnegative domain, coordinates are also constrained together. Even though the quadratic is a sum of one-coordinate terms, the actual stocks cannot vary independently: their sum is fixed. A product of Gaussian shapes in an unconstrained mathematical space is not automatically a product probability law for the physical system.

12.3 The bowl and its centre

Each local potential is a smooth quadratic bowl. It is zero at x_i^* , positive elsewhere, and symmetric about that centre. Moving one scale above or below the reference gives $V_i = 1/2$. Moving two scales gives $V_i = 2$. Moving four scales gives $V_i = 8$.

The symmetry is an explicit modelling choice. It does not assert that real deficit and excess have equal biological consequences. A clinic with too little water may face different problems from one with too much. The first Gaussian



Relative shape with peak 1; not an observed probability distribution.

Figure 12.2: The exponential shape associated with the quadratic. This relative shape does not itself supply a probability model or prove that observations are normally distributed.

model isolates a transparent geometry for studying an accounting and action process. A later asymmetric or multi-coordinate field would need its own declaration and review.

Definition and immediate consequence

With fixed positive scales, $V(x) = \frac{1}{2} \sum_i ((x_i - x_i^*)/\sigma_i)^2 \geq 0$. Equality holds exactly when every represented coordinate equals its reference. This follows from a sum of nonnegative squares. It is not a statement that every unrepresented human or ecological need is satisfied.

12.4 Equal total, unequal potential

Return to the main world. At (18, 2), standardized deviations are (4, -4) and potential is 16. At (14, 6), they are (2, -2) and potential is 4. At (10, 10), both are zero. Each state contains the same total stock, 20.

The landscape distinguishes configurations of a conserved quantity. It does not create or destroy the quantity it evaluates. This is why a conserved physical ledger can coexist with a changing potential. The total-mass invariant

concerns a sum of stocks; the potential concerns squared departures from a reference.

The reflected state (2, 18) again has potential 16. It is different from (18, 2), although the scalar potential is equal. A single potential value does not reconstruct the state. If an account records only V , it loses the information about which place is above and which below its reference.

12.5 Reading the quadratic along the physical domain

For two cells with total 20, every physical state can be written $(r, 20 - r)$, with $0 \leq r \leq 20$. Under equal references ten and scales two,

$$V(r, 20 - r) = \frac{1}{2} \left(\frac{r - 10}{2} \right)^2 + \frac{1}{2} \left(\frac{10 - r}{2} \right)^2 \quad (12.4)$$

$$= \frac{(r - 10)^2}{4}. \quad (12.5)$$

This one-dimensional curve is the potential restricted to the physically allowed line. Its centre is $r = 10$. The two endpoints $r = 0$ and $r = 20$ each have potential 25. Every state in this particular closed world therefore has a finite potential between zero and 25.

The boundedness follows from the domain before an EBU-guided process is tested. A comparison between two action rules cannot claim success merely because both remain bounded in this finite simplex. Useful questions concern how often they occupy large-deviation states, whether they recover after forcing, whether they cycle, or whether affordability makes them refuse useful opportunities.

12.6 The marginal: a local slope

The local marginal is the derivative of the potential with respect to its stock:

$$\mu_i(x_i) = \frac{dV_i}{dx_i} = \frac{x_i - x_i^*}{\sigma_i^2}. \quad (12.6)$$

Below reference, the marginal is negative: adding a small amount lowers local potential. Above reference, it is positive: adding more raises local potential. At the reference it is zero.

Derive the formula rather than memorize it. Write $V_i = (x_i - x_i^*)^2 / (2\sigma_i^2)$. The reference and scale are fixed. Differentiating the square gives $2(x_i - x_i^*)$; the two cancels the one-half. The remaining denominator is σ_i^2 .

The potential is dimensionless in this model. The marginal therefore has units X^{-1} , or potential units per stock unit. It is neither a stock nor a complete

action value. Multiplying by a small stock increment gives the first-order potential change. A finite increment also encounters curvature, which we derive in Chapter 14.

12.7 A gentle derivative laboratory

For one cell with reference ten and scale two, $V(x) = (x - 10)^2/8$. Starting at 14, add a small increment h . Direct expansion gives

$$V(14 + h) - V(14) = h + \frac{h^2}{8}. \quad (12.7)$$

Divide by $h \neq 0$: the ratio is $1 + h/8$. As h approaches zero, the ratio approaches one. That limit is the marginal at 14. The formula also shows what the derivative leaves out for a finite change: the quadratic term $h^2/8$.

At stock six, the same calculation gives marginal minus one. Adding a small positive amount lowers potential. At ten, the marginal vanishes, but a finite nonzero displacement raises the potential by its square divided by eight. Zero initial slope is therefore not permission to move a finite distance without effect.

This is the bridge between an intuitive landscape and calculus. A slope is a nearby rate of change. It describes the tangent at one point. The finite EBU account compares two actual endpoints within the declared model.

12.8 The gradient and the Hessian

Collect the marginals into the gradient vector,

$$\mu(x) = \nabla V(x) = \left(\frac{x_1 - x_1^*}{\sigma_1^2}, \dots, \frac{x_N - x_N^*}{\sigma_N^2} \right)^T. \quad (12.8)$$

The symbol ∇ , read “nabla,” denotes the vector of partial derivatives. A partial derivative varies one coordinate while holding the others fixed in the mathematical function. Physically allowed directions can then be built from combinations that respect conservation.

The Hessian records derivatives of those slopes:

$$H = \text{diag} \left(\frac{1}{\sigma_1^2}, \dots, \frac{1}{\sigma_N^2} \right). \quad (12.9)$$

“Diagonal” means the off-diagonal entries are zero. Each local slope changes at the constant rate $1/\sigma_i^2$ with its own stock. The Hessian is positive definite because all scales are positive. For any nonzero vector d , $d^T H d = \sum_i d_i^2 / \sigma_i^2 > 0$.

The matrix form of the potential is

$$V(x) = \frac{1}{2}(x - x^*)^T H(x - x^*). \quad (12.10)$$

It compresses the sum of standardized squares. It does not replace their physical interpretation. The row-vector, matrix, and column-vector multiplication expands back into the same local contributions.

12.9 Ellipses, distance, and unequal scales

In an unconstrained two-coordinate drawing, points with a fixed potential satisfy

$$\frac{(x_1 - x_1^*)^2}{\sigma_1^2} + \frac{(x_2 - x_2^*)^2}{\sigma_2^2} = 2V. \quad (12.11)$$

These curves are ellipses. With equal scales they are circles in the chosen physical coordinate units. A larger scale stretches the contour along its coordinate: a larger physical deviation is needed to produce the same contribution. This is the geometric meaning of scale weighting.

The physical fixed-mass states occupy only a slice of this drawing. For two cells they lie on a line of constant sum. The field evaluated along that line is the restriction of the same quadratic, not a second potential. A picture of a circular bowl should therefore not suggest that every direction in the plane is a permitted physical motion.

The quantity $\sqrt{2V}$ is the Euclidean length of the standardized deviation vector. It can be described as a distance in these standardized coordinates. It is not geographical distance, and it is not itself the signed action value. Taking its square root changes finite differences and derivatives. We retain V as the declared potential even when the distance picture helps intuition.

12.10 Diagonal geometry does not mean independent outcomes

The absence of off-diagonal Hessian entries says that the declared local marginal of one coordinate does not directly depend on another coordinate in the unrestricted function. Physical conservation still couples stocks, and the graph couples possible actions. If Ridge sends to Vale, both coordinates change together.

Two actions that share Ridge can therefore interact in their combined value. Their increments overlap in Ridge's squared deviation. This does not require a correlated multivariate Gaussian or a non-separable potential. It follows from applying a separable nonlinear function to a sum of increments.

A richer potential might use a symmetric positive-definite matrix with off-diagonal entries to represent declared cross-coordinate dependence. That would be another model. Its affected-factor neighbourhood, calibration, and interpretation would require review. We do not introduce it merely to obtain an attractive interaction effect. The diagonal model already has enough structure to distinguish local potential geometry from action overlap.

12.11 A bounded domain gives a bound before any experiment

Because nonnegative stocks sum to M , every coordinate lies between zero and M . Its absolute deviation from a fixed reference cannot exceed the larger of x_i^* and $|M - x_i^*|$. Hence

$$V(x) \leq \frac{1}{2} \sum_i \frac{\max\{(x_i^*)^2, (M - x_i^*)^2\}}{\sigma_i^2}. \quad (12.12)$$

For finitely many cells and fixed positive scales, the right-hand side is finite. It is a convenient bound, not necessarily the tight maximum because each coordinate cannot attain its individual extreme simultaneously under the common total.

This distinction matters to scientific interpretation. If a future comparison asks whether EBU prevents unbounded growth of this potential, the physical domain has already answered that question for every admissible policy. A meaningful study must ask something that the domain has not guaranteed: recovery after a shock, frequency of substantial deviation, distribution across cells, cycling, or refusal. The theorem is useful because it narrows the question honestly.

12.12 Convexity without mystery

A quadratic bowl with positive curvature is convex. Geometrically, the straight chord joining two points on its graph lies on or above the graph. Algebraically, the tangent at a starting point underestimates the potential at a finite displaced point by a nonnegative quadratic amount.

This will matter for valuation. If the initial slope predicts an improvement, multiplying that slope by a large action quantity overstates the finite field improvement. The potential curves upward relative to its tangent. Exact endpoint comparison automatically includes that curvature.

Convexity is a property of the potential, not a promise about action choices. A random actor can choose a direction that raises a convex potential. A capacity rule can permit cycles. A graph can block a path toward the minimum. A

simple bowl is mathematically helpful precisely because these separate mechanisms remain visible; the shape does not decide them by itself.

12.13 Why the potential is not a controller

Imagine drawing this landscape on paper and placing a marker at $(18, 2)$. The drawing tells us the value at every admissible state. It does not move the marker. To move it, one must specify a physical action and an actor or dynamic rule that chooses that action.

A gradient-flow law would be one possible added rule. A largest-value selector would be another. The prospective programme instead asks what happens under a separately declared random choice among feasible and affordable options. These mechanisms can use the same potential while producing different behaviour. This book does not introduce an optimizing or restoring law just because the picture resembles a bowl.

Even the phrase “downhill” should be read carefully. It identifies a decrease of the declared potential along a direction. It does not make the system slide there by itself. The distinction protects the later experiment from becoming a demonstration of a recovery rule placed into the plant in advance.

12.14 Do not replace the function while keeping its name

The bounded function $1 - \exp(-V)$ can be formed from the quadratic. It is not the same potential. Its derivative multiplies the quadratic gradient by $\exp(-V)$, and its finite differences give different values. Large deviations become compressed toward one.

Such a transformation might be studied under another declared model, but it cannot be substituted merely because both expressions use a Gaussian exponential. The active definition is the quadratic negative-log potential itself. Field, marginal, curvature, and finite account must all be derived from that same definition.

Likewise, dropping a coordinate, adding a weight, or changing a scale changes the functional. Precise notation prevents an appealing diagram from concealing a substantive amendment. Every quantitative claim in the main example traces back to Equations 12.1 and 12.2.

12.15 Guided problem: three cells

In a new three-cell illustration, choose reference $(4, 6, 10)$, scales $(2, 2, 4)$, and state $(2, 8, 10)$. Total stock and reference total are both 20. Standardized deviations are $(-1, 1, 0)$, so $V = 1$.

The marginals are $(-1/2, 1/2, 0)$. Moving a small lossless quantity from cell 2 to cell 1 has direction $(1, -1, 0)^T$. Its initial potential derivative is $(-1/2)(1) + (1/2)(-1) = -1$, so the direction initially lowers potential. A two-unit transfer reaches the reference, giving exact field reduction one. Initial slope times quantity would predict two, twice the available reduction.

This calculation uses a new reference and scale declaration, identified openly. It reinforces the same mathematical structure without quietly changing the parameters of Ridge and Vale. The reader can check each coordinate independently before adding the total.

12.16 One potential value, several physical descriptions

The scalar potential deliberately compresses a state vector. That compression is useful for finite identities, but it loses information. In the symmetric two-cell fixture, the same value four occurs at $(14, 6)$ and $(6, 14)$. These states place the shortage at different locations. A clinic at Vale experiences them differently even though the total standardized squared deviation is equal.

In a larger graph many more states can share one value. They can differ in which cell is far from reference, whether an available edge points toward relief, and how much stock a prospective actor can access. Equal potential does not imply equal feasible menus, equal service, or equal future behaviour.

For this reason, a future study should retain the state and relevant distributional diagnostics alongside V . The finite account uses the scalar because its changes have a precise meaning. Interpretation should not discard the coordinates that gave it that meaning. A contour of equal potential is a mathematical level set, not a class of socially or physically interchangeable worlds.

12.17 Practice and reflection

With reference ten and scale two, calculate local potential and marginal at stocks 4, 10, and 16. The potentials are 4.5, 0, 4.5; the marginals are $-1.5, 0, 1.5$. Explain why a nonnegative potential can have a negative derivative.

For the main two-cell state, verify $\mu = (2, -2)^T$ and $H = \text{diag}(1/4, 1/4)$. Then explain why equal total stock does not imply equal potential, and why equal potential does not imply equal state. Use $(18, 2)$, $(10, 10)$, and $(2, 18)$ to support the answers.

Finally, write one statement that this potential establishes and one statement it leaves open. A valid pair is: “Its value is zero exactly at the declared

reference” and “The proposed actor process may or may not spend much time near that reference.” The first follows from the definition; the second needs a specified process and further analysis.

From a marginal to an edge field

13.1 What the word field means here

A map of local potential values is not yet an action account. To understand a proposed transfer, we need to know how the potential changes in the direction of that transfer. The marginal at each cell tells us how its local contribution responds to a small addition. The edge field combines the source and destination responses with the physical signs of the action.

In this book, the edge field is a directional derivative of a declared potential. It is sometimes called a force by analogy with potential-based physics. Its unit here is potential per stock, not newtons. We derive its formula from the action column and the chain rule; no gravitational or electromagnetic law is assumed.

This careful use of the word field matters. It gives us a useful mathematical structure without claiming that the potential is a newly discovered physical interaction. A stock distribution and a declared Gaussian reference determine the values. An actor or physical law still has to specify what happens next.

13.2 The direction of one lossless transfer

For an edge $e = (i \rightarrow j)$, the state after a finite quantity q is $z + S_e q$. Its source coordinate is $z_i - q$; its destination coordinate is $z_j + q$. To find the initial response, imagine changing q by a very small amount while holding the initial baseline fixed.

The source changes at minus one stock unit per unit sent, and the destination changes at plus one. The chain rule gives

$$\left. \frac{d}{dq} V(z + S_e q) \right|_{q=0} = -\mu_i(z_i) + \mu_j(z_j). \quad (13.1)$$

Define the edge field with the opposite sign, so a burden-reducing direction has a positive field:

$$f_e(z) = \mu_i(z_i) - \mu_j(z_j) = -\mu(z)^T S_e. \quad (13.2)$$

This sign convention is the bridge to signed EBU. The potential change is negative when a small action improves the field. The corresponding field improvement is positive.

13.3 Deriving the equation one term at a time

The source contribution to the derivative is $\mu_i dx_i/dq$. Because $dx_i/dq = -1$, this is $-\mu_i$. The destination contribution is $\mu_j dx_j/dq = +\mu_j$. Every unaffected coordinate has derivative zero and contributes nothing. Adding these terms gives the potential derivative; negating it gives f_e .

At the main state (18, 2), the marginals are 2 and -2 . Removing a small amount from Ridge lowers its local potential, because Ridge is above reference. Adding that amount to Vale also lowers its local potential, because Vale is below reference. Thus

$$f_e = 2 - (-2) = 4. \quad (13.3)$$

The two beneficial endpoint effects add. An infinitesimal sent quantity dq changes potential to first order by $-4dq$.

The field does not mean that every finite quantity earns $4q$. It is the starting slope. As Ridge loses stock and Vale gains it, both marginals move toward zero. The exact finite calculation must include that change.

A small quantity and its changing slope

Starting at (18, 2), sending $q = 0.1$ gives exact field reduction $4(0.1) - (0.1)^2/4 = 0.3975$. The initial-slope estimate is 0.4. Their difference, 0.0025, comes from curvature. It is a known algebraic term in this model, not an unexplained numerical discrepancy.

13.4 Four endpoint patterns

The signs of the endpoint marginals help interpret the direction, but the difference determines the result. An above-reference source has positive marginal. A below-reference destination has negative marginal. Together they give positive field. If both are below reference, removing source stock is locally harmful while adding destination stock is locally helpful; their magnitudes compete.

If both are above reference, withdrawal helps the source while delivery worsens the destination. Again, the field compares the two effects. A below-reference source sending to an above-reference destination has negative field because both endpoint contributions raise burden.

Source	Destination	Initial effect
Above reference	Below reference	Both changes lower local burden.
Below reference	Below reference	Source harm competes with destination relief.
Above reference	Above reference	Source relief competes with destination harm.
Below reference	Above reference	Both changes raise local burden.

These statements concern the declared potential. They do not decide whether the transfer serves an urgent need or violates another constraint. An application can require additional measurements of service and feasibility.

13.5 Raw stock is not the field

Two cells can contain the same stock and have different marginals because their references or scales differ. In a separate two-cell illustration, let both stocks be eight, references be ten and six, and scales both two. The marginals are $-1/2$ and $+1/2$. Sending from the second cell to the first initially lowers potential, although the raw stocks are equal.

The reference totals and stock totals in this example are both 16, so the comparison is compatible with the closed domain. The field expresses deviation relative to function as declared by the model. It is not simply a pressure to equalize every raw number.

Unequal scales provide another distinction. Keep the main stocks and references but change scales to $(2, 4)$ for this sensitivity illustration. Marginals become 2 and $-1/2$, so the field is 2.5, rather than four. The physical action column is unchanged. The model's valuation geometry has changed, and the record must identify that new declaration.

13.6 Field along a finite path

For the Gaussian potential, the marginal changes linearly with stock. Along $z + S_e q$,

$$\mu_i(q) = \frac{z_i - x_i^* - q}{\sigma_i^2}, \quad (13.4)$$

$$\mu_j(q) = \frac{z_j - x_j^* + q}{\sigma_j^2}. \quad (13.5)$$

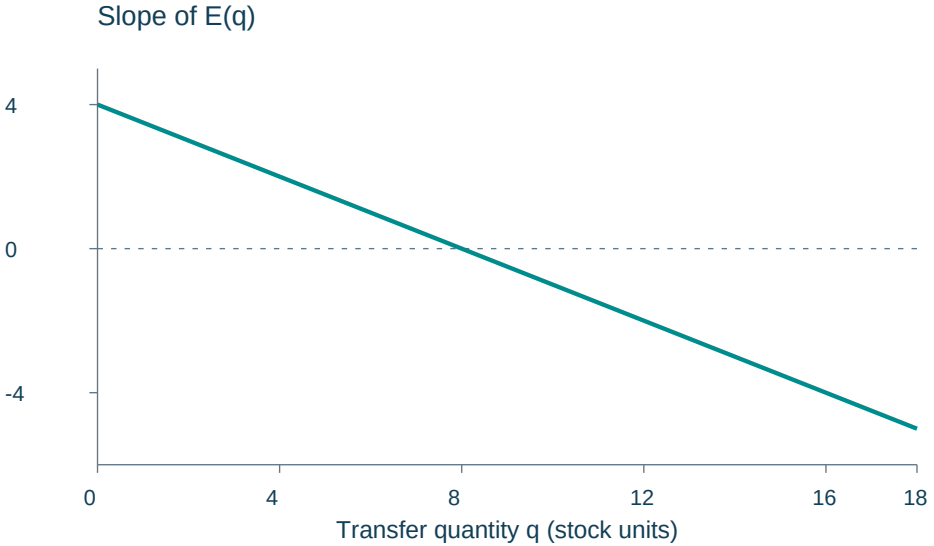
Their difference is

$$f_e(q) = f_e(0) - q \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (13.6)$$

The slope of the field along the action is negative because both positive curvature contributions add. In the main example this becomes $f_e(q) = 4 - q/2$.

At $q = 0$, the field is four. At $q = 4$, it is two. At $q = 8$, it is zero. Beyond eight, it is negative. These are field values at points along the same declared path, not different schedules chosen after the result. The whole function follows from the initial state, reference, scales, and action direction.

The quantity eight reaches the reference for both cells in the symmetric fixture. That coincidence is special to the compatible two-cell setup. In a larger system, equality of marginals on one edge does not imply that every coordinate equals its reference.



The slope crosses zero at $q=8$; it does not select an action.

Figure 13.1: The instantaneous field along the running transfer falls as the quantity increases. Integrating it, rather than multiplying its starting value by every finite quantity, gives the exact finite difference.

13.7 Zero field deserves careful reading

A zero edge field says the initial directional derivative is zero. It does not say the direction has zero finite cost. In the Gaussian model the curvature along every nonzero transfer direction is positive. Therefore a finite move from a

point with $f_e = 0$ increases the potential, unless the net physical increment is zero.

At the reference (10, 10), both marginals vanish. Sending two units gives (8, 12). Each endpoint is one scale from reference and contributes one-half, so total potential rises to one. The initial field was zero; exact finite EBU is minus one. The difference is fully explained by the quadratic term.

A zero field can also arise from equal nonzero endpoint marginals in a larger or differently constrained world. It means there is no first-order improvement along that edge. It does not certify that the complete represented world is at its minimum or that its real needs are satisfied.

13.8 The graph form of the field

Collect the edge fields into a vector f . With the incidence matrix S , the compact expression is

$$f = -S^T \mu. \quad (13.7)$$

The transpose turns each edge column into a row that selects the source and destination marginals. If there are N cells and m edges, S^T has shape $m \times N$, so the result has one field value per edge.

For a separately declared three-cell chain with marginals $(2, 0, -1)^T$ and edges $1 \rightarrow 2$, $2 \rightarrow 3$, the fields are two and one. The field vector reports the two directional slopes. It does not choose the first edge, allocate stock, or decide whether both actions can occur together.

This is an example of local computation with global notation. The formula is compact for a proof, but each edge needs only the relevant endpoint marginals under the separable model. A manager may assemble the whole vector for diagnostics; that does not authorize an actor to read a global score or a future outcome.

13.9 Reverse directions and cycles

If the reverse lossless edge exists, its column is $-S_e$ and its field at the same state is $-f_e$. At the main baseline, Ridge-to-Vale has field four and Vale-to-Ridge has field minus four. At a later state their values must be reevaluated. A reverse action performed after a forward action does not use the old reverse marginal.

Around a directed cycle of lossless edges evaluated at one common state, marginal differences cancel. For a three-cell cycle,

$$(\mu_1 - \mu_2) + (\mu_2 - \mu_3) + (\mu_3 - \mu_1) = 0. \quad (13.8)$$

This is an identity for initial fields at that state. It does not claim that arbitrary unequal finite quantities around the cycle have zero value, nor that the physical cycle is free of real process burden. The complete finite state increment and any declared process costs still determine the account.

If equal simultaneous quantities circulate around a lossless closed cycle, the net stock increment can be zero. The field component is then zero because the endpoint equals the baseline. Actual circulation may use energy, time, or equipment in a fuller model. Those carriers must be represented before claims about their burden are made.

13.10 The field around a shared source

Consider a separate three-cell fixture with state $(14, 3, 3)$, reference $(8, 6, 6)$, and equal scales two. Both state and reference sum to twenty. The marginals are $(1.5, -0.75, -0.75)$. The source-to-each-destination field is therefore 2.25.

The equality of the two starting fields says the destinations are symmetric under this declaration. It does not allocate a source quantity to either actor. If one actor sends two units, the source marginal falls by one-half and its destination marginal rises by one-half. Recomputing the other edge at the resulting state gives a smaller source contribution even though its own destination has not changed.

This is why several actors cannot each treat a shared initial field as an unlimited personal resource. Their actions use one physical coordinate. The correct joint finite value includes that shared curvature. A simultaneous attribution convention can divide a joint value under declared path semantics, but it cannot create several independent copies of the source's starting state.

An experiment should therefore distinguish equal local fields, equal finite quotes, equal physical permissions, and equal selection probabilities. They coincide only under additional symmetry conditions. For example, unequal quantity menus can produce unequal group frequencies even when endpoint parameters are identical. These are protocol properties, not new forces in the potential.

13.11 A field can be useful even when it is not followed

A thermometer remains informative when nobody turns the heater on. Similarly, the derivative of a potential can diagnose the initial effect of a candidate action even when the actor chooses differently. The value's explanatory role does not require an automatic law of obedience.

This makes negative outcomes interpretable. If a permitted and affordable random action moves uphill, the field account can record the increase without treating it as a malfunction of the equation. The question for later research is how the complete policy changes the distribution of such events over time. That question needs the actor process, forcing, and account rules in addition to the local derivative.

13.12 An explicitly separate loss-aware boundary illustration

For a declared lossy transfer, the source changes by $-q$ and the usable destination by $+\eta q$. If the local potential covers those two usable-stock coordinates, the chain rule would give $f = \mu_i - \eta\mu_j$. If a loss coordinate enters the potential, its derivative contribution must also be included. The general formula is always minus the gradient dotted with the complete action direction.

For example, with $\mu_i = -1$, $\mu_j = -3$, and $\eta = 1/4$, the two-coordinate expression is $-1 + 3/4 = -1/4$. Source harm exceeds delivered relief. A lossless expression would give two and answer the wrong physical question. This illustration changes the transformation, so it is outside the main lossless domain and does not authorize an implemented lossy study.

The lesson is structural. Derive the field from the actual declared state change. Do not derive a convenient lossless slope and then add an arbitrary adjustment elsewhere to compensate. Physics determines the directional derivative's coefficients.

13.13 Where the Onsager connection comes from

Lars Onsager's two 1931 papers studied coupled irreversible processes, including heat conduction, electrical transport and diffusion. Their reciprocal relations connect response coefficients under microscopic-reversibility assumptions. The theory concerns thermodynamic forces and fluxes in a linear-response setting; it is not a general statement that every system follows a gradient of any function one chooses [9, 10].

The useful idea for EBU is to distinguish a driving difference from the strength of response. Think of two otherwise similar routes with the same endpoint marginal difference. One route can carry a large flow; the other responds slowly. The field difference and the route's responsiveness describe different things. This is why the original EBU architecture introduced an Onsager-type force–mobility–flux structure rather than treating the marginal itself as a transported quantity.

13.14 Force, mobility and flux in the EBU model

For a lossless edge, the already derived field force is $f_e = \mu_i - \mu_j$. The earlier EBU controller used

$$J_e = M_e[f_e - \theta_e]_+, \quad M_e > 0, \quad \theta_e \geq 0. \quad (13.9)$$

Here J_e is a flow rate and M_e is mobility: the declared response per unit force. The positive-part operation $[a]_+ = \max(a, 0)$ permits flow only in the declared edge direction and only above the activation threshold θ_e . This edge activation threshold is not an Allee threshold or a lower/upper homeostatic stock band. It belongs to this optional controller, not to the Gaussian potential.

In the dimensionless Gaussian normalization, force has units X^{-1} , mobility has units X^2/time , and their product has units of stock per time. A finite duration gives quantity $q = \Delta t J_e$. Physical permission still has to check that quantity; high mobility cannot create stock that the source does not hold.

At the main baseline the force is four. As a separate controller illustration, choose mobility one-half and activation threshold zero. The requested rate is two stock units per time unit. Over half a time unit it requests one stock unit. The exact finite field value of that quantity is $15/4$, not the initial-force estimate four. Mobility sets the response rate; the finite equation values the complete accepted change. These illustrative choices are not a registered experimental configuration.

13.15 Why borrow this structure, and where does it stop?

The structure makes a conditional descent argument possible. If there is no external drive and the physical evolution is exactly $\dot{x} = SJ$, then the chain rule gives $\dot{V} = -\sum_e f_e J_e$. For the declared threshold-mobility response, every active product is nonnegative, so that particular continuous-time law cannot increase the potential. Part II derives this carefully and explains the extra conditions needed for finite timesteps.

This is a reason to study the controller, not evidence that economic actors obey it. The EBU potential is not automatically thermodynamic free energy, and its decrease is not automatically measured entropy production. Nor does using diagonal mobilities demonstrate Onsager reciprocity for a real coupled social or ecological system. The original physical analogy motivates a mathematical architecture; the declared model still has to stand on its own.

13.16 Why the actor law remains an explicit choice

It would be possible to propose a flow proportional to positive field, or to choose whichever edge has the largest field. Either choice would add a con-

troller. It might be useful for a different research question, but it would not follow from Equation 13.2 alone.

The present prospective programme keeps the valuation rule distinct from the actor's choice. Feasible candidate actions can receive positive, zero, or negative values. A separately declared capacity policy can filter affordability. Random selection among remaining options is another declared step. Replacing that selection by largest field or largest finite EBU would change the experiment's subject.

This distinction also prevents an easy but circular success claim. If a plant is given autonomous restoring flow, observing movement toward the reference does not by itself demonstrate an economic accounting mechanism's contribution. The field may inform analysis without being a hidden force that physically compels the actors.

13.17 Guided problem: equal field, unequal finite behaviour

Two separate hypothetical edges have the same initial field four. Edge A connects cells with scales two and two, so its curvature along quantity is $1/4 + 1/4 = 1/2$. Edge B connects cells with scales one and one, so its curvature is two. Their initial tangents match, but their fields fall at different rates as quantity increases.

For a quantity of two, the exact field reductions, derived fully next chapter, are

$$F_A(2) = 4(2) - \frac{1}{2}(1/2)(2)^2 = 7, \quad (13.10)$$

$$F_B(2) = 4(2) - \frac{1}{2}(2)(2)^2 = 4. \quad (13.11)$$

Both calculations assume compatible declared states that give initial field four and permit the quantity. The example isolates curvature; it is not a recommendation between two real services. It shows why the starting field cannot replace a finite schedule.

Now ask what would happen at quantity four. Edge A gives $16 - 4 = 12$, while B gives $16 - 16 = 0$. The same initial field is consistent with very different finite values. A complete quote requires the potential along the action direction, not merely one slope sample.

13.18 A sign routine for every new example

When a formula feels abstract, first state the physical change: the source loses and the destination gains. Next ask how each change affects its local potential. Translate those answers into derivative signs. Only then combine the terms.

For an above-reference source, $\mu_i > 0$, so its negative stock increment contributes a negative potential change. For a below-reference destination, $\mu_j < 0$, so its positive stock increment also contributes a negative change. The negative of their sum is positive field. This routine is slower than memorizing a sign, but it is much harder to misuse when the carrier or action direction changes.

The routine also exposes ambiguous action descriptions. If one cannot say which coordinate decreases, it is too early to calculate the field. A transformation involving several inputs and outputs needs a correspondingly complete column before the dot product has physical meaning.

13.19 Practice with solutions

In the main model, calculate the field after sending four units from Ridge to Vale. The endpoint is $(14, 6)$, marginals are $(1, -1)$, and the field is two. After a total of eight units, both marginals and the field are zero. After a total of twelve, marginals are $(-1, 1)$, and the field is minus two.

For a separate compatible state $(9, 7)$, reference $(10, 6)$, and equal scales two, marginals are $(-1/4, 1/4)$. The field from the second cell to the first is one-half. A transfer in that direction initially improves the potential even though the sending cell has the smaller raw stock.

Finally explain why a reported field of $4X^{-1}$ cannot be entered as four stock units in a physical ledger. Its unit is different and its meaning is a derivative. To form a first-order field change one multiplies by quantity; to form exact finite EBU one evaluates the complete finite difference and any declared process burden.

What to carry forward

The edge field is the negative directional derivative of the declared potential. For a lossless transfer it is $\mu_i - \mu_j$. It combines endpoint slopes, changes along finite motion, and supplies no actor policy by itself. Exact finite value requires the curvature that the starting slope omits.

Finite changes and curvature

14.1 The whole movement matters

A slope describes the landscape at one point. A finite action travels a distance through that landscape. If the slope changes along the way, its starting value multiplied by the distance cannot give the exact endpoint difference. The Gaussian model makes this distinction unusually transparent because the omitted term can be written exactly.

Return to Ridge and Vale. The starting edge field is four. A four-unit transfer therefore has initial linear estimate $4 \cdot 4 = 16$. Its exact field reduction is twelve. An eight-unit transfer has linear estimate 32 but exact reduction sixteen. At sixteen units the linear estimate is 64, while the exact reduction is zero. No measurement noise or numerical integration is needed to explain these differences. They follow from the curvature of the declared potential.

This chapter retains the original book's lesson about overshoot while deriving it directly for finite quantities. We are not choosing a timestep for a restoring controller. We are evaluating a declared physical increment exactly.

14.2 Expand one square first

For one coordinate, write its initial deviation as $a = x_i - x_i^*$ and its physical increment as d . Then

$$(a + d)^2 = a^2 + 2ad + d^2. \quad (14.1)$$

Dividing by $2\sigma_i^2$ gives

$$V_i(x_i + d) - V_i(x_i) = \frac{x_i - x_i^*}{\sigma_i^2} d + \frac{d^2}{2\sigma_i^2}. \quad (14.2)$$

The first term is marginal times increment. The second is the curvature contribution. It is nonnegative and vanishes only when $d = 0$, given a finite positive scale.

This is exact expansion, not an approximation truncated after a convenient term. A quadratic has no cubic or higher terms to omit. That fact makes the Gaussian domain a clear introductory setting: the finite correction can be derived with the algebra of a square before using vector calculus.

14.3 Add the coordinates

Let δ be any declared finite state increment. Sum the one-coordinate expansions:

$$V(z + \delta) = V(z) + \mu(z)^T \delta + \frac{1}{2} \delta^T H \delta. \quad (14.3)$$

The dot product $\mu^T \delta$ is the initial linear contribution. The quadratic form $\delta^T H \delta$ equals $\sum_i \delta_i^2 / \sigma_i^2$. The word “form” simply describes this matrix expression producing a scalar.

Define the field reduction before any separate process burden as $F = V(z) - V(z + \delta)$. Then

$$F = -\mu(z)^T \delta - \frac{1}{2} \delta^T H \delta. \quad (14.4)$$

The sign reverses because improvement means lower potential. A beneficial initial direction makes the first term positive. Curvature subtracts from that initial linear estimate.

Exact quadratic finite-change identity

For fixed references, fixed positive scales, and the declared additive Gaussian potential, every finite increment satisfies Equation 14.3. Therefore exact field reduction equals the initial linear improvement minus one-half the squared standardized increment. Physical feasibility is a separate requirement on the increment and event.

14.4 Specialize to one lossless edge

For $i \rightarrow j$, the increment is $\delta_i = -q$, $\delta_j = q$, with every other entry zero. Substituting into the exact identity gives

$$F_e(q) = q(\mu_i(z_i) - \mu_j(z_j)) - \frac{q^2}{2} \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (14.5)$$

Name the positive directional curvature

$$k_e = \frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} > 0. \quad (14.6)$$

Then the schedule is the simple parabola $F_e(q) = f_e(0)q - k_e q^2/2$. This notation makes the roles visible: $f_e(0)$ is the initial benefit per stock unit, while k_e tells how quickly that benefit falls as more is sent.

For the main fixture, $f_e(0) = 4$ and $k_e = 1/2$. Thus

$$F(q) = 4q - \frac{q^2}{4}. \quad (14.7)$$

The physical source limit for this isolated lossless transfer is $0 \leq q \leq 18$. The algebraic parabola can be evaluated outside that interval, but such values are not physically executable points of the declared event.

14.5 Read the complete schedule

q	Ridge	Vale	After-potential	Field reduction
0	18	2	16	0
2	16	4	9	7
4	14	6	4	12
6	12	8	1	15
8	10	10	0	16
12	6	14	4	12
16	2	18	16	0
17	1	19	20.25	-4.25
18	0	20	25	-9

Every row is an independent counterfactual from the same baseline, not the next step of a trajectory. That distinction is essential. The $q = 6$ row means send six units from (18, 2), not send six after the $q = 4$ row has already happened.

Potential initially falls as the action moves toward the reference. It reaches its minimum at eight. Further movement crosses the reference and climbs the other side. The total field reduction remains positive until sixteen because the final potential is still below the original 16. Positive total value therefore does not imply that every part of the path is downhill.

14.6 The maximum of a schedule is not a selector

Differentiating the cost-free field schedule gives

$$F'_e(q) = f_e(0) - k_e q. \quad (14.8)$$

If $f_e(0) > 0$, its unconstrained vertex is at $q_v = f_e(0)/k_e$. In the main fixture this is eight. The positive root other than zero is $q_0 = 2f_e(0)/k_e = 16$.

These quantities describe the function. They do not instruct the actor to choose q_v . A physical cap may make the vertex unavailable. A candidate menu

may omit it. A prospective random selector may choose another affordable quantity. A service requirement may refer to a different measure. Naming the maximum allows us to understand the geometry without quietly installing optimization.

If the initial field is nonpositive, every positive quantity has negative field reduction in this quadratic lossless model. At zero initial field, curvature alone gives $-k_e q^2/2$. At negative initial field, both terms are negative. This is a conditional mathematical statement about one fixed direction, not a claim that the action must be morally prohibited.

14.7 An integral gives the same finite difference

The field along the path is $f_e(s) = f_e(0) - k_e s$. Integrating from zero to the finite quantity gives

$$\int_0^q f_e(s) ds = f_e(0)q - \frac{1}{2}k_e q^2 = F_e(q). \quad (14.9)$$

This is the fundamental theorem of calculus in a particularly simple form. The area under the changing field equals the endpoint potential reduction. It is not a different competing definition of single-action value.

For $q = 4$, the field falls from four to two. Its average is three, so the integral is $3 \cdot 4 = 12$. For $q = 8$, the average is two and the integral is sixteen. For $q = 16$, the field falls from four to minus four; positive and negative signed areas cancel.

The integral uses a mathematical path through state space. For an isolated linear transfer whose physical model realizes constant-rate motion, that path can also describe the modeled physical progression. If only endpoints have been declared, the path is an interpolation used to derive the identity. A mathematical identity alone does not establish which physical timing occurred.

14.8 Average, initial, and final slopes

For $q > 0$, the average field is

$$\bar{f}(q) = \frac{F_e(q)}{q} = f_e(0) - \frac{1}{2}k_e q. \quad (14.10)$$

It lies halfway between the initial and final fields because the field is linear along this quadratic path. The exact total is quantity times this average, not quantity times the initial value.

At the main $q = 12$ point, initial field is four, final field is minus two, average field is one, and total reduction is twelve. The final field is already

negative even though the total remains positive. This distinction helps interpret an action that has passed the potential minimum: the action as a whole still improves on its starting state, but its last increments undo some earlier improvement.

An intensity diagnostic such as $F(q)/q$ can be informative, but it is not the total finite value. Ranking by intensity and ranking by total answer different questions. Neither ranking is the action-selection rule of the prospective random programme unless explicitly adopted.

14.9 At the reference every nonzero net change costs field value

At $z = x^*$, the gradient is zero. Equation 14.4 becomes

$$F = -\frac{1}{2}\delta^T H \delta. \quad (14.11)$$

It is negative for every nonzero net increment. This follows from positive definiteness of H . The reference is a strict minimum of the potential.

The phrase “net increment” matters for groups. Several nonzero actions can cancel so that their sum is zero. In an ideal cost-free model the group’s endpoint then equals its baseline and its field value is zero. It would be false to infer negative field value merely from the presence of nonzero child actions. Actual physical process burdens, if represented and declared, remain separate.

This identity also explains a possible initial refusal under a prospective zero-genesis capacity policy. If a policy disallows every negative account change and the world begins exactly at the reference with no balances, nonzero net departures may be unaffordable. That conclusion depends on the policy in addition to the quadratic. External forcing or a different permitted group can change the situation. The potential alone is not the full process.

14.10 Two coordinates with unequal scales

Use a separate sensitivity example with the main stocks and references but scales (2, 4). Initial potential is ten, field is 2.5, and directional curvature is $1/4 + 1/16 = 5/16$. Therefore

$$F(q) = \frac{5}{2}q - \frac{5}{32}q^2. \quad (14.12)$$

The vertex remains eight in this two-cell compatible setup, because a transfer of eight reaches both references simultaneously. The maximum is ten, the entire starting potential. At four units, the reduction is $10 - 2.5 = 7.5$, and the after-potential is 2.5.

The unchanged vertex should not be generalized to all networks. Here there is only one mass-conserving degree of freedom, and both deviations are opposite amounts. In a multi-cell world an action can improve one pair while leaving other deviations untouched. Different scales then affect tradeoffs among available directions and finite quantities.

14.11 Several actions introduce cross terms

For two additive increments δ_a and δ_b , the net increment is their sum. Expanding its quadratic term gives

$$(\delta_a + \delta_b)^T H(\delta_a + \delta_b) = \delta_a^T H \delta_a + 2\delta_a^T H \delta_b + \delta_b^T H \delta_b. \quad (14.13)$$

Consequently the joint field reduction differs from the sum of standalone reductions evaluated at one baseline by

$$F_{ab} - F_a - F_b = -\delta_a^T H \delta_b. \quad (14.14)$$

This previews the group chapters. The cross term is present even though the potential is separable. The overlap occurs because both actions change a shared coordinate.

If both remove stock from the same source and change disjoint destinations, their source components have the same sign, giving a positive inner product and a negative correction. If one adds to a coordinate while the other removes, that shared contribution can have the opposite sign. No universal assertion of positive synergy follows from the existence of a group.

This is a comparison of fixed increments at the same baseline. If each hypothetical subset reruns a resolver and gets new quantities, it defines another set of comparisons. The simple fixed-increment expansion must not be used to describe it without checking those new dependencies.

14.12 Worked shared-source correction

Use a new three-cell declaration: state (14, 3, 3), reference (8, 6, 6), scales two at every cell, and lossless edges from the source to both destinations. Initial potential is

$$\frac{1}{2} \left(3^2 + (-1.5)^2 + (-1.5)^2 \right) = 6.75. \quad (14.15)$$

Each actor proposes quantity two. Alone, either action has initial field 2.25 and directional curvature $1/2$, so its standalone reduction is $2.25(2) - (2)^2/4 = 3.5$. The two standalone values sum to seven.

Together the increment is $(-4, 2, 2)$, giving endpoint $(10, 5, 5)$. Standardized deviations there are $(1, -0.5, -0.5)$, so potential is 0.75. Exact group reduction is $6.75 - 0.75 = 6$, not seven.

The cross-term calculation gives the same correction. The action increments are $\delta_a = (-2, 2, 0)$ and $\delta_b = (-2, 0, 2)$. With $H = I/4$, their weighted inner product is one, entirely from the common source. Thus $F_{ab} = 3.5 + 3.5 - 1 = 6$. No potential coupling between different cells was needed.

This group is physically possible under an ordinary frozen-source availability rule because aggregate withdrawal four does not exceed fourteen. Physical feasibility therefore does not solve the separate overcounting problem. Each check has its own job.

14.13 Worked cancellation correction

At the main baseline, suppose both directed lossless edges exist. Let action A send four from Ridge to Vale, and action B send four in the reverse direction. The reverse standalone action is not feasible from Vale's two units under a frozen-source rule, so its same-baseline value must not be presented as an admissible physical comparator there.

To make the comparison legitimate, declare a separate baseline $(10, 10)$, with the same reference and scales. Both four-unit standalone transfers are now physically feasible. Each alone raises potential from zero to four, so each has field value minus four. The joint cancelling group has net increment zero and field value zero.

The same-baseline correction is therefore $0 - (-4 - 4) = 8$. It is positive because the hypothetical standalone endpoints each depart from the reference while the group endpoint does not. This does not identify an extra eight units to issue in addition to the group value. It is a diagnostic difference between two different accounting comparisons.

The explicit baseline change in this example is essential. Without it, an appealing algebraic illustration would depend on a standalone action that cannot be performed in the declared physical world. Mathematical formulas must retain the admissibility assumptions of the comparisons they claim to represent.

14.14 A physical process term changes the total schedule

The finite field reduction is one part of signed EBU. If a separately declared nonnegative process burden $C(q)$ is justified and not already represented, the total is $E(q) = F(q) - C(q)$. The ideal main model sets $C = 0$.

For a purely illustrative extension, suppose a declared process burden is linear, $C(q) = cq$, with a physically justified coefficient in potential units per stock unit. The schedule becomes $(f_e(0) - c)q - k_e q^2/2$. The field curvature has not changed, but the net value and its positive interval have.

A nonnegative cost function need not be convex. Therefore nonnegativity alone does not establish concavity of the full schedule. The field component is concave because the potential is convex along the action. Adding $-C$ preserves concavity when C is convex on the domain. A fixed activation burden creates a separate zero branch and needs special care, discussed in Chapter 16.

14.15 Why dividing the action does not erase curvature

Take the main four-unit transfer and split it into two consecutive two-unit actions under one fixed potential and no intervening external change. The first action goes from potential 16 to nine, giving seven. The second begins at the new state (16, 4) and goes from nine to four, giving five. Their sum is twelve, the unsplit four-unit field reduction.

If both pieces were instead valued independently from the original state, each would claim seven, for a total of fourteen. That is a different calculation and overstates the actual four-unit endpoint reduction. The second piece has been allowed to reuse the first piece's initial field.

This example distinguishes sequential telescoping from independent same-baseline claims. The correction is not an arbitrary penalty for using two names. It is the mathematical consequence of evaluating the second physical change where it actually begins. Simultaneous groups require a joint value and a separately declared attribution convention; those are developed later.

14.16 Guided problem: derive an entire schedule

Use a separate lossless two-cell example with total 16, state (12, 4), reference (8, 8), and equal scales two. Initial marginals are one and minus one, so the field is two. Directional curvature is one-half. The exact schedule is

$$F(q) = 2q - \frac{q^2}{4}, \quad 0 \leq q \leq 12. \quad (14.16)$$

At $q = 2$, field reduction is three; the endpoint (10, 6) has potential one, compared with initial four. At $q = 4$, the reference is reached and reduction is four. At $q = 8$, the endpoint is (4, 12) and reduction is zero. At $q = 10$, reduction is $20 - 25 = -5$.

Now perform the same exercise by endpoint evaluation alone. This independent form of arithmetic verifies signs and normalization. Finally integrate the field $2 - q/2$ from zero to each quantity. The three methods agree because they are three expressions of the same quadratic identity, not three separate scientific findings.

14.17 Practice and reflection

For the main schedule, solve $F(q) = 12$. The two roots are four and twelve. Both provide the same total field reduction but leave different endpoint distributions, $(14, 6)$ and $(6, 14)$. Explain why a value alone cannot identify what physically happened.

Calculate the difference between the linear estimate and exact field reduction at $q = 3$. It is $q^2/4 = 2.25$; the linear estimate is twelve and the exact reduction is 9.75. This difference is a curvature contribution, not an optional audit adjustment.

At the reference, let $\delta = (-1, 1)^T$ with scales two. The quadratic term is $\frac{1}{2}(1/4 + 1/4) = 1/4$, so the field value is minus one-quarter. Explain how a zero gradient and a negative finite value coexist without contradiction.

Permission before value

15.1 A positive number cannot create a physical possibility

An EBU calculation can evaluate a hypothetical endpoint. It cannot make that endpoint physically attainable. A source with three units cannot supply an isolated four-unit lossless transfer. A route that does not exist cannot become available because its imagined value is high. A tank's ownership, access rules, and equipment limitations are also not determined by a potential difference.

Permission therefore comes before treating a quote as an executable offer. The phrase does not mean that every physical permission rule has already been settled. It means the record must identify a declared admissible action before a value can promise anything about its accepted execution.

The first model represents internal relocation of a conserved scalar. It does not import a regenerative extraction rule merely because both rules use source stocks. Moving resource between represented stores and withdrawing a resource from a regenerating ecological stock are different physical questions. Their permissions must be justified in their own domains.

15.2 Three questions with three answers

The physical question asks whether the action can occur under the declared world and event rules. The valuation question asks how it changes the fixed potential, with any justified process burden. The account question asks whether a proposed owner or actor can afford the signed change under a separately declared capacity policy.

A legal or ethical permission question can add another layer in an application. This textbook does not collapse it into physical feasibility. A mechanically feasible transfer can still be unauthorized by the people responsible for the resource. Conversely, institutional approval cannot supply missing physical stock.

Layer	Example of its question
Physical feasibility	Does the source contain the required stock, and is the declared route available?
Finite valuation	Does the accepted transformation lower or raise the declared potential?
Capacity policy	Does the signed account update satisfy the proposed balance rule?
Institutional authority	Who may authorize the action, and what rights or duties apply?

The layers can interact through a specified protocol. Their interaction does not make them the same object. Keeping their answers separate is especially important when an action is refused: one should be able to say which rule failed.

15.3 The simplest isolated transfer domain

For an isolated lossless edge $i \rightarrow j$ with no additional storage bound, no intervening flow, and nonnegative stock, a finite quantity must satisfy

$$0 \leq q \leq z_i. \quad (15.1)$$

The lower bound respects the declared direction. The upper bound ensures the source successor $z_i - q$ remains nonnegative. The destination gains a nonnegative quantity, and total stock remains unchanged.

This interval is the physical domain of that simple event. In the main fixture it is $[0, 18]$. Every point in it is physically possible under these particular assumptions, even though its EBU may be negative. The $q = 17$ point is feasible and has field value -4.25 . The $q = 19$ point is outside the domain and cannot be accepted as this transfer.

Real infrastructure may impose a smaller maximum quantity, a finite receiving capacity, a time-dependent access restriction, or a quality condition. Those limits must be stated as physical inputs. The Gaussian scale is not a substitute for them. A convenient potential value cannot waive them.

15.4 One source, several requests

Suppose Ridge holds eight units and two actors each propose to send five units to different destinations. Each request alone is no larger than eight. Together they require ten. If the event supplies both from the same frozen source without incoming stock available during execution, the group is infeasible.

The group must be tested as a group. For that declared event semantics, a source-availability condition is

$$\sum_{a: \text{src}(a)=i} q_a \leq z_i. \quad (15.2)$$

This condition is not an allocation rule. It says which sets of quantities pass the source check. Choosing which actor receives a reduced quantity, whether all requests are proportionally scaled, or whether a different group is selected requires a further declared mechanism.

There is no scientific need to pretend those choices are determined by conservation. Conservation constrains the total; it does not decide fairness, priority, or actor sampling. A later protocol can choose among alternatives before outcomes are seen and test the resulting process honestly.

15.5 Incoming stock and event semantics

Consider a three-cell chain. The middle cell begins empty. One action sends four units into it, while another sends four out. The net endpoint leaves the middle cell empty, with no negative stock anywhere. Does that prove the group is physically feasible?

Only if the declared physical event permits this synchronized passage. A continuous pipe can transmit incoming flow while releasing outgoing flow, provided its complete physical constraints are satisfied. A warehouse dispatch protocol may require goods to be received and verified before they can be sent onward. In that second model, an empty middle store cannot fund the outgoing action at the same frozen event.

The distinction cannot be resolved by adding or subtracting the same columns. The endpoint calculation is common to both stories. A protocol must additionally specify the realized path or availability rule. Until then, the right answer is that endpoint feasibility has been checked while event feasibility remains unspecified.

A necessary check is not a complete permission

Nonnegative endpoints and conserved totals are necessary in the first domain. They may be insufficient when a group depends on timing, shared equipment, transient availability, or other declared physical constraints. A complete acceptance rule must name those constraints instead of inferring them from the final sum.

15.6 Joint feasibility should not be smuggled into valuation

Suppose a group has an attractive joint field reduction but violates a source limit. One cannot rescue it by dividing its value among actors differently. The physical quantities are infeasible before any attribution is considered.

Conversely, a feasible group may have a joint value that differs from the sum of independent quotes. That is a valuation issue after the physical increment is established. The remedy is to calculate the joint field value from its common baseline and endpoint, then apply a declared attribution convention if one is needed. It is not to modify stocks until a preferred sum appears.

This separation makes debugging and scientific review more precise. A rejected group can fail because it double-used stock, because its receipt split was undefined, or because an owner could not afford a negative attribution. These are different propositions, each with different evidence and remedies.

15.7 Candidate menus are restrictions, not all of physics

The scalar resource is continuously divisible in the ideal model. A finite candidate menu might nevertheless offer only quantities one, two, and four on an edge. That restriction can make a future experiment finite and reproducible. It does not prove that intermediate quantities are physically impossible.

If the best recovery quantity is three and the menu offers only one and four, an observed outcome can depend on the menu. A comparison must state that limitation. Changing the menu after seeing that EBU performs poorly would change the registered problem. So would including many duplicates of one action if selection is uniform over menu entries.

The menu must also distinguish a no-action option from an empty feasible set. A nonempty menu containing rest can select rest by policy. An empty menu may trigger a separately declared no-action event. Both leave the physical state unchanged in the simplest case, but they have different explanations and can have different probabilities in a stochastic process.

15.8 Permission is not a preference ranking

An admissible set tells us which options are allowed. A selector chooses among them. The set may contain actions with high, low, zero, and negative EBU. Removing every negative action would add a sign-based filter. Choosing the largest positive action would add an optimizer. Neither is implicit in the definition of physical feasibility.

The prospective programme proposes an EBU-independent construction of candidates, joint physical checks, exact value and attribution, owner afford-

ability, and then unweighted random selection among the surviving options. This order is a design direction with remaining protocol decisions. It must not be rewritten in explanatory prose as “the field chooses the best action.”

Why random selection? It makes a particular research question possible: whether a declared signed capacity mechanism changes behaviour without directly rewarding the actor with an optimizer that always picks the largest field improvement. Whether that mechanism is useful remains open. The potential supplies values; the stochastic rule determines how those values affect available choices.

15.9 The prospective capacity boundary

The programme describes nonnegative persistent balances and signed updates assigned through a declared ownership map. For an affected owner i , an affordability check would require

$$B_i + \Delta B_i \geq 0. \quad (15.3)$$

Here B_i is proposed action capacity, not a physical stock and not a legal monetary entitlement. The update ΔB_i depends on the separately specified group attribution and owner mapping. Its meaning does not follow from the symbol B used in another historical context.

For an illustrative arithmetic check, an owner with balance five and proposed update minus four remains at one. With update minus six, the proposed event fails that balance condition. Neither calculation changes the physical water in Ridge. If a transfer is physically impossible, a large balance cannot make it possible.

The fact that child attributions sum to a group field reduction does not determine who owns them or whether simultaneous positive and negative entries may net within one owner. Those are policy choices. The prospective programme identifies them as decisions for registration; this chapter does not adopt them merely by presenting the inequality.

15.10 Why preservation of every local stock is the wrong question here

An internal transfer deliberately reduces its source stock and increases its destination stock. Requiring every source to remain unchanged would eliminate every nonzero transfer. That may be appropriate for a particular protected resource or a distinct extraction policy, but it is not a consequence of global mass conservation.

Our Gaussian reference can value source reduction positively when the source is above its reference. It can also value it negatively when it is below. The value is an account of deviation, not an extraction permission. A hard rule protecting a source for some domain reason must be declared separately and assessed for that domain.

This distinction avoids reusing a historically valid theorem outside its assumptions. A regenerative source certificate can depend on growth, reserves, and a no-export successor. The closed internal-transfer world contains none of that regenerative law. Importing the certificate's arithmetic while omitting its eligibility assumptions would not inherit its authority.

15.11 What a permission record should expose

A reader should be able to reconstruct the proposed event from the permission record. It names the state epoch, physical action identifiers, finite quantities, shared resources, relevant physical limits, and the acceptance result. If accepted quantities differ from requested quantities, the mechanism producing that change must be stated.

The record also identifies what was not checked. If it verifies only endpoints, it must not claim a time-resolved physical path. If it uses a menu restriction, it must not claim exhaustive continuous feasibility. If a resource-quality issue lies outside the model, it should remain visible as a limitation.

For a real application, measurement provenance and institutional authority would accompany these data. In our textbook arithmetic, the inputs are declared illustrative values. The structure is still worth learning because it makes the difference between a physical statement and an inferred permission explicit.

15.12 Guided problem: four candidate actions

At the main baseline, consider the candidate quantities 0, 4, 17, 19 on Ridge-to-Vale. Under the simple isolated transfer rule, zero, four, and seventeen are feasible; nineteen is not. Their field values are respectively zero, twelve, and minus 4.25. The value of nineteen can be evaluated as a formal polynomial, but it is not an admissible schedule point for this event.

Now add, for this exercise only, an independently declared route quantity limit of six. The feasible listed candidates become zero and four. The cap comes from the hypothetical route declaration, not from the potential. Do not change the field formula when applying it.

Finally suppose a separately declared capacity policy requires the actor to fund negative signed changes and the actor's capacity is three. Under the original eighteen-unit physical limit, the seventeen-unit action would be physically feasible but fail this particular affordability check. With capacity five it would pass the illustrative account condition. Its physical quantities and its EBU remain the same in both cases.

15.13 Guided problem: two actors share eight units

Source stock is eight. Actor A requests five for destination A, and actor B requests five for destination B. Three possible groups are $\{A\}$, $\{B\}$, and $\{A, B\}$, with the requested quantities fixed. Under a no-incoming source-availability rule, the single-action groups pass and the two-action group fails.

One possible future allocation mechanism could reduce each request to four. Another could choose one request unchanged. A third could reject the contested group and select from the passing groups. These mechanisms produce different physical increments and potentially different field values. No choice is adopted by this example.

A correct comparison would define the mechanism before computing and interpreting outcomes. If a subset resolver changes quantities, its subset value must be calculated from the quantities that resolver actually declares. It cannot be substituted into a fixed-increment interaction formula without acknowledging the change of object.

15.14 Failure is useful information

Refusal need not mean the theory has broken. It can mean a physical constraint is doing exactly what it says. The scientific question is whether the permitted process can achieve useful outcomes under its declared conditions, and how its failures compare with a suitable alternative.

A permanently empty menu can also expose a badly posed study. If every action is prohibited by an imported policy unrelated to the new domain, no accounting mechanism is being meaningfully tested. Similarly, a world with no disturbances and every stock already at reference can make a zero-genesis affordability process inert. These are design facts to identify before attributing success or failure to EBU generally.

The remedy is to state the research question and the physical world consistently. It is not to choose permissions after inspecting outcomes until a favoured method acts. Prospective design can be revised openly before registration; reported evidence must retain the protocol under which it was produced.

15.15 Practice and reflection

A source begins with six units and two simultaneous outgoing quantities are four and three. Under the frozen-source rule used in the exercise, aggregate withdrawal seven fails even though each request passes alone. If an independent incoming transfer supplies two during the same event, explain why its presence still does not settle feasibility until timing or flow-through semantics are declared.

For an isolated main-fixture transfer of twelve units, the endpoint is (6, 14) and field reduction is twelve. It is physically feasible despite passing the reference. Explain why the fact that eight units would maximize the field reduction does not make twelve physically forbidden.

Finally list the information needed to diagnose a refusal. A complete answer distinguishes absent edge, insufficient stock, joint resource conflict, menu omission, undefined attribution, and failed affordability. Treating all these as “the EBU says no” would conceal the actual mechanism.

The exact local signed EBU quote

16.1 The pieces now fit together

We have a physical state, an action map, a compatible reference, a positive scale at each cell, a potential, and a declared admissible action. We can now state the signed finite EBU quote without asking any symbol to perform an undeclared task.

The quote compares the represented field before and after the action, from one frozen baseline. It subtracts only a separately declared physical action-process burden that is not already represented in the state change or potential difference. The result is a signed total for a finite quantity. It can be positive, zero, or negative.

The ideal main model sets the separate process burden to zero. That choice simplifies the first domain and must remain visible. It does not claim that real transport consumes no energy or causes no wear. It says those carriers are outside this illustrative lossless scalar account, whose adequacy is limited accordingly.

16.2 The shared baseline z

The baseline z is the state from which the accepted event is evaluated. It is shared by both sides of the quote. If an external change has occurred before the action epoch, it is already reflected in z and separately recorded. If no external change occurs in the example, z is simply the current state.

For a finite source-side quantity q on edge e , the action successor is $z + S_e q$. The notation uses quantity directly. A rate-based description must first convert $q = \Delta t j$. The current finite formula does not multiply this quantity by time again.

The no-action comparison at this event is z itself under the declared action-only interval. More elaborate event models can have shared drive and different chronology, but they must explicitly define the baseline. A historical no-action successor computed from a drive law cannot be imported into a forcing-first event by reusing the letter z alone.

16.3 The local equation

Let A be the set of coordinates whose potential terms are affected by the action. In the separable model, a single edge has $A = \{i, j\}$. Define $V_{\text{loc}} = \sum_{k \in A} V_k$. The quote is

$$E_a(q) = V_{\text{loc}}(z) - V_{\text{loc}}(z + S_e q) - C_a(q). \quad (16.1)$$

For a lossless edge this is

$$E_a(q) = V_i(z_i) + V_j(z_j) \quad (16.2)$$

$$- V_i(z_i - q) - V_j(z_j + q) - C_a(q). \quad (16.3)$$

Every unchanged local term would cancel if we wrote the global potential instead. Thus the field component is exactly the global potential difference while requiring only the affected local terms. Chapter 19 develops that identity and its limits in more detail.

The exact signed finite account

Signed EBU equals before-potential minus after-potential, evaluated under one declared field and baseline, minus any justified non-duplicated process burden. Positive means net reduction of that represented burden. The definition does not itself choose the action or assign institutional rights to the resulting number.

16.4 The Gaussian form

Substituting the exact quadratic expansion gives

$$E_a(q) = -\mu(z)^T S_e q - \frac{1}{2} q^2 S_e^T H S_e - C_a(q). \quad (16.4)$$

For one lossless transfer,

$$E_a(q) = q(\mu_i - \mu_j) - \frac{q^2}{2} \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right) - C_a(q). \quad (16.5)$$

The marginals in this expression are evaluated at z . Writing them without their arguments is a typographical convenience, not permission to substitute values from another epoch.

The equation has a useful division of labour. The linear term is the initial field contribution. The quadratic term accounts for its change over the finite displacement. The final term, when present, belongs to a separately justified process category. The first two terms already include the source and destination consequences represented by the potential.

16.5 What one EBU measures here

Because each standardized deviation is dimensionless, V is dimensionless. We label a unit of its signed finite account EBU. This label helps distinguish the value from stock units, but it does not make the value a joule, litre, or legal currency unit.

The scale declarations determine the size of the numbers. If all scales were multiplied by two while physical stocks and references remained unchanged, every potential and cost-free field value would be divided by four. That would be a different valuation geometry. Cross-model numerical comparisons therefore require compatible definitions, not merely the same printed acronym.

Quantity	Unit	Interpretation
q, z_i, σ_i	X	Physical quantity or scale
μ_i, f_e	X^{-1}	Potential change per stock
H_{ii}	X^{-2}	Curvature per squared stock
qf_e, q^2k_e	dimensionless	Finite field-account terms
C_a, E_a	potential-account unit	Declared process burden and signed EBU

The table is a practical way to catch a repeated timestep or a rate accidentally used as a quantity. Every term subtracted from a potential must have the same unit.

16.6 The main quote, step by step

Freeze the baseline (18, 2), reference (10, 10), scales (2, 2), lossless edge Ridge-to-Vale, and $C = 0$. The source contains eighteen units, so the isolated physical interval is $0 \leq q \leq 18$. The complete schedule is

$$E(q) = 4q - \frac{q^2}{4}. \quad (16.6)$$

At $q = 4$, compute before-potential 16. Apply the actual finite changes $(-4, +4)$, producing (14, 6). Compute after-potential four. Subtract to obtain 12EBU.

At $q = 8$, the endpoint is the reference and after-potential is zero, giving 16EBU. At $q = 16$, the reflected state has potential 16, giving zero. At

$q = 17$, after-potential is 20.25, giving -4.25EBU . The formula records all signs with one rule.

No separate restoration bonus is added to the positive cases. No separate moral penalty is added to the negative case. A single finite difference expresses the state-dependent valuation. That symmetry is essential for the later accounting identities for splitting and closed cycles.

16.7 Process burden: classify before subtracting

A proposed C_a must identify a physical action-process burden that is not already represented through the state transition or evaluated field. It must use compatible potential units, be declared before the action, and satisfy the relevant non-duplication rule. In the standard nonnegative convention, $C_a(q) \geq 0$ and $C_a(0) = 0$.

Source drawdown is already in $V_i(z_i) - V_i(z_i - q)$. Destination receipt is already in its corresponding term. Adding another deduction called “stock depletion” for the same represented consequence would count it twice. Similarly, if an equipment-wear coordinate already enters the field, its burden cannot be charged again as if absent.

Monetary price, wages, inconvenience, and protocol penalties are not automatically physical burden. They may be important institutional or economic quantities, but converting them into C_a requires a separately justified model. The physical account must not quietly import a market price as a law of environmental value.

Candidate item	Treatment
Source deviation already in V_i	Included in the finite field change; no second subtraction.
Destination deviation already in V_j	Included in the finite field change; no second subtraction.
Distinct unrepresented physical process	Candidate for a declared, calibrated process term.
Purchase price or wage	Belongs to an institutional/economic account unless a separate theory justifies another role.
Fraud or access penalty	Belongs to governance and protocol enforcement.

16.8 A physical ledger does not settle the cost classification

An extended model may include a loss accumulator to close its physical conservation account. Whether a related physical burden belongs in its potential, in a justified process term, or in neither is a further definition. One cannot infer permission for a cost merely because the accumulator is absent from V , nor assume that every bookkeeping coordinate already prices every consequence of loss.

The classification must identify the physical carrier and the particular burden being represented. If an old rationale assumed that no destroyed-stock coordinate existed, adding such a coordinate changes the premise of that rationale. The wording and authority need reconciliation before reusing the cost formula. This book's ideal lossless core avoids asserting that this open extension has been resolved.

There is also no justification for inventing extra burdens merely because upstream and downstream actions exist. A fuel-to-electricity-to-water chain needs linked physical transformations and complete accounts. It does not need arbitrary surcharges at every label. Each actual burden is represented once in its declared place.

16.9 An illustrative cost extension

To understand the algebra, suppose a separate, explicitly hypothetical physical process has already been justified with $C(q) = q/2$. This changes the ideal example's cost assumption and is not a fitted real-world calibration. The total schedule becomes

$$E_C(q) = \frac{7}{2}q - \frac{q^2}{4}. \quad (16.7)$$

At four units the field reduction remains twelve, while the process burden is two, giving total ten. The physical source and destination stocks remain (14, 6). The process declaration changes the extended account, not the water transfer already specified.

The ledger identity is

$$E_C(q) + [V(z + S_e q) - V(z)] + C(q) = 0. \quad (16.8)$$

For this example it is $10 + (4 - 16) + 2 = 0$. The identity is a direct rearrangement of the definition. It shows where the signed value balances, but it does not validate the physical calibration of C .

If the candidate cost has no defensible carrier or duplicates field burden, the correct response is to reject the declaration, not to praise the arithmetic be-

cause it closes. Any subtraction closes its own rearranged equation; scientific meaning requires more.

16.10 The zero branch and an activation burden

In the main cost-free model, $E(0) = 0$ directly. If a distinct physical process has a fixed activation burden $c_0 > 0$, a possible declared function is $C(0) = 0$ and $C(q) = c_0 + C_{\text{cont}}(q)$ for positive q . The exact no-action branch remains zero, while an arbitrarily small positive action can incur the fixed activation burden.

This creates a downward jump in the total value as one leaves zero. It must not be interpolated away. A quote representation that joins the zero point smoothly to a positive-quantity branch would describe another cost law.

The field schedule is concave. The total schedule is concave on a domain when the subtracted cost is convex there. A fixed activation jump is not convex at the included rest point, so full-domain concavity cannot be inferred. None of these shape facts replaces the requirement that the complete schedule be defined before execution.

16.11 A quote is a function, not one optimistic number

An action may execute less than its offered quantity. A committed schedule specifies what the value would be at every allowed measured point. In the Gaussian core the function is a simple polynomial with a stated domain and parameter version. The actor and auditor need not invent a new valuation rule if the measured quantity differs within that domain.

Suppose the main schedule allows up to six units, but the action delivers four. Evaluating the predeclared function at four gives twelve. Replacing the function after delivery with a more favourable scale or reference would be revaluation. The distinction concerns when the rule was fixed, not whether the final measured point was known in advance.

If execution exceeds the accepted physical domain, the original quote does not silently expand. The physical deviation and protocol violation must be recorded. Any richer debit or remedy needs its own prior rule; an auditor cannot manufacture an arbitrary punishment after seeing the outcome. The detailed lifecycle is the subject of Chapter 17.

16.12 Local information and complete affected terms

For one lossless edge under the separable potential, the quote reads endpoint stocks, endpoint references and scales, the action quantity, edge identity, ac-

cepted domain, and any justified local process declaration. It can evaluate four local potential numbers and the cost. It need not inspect distant stocks or calculate a global sum.

A researcher may calculate global diagnostics for analysis, but those diagnostics are not thereby actor inputs. Likewise, a prospective capacity check may read balances as part of its own layer; that does not make balances inputs to the physical field valuation. Keeping the computational boundaries explicit helps avoid a hidden optimizer or information advantage.

If a richer potential adds a factor involving several coordinates, every affected factor must be included. Omitting it while retaining the claim of exact locality would be wrong. Locality means the complete affected dependency set, not a fixed two-cell rule regardless of the actual potential.

16.13 Single action and group value

For a group with net increment δ_G , the exact joint account is

$$E_G = V_{\text{loc}}(z) - V_{\text{loc}}(z + \delta_G) - C_G. \quad (16.9)$$

This evaluates the group endpoint once from the common baseline. It does not establish each child's entitlement or causal contribution. A declared straight-path decomposition can produce child numbers that sum exactly to the field reduction, but the interpretation of that decomposition requires its own assumptions.

It is therefore possible to know the exact joint value while a proposed capacity policy remains unregistered because ownership or attribution semantics are open. That is an honest boundary between an established algebraic object and an incomplete institutional or experimental rule. The later group chapters develop the distinction without forcing a one-action-only world.

16.14 Guided problem: audit a misleading quote

A report for the main four-unit action says: “Starting field four, quantity four, so EBU sixteen. The destination received four units; subtract four more for source depletion, leaving twelve.” The final number happens to equal the correct answer. Is the derivation acceptable?

No. The initial sixteen is a linear overestimate. The subtracted four happens to match the curvature term for this fixture, but labelling it source depletion misclassifies its meaning and risks double counting. At another quantity, the accidental correction would fail. The exact derivation is $16 - 4 = 12$ from endpoint potentials, or $4q - q^2/4$ from quadratic expansion.

This exercise shows why a correct final number is insufficient evidence of a correct account. Units, categories, baseline, and derivation must also match the definition. Otherwise a fixture can pass by cancellation of two mistakes.

16.15 Guided problem: compare totals and intensities

For the main schedule, $q = 2$ has total seven and intensity $7/2 = 3.5$ EBU per stock unit. The four-unit action has total twelve and intensity three. The eight-unit action has total sixteen and intensity two.

The smaller action has the highest intensity among these options, while the largest listed action has the highest total. Neither observation defines the programme's selector. They are different diagnostics. A report should state which quantity it uses and what question that quantity answers.

If the practical concern is a specified delivered service, comparisons may also need equal service or an explicit service metric. Potential change alone does not ensure comparable clinical outcomes. That issue belongs in research design rather than being resolved by whichever ratio favours one action.

16.16 Practice and reflection

Compute the main quote at $q = 6$. The after-state is (12, 8), after-potential one, and EBU fifteen. Verify the exact quadratic formula gives the same result. Then apply the explicitly hypothetical $C(q) = q/2$: the field component remains fifteen, process burden is three, and total is twelve.

Explain why a cost called “wage” cannot simply be subtracted from a dimensionless physical potential. A complete answer identifies both the unit mismatch and the absence of a declared physical interpretation. This does not deny the importance of wages; it keeps distinct accounts meaningful.

Finally describe the minimum information needed to reproduce a quote: a fixed potential version, affected baseline coordinates, complete action increment, admitted quantity domain, and any process declaration. Add a sentence distinguishing a reproducible mathematical quote from evidence that the field represents all relevant consequences.

What to carry forward

The exact local quote is a finite signed potential difference minus declared, non-duplicated physical process burden. The Gaussian form includes curvature exactly. Its schedule and domain precede execution. It supplies a valuation object, while physical permission, group attribution, capacity policy, and real-world usefulness retain their own requirements.

From a pre-action schedule to a measured record

17.1 A promise is not yet a physical event

Ridge agrees to send four stock units to Vale. Before anything moves, we can calculate what that action would mean under the frozen model. Afterward, a meter may show that only three units moved. The proposed quantity, permitted quantity and observed quantity answer different questions. Keeping all three is part of a useful account.

The preceding chapter gave an exact value for a finite action. Here we ask what information makes that equation reproducible across time. The object of interest is still a physical change. A message requesting a transfer does not move the water. A signed agreement does not prove delivery. A correct calculation of a planned successor does not prove that the successor occurred.

The original book called this progression a quote-to-settlement lifecycle. We retain that teaching idea, while separating a physical value record from any later rule that changes a spendable balance. This chapter specifies the information relationships needed to understand an account. It does not adopt a complete operating protocol for the prospective Gaussian study.

Three quantities in one story

The request says what an actor asks to do. The accepted domain says what the declared rules permit. The measured record says what the evidence supports as having occurred. A scientifically useful record never silently substitutes one for another.

17.2 Freeze the whole function, not one attractive point

For the recurring Ridge–Vale example, let

$$z = (18, 2), \quad x^* = (10, 10), \quad \sigma = (2, 2), \quad C = 0.$$

A lossless transfer of quantity q changes the state to $(18 - q, 2 + q)$. Its exact schedule is

$$E(q) = 4q - \frac{q^2}{4}. \quad (17.1)$$

If the accepted domain is $0 \leq q \leq 4$, the quoted value at full execution is $E(4) = 12$. The equation also specifies every smaller accepted quantity. At $q = 3$ it gives $12 - 9/4 = 9.75$. That is the value of the three-unit event on this schedule.

Multiplying the full quote by the fraction executed would give $12(3/4) = 9$, which is different. The first three units and the fourth unit do not have the same marginal effect. We can see this without calculus: the three-unit successor is $(15, 5)$, its potential is $25/4 = 6.25$, and $16 - 6.25 = 9.75$. The direct endpoint calculation agrees with the function.

This is why a pre-action account should identify its whole domain and formula. A few selected values may illustrate the formula, but they do not replace it. A table can represent an exact schedule only if its interpolation rule is also justified. Connecting points with straight lines usually changes a curved function.

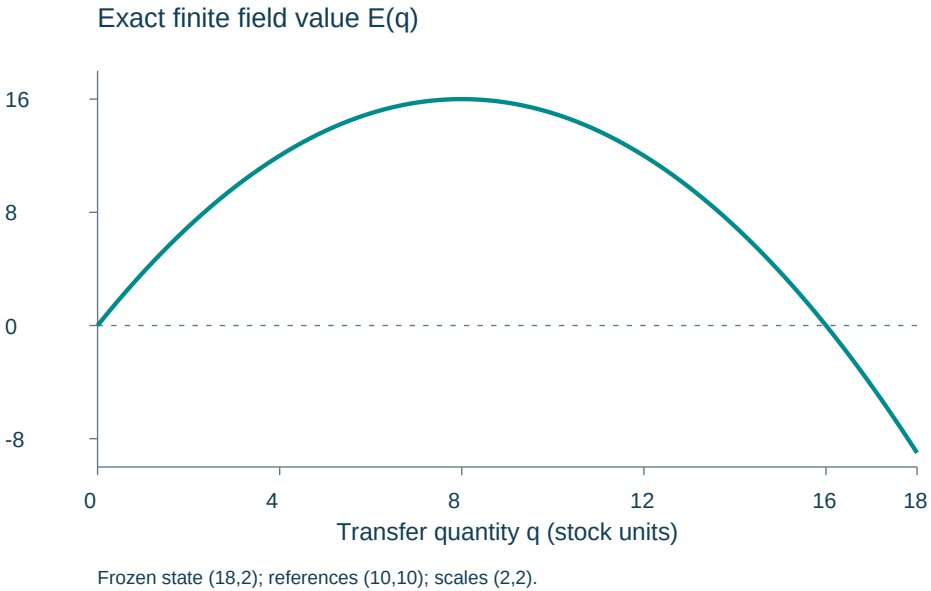


Figure 17.1: The full finite-value curve for the running two-cell example. A positive initial slope does not make every finite quantity beneficial.

17.3 The information held fixed

To reproduce the schedule, the reader needs the baseline state, coordinate ordering, action map, reference vector, positive scales, process-burden declaration, units and accepted domain. A version label should identify the equation and those definitions. A timestamp alone is insufficient: two records at the same time can describe different fields.

The scope must also say whether q is quantity or rate. Here q is a quantity. If a meter reports a rate j over duration Δt , then $q = \Delta t j$ when the rate is constant. A variable rate requires its time integral. Replacing a reported average rate with an instantaneous endpoint reading can change the quantity, even if the numerical values look similar.

For a group, the record needs the accepted vector, the joint successor and the path convention if child attributions are displayed. A child's measured short-fall can change the group's field value and the attributions of other children. It cannot always be handled by independently scaling one child's old number. Chapter 22 explains the mathematical reason.

17.4 The shared baseline removes an external change

Suppose an external, lossless redistribution changes the physical state from (16, 4) to (18, 2) before the actor's epoch. Under our fixed potential,

$$V(16, 4) = 9, \quad V(18, 2) = 16.$$

The external change increased represented deviation by seven. The actor then transfers four units, ending at (14, 6) with $V = 4$. The actor's field value is $16 - 4 = 12$.

Using the earlier state as the actor's baseline would give $9 - 4 = 5$. It would mix the external worsening with the actor's improvement. Neither number is arithmetically mysterious; they answer different questions. The external record is +7 in potential change, while the action record is +12 in avoided potential. Together they explain why the complete interval changed potential by $4 - 9 = -5$.

Write the general relationship with pre-forcing state x , frozen post-forcing baseline z and action successor y :

$$D_{\text{ext}} = V(z) - V(x), \tag{17.2}$$

$$E = V(z) - V(y) - C, \tag{17.3}$$

$$V(x) - V(y) - C = E - D_{\text{ext}}. \tag{17.4}$$

The subtraction makes the attribution boundary explicit. Calling the external change a disturbance does not require it to increase burden. It can also reduce burden, in which case D_{ext} is negative.

No-action neutrality

At the shared baseline, $E(0) = V(z) - V(z) - C(0) = 0$ when $C(0) = 0$. An idle actor receives no field reduction from an external event merely because that event preceded the observation. This is an algebraic statement within the declared comparison.

17.5 An epoch has a beginning and an expiry

Imagine a pump quote issued at 14:00 for a short movement of water. At 14:03 another pipe fails. If that failure changes a required precondition, the original quote may no longer describe the available action. A protocol must say whether the action can continue, whether its authorization expires, or whether a new epoch is opened.

What cannot be justified by the original formula is silently choosing the most favourable baseline after seeing the outcome. An actor might prefer the high-burden state immediately after failure; an institution might prefer the lower-burden state before it. Reproducibility requires the comparison rule to be specified before those preferences can select it.

A new epoch is not an admission that earlier arithmetic was wrong. It records that the facts or applicable model have changed. A useful archive keeps the earlier proposal, its expiry reason and the new proposal. This lets a later reviewer distinguish a legitimately revised opportunity from an outcome-dependent rewrite.

The exact timing and expiry mechanics belong to a registered protocol. We need not invent them here to understand the boundary. The general requirement is consistency: the physical interval, state commitment, accepted domain and measurement claim must refer to the same event.

17.6 Shortfall, overexecution and missing evidence

For a measured quantity within the valid domain, the committed schedule can be evaluated at that quantity when its other assumptions remain satisfied. Our three-unit example is such a case. It does not require a new potential or a proportional approximation.

If the accepted domain ended at four but the meter reports five, the physical report must retain five. A rule limiting accepted quantity cannot erase an extra unit from the reservoir. At the same time, the earlier commitment did not authorize five. Describing the physical consequence and deciding the institutional response are separate tasks.

The mathematical function may be evaluable beyond its accepted domain, but mathematical evaluability is not permission. A retrospective diagnostic can explicitly calculate the consequence of overexecution under the frozen field. It must not be mislabeled as an authorized receipt or a precommitted positive credit.

If the sensor is missing or inconsistent, “not established” is a possible outcome. Zero is a measured or derived value, not a universal substitute for uncertainty. Crediting the full request would invent execution; charging an arbitrary negative amount would invent a physical burden. Any sanctions or compensation need their own declared basis.

17.7 Identity prevents counting the same change twice

One transfer can appear in the source meter, the destination meter, a pump controller and a transporter’s report. Those records can corroborate one another. They do not automatically describe four actions. A stable event identity connects them to one physical transformation.

Conversely, a chain can contain several physical transformations under one invoice. Fuel use, electricity generation, delivery and water pumping must retain their physical boundaries where they are in scope. A commercial reference number can link the chain without collapsing its quantities or converting its resource units into one scalar.

Repeated actions can be distinguished by their execution intervals and predecessor states. Splitting an actual continuous movement into bookkeeping pieces must not let each piece claim the same original improvement. If the pieces are sequential, each uses the preceding piece’s successor. If they are simultaneous, the group account must reconcile their joint effect.

17.8 A hash protects a record, not a sensor

A digital fingerprint can help establish that the equation and inputs shown after an event match the committed bytes. It does not establish that the water existed, that a meter was calibrated, or that the declared reference was appropriate. Integrity of a file and truth of a measurement are different properties.

For a physical account, the evidence chain can include device identity, calibration records, location, time interval, unit conversion, independent observations and documented corrections. The appropriate set depends on the domain. A teaching tank does not prescribe a surveillance system for a hospital.

Records should explain uncertainty without burying it in the potential's scale. In the Gaussian model, σ_i defines the reference geometry. A sensor's uncertainty concerns knowledge of x_i . Treating those as the same parameter would change both the meaning of the field and the meaning of confidence in the observation.

17.9 Corrections preserve the earlier account

Suppose a later calibration finds that a meter overstated delivery. A correction should identify the earlier record, the evidence for the correction and the authorized correction procedure. Deleting the earlier line conceals what participants actually knew and acted on at the time.

A physical reversal is different. Returning water from Vale to Ridge is a new action with a new predecessor state. It may reverse the earlier endpoint change, but it is not a correction to the fact that the first action occurred. Its field value must be computed under the applicable fixed-field interval and any declared process burden.

Versioning the potential is also different from correcting a measurement. If new evidence justifies a different reference, the same unchanged state can acquire a different potential. That is a model revision. It needs a separate reconciliation term when accounts span the change; it is not physical restoration by the actor.

17.10 A complete worked action record

Consider the three-unit shortfall again. A readable record contains the following facts:

Item	Declared or observed content
Physical object	Lossless Ridge-to-Vale transfer of one homogeneous stock
Baseline	$(18, 2)X$, ordered Ridge then Vale
Field	Reference $(10, 10)X$; scales $(2, 2)X$
Domain	Accepted quantity between zero and four stock units
Schedule	$E(q) = 4q - q^2/4$, with $C = 0$
Observation	Three stock units transferred; endpoint $(15, 5)X$
Calculation	Before 16; after 6.25; signed field value 9.75
Status	Illustrative arithmetic; no real event or spendable credit claimed

The record is longer than the number 9.75 because it explains what the number means. Someone can check the arithmetic without knowing the actor's income, preferences or occupational status. Whether that person is entitled to compensation remains a separate question.

17.11 Guided problem: a changed observation

Keep the same frozen field and accepted domain. The source reports that $2.5X$ left Ridge; the destination reports that $2.5X$ arrived. Calculate the successor, its potential and the exact value. Then compare the result with scaling the full four-unit quote by the delivered fraction.

The successor is $(15.5, 4.5)X$. Each endpoint differs from its reference by $5.5X$ in magnitude. Thus $V = 2(1/2)(5.5/2)^2 = 7.5625$, and the field reduction is $16 - 7.5625 = 8.4375$. Equivalently, the schedule gives $4(2.5) - (2.5)^2/4 = 8.4375$.

Proportional scaling would give $12(2.5/4) = 7.5$. It is smaller because it assigns the fourth unit's lower marginal contribution to earlier units as well. The endpoint equation avoids that substitution. The two meters agree on conservation in this illustration; agreement alone does not establish their calibration in a real application.

17.12 Practice and reflection

First, evaluate $E(1)$, $E(2)$ and $E(4)$. The answers are 3.75, 7 and 12. Explain why doubling quantity from one to two does not double value. Next, suppose an external event takes $(18, 2)$ to $(16, 4)$ before a two-unit action. The new baseline has $V = 9$; the action ends at $(14, 6)$ with $V = 4$, so the action value is 5. The external potential change was -7 . Neither should be replaced by the other.

Finally, write an example of a missing measurement, a physical reversal and a field-version change. For each, name the record that must be retained and the new information that must be added. The purpose is to recognize distinct events before assigning numbers to them.

What to carry forward

A schedule is a conditional account over a declared domain. Verified execution supplies the point to evaluate. External changes, expired conditions, missing evidence and later corrections remain visible. The equation determines the field difference; an institution must separately define the consequences of recording it.

The next chapter follows these signed values into the real-needs examples that motivated the book. Understanding a positive number is incomplete until we can also explain a zero or a negative number without confusing the action with the person it serves.

Positive, zero and negative EBU in real needs

18.1 The sign belongs to an action in a state

The same bottle of medicine can arrive where it is urgently needed, where stock is already well placed, or where delivery displaces something else. The name of the product does not determine the sign of its EBU. The declared before-state, physical transformation, reference, scale and process burden determine the calculation.

This statement is sometimes uncomfortable because everyday language uses “valuable” for several different things. A treatment can be valuable to a patient. A delivery can have a high market price. A redistribution can lower a particular stock potential. These statements may all be true, but they do not express the same relationship.

The examples in this chapter keep the physical account visible alongside service and human need. All numbers are invented teaching inputs. They are not clinical guidance, a valuation of a life or a claim that medicine can be represented adequately by two stocks.

Read the sign with its noun

Positive net EBU means potential reduction exceeds declared residual process burden. Field value alone is the potential reduction before that subtraction. Neither means a good person, an adequate wage, a complete cure or a socially optimal decision. Negative values are equally specific.

18.2 One delivery, three starting states

Use two medicine-storage locations with compatible references $(10, 10)X$, scales $(2, 2)X$ and zero residual process burden. The locations are a regional store and a clinic. A delivery sends $2X$ from the first to the second, with perfect quantity retention in this ideal illustration.

Start first from (18,2)X. The potential is 16; the successor (16,4)X has potential 9. The delivery's field value is +7. The clinic still has less than its declared reference, so a positive result does not mean all relevant needs have been met.

Now start from (11,9)X. The potential is 1/4. The same delivery produces (9,11)X, also with potential 1/4. Its field value is zero. Two physical stocks changed, and medicine reached the clinic, but the chosen quadratic assigns equal deviation to the endpoints.

Finally start from (10,10)X. Potential is zero. The same delivery produces (8,12)X, whose potential is 1. The field value is -1. This is the simplest example of a permitted physical change that moves the modeled system away from its reference.

Before	After	V before/after	E
(18, 2)	(16, 4)	$16 \rightarrow 9$	+7
(11, 9)	(9, 11)	$1/4 \rightarrow 1/4$	0
(10, 10)	(8, 12)	$0 \rightarrow 1$	-1

The common action description is not enough to reproduce any of the three answers. The starting state is indispensable. So is the fixed field: using different references would ask a different question.

18.3 A zero is not an empty event

In the middle example, the source and destination exchange their deviations. The endpoint states differ, although their potentials agree. A scalar function assigns one number to a state containing several numbers; equal potential generally does not imply equal state.

This matters for interpretation. One endpoint may have a different available route, future need or exposure to failure. Those differences can change later opportunities even when current V is equal. They cannot be recovered from the scalar alone. A complete action record keeps the state, not merely its potential.

Zero can also arise when the total increment of a group cancels. Two opposing ideal transfers may leave the net stocks unchanged. That does not establish that real pumps consumed no electricity. If pumping energy is in scope, its transformation must be represented through the appropriate carrier or boundary. The ideal $C = 0$ example is deliberately narrower.

Finally, zero field reduction can combine with positive residual process burden to produce a negative total. Reporting the field component and C separately

allows the reader to distinguish unchanged represented deviation from an unrepresented-but-declared action burden.

18.4 The full quantity can reverse the initial sign

Return to (18,2)X. A small Ridge-to-Vale transfer improves the Gaussian field, but the exact schedule is

$$E(q) = 4q - q^2/4.$$

At $q = 8$, the endpoint is the reference and $E = 16$. At $q = 16$, the endpoint is (2,18) and $E = 0$. At $q = 17$, the endpoint is (1,19), whose potential is $81/4 = 20.25$, giving $E = -4.25$.

The positive initial slope did not promise that every amount in that direction would be beneficial. The transfer passed through the reference and continued until the full action increased deviation. A finite calculation includes the entire movement. A slope is a nearby statement.

One could design a controller that chooses $q = 8$ to maximize this particular schedule. That is an additional selection rule, not part of the definition of EBU. A different actor might choose an amount for a deadline, a random menu or another objective. This book teaches the measurement without silently building a maximizer into the word “field.”

18.5 An essential action can have negative field value

Suppose a patient needs a dose at the second location, while the simplified inventory reference currently places equal stock at both locations. The example starting at (10,10) says a two-unit transfer has field value -1 . It does not tell us whether the patient should receive treatment.

The potential may not represent clinical timing, patient condition, dose compatibility, expiry or access. A reader who turns -1 into automatic refusal has supplied a social rule and a claim of model completeness that the equation never contained. A reader who changes the result to positive because the patient needs help has replaced the declared physical comparison with a different judgment.

An honest account can state both: the represented inventory deviation increased by one, and the delivery met a separately described urgent need. If the scientific model is inadequate for the intended decision, its scope must be revised through explicit evidence and governance. The old event remains described by the field actually used.

An action is not a person

Do not say that a patient “is a burden of one EBU.” Say which delivery action changed which represented state under which field. A human need and a person’s claim to care are not coordinates in this two-tank demonstration.

18.6 Feasibility, value and service are separate tests

A movement can have positive hypothetical value while being physically impossible. If Ridge contains only one unit, an instruction to remove two cannot be made valid by a favourable clinic term. Conservation and source availability constrain the action before its potential difference is treated as an executable opportunity.

A feasible action can fail a service requirement. A delivery arriving after a deadline may preserve a stock balance while failing its intended use. Time can be part of the represented state, a constraint or a separately reported outcome, but it cannot be omitted and then assumed satisfied.

A negative action can be feasible and successful on a service dimension. The three dimensions therefore form a small report, not a single ranking:

Physical status	Field status	Service interpretation
Feasible	Positive	Can improve the field while leaving demand unmet
Feasible	Zero	Can move goods without changing the chosen potential
Feasible	Negative	Can meet a need while worsening represented deviation
Infeasible	Any hypothetical sign	No executed candidate follows from the sign

The report avoids calling every dimension “success.” A later scientific comparison must declare which outcomes matter and how conclusions are drawn. Choosing that interpretation after seeing results would change the question being tested.

18.7 A repair needs a represented physical consequence

A mechanic repairs a pump. A water-stock model that observes only the current litres may show no immediate change in x . Under that model, the stock field effect is zero. This is not evidence that repair is useless; the model

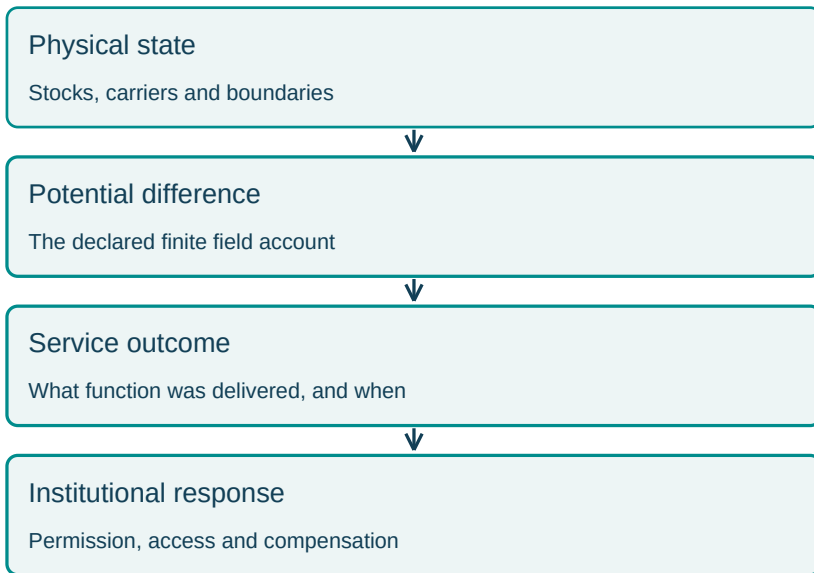


Figure 18.1: Physical stock, potential and an action account answer different questions. Their records must be linked without treating them as the same quantity.

lacks the equipment condition or other capacity coordinate needed to describe the benefit.

One option for future research is to represent a certified physical equipment state. That requires a definition, unit, measurement method and justified contribution to a potential. Merely writing “pump repaired” in a field does not supply these. Another option is to report the repair in a linked physical record while observing its later effects without issuing an invented current field reward.

The distinction also applies to diagnosis. Finding a leak provides useful information. If no physical coordinate in the model changes, a separate claim of positive physical field reduction needs an information or service model that actually defines such a change. Professional effort deserves consideration in compensation, but occupation and effort do not replace the state difference.

18.8 Five layers in an infrastructure decision

Consider building a shared cold store. First describe construction: materials, energy, land, transport and waste, with appropriate carriers and boundaries.

Second describe what capability has been established and how it is measured. Third identify who has authority to approve the construction. Fourth determine who bears its institutional responsibilities. Fifth specify how workers and providers are compensated.

These five questions can refer to one project without becoming one equation. A high future service expectation does not erase present construction burden. Conversely, a negative present physical account does not show that investment should never occur. Long-term planning can consider future outcomes, but its forecast is not an observed action receipt.

A changed route or machine can also alter future action maps and efficiencies. That is a versioned physical and modeling change. The old route's receipts should remain interpretable under the old description; the new route's receipts should identify the new description. Rewriting old efficiencies would make improvement impossible to audit.

18.9 The rural clinic as a linked case

The clinic needs medicine, refrigeration, water and a working backup supply. In ordinary conversation these form one service. Physically they contain different carriers and transformations. The medicine transfer changes dose stocks; refrigeration transforms energy while maintaining temperature; water delivery changes water stocks; repair changes equipment condition when that state can be measured.

Linking these records preserves dependence: medicine can become unusable without refrigeration, and a clinic can cease functioning without water. Adding their numerical EBU values without a declared common calibration would hide differences of meaning. A small positive water account cannot automatically compensate for an unrepresented failure of medicine storage.

The teaching scalar is useful because it allows complete arithmetic. It is not useful if we forget why it is small. For a real clinic, the scientific task includes deciding which states and dependencies are necessary for the intended claim. The potential must be evaluated as a model of that task, not as a universal vocabulary of health.

18.10 Recurring burdens can inform redesign

Suppose repeated emergency deliveries are physically expensive. The records can motivate questions about regional storage, local production, repair skills or transport reliability. They cannot by themselves prove which investment will succeed. Comparing a new store with existing deliveries requires consistent service, time horizon, boundary and uncertainty treatment.

This is a place where monetary and physical records can complement one another. A financially inaccessible improvement might reduce physical burden. A financially attractive improvement might shift burden outside the current boundary. An institutional study must examine access and distribution alongside physical consequences.

Such questions preserve the original book's ambition while making its status clear. EBU may provide information useful for adaptation. Whether people can act on it, whether benefits are captured, and whether outcomes improve are empirical and institutional questions. A sequence of exact receipts is evidence about the recorded events, not automatic proof of a just transition.

18.11 Guided problem: different needs at equal potential

Use the same field and compare $(14, 6)$ with $(6, 14)$. Both have potential 4. From each, consider a two-unit first-to-second transfer. The first ends at $(12, 8)$ with potential 1 and has value $+3$. The second ends at $(4, 16)$ with potential 9 and has value -5 .

The starting potentials were equal, but the local directions were different. This shows why recording only V loses decision-relevant state information. Now suppose the second location in both cases has a separate urgent deadline. That service fact does not change the arithmetic. It changes the complete interpretation and may reveal that the current reference model needs additional representation.

18.12 Practice and reflection

For $z = (12, 8)$, calculate a two-unit and a four-unit transfer. The initial potential is 1. The first transfer reaches $(10, 10)$ and gives $+1$; the second reaches $(8, 12)$ and gives zero. State a reason why the zero-value action might still have a separately measured service outcome.

Then describe a useful maintenance action whose immediate benefit is missing from a stock-only model. Identify the missing physical quantity, possible evidence for it, and the institutional question that would remain even after it was measured. Do not assign a number to the benefit without defining the new field.

Finally rewrite the sentence "This household caused five negative EBU" as an action-specific report. A good answer identifies the device or transport event, model version, physical cause, service delivered and measurement status. It avoids converting a receipt into a judgment of the household's worth.

Local actions and exact global identities

19.1 The apparent paradox

An action at Ridge and Vale can be valued using Ridge and Vale data. Yet we have described its field effect as a change in a potential summed across a whole network. How can a calculation that does not read the other locations give the complete result?

The answer is cancellation. Under the declared separable potential, every untouched location contributes exactly the same term before and after the action. When those terms are subtracted, they disappear. The local calculation is not a statistical estimate of the remaining world; it is the part of an exact difference that survives.

This distinction is central to the project. It explains how a local action account can be mathematically complete within a declared model. It also identifies the boundary of that claim: the model's potential must include the relevant effects, and the action's evaluation must include every term that actually changes.

19.2 Support means the coordinates changed

Let δ be an action's finite increment. Define its write support by

$$W = \{i : \delta_i \neq 0\}. \quad (19.1)$$

For a positive lossless transfer along $i \rightarrow j$, this set contains i and j . A group can change more coordinates. A proposed edge of zero quantity changes none, even though its name still refers to two endpoints.

Support is an algebraic description of the state change, not a prediction of every future consequence. Water arriving at Vale may enable a later transfer to another place. That later event has its own increment and baseline. It does not automatically appear in the present endpoint difference.

For the Gaussian model,

$$V(x) = \sum_{i=1}^N V_i(x_i), \quad V_i(x_i) = \frac{(x_i - x_i^*)^2}{2\sigma_i^2}.$$

Each local term depends on one coordinate and fixed local parameters. This is what makes endpoint support sufficient for the present model.

19.3 The cancellation proof, one line at a time

Write the complete difference as a sum:

$$V(z) - V(z + \delta) = \sum_{i=1}^N [V_i(z_i) - V_i(z_i + \delta_i)]. \quad (19.2)$$

For an index outside W , the increment is zero. Its summand is therefore $V_i(z_i) - V_i(z_i) = 0$. Removing those zero summands gives

$$V(z) - V(z + \delta) = \sum_{i \in W} [V_i(z_i) - V_i(z_i + \delta_i)]. \quad (19.3)$$

Define V_{loc} as the sum over W . Equation (19.3) says that its before/after difference is the same as the global difference. The proof does not need the values of the unchanged terms.

Exact local-to-global cancellation

Under one fixed separable potential, with a correctly declared action increment, the sum of the changed local terms equals the complete potential difference. Every unchanged term cancels. This equality concerns the declared model; it does not prove the model contains every consequence that matters.

Notice what the proof did not use. It did not require equal references, equal scales or a Gaussian shape. The same cancellation works for any fixed sum of one-coordinate functions. Gaussian geometry provides convenient explicit formulas for the changed terms; additivity provides locality.

19.4 A five-cell numerical account

Take five cells with reference 10X and scale 2X each. Let

$$z = (18, 2, 12, 8, 10)X.$$

The corresponding potential terms are $(8, 8, 1/2, 1/2, 0)$, so the global potential is 17. Transfer 4X from cell 1 to cell 2. The successor is $(14, 6, 12, 8, 10)X$, with terms $(2, 2, 1/2, 1/2, 0)$ and total 5.

The global reduction is $17 - 5 = 12$. The local reduction is $(8+8) - (2+2) = 12$. Reading the last three stocks would add no information needed for this action

value, provided the declared state transition really leaves them unchanged and the potential has the stated separable form.

The whole network still has potential 5 after the transfer. The action improved its own affected region; it did not certify that all cells reached their references. Local correctness and system-wide completion are separate propositions.

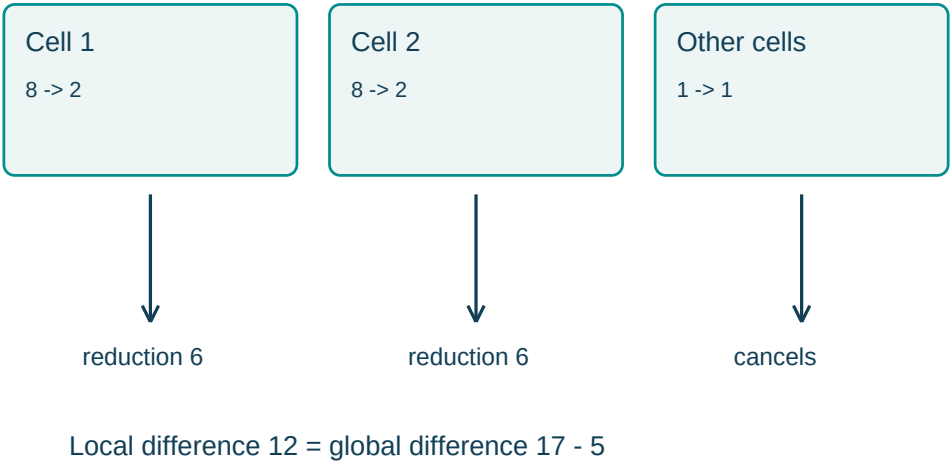


Figure 19.1: The unchanged cells jointly contribute one unit to both global totals and cancel. The local calculation includes both changed endpoints.

19.5 Global refers to the declared boundary

The adjective “global” can sound larger than the mathematics. Here it means all coordinates in the declared model. It does not mean all of Earth, all future time, every ecosystem, every financial consequence or every dimension of wellbeing.

Suppose the modeled network contains only five tanks. Equation (19.3) gives the exact change of that five-tank potential. If the pump also consumes electricity outside the represented boundary, the identity remains true for the tanks while the physical description is incomplete for a claim about the whole operation.

The proper response is to identify the missing carrier and intended scope. A linked electrical account can preserve the physical transformation. A calibrated extension might later place several effects in one declared functional. Neither extension follows merely from calling the five-tank sum global.

19.6 When a neighbour must also be read

A future potential might contain a factor depending on two coordinates, such as a declared interaction $w_{ij}(x_i, x_j)$. Changing x_i can then change that factor even if x_j stays fixed. Reading only the one-coordinate term at i would miss part of the difference.

To express the general rule, suppose

$$V(x) = \sum_{f \in \mathcal{F}} \phi_f(x_{D_f}), \quad (19.4)$$

where D_f is the set of coordinates used by factor f . The affected factors satisfy $D_f \cap W \neq \emptyset$. Every other factor cancels. Evaluation needs the union of coordinates read by the affected factors, which can be larger than the write support.

This is a conditional extension of the same cancellation argument. It does not adopt a new coupled potential for the present book's examples. It explains how to check locality if a later model needs such a potential.

19.7 A counterexample to endpoint-only reading

Consider an explicitly different teaching functional,

$$\tilde{V}(x) = \sum_{i=1}^3 V_i(x_i) + \frac{1}{2}(x_2 - x_3)^2,$$

with dimensionless numerical stock coordinates for this illustration, references $(5, 5, 3)$ and unit scales. Move one unit from cell 1 to cell 2, starting from $(6, 4, 3)$. The successor is $(5, 5, 3)$. Both the reference and physical state have total stock thirteen.

The separable part changes from $1/2 + 1/2 + 0 = 1$ to zero, a reduction of 1. The interaction changes from $1/2(4 - 3)^2 = 1/2$ to $1/2(5 - 3)^2 = 2$, an increase of $3/2$. The complete field reduction is therefore $1 - 3/2 = -1/2$.

An endpoint calculation that retained only V_1 and V_2 would report $+1$, including the wrong sign. A correct local calculation reads x_3 because the interaction factor involving x_2 depends on it. The direct write support is $\{1, 2\}$; the information neighbourhood is $\{1, 2, 3\}$.

The example also shows why the potential's dependency list is part of its scientific declaration. Locality cannot be imposed independently of the equation. Read too little and a changed term disappears. Read more than necessary and a selector may gain information that its design did not intend to expose.

19.8 Three different graphs

The physical graph connects locations by possible actions. The factor graph connects variables used together in the potential. The action-interaction graph connects action increments whose joint value contains a cross term. These graphs can differ.

Our main potential is separable, so its coordinate Hessian is diagonal. Nevertheless, two actions can both withdraw from Ridge. Their increments overlap in the Ridge coordinate, producing a quadratic cross term. A diagonal potential therefore does not imply independent action values.

Conversely, two physical edges can be far apart in a drawing but belong to the same modeled factor if the declared functional couples their variables. Geographic distance alone is not a reliable definition of a mathematical neighbourhood. The relevant question is dependency, followed by whether that dependency is physically justified.

Keeping the three graphs separate helps a reader avoid a common shortcut: “local” does not mean every kind of interaction is absent. It means the required information and support have been specified.

19.9 Global audit and actor information

A scientific evaluator may calculate full V to compare it with the sum of local effects. This is a useful arithmetic and implementation check. It can reveal a missing endpoint, duplicated factor, stale baseline or sign error.

That capability does not determine what an actor may inspect when selecting an action. If a future study defines local actors, its input boundary must be enforced independently of the evaluator’s larger view. The evaluator can observe a result without providing that result as a decision signal.

Likewise, an EBU-blind random proposal mechanism should not quietly read global potential to choose which candidates to offer. Local valuation and local selection are different design properties. Demonstrating the former by cancellation does not establish the latter.

This chapter uses paper calculations. It does not claim that a particular future software harness has already enforced every information boundary. A software claim needs its own implemented interface, tests and source identity.

19.10 Local computation does not choose governance

The theorem shows that some calculations can use small neighbourhoods. Different devices could compute them, and independent auditors could reproduce

them. These possibilities do not settle who may certify references, own sensors, approve model changes or resolve a dispute.

A centrally governed system can calculate local differences. A distributed institution can perform occasional global audits. Neither arrangement is made inevitable by a sum of potential terms. Political authority must be described in its own language rather than inferred from a graph drawing.

Privacy raises a related issue. A valid physical calculation may need quantities and dependencies while requiring little personal information. Whether a public record must identify an individual is an institutional decision. Mathematical support tells us which variables the calculation needs; it does not demand unlimited disclosure.

19.11 Aggregation can erase the effect

Suppose someone replaces Ridge and Vale by a single cell containing their total 20X. The internal four-unit transfer leaves that total unchanged. If the aggregate potential depends only on the total, it reports zero change. The fine account reports a reduction of 12.

There is no algebraic contradiction. The aggregate description discarded distribution between the two locations, precisely the information on which the fine potential depends. An aggregate account that is intended to preserve that effect needs additional state, such as internal deviation, or a justified summary of its potential.

This matters for hierarchical systems. A building can expose boundary flows to a district, but a district total alone does not reproduce every room's temperature or water distribution. A sound hierarchy must state what each level retains and which questions cannot be answered after aggregation.

The introductory locality theorem is not an automatic theorem of lossless compression. It gives a starting point for designing summaries and a test for checking them: does the summary preserve the specific field difference the application claims to compute?

19.12 Guided problem: determine the necessary reads

Take six independent Gaussian terms. An action changes coordinates 1 and 4. Which terms must be evaluated? Only V_1 and V_4 , using their before and after coordinates and fixed parameters. Every other term cancels.

Now add a factor $w_{45}(x_4, x_5)$ and another $w_{56}(x_5, x_6)$. The action changes x_4 but not x_5 . It changes w_{45} , so x_5 must be read. It does not directly change

w_{56} , whose two inputs remain the same. The necessary coordinates are 1, 4 and 5 for this declared instantaneous action.

The answer would change if the transformation also updated x_5 or if another factor depended on x_4 . This is why one must inspect the actual action map rather than propagate “affected” indefinitely through a diagram. Future influence and present algebraic dependence are different things.

19.13 Practice and reflection

In the five-cell numerical example, replace the four-unit transfer by two units. The first two potential terms change from (8, 8) to (4.5, 4.5); their reduction is 7. The remaining terms still sum to 1, so the global potential changes from 17 to 10, also a reduction of 7.

Next, draw a physical path of four locations. Identify one transfer’s write support. Add a hypothetical pair factor connecting one of its endpoints to a neighbour and write the resulting information neighbourhood. Explain why a causal prediction about the neighbour’s later behaviour would require more than this factor list.

Finally, explain the aggregation counterexample in words a new reader can understand. A good answer says that total quantity was preserved while internal distribution changed. It does not say that conservation failed or that the local-to-global theorem was approximate.

What to carry forward

Local exactness follows from the cancellation of unchanged factors. Its scope comes from the potential and action dependencies. A complete local calculation can equal a global model difference while leaving empirical completeness, selection, privacy and governance as separate questions.

Action receipts and exact closure

20.1 A receipt is a physical account line

A shop receipt records a commercial exchange. An EBU action record identifies a represented physical transformation, its baseline, its endpoint and the rule used to evaluate it. Calling both objects receipts should not make us assume that they have the same balancing entry.

In the ideal field account, a positive action value can be balanced by a reduction in potential. It need not be a transfer from a buyer's negative account. This is an algebraic relationship among values. Whether an institution issues a transferable claim on that basis is a further rule.

The preceding chapter showed why one action can be evaluated locally. We now connect many such lines. The basic calculation is simple: a state used as the endpoint of one action becomes the starting point of the next. Its potential appears once with a minus sign and once with a plus sign, and the two occurrences cancel.

20.2 Write the actual chain of states

Consider a sequence

$$x_0 \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_m. \quad (20.1)$$

The arrows describe authoritative physical transitions in a fixed accounting model. For the moment, no external change occurs between them. Define the signed value of line r as

$$E_r = V(x_{r-1}) - V(x_r) - C_r. \quad (20.2)$$

Each line can be computed from its complete affected factors. The local-to-global identity then lets us write the global expression without requiring every actor to read the whole field.

The order is not merely the order in which files happen to arrive. If two actions overlap physically, a software sorting rule does not prove that one observed state preceded the other. Such actions can be recorded as a joint transition

with their child identities retained. The telescoping calculation concerns successive authoritative transitions, which may be whole groups; individual group attributions require the separate treatment in Chapter 22.

20.3 The proof of receipt closure

Sum the lines and expand the first few terms:

$$\begin{aligned} \sum_{r=1}^m E_r &= V(x_0) - V(x_1) + V(x_1) - V(x_2) \\ &\quad + \cdots + V(x_{m-1}) - V(x_m) - \sum_{r=1}^m C_r \end{aligned} \quad (20.3)$$

$$= V(x_0) - V(x_m) - \sum_{r=1}^m C_r. \quad (20.4)$$

Equivalently,

$$\sum_r E_r + [V(x_m) - V(x_0)] + \sum_r C_r = 0. \quad (20.5)$$

The middle states did not become physically irrelevant. They are essential for checking each action. They simply disappear from this sum because their potential values are used consistently at adjacent boundaries.

Conditional receipt closure

Under one fixed potential and a correctly linked sequence of complete state transitions, exact endpoint differences telescope. Signed action values equal endpoint potential reduction minus the declared total process burden. This proves an accounting identity, not a theorem of fair distribution or eventual recovery.

The historical single-action theorem retains the execution and audit assumptions under which it was stated. Writing the general endpoint identity here does not retroactively replace those assumptions with a new operational protocol. The distinction lets us teach the algebra fully without changing scientific history.

20.4 A complete Gaussian example with three lines

Start from (18, 2)X, with references (10, 10)X and scales (2, 2)X. Three lossless transfers send quantities 2, 3 and 1 in that order. The states and potential values are

$$(18, 2) \rightarrow (16, 4) \rightarrow (13, 7) \rightarrow (12, 8), \quad 16 \rightarrow 9 \rightarrow 9/4 \rightarrow 1.$$

With $C_r = 0$, the three lines have values

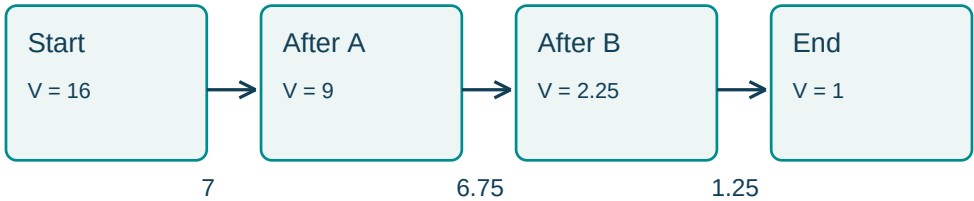
$$E_1 = 7, \quad E_2 = 27/4 = 6.75, \quad E_3 = 5/4 = 1.25.$$

Their sum is 15, exactly $16 - 1$. We can check every endpoint with $V(x) = ((x_1 - 10)^2 + (x_2 - 10)^2)/8$.

Notice that the three-unit middle action has a smaller value than a three-unit action considered alone at the original state. After the first action, the system is already closer to its reference. Reusing the original baseline for the middle line would describe a different counterfactual and break the chain.

Line	Quantity	V before	V after	E_r
1	2	16	9	7
2	3	9	$9/4$	$27/4$
3	1	$9/4$	1	$5/4$

This table is an illustrative calculation on supplied states. It does not report a sequence executed by the research software. Paper arithmetic and empirical execution have different evidential roles even when they use the same equation.



Action values: $7 + 6.75 + 1.25 = 15$

Endpoint difference: $16 - 1 = 15$

Figure 20.1: Three consecutive finite differences telescope to fifteen. Every action uses the endpoint of the preceding action as its own baseline.

20.5 What if a declared process burden is added?

To isolate the algebra, suppose a separately justified extension assigns nonduplicated burdens $C_1 = 0.2$, $C_2 = 0.3$ and $C_3 = 0.1$. Then the net lines are 6.8, 6.45 and 1.15, summing to 14.4. Equation (20.5) gives

$$14.4 + (1 - 16) + 0.6 = 0.$$

The numbers show how the account would close; they are not physical estimates for pumps or couriers. A real use would need to identify the process, its carrier, its unit conversion and why it was not already valued through the represented state.

The first lossless Gaussian model sets $C = 0$. We introduce the hypothetical extension so readers can follow the general equation without importing a hidden transport penalty into every example. Mere distance or the presence of several actors is not a licence to invent extra burden.

20.6 From action lines to attributed totals

Suppose the first and third sequential lines are assigned to actor A and the second to actor B under an explicitly declared record-ownership rule. Let T_A and T_B denote these attributed history totals, reserving B_i for the prospective capacity account introduced later. With zero process burden,

$$T_A = 7 + 1.25 = 8.25, \quad T_B = 6.75.$$

Their sum is the same 15. This is a partition of existing lines. It does not create an additional physical effect or require a payer.

The partition must include each complete line exactly once if these totals are to close. Excluding negative lines, copying a positive line to two actors, or retaining only the largest observed values changes the sum. A useful audit checks the mapping as well as the arithmetic.

Assigning an entire sequential line is simpler than dividing a simultaneous group. Known quantities and identities do not automatically determine unique child field values in a nonlinear group. A group record may have a correct total even while its individual interpretation remains unresolved.

20.7 Why positive-only recording fails

Consider a first action that moves potential from 1 to 4, followed by one that moves it from 4 to 0. With no process burden, the values are -3 and $+4$. The net reduction is 1, agreeing with the endpoint difference.

If the negative entry is dropped and only the positive one retained, the account reports 4. Three units of represented worsening have disappeared from the history. No rounding or statistical explanation is needed: the omission is visible in the exact equation.

This example does not by itself choose how an institution should treat debt or support an essential negative action. It says that a physical history must retain the signed effect even if institutional responsibility is shared, guaranteed

or handled separately. Compensation rules cannot change what the represented state did.

20.8 External changes between recorded actions

Now let an external event first take x_t to z_t , and let the action take z_t to x_{t+1} . Define

$$D_{\text{ext},t} = V(z_t) - V(x_t). \quad (20.6)$$

The action value is $E_t = V(z_t) - V(x_{t+1}) - C_t$. Adding and subtracting $V(x_t)$ gives

$$E_t = V(x_t) - V(x_{t+1}) + D_{\text{ext},t} - C_t. \quad (20.7)$$

Sum over the recorded interval:

$$\sum_t E_t = V(x_0) - V(x_N) + \sum_t D_{\text{ext},t} - \sum_t C_t. \quad (20.8)$$

External worsening can replenish deviation that later actions reduce. External improvement can reduce it without actor contribution. Both cases are represented by the signed external term. Describing all external changes as positive would discard half of the possible accounting situations.

For example, start at (10, 10) with potential zero. An external redistribution produces (14, 6) with potential 4. A two-unit corrective transfer then produces (12, 8) with potential 1. The actor value is 3, and $0 - 1 + 4 = 3$. The external event has no actor receipt; its separate potential change explains the total.

20.9 A prospective capacity account

The Local Gaussian research proposal considers an additional rule: use declared child attributions to update persistent nonnegative action-capacity balances. The mathematical word capacity here names permission to make later modeled actions under that proposed rule. It is neither water stock nor installed pump capacity.

Let B_i be those balances and J the cumulative external potential change. If the adopted updates satisfy

$$\sum_i \Delta B_i = V(z_t) - V(x_{t+1}), \quad \Delta J = D_{\text{ext},t},$$

in the ideal $C = 0$ model, then

$$V(x_{t+1}) + \sum_i B_{i,t+1} - J_{t+1} = V(x_t) + \sum_i B_{i,t} - J_t. \quad (20.9)$$

This follows by substitution. It is a theorem about the stated update equations. It does not prove that they are the right permissions for an economy, that every actor can act, or that the physical state will return to its reference.

Owner mapping, simultaneous netting, path meaning, numerical boundaries and action selection remain parts of the proposed model specification. The group closure identity does not decide them. A balance may become a consequence of a rule; that does not make the rule a consequence of the identity.

The prospective zero-genesis rule starts capacity accounts at zero. It does not quietly authorize borrowing, an automatic refill or an external event credit. Any such mechanism would change the proposal and need its own explicit authority; physical stock is not initial account capacity.

20.10 Physical conservation and value closure

In a lossless stock transfer, $\sum_i x_i$ remains constant. In a beneficial transfer, V falls. The conserved stock and the changing potential therefore cannot be the same quantity.

Our repeated example makes this concrete. Both (18, 2) and (14, 6) contain twenty stock units. Their potentials are sixteen and four. A field reduction of twelve does not mean that twelve stock units were produced or removed. It is the change of a dimensionless standardized discrepancy.

Likewise, equation (20.5) is an exact reconciliation of signed numbers derived from a potential. It is not a new physical conservation law for energy, matter or money. Calling its terms “balanced” is justified; calling the EBU number a conserved material token would change the meaning.

20.11 An audit begins with the links

To check a sequence, first compare adjacent state identities. Does each successor really match the next predecessor within the declared action-only interval? If not, determine whether an external event, field revision or missing action explains the gap.

Next check the potential version and complete affected factors for each line. Recalculate its before and after values and any process burden. Confirm that one physical event has not generated several independent settled copies. Finally compare the sum with the endpoint and external terms.

A remaining residual is information. It can indicate arithmetic error, omitted state, an unmatched version, duplicate accounting or a numerical approximation needing its own bound. In a real implementation the declared numerical

policy governs interpretation. A reviewer should not simply insert a balancing entry whose physical meaning is unknown.

20.12 Guided problem: find the missing line

A record begins at potential 16 and ends at 1. It lists values 7 and 1.25, with no external event or process burden. What value is missing if the history is otherwise complete?

The endpoint reduction is 15. The visible lines sum to 8.25, so the missing value is 6.75. This identifies the required residual, not the missing physical event. In the earlier three-line example it belongs to the transfer from (16, 4) to (13, 7). Without those state records we cannot infer its source, destination or performer merely from 6.75.

This distinction is useful in audit: a closure discrepancy can locate a problem numerically without reconstructing its cause. The missing evidence must still be obtained.

20.13 Practice and reflection

Choose three intermediate potentials between an initial 20 and final 5. Calculate four exact action differences. Their sum must be 15 even if one intermediate step increases potential. Add nonnegative process burdens totaling 2 and verify a net of 13. Explain why this exercise demonstrates telescoping but does not demonstrate physical feasibility of the chosen states.

Now insert an external worsening of 3 during the interval. If the physical endpoints and total process burden remain as stated, the action sum required by equation (20.8) is 16. An explicit chain must identify where the external change occurred. A numerical total alone cannot establish that history.

What to carry forward

Complete, consistently linked endpoint differences telescope. A partition of valid lines preserves their sum; duplicate or missing lines do not. External change and process burden remain explicit. A capacity or institutional account can use these records only under an additional declared rule.

The next chapter uses the same identity to examine cycles and splitting. It will show exactly what a closed account rules out, and why that conclusion is narrower than a guarantee against every possible incentive problem.

Closed cycles, splitting and external change

21.1 The tempting damage-and-repair loop

Imagine moving water away from a useful distribution and then moving it back. The return movement can have positive field value. Could an actor record only the return and claim a reward forever? The exact account answers this by retaining both movements and their true baselines.

The question is not whether repair can be helpful. Repair can reduce real represented damage. The question is whether a closed sequence can create a positive *net* field account while returning every represented coordinate to where it started under the same field.

This chapter develops the original book's closed-cycle and splitting ideas using Gaussian examples. It keeps the theorem's assumptions visible. A closed physical sequence, a driven service cycle and a change of model definition are different cases; treating all of them as one kind of loop obscures the calculation.

21.2 What closes in a closed cycle?

Let an action sequence start at x_0 and end at $x_m = x_0$. Assume a fixed potential, complete authoritative states, no unrecorded external event, unique action records and declared nonnegative process burdens. Telescoping gives

$$\sum_{r=1}^m E_r = - \sum_{r=1}^m C_r \leq 0. \quad (21.1)$$

In the ideal $C = 0$ example, the net is exactly zero. With genuine nonduplicated process burden it is negative.

Closed-cycle identity

For a consistently recorded, undriven closed state sequence under one fixed potential, the field differences sum to zero. Net EBU is therefore minus the declared total process burden. The result does not require Gaussian geometry, but it does require the same function at both appearances of every intermediate state.

If successive transitions are joint groups, the same endpoint calculation applies to their group totals. This does not establish arbitrary individual settlement rules inside each group. Nor does it silently replace the narrower operational assumptions of historical no-overissue theorems.

21.3 A lossless Gaussian loop

Begin at the reference $(10, 10)X$, with scales $(2, 2)X$. Transfer $2X$ from Ridge to Vale. The successor $(8, 12)X$ has potential 1, so the action has value -1 . Transfer the same quantity back. The endpoint returns to $(10, 10)X$ and potential zero, producing value $+1$.

The sum is zero. The return action really did lower the represented deviation. Its positive sign was not false. What would be false is calling that sign the net result of the whole loop while hiding the preceding negative action.

Suppose a separately declared physical extension charges 0.1 per actual pump activation, with no duplication in the state potential. The two values become -1.1 and $+0.9$, giving -0.2 . The extra terms are justified only if there were two physical activations of the declared kind. They cannot be invented merely to make a loop unattractive.

21.4 Return of a visible variable is not full return

A battery can return to the same charge while its health has changed. A tank can return to the same volume while its water quality has worsened. A vehicle can return to its depot after consuming fuel elsewhere. Closing one displayed coordinate does not establish that the complete physical state is unchanged.

If the omitted variable is outside the declared model, the mathematical cycle may be closed within that smaller model. The limitation is then representational: the account cannot support a broader statement about the whole process. Adding a loss coordinate or boundary flow can make the physical record more complete without automatically adding a new term to the value potential.

This is why conservation and valuation were introduced separately. An explicit sink may explain where quantity went. Whether the sink is valued by V requires its own declared definition. A physical loss must never be made to disappear simply because its valuation remains open.

21.5 Sequential splitting does not create field value

Return to the starting imbalance $(18, 2)X$. An unsplit four-unit transfer ends at $(14, 6)X$ and has field value 12. Now split it into two genuine sequential

two-unit transfers. The first ends at $(16, 4)X$ and gives 7. The second begins there, ends at $(14, 6)X$ and gives 5. Thus

$$7 + 5 = 12.$$

The second action cannot also claim 7. That number belongs to a two-unit counterfactual starting from the original state. Using it twice would sum two different hypothetical accounts, not the actual sequential history.

In general, if an intermediate state is y ,

$$[V(x) - V(y)] + [V(y) - V(z)] \quad (21.2)$$

$$= V(x) - V(z). \quad (21.3)$$

The equality says nothing about the number of real activations, measurement costs or physical losses. For equality of *net* EBU, the split and unsplit process burdens must also match.

21.6 Splitting an identity differs from splitting a process

An actor may divide a name or record into smaller pieces without changing the physical process. That should not create additional field reduction. A correctly linked account preserves the endpoint identity.

Physically stopping and restarting a machine can be different. It may change energy use, wear, temperature or another represented state. If those consequences are declared and measured, the split and unsplit actions no longer have the same complete endpoint and burden. A difference in value then records a physical difference rather than a reward for paperwork.

The same distinction applies to a group attribution. Dividing a fixed increment into proportional child increments along the same path preserves their summed path attribution by linearity. It does not prove that a strategic actor cannot change the accepted quantities, group membership or feasible menu through a different declaration. Those are mechanism questions beyond this arithmetic.

21.7 An undriven net bound

For a nonnegative potential and nonnegative process burdens, telescoping also gives

$$\sum_r E_r = V(x_0) - V(x_m) - \sum_r C_r \leq V(x_0). \quad (21.4)$$

An undriven history cannot produce a cumulative net greater than its initial represented potential under these assumptions. In the recurring example, that upper bound is sixteen.

The word net matters. Positive and negative entries can alternate. Their sum can be bounded while the sum of positive entries alone grows if repeated damage is separately retained as negative entries. The theorem is not a bound on gross activity, total wages, physical throughput or useful care.

It also does not imply that the bound is attainable. The action graph may prevent reaching the reference; permissions may exclude useful movements; an actor may choose poorly or not act. An upper bound on an account does not supply a mechanism that achieves it.

21.8 Equal potential is weaker than equal state

States (18, 2) and (2, 18) have the same potential. A lossless sixteen-unit transfer connects them and gives zero field value. That is an equal-potential transition, but it is not a closed state cycle.

The distinction matters for future opportunities. An edge leaving Ridge may now have much less available stock; an edge leaving Vale may have much more. A later action can therefore differ even though the prior scalar potential was equal.

The algebra of zero endpoint difference only requires equal potentials. Calling the process a closed *physical* cycle requires the stronger state equality, including all relevant coordinates. The book uses the stronger wording when claiming a return of the represented system.

21.9 External forcing can recreate deviation

Start at (10, 10). An external lossless redistribution changes the state to (14, 6), increasing potential by four. An action returns it to (10, 10), reducing potential by four. Repeating this externally driven pattern can produce repeated positive action values.

There is no violation of equation (21.1), because the interval is not undriven. The external record supplies the missing term:

$$\sum_t E_t = V(x_0) - V(x_N) + \sum_t D_{\text{ext},t} - \sum_t C_t.$$

After ten illustrative repetitions with zero process burden, the action values sum to forty and the external potential increases sum to forty. This is arithmetic on a specified pattern, not a simulated trajectory or an assertion that a particular actor would repeat the correction.

A real recurring need may create such a pattern, but it must be modeled through physical consumption, transformation or boundary exchange. In a

closed homogeneous stock demonstration, saying “demand consumed four” without a carrier or boundary would break conservation. A desire, request and physical sink are not interchangeable.

21.10 Who caused the purported external event?

The accounting classification must be honest. If an actor deliberately caused a damaging movement and later repaired it, describing the first movement as an unrelated natural disturbance can hide the actor’s negative line. The closed-cycle theorem cannot repair a falsified event classification.

This is a causal and verification question. A chronology alone does not establish responsibility, and a physical balance alone does not identify intention. A deployed account would need a declared event model and evidence for its attribution decisions.

The correct scientific conclusion remains narrow: separating an external term makes the algebra explicit. It does not prove that every real-world external term can be identified without uncertainty or strategic manipulation. Those problems require later methods and evidence.

21.11 Changing the ruler creates a model-change term

Suppose the state is $(14, 6)$ with reference $(10, 10)$. Under scales $(2, 2)$, its potential is 4. Under scales $(4, 4)$, its potential is 1. No water moved. The difference of three came from changing the ruler.

If an account evaluated earlier actions with the first scale and later actions with the second, the intermediate potential terms would no longer cancel without an additional term. Let V^{old} and V^{new} be the two definitions. At a fixed revision state x , define

$$D_{\text{model}}(x) = V^{\text{new}}(x) - V^{\text{old}}(x). \quad (21.5)$$

This term records a change of representation. It is neither an actor’s physical restoration nor a newly discovered physical loss. How an institution handles existing balances after revision is a separate policy question.

References and scales may need revision as knowledge improves. The lesson is not to freeze an inadequate model forever. It is to preserve the distinction between learning about the world and changing the world, so that an account remains auditable through both.

21.12 Bounded deviation does not prove recovery

In the first Gaussian physical domain, each nonnegative stock is at most the fixed total M . With finitely many fixed references and positive scales, V has a finite bound on that domain. This is true even for a mechanism that never approaches the reference.

A system can remain trapped away from its reference, alternate between two states or refuse every available action while remaining bounded. Therefore reporting only that the potential did not diverge would not establish homeostatic recovery in such a world.

The prospective capacity invariant in Chapter 20 adds an accounting constraint. It still does not choose the next action. To establish attraction, a proof would need assumptions on the transition mechanism; to demonstrate recovery empirically, a registered study would need appropriate observations and controls. Neither task is completed by boundedness alone.

21.13 Guided problem: audit a three-line loop

Let the successive potential values be 2, 7, 3, 2, with process burdens 0.2, 0.1, 0.3. Compute each line:

$$E_1 = 2 - 7 - 0.2 = -5.2, \quad E_2 = 7 - 3 - 0.1 = 3.9, \quad E_3 = 3 - 2 - 0.3 = 0.7.$$

The sum is -0.6 , exactly minus total process burden. These supplied potential values illustrate closure; a complete physical example would also need admissible states and actions producing them.

Now split the middle line through potential 5 with burdens 0.04 and 0.06. The new values are 1.96 and 1.94, summing to the same 3.9. If both pieces were instead valued from potential 7, they would not describe that sequential split.

21.14 Practice and reflection

Use the Gaussian reference state $(10, 10)$, move to $(7, 13)$ and then return. The intermediate potential is $9/4$, so the two cost-free values are $-9/4$ and $+9/4$. Add one actual activation burden of c to each action and express the net as $-2c$.

Next, describe a process returning a tank's volume while changing its quality. State which additional coordinate or boundary record would be necessary to assess closure of the intended physical scope. Finally, explain why a trapped state with constant nonzero potential satisfies boundedness while failing recovery.

Several actors on one nonlinear field

22.1 Known quantities do not make value linear

Two couriers each deliver two units from Ridge to Vale. We know their identities, quantities and directions. Yet valuing each delivery as though the other did not exist gives a different sum from valuing the combined four-unit movement. The difficulty is not missing quantity data. It is the curvature of the potential.

Physical increments can add linearly while the value of their sum remains nonlinear. This distinction is already present in a diagonal Gaussian potential. It does not require a mysterious nonlocal force or a new physical interaction between the couriers.

This chapter retains the original shared-source lesson and updates its arithmetic. It also makes the common-path decomposition explicit. A decomposition that closes mathematically is useful, but its interpretation must be stated: path accounting, physical realization, causal contribution and institutional entitlement are different claims.

22.2 The two-courier counterexample

Use the repeated state $(18, 2)X$, references $(10, 10)X$, scales $(2, 2)X$ and $C = 0$. Let the increments be

$$\delta_A = (-2, 2)X, \quad \delta_B = (-2, 2)X.$$

Either courier alone would produce $(16, 4)X$, reducing potential from 16 to 9. The two same-baseline singleton values therefore sum to $7 + 7 = 14$.

Their actual joint increment is $(-4, 4)X$, producing $(14, 6)X$ with potential 4. The joint field value is 12. The difference of two is not a rounding error or an unobserved physical loss. It is the discrepancy between a joint finite difference and a sum of different counterfactual finite differences.

A counterfactual sum is not the joint account

Each singleton value asks what would happen if that child acted alone at the frozen baseline. The joint value asks what happens when the accepted children act together. On a nonlinear field these are different comparisons, even when the child quantities are known exactly.

If the couriers actually act sequentially, the values are 7 and 5, which sum to 12. A physically meaningful sequence supplies different baselines. If execution is simultaneous, inventing that sequence afterward gives an allocation convention, not an observed intermediate history.

22.3 The exact group formula

Let the accepted increments be δ_a , and define $\delta_G = \sum_{a \in G} \delta_a$. Under fixed Gaussian geometry,

$$E_G = -\mu(z)^T \delta_G - \frac{1}{2} \delta_G^T H \delta_G \quad (22.1)$$

in the ideal cost-free field account. Here $H = \text{diag}(1/\sigma_i^2)$ and $\mu(z) = H(z - x^*)$.

The same-baseline singleton sum uses δ_a separately in each quadratic term. Expanding the joint square gives

$$E_G - \sum_a E_a^{(0)} = - \sum_{a < b} \delta_a^T H \delta_b. \quad (22.2)$$

The superscript (0) marks the common frozen baseline. It does not denote an observed first step. The right-hand side is a signed cross-term diagnostic for these fixed increments and this fixed potential.

For the two couriers, $H = \text{diag}(1/4, 1/4)$ and $\delta_A^T H \delta_B = 2$, giving $12 - 14 = -2$. The shared coordinates and directions determine the sign.

22.4 The discrepancy can have either sign

Convexity does not make every group discrepancy negative. Consider opposing increments from the reference state (10, 10):

$$\delta_A = (-2, 2), \quad \delta_B = (2, -2).$$

Each action alone increases potential to 1, so each singleton value is -1 . Their joint increment is zero and the ideal group field value is zero. Thus $E_G - (E_A^{(0)} + E_B^{(0)}) = 2$.

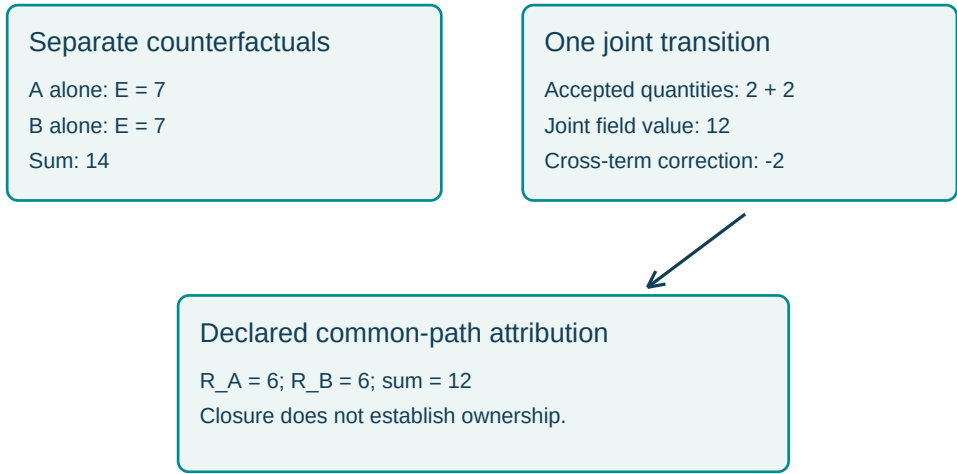


Figure 22.1: Two separately evaluated two-unit actions each have value seven at the frozen baseline. The joint four-unit action has value twelve, not fourteen. A declared straight-path decomposition assigns six to each.

The cross product is -2 , agreeing with equation (22.2). Two same-direction increments gave a negative discrepancy; opposing increments give a positive one. The physical feasibility of the combined flows must still be declared. Endpoint cancellation does not automatically permit unlimited circulation or establish that the real process is free.

For disjoint coordinate supports under diagonal H , the cross product is zero. The field values then add. Shared physical constraints can nevertheless couple the permitted quantities. An arithmetic additivity result does not remove a common pump-capacity or source-availability constraint.

22.5 One declared path through the joint change

Consider the straight path

$$\gamma(s) = z + s\delta_G, \quad 0 \leq s \leq 1. \quad (22.3)$$

The parameter s says what fraction of the joint increment has been included in this mathematical path. If the physical model explicitly stipulates constant-rate simultaneous execution, that path can also be its realized state path. If only endpoints are specified, it is a declared interpolation convention.

For each child, define the simultaneous-path field attribution

$$R_a = - \int_0^1 \nabla V(z + s\delta_G)^T \delta_a ds. \quad (22.4)$$

The minus sign follows the book's convention: reducing potential gives positive field value. The integral sums the child's directional contribution along the common path. Every child uses that same path; choosing a different path for each would require a different argument for closure.

This construction has Aumann–Shapley-type common-path mathematics. Naming that relationship is an attribution of mathematical structure, not an adoption of a fairness theorem or a claim that physical matter follows a cooperative-game allocation rule.

22.6 Why the path contributions close

Add equation (22.4) over children. Finite summation and integration commute, and the child increments sum to δ_G :

$$\sum_a R_a = - \int_0^1 \nabla V(z + s\delta_G)^T \delta_G ds \quad (22.5)$$

$$= - \int_0^1 \frac{d}{ds} V(z + s\delta_G) ds \quad (22.6)$$

$$= V(z) - V(z + \delta_G). \quad (22.7)$$

The last step is the fundamental theorem of calculus, under the differentiability and path-domain assumptions. Our fixed quadratic satisfies the required smoothness.

Common-path closure

For fixed additive child increments and one declared differentiable common path of the stated form, the signed derivative integrals sum to the exact group field reduction. The theorem establishes the sum. It does not establish ownership, causal necessity, fair shares or permission to spend an attribution.

If process burdens are included, the group net also subtracts C_G . Dividing that burden among children is another declared accounting step unless independently measured child burdens already provide the division. The field integral alone should not be described as the complete net receipt in that case.

22.7 The Gaussian integral is explicit

Because the Gaussian gradient is affine,

$$\nabla V(z + s\delta_G) = \mu(z) + sH\delta_G.$$

Substitute into equation (22.4) and integrate 1 and s :

$$R_a = -\mu(z)^T \delta_a - \frac{1}{2} \delta_a^T H \delta_G. \quad (22.8)$$

No numerical quadrature is needed for this quadratic. The expression includes the way each child overlaps the full group increment. It does not merely divide the total by the number of names on the record.

For the symmetric two-courier example, each attribution is 6 and their sum is 12. The same-baseline singleton value was 7, while the first and second sequential values were 7 and 5. These are three different questions with compatible arithmetic.

22.8 Unequal quantities on the same edge

Let A send one unit and B send three from the same baseline (18, 2). The group still sends four units and has field value 12. The singleton values are

$$E_A^{(0)} = 4 - 1/4 = 3.75, \quad E_B^{(0)} = 12 - 9/4 = 9.75,$$

summing to 13.5. The cross-term correction is -1.5 .

Along the common path, A's increment is one quarter of the total and B's is three quarters, so their field attributions are 3 and 9. This proportionality follows here because the increments are collinear and use the same path. It is not a universal justification for dividing all group values by sent quantity.

If the actions use different destinations, different scales or different physical maps, equal sent quantities need not have equal directional field effects. A proportional policy can still be defined, but it answers a policy question distinct from the common-path calculation.

22.9 A fan-out example with distinct destinations

Use three locations with $z = (18, 2, 10)$, references (10, 10, 10) and scales all equal to 2. Let A send two units from location 1 to 2 and B send two from location 1 to 3. The increments are $(-2, 2, 0)$ and $(-2, 0, 2)$.

Initial potential is 16. The joint successor is (14, 4, 12), whose terms are 2, 4.5 and 0.5, totaling 7. Group value is 9. Acting alone, A gives 7 and B

gives 3, totaling 10. Their one-unit discrepancy comes from the shared source coordinate.

The common-path attributions are 6.5 to A and 2.5 to B. They sum to 9. Equal sent quantities did not yield equal attributions because one destination began below reference while the other began at reference. The arithmetic describes the declared path; whether these are appropriate economic shares remains a separate question.

22.10 Three comparisons that must stay separate

The group-versus-singletons discrepancy compares simultaneous execution with two or more counterfactuals that each start alone. A parallel-versus-sequential comparison uses a complete sequential history. An institutional allocation compares ways of distributing a given joint number.

With fixed additive increments and the same endpoint, each physically feasible sequential order and the group have the same total field value. A feasible joint endpoint alone does not guarantee feasible intermediate states in every order. The singleton sum can still differ. Thus a nonzero singleton discrepancy does not establish that simultaneity created a physical efficiency gain.

If a resolver changes quantities when actions are replayed sequentially, or a delay changes the state, the sequential endpoint can differ. The comparison then has to identify the ordering, resolver, timing and endpoint. Reporting an “interaction benefit” without naming this comparator leaves its meaning incomplete.

22.11 Grouping follows dependencies

A shared coordinate often requires joint valuation. Shared constraints can also require joint physical resolution even when the eventual increments happen to have disjoint support. Relevant dependencies include overlapping execution or reservation intervals, common capacities, and effects on another action’s feasibility or completion.

A conservative grouping design collects connected dependent actions within a compatible accounting boundary. This prevents an action indirectly linked through a third child from being treated as independent without examination. It is a design rule requiring declared supports and timing; it is not established merely by two actors belonging to the same organization.

Every group should preserve its child records. Joint evaluation does not erase who sent what, which edge was used or which evidence supports the quanti-

ties. It acknowledges that the group's nonlinear field total must be computed together.

22.12 Why an allocation remains an additional choice

Many numerical allocations can sum to 12. Equal shares, proportional shares and some marginal constructions all close in a symmetric example. Closure alone cannot distinguish them. An institution may also consider responsibility, measured effort, guaranteed access, bargaining power and risk.

Even attractive algebraic properties have precise scope. Relabeling fixed children leaves the common-path formula unchanged. Proportionally splitting a fixed child preserves its summed attribution. But changing requests can change physical acceptance, group composition and affordability. Those effects require analysis of the entire mechanism.

The prospective Gaussian programme proposes using a declared path attribution in an action-capacity rule. That proposal can be studied without pretending it has already solved economic fairness. A negative result about the mechanism would not invalidate the calculus identity; a correct identity would not rescue a poorly performing mechanism.

22.13 A limited interaction diagnostic

For a fixed set of increments and a shared baseline, define $F(A) = V(z) - V(z + \sum_{a \in A} \delta_a)$. Its pair interaction is $-\delta_a^T H \delta_b$, and higher-than-pair terms vanish for this quadratic set function. This is an algebraic property of the fixed-increment definition.

If each subset reruns an allocation rule, its accepted increments may change. Then the function is no longer simply the same quadratic evaluated on subsets of fixed vectors, and the degree argument need not apply. Boolean-lattice formulas also require every subset being used to be admissible.

The book needs none of these subset diagnostics to calculate the main group total. They are optional later analysis, not extra value that can be issued on top of the endpoint difference. Adding "interaction reward" after already recording the full group value would count part of the same change twice.

22.14 Guided problem and practice

At (18, 2), let A and B each send one unit. Each singleton value is 3.75; the two-unit joint value is 7. The discrepancy is -0.5 . Common-path attributions are 3.5 each. Sequential values are 3.75 then 3.25. Explain every number by identifying its baseline and comparison.

Next use the reference state and opposing one-unit increments. Each singleton has value -0.25 , the joint value is zero and the discrepancy is $+0.5$. The common-path gradient is zero throughout the constant joint state path, so both field attributions are zero. State why this does not show that two real pump operations consume no energy.

Finally, describe two different institutional shares that sum to the fan-out total 9. The fact that you can do so is enough to show that a sum identity does not uniquely specify ownership. It does not establish that either rule is desirable.

What to carry forward

Accepted increments determine a joint endpoint and an exact group value. Same-baseline singleton values need not add. A declared common path supplies an exact decomposition, whose physical and institutional interpretation must be stated separately. The group total is never enlarged by adding its own interaction diagnostic again.

The next chapter carries this discipline beyond one local group, to routes, different resource accounts and the slower decisions that might learn from repeated physical records.

Routes, resource accounts and learning

23.1 A local equation in a connected world

The antidote a clinic needs may be made far away. A pump can depend on a spare part, an electrical supply and a skilled repairer. A water transfer can depend on a chain of transformations that began with fuel. These examples were central to the original Part I because they prevent the local equation from becoming an isolated tank puzzle.

Local accounting does not mean that the world consists of disconnected actions. It means that each calculation uses a declared physical boundary and the complete factors affected within it. Links between actions retain sequence, dependence, quantities and provenance. They do not automatically make the whole chain one indivisible action.

This chapter explains what can already be said about composition and what would need a larger model. It preserves the practical questions about routes, infrastructure, cooperation, poverty and institutions while keeping them distinct from the Gaussian identities.

23.2 A route is an ordered physical record

Let a stock travel from Ridge to a hub and then to Vale. The path has two directed edges. The first action decreases Ridge and increases the hub; the second decreases the hub and increases Vale. Each action has a finite quantity, time interval and predecessor state.

If the actions occur successively under one fixed potential and no intervening external change, their field differences telescope. The hub may temporarily carry extra stock, but its intermediate potential cancels when the route total is added. This cancellation requires using the actual intermediate state, not evaluating both edges independently against the original one.

Suppose the hub already contains stock. The later outgoing action may use stock that was present earlier rather than a physically distinguishable parcel just received. A quantity ledger can still track flows. A provenance claim about which shipment served which patient needs an explicit tracing conven-

tion or evidence. Algebraic endpoint equality does not identify the history of individual molecules.

23.3 A complete two-edge Gaussian route

Use three cells with references $(10, 10, 10)X$, scales $(2, 2, 2)X$ and initial state $(18, 10, 2)X$. The order is Ridge, hub, Vale. Potential is $8 + 0 + 8 = 16$.

Send four units from Ridge to the hub. The state becomes $(14, 14, 2)X$, with potential $2 + 2 + 8 = 12$. The first action's value is 4. Now send four from the hub to Vale. The state becomes $(14, 10, 6)X$, with potential $2 + 0 + 2 = 4$. The second action's value is 8. Their route total is 12.

A direct lossless transfer of four from Ridge to Vale has the same endpoint and the same field reduction. Under this ideal field alone, adding an intermediate hub has not created a geographical fee. If the route has additional energy use, time effects or material losses, those are additional physical facts that must be represented and compared.

Stage	State	V	Field value
Before	$(18, 10, 2)$	16	–
Ridge to hub	$(14, 14, 2)$	12	4
Hub to Vale	$(14, 10, 6)$	4	8

The example provides a calculation, not a recommendation that every delivery should pass through a hub. It omits time, handling and equipment. The scope is stated precisely so that a later extension can identify what changed.

23.4 Distance enters through physical mechanisms

A longer route may require more energy, more storage, a different transport mode or greater exposure to delay. Distance matters through those physical relationships. The present equation does not contain a universal inverse-square distance law or an automatic per-kilometre burden.

Two routes of the same length can have different grades, loads, efficiencies and reliability. Two routes of different lengths can deliver the same service with different total physical effects. A route comparison must therefore state the transformations and constraints rather than use distance as a substitute for them.

For a time-sensitive medicine, a shorter delay can matter even when transported quantity is equal. If the model ignores expiry or temperature, it cannot certify that the two deliveries are equivalent. A travel-time label in a table is not enough unless the intended service comparison actually uses it.

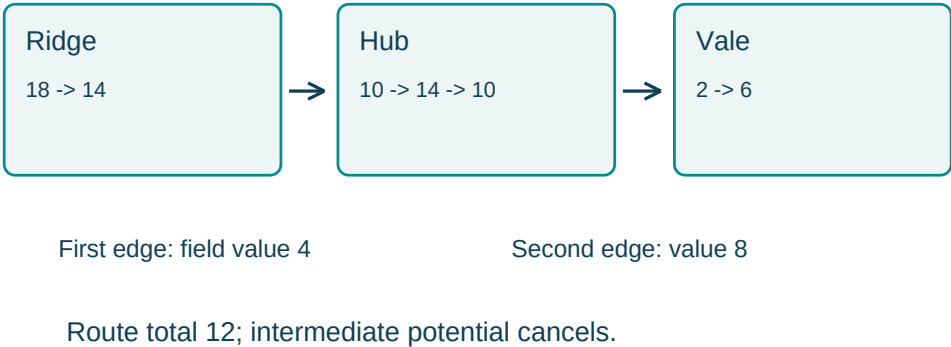


Figure 23.1: The relay route has two real transformations and two consecutive baselines. Their signed values add to the endpoint difference.

23.5 A loss ledger is a physical extension

The main Gaussian examples are lossless. To understand the boundary, consider a separate quantity-only route where ten units pass through efficiencies 0.9, 0.8 and 0.95. Delivered amounts are 9, 7.2 and 6.84. The successive missing quantities are 1, 1.8 and 0.36, totaling 3.16.

Those quantities need named carriers, sinks or boundary crossings. The physical ledger can then show $10 = 6.84 + 3.16$ in the relevant stock units. Whether any “loss” remains useful somewhere else depends on the carrier and boundary. Water entering another basin is not annihilated water; usable medicine becoming waste is a transformation of condition or classification.

The product $0.9 \times 0.8 \times 0.95 = 0.684$ is an end-to-end quantity retention ratio. EBU values are not multiplied that way. Each finite value depends on its own before and after field. The lossless Gaussian controller or attribution formulas cannot simply inherit every interpretation from a heterogeneous-efficiency world without re-derivation.

Scope of the loss illustration

The route calculation above closes physical quantities only. It adopts no loss-aware runtime, no valuation of the sink, and no residual dissipation charge. Physical accounting and valuation need separate declarations, especially where an older process-burden rationale assumed a missing loss coordinate.

23.6 Fuel, electricity and water stay linked and distinct

A generator converts fuel energy into electricity and other outputs. A delivery system conveys electricity. A pump uses electricity to move water and generates heat. These are linked transformations, but the fuel quantity, electrical energy and water quantity do not share one stock unit.

An adequate physical description names each input and output, including boundary exchange and losses of useful form. It does not call the entire chain one water action while treating the upstream transformation as invisible. Nor does it subtract an additional burden at every arrow merely because the arrows exist.

The accounting boundary determines which effects are valued through state differences and which residual physical burden, if any, is separately declared. Double-counting can arise if the same electrical consumption lowers a represented energy account and is also subtracted unchanged as a generic water-process cost without an explicit cross-account interpretation.

The original book's practical lesson survives: decompose enough to measure physical transformations, and preserve enough links to understand the service. Neither a disconnected list nor an all-purpose mega-action provides both.

23.7 Several resource accounts form a vector

Suppose one project reports +8 in a declared water field and −2 in a declared energy field. The honest record can display

$$\mathbf{E}_a = \begin{pmatrix} +8 \text{ EBU}_{\text{water}} \\ -2 \text{ EBU}_{\text{energy}} \end{pmatrix}. \quad (23.1)$$

Even if both numbers are mathematically dimensionless after standardization, their scales and interpretations can differ. Adding them to obtain 6 chooses a conversion or weighting rule. Dimensionless notation alone does not justify that choice.

A scalar combination may be useful for a specified decision, but its weights need an explicit meaning and sensitivity analysis. If one resource is non-substitutable for an essential function, an easy gain in another account cannot automatically compensate for its loss. Physical feasibility and minimum service conditions may remain separate constraints.

Within each resource account, the relevant endpoint and process terms can close. The vector preserves those separate reconciliations. It allows comparison without pretending that one universal token has already solved the problem of valuing heterogeneous systems.

23.8 Planning is a different use of the records

A planner can compare feasible routes, examine repair strategies or forecast future needs. Such planning may use information that a local action quote does not: deadlines, inventories elsewhere, expected failures and future scenarios. This is acceptable when the layer and its assumptions are declared.

The planner should not be smuggled into the EBU formula. The formula evaluates an action given its state and field. A route optimizer chooses among possible sequences using an objective. A random actor chooses by a different rule. Both can use the same value calculation without becoming the same experiment.

If a planner's prediction later differs from reality, the forecast and actual record should be compared. Replacing the actual physical result with the predicted favourable one would destroy the account's purpose. A forecast can justify a decision under uncertainty while remaining a forecast.

23.9 Receipts as evidence for maintenance

A series of well-specified records may reveal that a pump repeatedly uses more electricity for the same delivered water, that a route often misses a deadline or that emergency parts come from one fragile source. These patterns can motivate investigation.

They do not automatically identify a cause. A higher apparent burden may reflect changing loads, altered references, better measurement or a genuine equipment problem. Comparisons need consistent conditions and versioned definitions. A report should retain these explanatory variables instead of ranking devices by one unqualified average.

Maintenance itself has physical consequences. It can consume parts now and improve later performance. Its present construction or repair record and the later operating records are connected, but the later improvement cannot be issued in advance as an observed current field reduction unless a justified present capacity coordinate already represents it.

23.10 Infrastructure changes the menu of possible actions

Building a bridge, laying a pipe or adding a cold store can change which actions are feasible. This is a deeper change than selecting a different quantity on an existing edge. The action menu, physical map and possibly the represented state must be updated with explicit versions.

The field can describe a consequence only through its declared variables. If a new bridge changes travel time but the model contains neither time nor service

constraints, the benefit is not captured by merely adding an edge label. The limitation should be reported and the representation revised where justified.

Once the new infrastructure operates, future receipts can supply evidence about actual use and consequences. That evidence may differ from the investment forecast. A good archive lets readers study the difference, including unused capacity, maintenance demands and unexpected access effects.

23.11 Cooperation is a hypothesis to examine

Two clinics might share a cold store, transport capacity or specialist repair service. If doing so meets the same needs with lower total represented burden, the group account can show that difference. It still does not determine how to distribute costs, access, responsibilities or gains.

Shared provision can create dependencies and vulnerabilities as well as efficiencies. A failure at a common store can affect both clinics. A powerful participant can control access. Geographic concentration can disadvantage a remote community. These concerns require service and institutional analysis alongside the physical account.

Thus the reasoned hypothesis is conditional: better physical information may help identify beneficial cooperation. Whether it produces stable, fair cooperation depends on mechanisms and behaviour that must be studied. A calculation showing a lower potential is not a proof of voluntary participation or equitable bargaining.

23.12 Poverty can constrain the choice set

A household may use an inefficient device because it cannot obtain a replacement. Another household may have capital, secure housing and access to shared infrastructure. Their physical records can differ because their feasible opportunities differ, not because one values the environment more.

An institution that attaches consequences to EBU records must confront that asymmetry. It could provide access, collective investment, support for essential services or other arrangements, but the equation does not choose among them. A physical burden should remain visible while responsibility and assistance are considered separately.

This is why socioeconomic questions were placed at the beginning of the revised book. They are not a decorative appendix to an otherwise finished currency. An account intended to serve life must be assessed for how people actually encounter its rules, including people with disability, limited resources or unavoidable high-burden needs.

23.13 Observation does not require unlimited surveillance

A record can identify a transformation without publicly exposing every personal detail of its beneficiaries. The information needed to verify a physical event depends on its quantities, timing, boundaries and evidence. Some identity links may be held by authorized auditors; other information may be aggregated or withheld.

The choices need explicit governance: purpose, access, retention, correction and appeal. Mathematical locality offers a way to describe necessary physical inputs. It does not itself design secure storage or justify collecting personal data unrelated to the calculation.

More observations can improve an estimate, but more data can also multiply inconsistent definitions and opportunities for misuse. The quality of a physical account depends on what its measurements mean and how they can be challenged, not simply on how many records exist.

23.14 Money and physical accounts can coexist

A payment can finance a repair while a physical account describes its effects. The two ledgers may inform the same decision without being interchangeable. Money can redistribute claims or fund access; it cannot retroactively change the litres moved or the field used in an observed account.

One possible research sequence begins with observational physical accounts in a narrow domain, examines measurement and interpretability, and only later studies institutional consequences. This is a possible path, not an adopted deployment plan. Broader claims require broader evidence.

The prospective synthetic capacity rule is also narrower than a real economy. A nonnegative balance controlling a simulated action menu does not supply property rights, income support, appeals, labour agreements or democratic authority. Those topics remain central to the later institutional programme precisely because the physical equation cannot settle them alone.

23.15 Guided problem: compare complete routes

Two routes have the same complete represented endpoint and use one fixed field. Suppose an explicitly declared physical extension assigns residual process burdens 7 and 18 to the routes. Their field components are equal, so the first route's net value exceeds the second's by 11. This difference follows without knowing the common endpoint reduction.

Now suppose the lower-burden route misses a required deadline. It no longer satisfies the same service problem. A report should say that it fails the dead-

line, not rank it as the better solution by ignoring time. If time-dependent degradation changes the physical endpoint, even equality of the field components must be reconsidered.

23.16 Practice and reflection

Recalculate the two-edge Gaussian route with quantity two. The first state becomes (16, 12, 2) with potential 13; the second becomes (16, 10, 4) with potential 9. The values are 3 and 4, totaling 7, equal to the direct endpoint reduction.

Then choose a familiar service, such as drinking water or a refrigerated meal. Draw three linked physical transformations and name the carrier at each. Identify one quantity ledger, one possible service constraint and one institutional question that cannot be read from the physical ledger.

Finally describe an infrastructure improvement and separate its present construction effects, measured capability, future forecast, authorization and compensation. State which claim can be observed now and which must wait for later evidence.

What to carry forward

Routes compose physical actions under explicit states and times. Resource accounts retain their meanings as a vector unless a conversion is justified. Records can inform learning and cooperation, but adaptation, access and governance require their own evidence and choices.

The eight practice studios now rebuild the complete account. Their repeated scenes are deliberate: understanding grows when the same physical quantity can be traced through a drawing, a vector, a potential, an exact value and an auditable record.

Studio: reading the Ridge–Vale state

Physical picture: begin with the objects

Ridge has a store. Vale has a clinic. At the instant we inspect them, the store contains eighteen units of one usable stock and the clinic contains two. Before asking what a transfer is worth, we must learn to say exactly what those two numbers describe. A reliable calculation begins with names, units, and an order of coordinates.

This studio returns to the small world that has accompanied the main chapters. Its pace is deliberately slower. Reading a definition is one skill; rebuilding a complete calculation from a blank sheet is another. Work with a pencil, leaving enough space to write a sentence beside each equation. The sentence should say what changed physically. If the sentence and the equation disagree, stop at that line. A later correct multiplication cannot repair a mistaken description of the action.

Our numbers are teaching data. They are not measurements from a real clinic, nor recommended stocks for any medical service. In the closed illustration, X denotes the unit of a single homogeneous, divisible resource. We use one resource throughout a calculation. We do not combine water, medicine, electricity, and health into an unnamed stock merely because each can be represented by a number.

24.1 Make a state card

Write the following card before doing arithmetic:

Index	Place	Symbol	Present stock
1	Ridge store	x_1	18X
2	Vale clinic	x_2	2X

The state is the ordered column

$$z = \begin{pmatrix} 18 \\ 2 \end{pmatrix} X. \quad (24.1)$$

The letter z identifies the frozen state from which the actions in this epoch will be compared. “Frozen” means that all candidate calculations use these same recorded inputs. It does not mean that time has physically stopped, or that observations cannot become stale. A practical system needs a rule for recognizing a new epoch when relevant conditions change.

The subscript in x_2 names the second entry. It does not multiply x by two. The brackets gather two measurements into one mathematical object without mixing their locations. Ridge’s eighteen remain at Ridge. Vale’s two remain at Vale. The vector records their relation to the model’s coordinate list.

Now deliberately reverse the list of places. If Vale is placed first, the same physical situation is represented by $(2, 18)^T$. That is a harmless change of notation only if every associated vector and matrix is reordered too. Reversing the state while leaving an action column unchanged describes a different event. This is why names belong beside numbers even when the numerical exercise is small.

24.2 Separate a state from a change

A stock tells us what is present. An increment tells us how that stock changes. A lossless transfer of four units from Ridge to Vale has increment

$$\delta = \begin{pmatrix} -4 \\ +4 \end{pmatrix} X. \quad (24.2)$$

The minus sign is attached to Ridge because Ridge gives up stock. The plus sign is attached to Vale because Vale receives stock. Add corresponding entries:

$$z + \delta = \begin{pmatrix} 18 - 4 \\ 2 + 4 \end{pmatrix} X \quad (24.3)$$

$$= \begin{pmatrix} 14 \\ 6 \end{pmatrix} X. \quad (24.4)$$

Read this aloud: the successor contains fourteen units at Ridge and six at Vale. Do not read it as “the system gained four.” One place gained four, another lost four, and the complete system gained zero.

The present chapter performs arithmetic on declared before and after states. It does not execute a research model. In an empirical application, the first vector would come from observations, the proposed increment from a transformation model, and the successor would require confirmation. Those three sources of information should remain visible in a receipt. A predicted successor is not made observed merely by writing an equality in a planning sheet.

24.3 Use the vector of ones to ask a physical question

Let

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (24.5)$$

Its transpose is the row $\mathbf{1}^T = (1, 1)$. Multiplying this row by a state column adds the entries:

$$\mathbf{1}^T z = 18 + 2 = 20X. \quad (24.6)$$

For the change,

$$\mathbf{1}^T \delta = -4 + 4 = 0X. \quad (24.7)$$

For the successor,

$$\mathbf{1}^T (z + \delta) = 14 + 6 = 20X. \quad (24.8)$$

These lines agree. We have checked conservation of the represented resource for this transformation. We have not checked whether the distribution meets a clinic’s actual needs, whether transport is authorized, or whether a route exists. Each of those questions requires information beyond the total.

Suppose a copied receipt instead says $z' = (14, 5.8)^T X$. Its total is $19.8X$. In the present lossless two-cell model this is an inconsistency. Do not relabel the missing $0.2X$ as “burden” and continue. Either the copied numbers are wrong, the transformation was not lossless, or the boundary omits a carrier or loss destination. A more detailed physical model can represent losses, but it must declare where the missing quantity went. The scalar potential does not perform that physical bookkeeping.

24.4 Transpose and dot product, without mystery

The superscript T changes a column into a row. It changes orientation, not contents:

$$\begin{pmatrix} a \\ b \end{pmatrix}^T = (a, b). \quad (24.9)$$

A row times a column is a dot product. Multiply entries that share a position, then add. For example,

$$(3, -2) \begin{pmatrix} 5 \\ 4 \end{pmatrix} = 3(5) + (-2)(4) = 7. \quad (24.10)$$

The result is one number. It is not the vector $(15, -8)^T$, although that vector contains the two contributions before they are summed.

Later we will derive the marginal vector at our Ridge–Vale state. With reference stocks $(10, 10)^T X$ and scales $(2, 2)^T X$, it is

$$\mu(z) = \begin{pmatrix} 2 \\ -2 \end{pmatrix} X^{-1}. \quad (24.11)$$

Use it here simply to practise the operation:

$$\mu(z)^T \delta = 2(-4) + (-2)(4) = -16. \quad (24.12)$$

The units cancel: $X^{-1}X = 1$. This is the linear change predicted by the starting marginal. It is not the exact finite potential change. Curvature will add four, giving -12 overall. The distinction is introduced now so that a familiar-looking dot product never quietly becomes a complete receipt.

24.5 A quantity and a rate need different names

In this edition, q is a finite source-side quantity. A rate is written j . If a constant rate lasts for duration Δt , then

$$q = \Delta t j. \quad (24.13)$$

For $j = 8X/T$ and $\Delta t = 0.5T$,

$$q = (0.5T)(8X/T) = 4X. \quad (24.14)$$

The units of time cancel. The increment is therefore $(-4, 4)^T X$. Adding the numerical rate eight directly to Vale's stock would be wrong: it would add a quantity per time to a quantity. Multiplying the finite quantity four by the duration again would also be wrong. The conversion is made once.

The same finite transfer could occur at rate four for one time unit, or rate sixteen for one quarter of a time unit, if those rates were physically permissible. In this ideal stock-only illustration they have the same endpoint. This does not establish that their real energy requirements, risks, or process burdens are equal. Duration can matter through additional declared physical mechanisms. We have only held those mechanisms outside this elementary account.

24.6 References and scales are data of another kind

Place two more columns beside the state card:

$$x^* = \begin{pmatrix} 10 \\ 10 \end{pmatrix} X, \quad \sigma = \begin{pmatrix} 2 \\ 2 \end{pmatrix} X. \quad (24.15)$$

The star marks a declared reference. It does not mean multiplication, an especially accurate measurement, or a prediction of the next state. The reference says where this model places zero standardized deviation. The scale says how a stock displacement is converted into a dimensionless displacement. Neither is inferred from the numbers eighteen and two alone.

Compute the deviation entry by entry:

$$z - x^* = \begin{pmatrix} 8 \\ -8 \end{pmatrix} X. \quad (24.16)$$

Divide each entry by its own scale:

$$\left(\frac{z_i - x_i^*}{\sigma_i} \right)_{i=1,2} = \begin{pmatrix} 4 \\ -4 \end{pmatrix}. \quad (24.17)$$

The standardized vector has no stock unit. Ridge is four declared scales above its reference; Vale is four scales below. Squaring these entries will remove the signs, which is useful for a nonnegative potential but loses information about direction. Keep the signed deviations in the worksheet. A scalar score should not replace the state that gave rise to it.

The reference total is twenty, matching the actual total. Thus the reference is compatible with conservation in this small world. Compatibility is necessary for reaching it by internal transfers, but not sufficient. A closed gate, an absent route, a forbidden direction, or an action menu with unsuitable quantities may still prevent access to the reference.

24.7 Labels for observation, definition, and consequence

Give each line of the worksheet a source label. The stocks eighteen and two are *illustrative inputs*. The coordinate order and the potential family are *definitions*. The sum twenty is an *arithmetic consequence*. The claim that an actual tank holds eighteen units would be an *observation* requiring an instrument and a time. The claim that ten units is a suitable functional reference would require a calibration argument.

These labels are not ceremonial. Consider two errors that produce the same attractive final number. In one, the stock was mistyped. In the other, the chosen reference did not describe the relevant function. The first may be fixed by correcting a record. The second requires reconsidering a model assumption. If every number is simply called “the data,” the two problems become difficult to distinguish.

24.8 Guided practice

Work through the following linked tasks before reading the solutions. Each task changes only the facts it explicitly names. Otherwise retain the lossless homogeneous resource and the coordinate order Ridge, Vale.

1. Start from $(12, 8)^T X$. Transfer $3X$ from Ridge to Vale. Write the increment, successor, and both totals.
2. A clerk writes the same action as $(+3, -3)^T X$. What physical event would that column describe? Is it merely an alternative notation?
3. Write $(7, -2, 4)^T$ as a row and take its dot product with $(1, 3, -1)^T$.
4. A constant rate of $6X/T$ lasts $0.25T$. Find the finite quantity and the successor from our original state.
5. For that successor, calculate deviations from $(10, 10)^T X$ and standardized deviations using scale two at both cells.
6. The reference is changed to $(12, 8)^T X$. Does the actual stock change? Does the reference remain mass-compatible? Which entries on the worksheet must be recalculated?
7. A report gives only the total $20X$. Can you reconstruct the state? Supply two nonnegative states with that total.
8. Explain why a number labelled EBU cannot be added to twenty stock units to obtain a larger stock.

24.9 Solutions and reading the mistakes

For the first task, the increment is $(-3, 3)^T X$, so the successor is $(9, 11)^T X$. Both totals are twenty. This action passes the elementary nonnegative-stock check. Whether it is otherwise permitted depends on the route's declared conditions.

The clerk's column in the second task describes movement from Vale to Ridge. It reverses the physical direction under the fixed coordinate order. It can represent the original action only if the coordinate list and every associated object are also reversed. A sign error is not repaired by saying that vectors are abstract.

The third dot product is $7(1) + (-2)(3) + 4(-1) = 7 - 6 - 4 = -3$. For the fourth task, the finite quantity is $1.5X$. The successor is $(16.5, 3.5)^T X$. In the fifth, the signed deviations are $(6.5, -6.5)^T X$, and the standardized deviations

are $(3.25, -3.25)^T$. A decimal quantity is allowed here because divisibility is part of the teaching model; a model of indivisible packages would need another action domain.

Changing the reference in task six moves no stock. Its total remains twenty. Deviations, potential, marginals, and subsequent EBU comparisons must be recomputed under the new definition. Historical receipts should retain their old definition rather than being silently rewritten. A reference change is itself something the audit record must make visible.

For task seven, $(18, 2)^T X$ and $(10, 10)^T X$ both have total twenty, yet describe different distributions. There are many more. Conservation alone does not locate stock. For task eight, stock and EBU are different mathematical objects: one measures a physical quantity, while the other records a signed result derived from a declared potential and process account. A positive result does not fill a tank.

24.10 A three-cell reading exercise

Add a local store called Hill with stock $5X$. For this exercise the coordinate order is Ridge, Vale, Hill, and the state is $(18, 2, 5)^T X$. Its total is twenty-five. An increment $(-4, 3, 1)^T X$ would produce $(14, 5, 6)^T X$. The increment sums to zero, so the endpoint total is again twenty-five.

The new column describes a distribution of one withdrawal across two destinations. It does not specify whether that distribution occurs in one simultaneous event or two ordered actions. The endpoint arithmetic alone cannot tell us the timing. To know it, we need the action record. This is a first example of the difference between information contained in a vector and information needed to explain the vector.

Try reading the same column as two child increments, $(-3, 3, 0)^T X$ and $(-1, 0, 1)^T X$. Their sum is the original increment. The decomposition identifies amounts on two routes but still leaves physical permission and the event order to be declared. If an intermediate source restriction applies, the order may matter even though addition of the final increments is commutative.

To evaluate a potential in this larger world, we must add a Hill reference and scale. We cannot append its stock to the old state while silently omitting its contribution if Hill changes. One mass-compatible illustrative reference is $(10, 10, 5)^T X$, with scales two at all three places. Another compatible reference could distribute twenty-five differently. Choosing between them needs a functional argument; conservation alone does not choose it.

There is a useful lesson about matrices here. A two-entry marginal cannot be multiplied against a three-entry increment to obtain a valid complete dot

product. The dimensions disagree because the descriptions disagree. A dimension check is therefore more than a rule learned in linear algebra. It can reveal a missing part of the physical account.

24.11 Keep an observation time on the card

Imagine that the stock card was copied at nine o'clock and the action was considered at ten. A label saying "eighteen" is not enough to establish that eighteen remained available. The card needs an observation time or an event identifier, together with the rule that says when a record remains valid. If another carrier used the source meanwhile, the old state is a historical observation rather than the current baseline.

The book's frozen examples avoid this complication by declaration. A real measurement system cannot avoid it by typography. It must connect the action to the observations and commitments that support it. That requirement does not turn the arithmetic into a numerical uncertainty controller. It simply preserves the meaning of the word "present" when the record is used.

24.12 What to carry forward

You can now read the state without treating notation as a barrier. A coordinate has a name, a unit, and a fixed position. An increment has a physical direction. A total asks a conservation question. A reference and a scale describe the evaluation model. A dot product combines matched coordinates but acquires its meaning from the quantities being multiplied. These habits will remain useful when there are hundreds of nodes instead of two.

In Chapter 25, we will put a graph around the numbers. The key task will be to write the complete physical transformation before calculating its value. If the transformation is incomplete, the cleanest later potential calculation will still answer the wrong question.

Studio: draw the action before calculating it

Physical picture: an arrow needs a physical meaning

An arrow from Ridge to Vale can stand for a permitted movement of stock. It must say which stock moves, which way it moves, how much leaves, and where that quantity arrives. A line drawn on a map supplies none of those facts by itself. This studio builds an action record whose conservation can be checked before any EBU appears.

Keep the state card from Chapter 24: Ridge holds 18X, Vale holds 2X, and the represented total is 20X. This is the same homogeneous lossless teaching world. We begin with only one directed route. No regeneration, consumption, leakage, or unrecorded import acts during the action. If another mechanism is later added, it will receive its own explicit line.

The purpose is not to deny that real delivery needs vehicles, labour, energy, and time. It is to make the physical scope readable. A simple model should show what it represents and where its conclusions stop. Hiding a complication is dangerous; declaring a limited question is often useful.

25.1 Name the graph

Let the node set be $N = \{R, V\}$, where R means Ridge and V means Vale. The directed edge is $e = (R, V)$. Its ordered pair says that Ridge is the source and Vale is the destination. A reverse route (V, R) , if available, is a separate member of the edge set. It may have a different physical capacity or require different equipment.

For the present edge, declare a lossless transformation and a finite route cap 6X per action epoch, shared by all simultaneous actions using this route. This cap is illustrative physical input. It is not derived from the reference stock, the potential, or the amount of EBU a transfer might produce. An engineer would need to justify an empirical cap through the actual equipment and operating conditions.

The model admits quantities satisfying

$$0 \leq q \leq 6X, \quad q \leq z_R. \quad (25.1)$$

The first restriction is the route limit. The second prevents withdrawing more than the source contains. A destination capacity, if relevant, would add a further constraint; none is needed for the small quantities used here. Absence from this example does not establish absence from real clinics.

25.2 Build the action column

The coordinate vectors

$$e_R = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_V = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (25.2)$$

select Ridge and Vale. A lossless transfer has column

$$S_e = -e_R + e_V = \begin{pmatrix} -1 \\ 1 \end{pmatrix}. \quad (25.3)$$

Multiply this dimensionless column by the finite quantity q . For $q = 4X$,

$$\delta = S_e q = \begin{pmatrix} -4 \\ 4 \end{pmatrix} X. \quad (25.4)$$

There is no extra duration in this expression. If the input had instead been a rate, we would first calculate $q = \Delta t j$. The action column describes a change per source-side unit; it does not know whether that unit was moved slowly or quickly.

The action support is $\{R, V\}$, because those are the stock coordinates changed by this simplified action. A receipt can be local because the unaffected coordinates cancel from the potential difference. Locality does not allow us to omit an affected coordinate. If a battery is part of the physical model and the delivery changes it, the support must expand.

25.3 Make a stock receipt before a value receipt

For the four-unit proposal, write

Physical line	Quantity	Role
Ridge withdrawal	4X	Source decrease
Vale delivery	4X	Destination increase
Loss in this model	0X	Declared idealization
Unmoved Ridge stock	14X	Remaining stock
New Vale stock	6X	Successor stock

The transfer closure is $4 = 4 + 0$. The system closure is $18 + 2 = 14 + 6$. These are related checks with different scopes: one follows the moved quantity, the other follows the whole represented stock.

Now imagine a transcription error changes Vale's delivery to five while leaving withdrawal four. The system would gain one unit. No positive value calculation can validate that transformation. Correct the physical record first. Conversely, a transfer that conserves stock can still move it into an unsuitable place. Conservation is a prerequisite of this model, not a complete evaluation of its consequences.

25.4 Add a hub without pretending it is a direct edge

Replace the two-node route with a three-node route for this section only. The coordinate order becomes Ridge, hub, Vale, and the initial state is

$$x^0 = \begin{pmatrix} 18 \\ 0 \\ 2 \end{pmatrix} X. \quad (25.5)$$

There are two directed edges, $R \rightarrow H$ and $H \rightarrow V$, with columns

$$S_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}. \quad (25.6)$$

Four units first move from Ridge to the hub:

$$x^1 = x^0 + 4S_1 = (14, 4, 2)^T X. \quad (25.7)$$

Then three units move from the hub to Vale:

$$x^2 = x^1 + 3S_2 = (14, 1, 5)^T X. \quad (25.8)$$

The warehouse released four, the clinic received three, and the hub retained one. The account closes without loss: $4 = 3 + 1$. Stock left at an intermediate node has not vanished.

The second action is feasible from x^1 , where the hub holds four. It is not feasible as a standalone withdrawal from x^0 , where the hub holds zero. A drawing of the whole route should preserve this dependency. Writing the final net increment $(-4, 1, 3)^T X$ is useful for checking endpoints, but it does not make the two physical acts interchangeable or erase the need to supply the hub first.

25.5 A simultaneous sum requires a simultaneous contract

Return to the two-node state. Two carriers propose quantities three and two on the same edge. If the physical contract permits their joint action, its net increment is

$$\delta_G = S_e(3 + 2) = (-5, 5)^T X. \quad (25.9)$$

The aggregate quantity five is within the illustrative route cap six and the source stock eighteen. Both child records can be preserved even though the endpoint is calculated once.

If the carriers each propose four, each proposal individually fits the six-unit cap, but their sum eight does not. Testing each request alone is insufficient when both use the same resource. The group must be assessed against the aggregate constraint. How an over-capacity group is rejected, reduced, or reassembled is a declared resolver policy. We do not choose one merely to obtain a convenient total.

Notice that the word “simultaneous” adds more than a plus sign. It requires an account of overlapping use, accepted quantities, common inputs, and whether intermediate constraints matter. The endpoint arithmetic can be simple while the physical permission question remains substantial. Chapter 30 will return to the distinct question of valuing a permitted group.

25.6 Represent external change as its own event

Suppose an external mechanism transfers one unit from Ridge to Vale before the actor’s epoch. The old state is $x = (18, 2)^T X$, and the post-forcing baseline is

$$z = x + (-1, 1)^T X = (17, 3)^T X. \quad (25.10)$$

The actor then proposes four units, yielding $(13, 7)^T X$. The external increment and actor increment are both conservative, but they belong to different records. The actor’s quote compares $(17, 3)$ with $(13, 7)$. It does not claim the earlier movement as its own.

With the base reference and scales, the old potential is sixteen and the post-forcing potential is 12.25. Thus forcing lowered potential by 3.75. That fact belongs to the external audit. The actor’s later field reduction is $12.25 - 2.25 = 10$. The complete change of thirteen and three quarters is the sum of two identified causes. Keeping the causes separate is useful even when they happen to point in the same direction.

An external inflow would require an explicit boundary crossing and change the closed-system mass. We have used conservative forcing here so that the total

remains twenty. The phrase “external” identifies who or what determines the event, not necessarily a physical inflow from outside the represented boundary.

25.7 When one action touches another resource

Imagine a real carrier uses a battery while moving medicine. A physical action record might contain medicine increment $(-4, +4)X$ and battery withdrawal $0.2kWh$. These are linked effects of a declared transformation. They do not become comparable by placing them in one column.

The medicine total still closes in medicine units. The energy account needs its own carriers, conversion mechanism, and explicit dissipation or boundary. If a separately declared energy potential exists, it can produce an energy-field result. We should retain the two results as named components until a justified common valuation has been supplied. The fact that both results are numerically dimensionless after normalization does not establish ecological or ethical interchangeability.

A process burden can represent a declared physical action burden not already counted through evaluated state effects. It must name what it covers. “There was a vehicle” is not a calibration. Nor may the same battery depletion be counted fully in an energy potential and again under another label without an explicit non-overlapping definition. In the ideal stock calculations of this studio, $C = 0$ is declared for simplicity; real transport has not thereby been proved free.

25.8 Audit a flawed action drawing

Consider this report: “Ridge has eighteen units. Two carriers each take four. The route cap is six per epoch, so each is below the cap. Vale receives nine after efficient handling. The source is recorded as twelve because that is a preferred operating level.”

The first defect is aggregate permission: eight exceeds six. The second is physical closure: eight withdrawn cannot produce nine delivered in the declared lossless transfer. The third is an invented source correction: eighteen minus eight is ten, not twelve. A preferred state cannot replace an endpoint. This report contains several distinct errors, and identifying the first does not make the rest irrelevant. A useful review records each defect while refusing to certify the event.

It is tempting to solve such a report by adjusting one number until the totals agree. Resist that temptation. A corrected receipt needs evidence about what happened or a corrected prospective transformation. Conservation gives a consistency condition, not permission to invent observations.

25.9 Practice, with complete answers

1. A route admits at most $5X$, and Ridge contains $4X$. Is a proposal of $4.5X$ admissible under the two declared restrictions?
2. From the original two-node state, write the action column for a permitted reverse transfer of $1X$, and give its successor.
3. In the hub example, send four on the first edge and all four on the second. Give both live states and verify total stock.
4. Two proposals of $3.5X$ share the six-unit route. What does the aggregate check establish, and what does it leave undecided?
5. A delivery record lists source withdrawal four, clinic delivery three, and hub remainder one. Does that prove a loss of one? Explain using the complete boundary.
6. Draw a linked fuel, electricity, pumping, and water movement account. Which quantities can be added physically, and which must remain in distinct units?

The first proposal fails the stock restriction even though it passes the route cap. Its hypothetical source would be $-0.5X$. The joint admissible maximum under just these two restrictions is four, but that observation does not automatically resize a submitted proposal. Resizing requires a declared procedure and a new accepted quantity.

For the second answer, the reverse column is $(1, -1)^T$. The successor is $(19, 1)^T X$. Both entries remain nonnegative, and the total remains twenty. Reverse permission was supplied explicitly by the question; it did not follow from the existence of the forward arrow.

For the third, the states are $(14, 4, 2)^T X$ and $(14, 0, 6)^T X$. Both sum to twenty. The hub empties without becoming negative. It was necessary as an intermediate location even though its initial and final stocks are both zero. Endpoint equality at one coordinate does not prove that nothing happened there.

For the fourth, the requested aggregate is seven, so the submitted group fails the cap. This establishes refusal of that exact joint proposal under the stated contract. It does not decide whether one carrier should proceed, whether both should reduce quantities, or whether the group should wait. Those are additional choices.

For the fifth, the apparent difference is inventory retained at the hub. The complete route closes as $4 = 3 + 1$, with zero declared loss. A clinic-only

boundary might record three units entering it, but that narrower boundary cannot explain the whole source withdrawal without the route record.

The final drawing should retain at least four linked transformations. Fuel conversion changes a fuel store and energy carriers; electricity delivery changes the relevant electrical account; pump operation converts delivered energy into mechanical action with declared dissipation; water movement changes the source and destination water stocks. Litres of fuel, kilowatt-hours, and cubic metres of water cannot be summed as if they were one material. The chain identifies dependency without inventing a universal scalar. Each actual plant needs far more specific mechanisms than this sketch supplies.

25.10 Endpoint feasibility and path feasibility

In the present additive lossless model, if both endpoint stocks are nonnegative, the straight interpolation between them is also nonnegative. Each intermediate coordinate is a weighted average of its before and after values:

$$x_i(s) = (1 - s)z_i + sz'_i, \quad 0 \leq s \leq 1. \quad (25.11)$$

Since both weights and both endpoint values are nonnegative, the intermediate value is nonnegative. This is a small conditional proof that can simplify a stock-only path check.

It does not prove that every physical mechanism connecting those endpoints is possible. A valve might be closed for part of the interval, two carriers might compete for a loading bay, or a medicine might require a temperature condition not represented in the stock vector. Those constraints are not addressed by nonnegative interpolation. The scope of the proof should stay as small as its premises.

The hub example shows another limitation. Its final net increment has a nonnegative endpoint, but applying the second child before the first fails. A net path and a sequence of atomic actions are different physical descriptions. If the model realizes the two-stage route, its intermediate state is (14, 4, 2), not an arbitrary interpolation chosen for convenience.

This is also why an action record should distinguish a mechanism from an endpoint. Two different mechanisms can leave the same stocks while using different resources. A medicine shipment by insulated road vehicle and a shipment by aircraft might deliver the same quantity to the same clinic, yet have very different energy and timing accounts. The stock potential would give the same stock-field difference, while a fuller account could distinguish the routes.

25.11 A naming exercise for the action support

Take a pen and circle every noun in the proposed sentence: “A refrigerated carrier takes medicine from the warehouse through the hub to the clinic.” The nouns do not all correspond to the same kind of model object. Warehouse, hub, and clinic can name stock locations. Carrier can name an actor and a physical vehicle. Refrigeration can name a transformation involving energy and temperature. The action support should be built from the affected state variables, not merely from this grammatical list.

For example, naming the carrier does not automatically create a new stock coordinate. Conversely, failing to mention heat in the sentence does not make actual heat production disappear from a complete energy account. The state schema determines what is measured, while the physical mechanism determines which of those measurements change. Keeping both explicit is a stronger habit than assuming that a familiar everyday noun already defines a scientific variable.

25.12 The drawing is part of the mathematics

An action column is a compact physical statement. Its negative entries name withdrawals and its positive entries name arrivals. A graph declares which transformations are available. Live route states reveal causal order. Aggregate checks prevent several actions from spending the same physical capacity twice. External events remain separately identified, and additional resources expand the account instead of disappearing into an attractive value number.

The next studio takes this explicit physical world and builds its Gaussian potential. Because the state and transformation are now clear, changes in the potential can be understood as changes in evaluation rather than mysterious changes in the stock itself.

Studio: build the Gaussian landscape

Physical picture: give distance a declared scale

Ridge has eighteen units where the teaching reference is ten. Vale has two where its reference is also ten. These are equal departures in opposite directions. A Gaussian landscape begins by measuring each departure in a declared local scale, then squaring and adding. The shape is simple enough to inspect completely, which makes its assumptions especially visible.

The stock account has already told us what exists and what can move. It has not said how a distribution should be evaluated. This studio supplies a declared evaluation function and shows what follows from it. The reference is part of the model's functional hypothesis; a real application would need evidence and public justification for that hypothesis. Nothing in the name Gaussian chooses the right reference for a clinic.

We retain the two-cell world with total $20X$, reference $x^* = (10, 10)^T X$, and positive scales $\sigma_R = \sigma_V = 2X$. These numbers are fixed throughout a comparison. When an exercise changes one of them, it will say so. We assume no independent process burden for the ideal transfers, so $C = 0$.

26.1 Construct one local contribution slowly

For cell i , define

$$V_i(x_i) = \frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2. \quad (26.1)$$

Read the operations in their physical order. First subtract the reference from the present stock. Next divide by a stock scale. Then square the dimensionless displacement. Finally multiply by one half. The order matters: squaring the stock before subtracting the reference would define another function.

At Ridge,

$$x_R - x_R^* = 18 - 10 = 8X, \quad (26.2)$$

$$\frac{x_R - x_R^*}{\sigma_R} = \frac{8X}{2X} = 4, \quad (26.3)$$

$$V_R(18) = \frac{1}{2}(4)^2 = 8. \quad (26.4)$$

At Vale,

$$x_V - x_V^* = 2 - 10 = -8X, \quad (26.5)$$

$$\frac{x_V - x_V^*}{\sigma_V} = -4, \quad (26.6)$$

$$V_V(2) = \frac{1}{2}(-4)^2 = 8. \quad (26.7)$$

The local contributions are equal, but the physical conditions are opposite. Ridge is above reference and Vale below. The sum

$$V(z) = V_R(18) + V_V(2) = 16 \quad (26.8)$$

records total standardized squared deviation. It is dimensionless. It is not twenty stock units, sixteen joules, or a direct measurement of sixteen units of suffering.

26.2 Why the square is there

A signed sum of deviations would be zero at our starting state: $8 + (-8) = 0$. That cancellation would conceal the distributional mismatch that this model was designed to represent. Squaring makes both deviations contribute positively. A state has zero potential only when every included deviation is zero, since every scale is strictly positive.

Squaring also makes larger departures more costly at an increasing rate. With scale two, a displacement of two gives contribution $1/2$, a displacement of four gives two, and a displacement of eight gives eight. Doubling the departure multiplies its contribution by four. This is a modelling choice with mathematical consequences; it is not an experimentally established universal response curve for every living system.

The factor one half simplifies derivatives. It is not essential to the location of the minimum. Removing it would double every potential, marginal, and finite field value. That change could be handled consistently, but a receipt must identify which convention it used. Mixing conventions between the before and after sides would destroy the intended comparison.

26.3 Calculate several complete states

Keep the mass twenty and evaluate the following distributions:

State	Ridge term	Vale term	Total V
(18, 2)	8	8	16
(16, 4)	4.5	4.5	9
(14, 6)	2	2	4
(12, 8)	0.5	0.5	1
(10, 10)	0	0	0
(2, 18)	8	8	16

Every row describes the same total stock. The potential changes because distribution changes. The final row has the same potential as the first, although Ridge and Vale exchange their positions relative to reference. Equality of scalar value is weaker than equality of state.

Do not interpret the table as a sequence that a controller has produced. It is a set of declared states evaluated under one formula. The table tells us how the landscape is shaped. To obtain a history, we would need available actions, a selection rule, permission, and actual events. The potential does not move any stock by being evaluated.

26.4 Follow the conservation line

Write a two-cell state with total twenty as $(10+d, 10-d)$, where $-10 \leq d \leq 10$ in stock units. Substituting gives

$$V(d) = \frac{(d)^2}{8} + \frac{(-d)^2}{8} \quad (26.9)$$

$$= \frac{d^2}{4}. \quad (26.10)$$

The minimum lies at $d = 0$, which is the reference. Both ends of the physical line, $(20, 0)$ and $(0, 20)$, have potential twenty-five. On this closed line the potential cannot diverge to infinity: the domain itself is bounded.

This matters for later scientific claims. Showing that a simulated history remains bounded in this finite closed model would not by itself establish a special stabilizing benefit of EBU. The model's nonnegative fixed-mass domain already supplies boundedness. More informative questions concern where time is spent, whether the reference is approached, how often actions are refused, and what cycles occur. None is answered by the table alone.

26.5 Why the word Gaussian is appropriate, and limited

Consider the positive shape

$$g_i(x_i) = \exp \left[-\frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2 \right]. \quad (26.11)$$

It is highest at the reference and decreases symmetrically away from it. Taking its negative logarithm gives $-\log g_i = V_i$. The potential is thus the relative negative logarithm of a Gaussian-shaped reference function, with its minimum set to zero.

This observation explains a geometry. It does not prove that actual clinic stocks are normally distributed. A probability density would need a domain and normalization, and an empirical distribution would need observations. In our conserved model the coordinates are coupled by their fixed sum. One cannot independently draw an arbitrary Ridge stock and an arbitrary Vale stock while also insisting that they sum to twenty.

Nor does the shape imply a restoring process. A ball on a physical surface has dynamics involving forces and constraints. Our landscape is an evaluation function until a separate actor or plant law is supplied. Reading a Gaussian curve as an automatic homeostatic mechanism would insert precisely the behaviour that the research programme needs to investigate.

26.6 Change a scale and watch what it changes

For this section, retain reference ten at each cell but change Ridge's scale to four while Vale's remains two. The starting state stays (18, 2). Ridge's contribution becomes

$$V_R(18) = \frac{1}{2}(8/4)^2 = 2, \quad (26.12)$$

while Vale's remains eight. The total is ten. After a four-unit transfer, the state is still (14, 6), because changing a scale moves no stock. Its potential is

$$\frac{1}{2}(4/4)^2 + \frac{1}{2}(-4/2)^2 = 0.5 + 2 = 2.5. \quad (26.13)$$

The field reduction is therefore 7.5, rather than twelve in the equal-scale example.

A wider scale assigns less potential to a given displacement and a smaller marginal magnitude at that displacement. This is why scales must be justified instead of chosen to make a preferred action look good. Calling σ a standard deviation without evidence does not provide that justification. Here it is a

declared stock scale. Measurement uncertainty is another issue and should not be silently substituted for functional sensitivity.

Return to scale two at both cells after this comparison. A clear worksheet writes the changed assumptions beside each example, so that arithmetic from one landscape is not carried into another unnoticed.

26.7 Change a reference without changing the world

Now consider reference $(12, 8)^T X$, again with both scales two. It is mass-compatible. At actual state $(18, 2)$, the deviations are six and minus six, giving total potential nine. At $(14, 6)$, the deviations are two and minus two, giving total potential one. The same four-unit physical transfer has field reduction eight under this changed reference.

The difference from twelve is not an arithmetic conflict. Different functional references define different questions. A responsible account must disclose the reference and preserve its version. If society later revises what the reference should be, the change should be documented as a change of evaluation, with its own consequences for historical comparisons.

An incompatible reference creates another issue. If references sum to twenty-two while the closed system holds twenty, all coordinates cannot reach their references simultaneously. One could still define a mathematical potential, but the claim that zero is reachable by internal redistribution would fail. Our main model avoids that conflict by requiring compatible total reference mass.

26.8 A soft landscape and a physical boundary

The Gaussian potential assigns values even to some numerical vectors that violate the physical domain. For instance, substitution at $(-1, 21)$ is algebraically possible, but the negative stock is forbidden in our world. A formula's ability to return a number does not authorize its input as a physical state.

The same principle holds for route caps. A transfer of eight from $(18, 2)$ reaches the reference and gives the largest field reduction along that edge. If the route can carry only six in an action, eight is unavailable in that action. The potential does not enlarge the pipe. Keep domain constraints outside the evaluation formula and check them before interpreting a quoted candidate as selectable.

A small potential is a scoped statement

Small V means small standardized deviation in the coordinates and references that have been declared. It does not establish that every relevant need is represented, that a state is clinically safe, or that the reference is ethically sufficient. Those claims need their own evidence. The benefit of an explicit landscape is that its scope can be examined.

26.9 Practice problems

1. Using the base reference and scales, calculate both local terms and total potential at (15, 5), (11, 9), and (0, 20).
2. At a single cell with reference ten and scale two, compare the contributions at stocks seven and thirteen. What information is lost when their equal values replace their states?
3. Keep (18, 2), but use scales one and two at Ridge and Vale respectively. Calculate the total and compare it with sixteen.
4. With base scales, use reference (14, 6). Calculate the field reduction of the transfer from (18, 2) to (14, 6).
5. A modeller replaces V by $1 - \exp(-V)$. Is this merely another notation for the same finite field differences?
6. Explain why $V = 0$ at (10, 10) neither consumes nor creates the total twenty units.
7. A report says, “The bell shape proves that repeated random action returns the clinic to ten.” Identify the missing argument.

26.10 Worked answers

At (15, 5), each displacement has magnitude five. Each contribution is $25/8 = 3.125$, so the total is 6.25. At (11, 9), each contribution is $1/8 = 0.125$, total 0.25. At (0, 20), each is $100/8 = 12.5$, total twenty-five. All three states conserve twenty units; their potentials are different evaluations of their distributions.

Stocks seven and thirteen are three units below and above reference. Both contributions are $9/8 = 1.125$. Their signs and therefore their local directions of improvement are lost if the state is discarded. A receipt should retain enough information to recover the distinction.

With Ridge scale one, its contribution at eighteen is $64/2 = 32$. Vale still contributes eight, total forty. This is a stronger declared sensitivity to Ridge deviation, not additional physical stock. With reference (14, 6), the original state has two displacements of magnitude four, total four. The successor is the new reference and has zero, giving field reduction four.

The exponential replacement changes the finite differences. From potential sixteen to four, the original reduction is twelve. The transformed difference is $\exp(-4) - \exp(-16)$, which is a different number and has different marginal sensitivity. A monotone transformation preserves state ordering but generally changes differences. EBU is built from differences, so the choice matters.

Zero potential in the sixth question means that both present stocks equal their references. The physical total remains twenty. In the final question, a shape has been mistaken for a dynamics theorem. A law of action, a feasible menu, and a proof or registered study of the resulting process are needed. Random choice can include movements away from the reference, and permission may restrict the available return path.

26.11 A reference is not necessarily an empirical average

Suppose repeated observations at a clinic had arithmetic mean six units. It would not follow that six is the correct functional reference. The clinic might repeatedly be under-supplied. Conversely, a desired reference ten would not prove that ten is its historical average. An observational summary and a functional specification can agree, but they answer different questions.

The same care applies to the scale. A measured standard deviation of two units describes variation in an observed series under its sampling conditions. A declared functional scale of two units describes how the potential weights displacement. Using the same number for both purposes would need an argument. Without one, the shared symbol could conceal a change of meaning.

This does not make reference selection arbitrary. It makes its justification part of the model rather than an automatic consequence of a familiar probability curve. One may use physiological evidence, service requirements, engineering analysis, or a public decision procedure where appropriate. The resulting assumptions should be recorded in a form that another reader can challenge and revise transparently.

26.12 Equal raw stocks can have unequal marginal significance

Consider a changed three-cell example with stocks (8, 8, 8), reference (10, 6, 8), and scales two throughout. Both stock and reference totals are twenty-four.

The actual stocks are equal, but the deviations are $(-2, 2, 0)$. Local potentials are $0.5, 0.5, 0$, and the marginals are $(-0.5, 0.5, 0)$ in inverse-stock units.

A lossless two-unit transfer from the second cell to the first reaches the reference and lowers potential by one. A rule based only on equality of raw stocks would see no imbalance. The declared functional reference sees a difference because the cells have different roles. Whether those roles justify references ten and six is a separate substantive question, but the arithmetic is clear once they are supplied.

If the first cell's scale changes from two to four, its initial contribution falls from 0.5 to 0.125 . The second still contributes 0.5 , total 0.625 . Reaching the same reference now has field value 0.625 . Once again the physical transfer is unchanged and the evaluation differs. This is why a report should publish the field definition rather than only the final EBU number.

An important limit remains. Unequal references can encode functional differences, but they can also encode poor assumptions or unjustified preferences. Mathematical consistency does not certify their legitimacy. The Gaussian construction makes such choices inspectable; it does not relieve the modeller of responsibility for them.

26.13 Carry the construction into the next studio

The landscape is now more than a name. You have constructed its local contributions, checked its units, followed the fixed-mass line, and seen how scales and references change its meaning. The next task is to differentiate it and compare a starting slope with a complete finite movement. That comparison will show why the exact EBU equation contains curvature even when the physical increment is as simple as four units along one edge.

Studio: from slope to finite field effect

Physical picture: the first slope is not the whole journey

At Ridge eighteen and Vale two, a small transfer toward Vale reduces both standardized deviations. But the advantage of the next unit changes as the transfer proceeds. A finite account must follow that change. The quadratic landscape lets us see the complete relation between local slope and exact endpoint value without approximation.

Use the base state $z = (18, 2)^T X$, reference $(10, 10)^T X$, scales two, and the lossless edge Ridge to Vale. For this studio, suppose quantities from zero to eighteen are available for algebraic inspection. Actual route permission can be narrower; the extended range is a diagnostic of the equation. It does not override a cap declared in a different example.

27.1 Differentiate one local contribution

Write the local function in a form that makes its constants visible:

$$V_i(x_i) = \frac{(x_i - x_i^*)^2}{2\sigma_i^2}. \quad (27.1)$$

Differentiate with respect to x_i , holding the reference and scale fixed:

$$\mu_i(x_i) = \frac{1}{2\sigma_i^2} 2(x_i - x_i^*) \quad (27.2)$$

$$= \frac{x_i - x_i^*}{\sigma_i^2}. \quad (27.3)$$

The derivative is called the marginal. Its unit is inverse stock because the potential is dimensionless. At Ridge it is $(18 - 10)/4 = 2X^{-1}$. At Vale it is $(2 - 10)/4 = -2X^{-1}$.

The signs have direct meaning. Adding a small amount to Ridge at eighteen increases the potential; removing a small amount decreases it. Adding a small amount to Vale at two decreases the potential. These statements concern local changes under this declared function. They do not tell a carrier how to act or establish that all real needs have been represented.

27.2 Derive the edge field from the physical column

Along a finite transfer coordinate r , the physical state is

$$x(r) = z + S_e r = (18 - r, 2 + r)^T \mathbf{X}. \quad (27.4)$$

When r increases, Ridge decreases at unit rate with respect to r , while Vale increases at unit rate. The chain rule gives

$$\frac{dV(x(r))}{dr} = -\mu_R(x_R(r)) + \mu_V(x_V(r)). \quad (27.5)$$

Define the burden-reducing edge field as its negative:

$$f(r) = \mu_R(x_R(r)) - \mu_V(x_V(r)). \quad (27.6)$$

For our numbers,

$$\mu_R(x_R(r)) = (8 - r)/4, \quad (27.7)$$

$$\mu_V(x_V(r)) = (-8 + r)/4, \quad (27.8)$$

$$f(r) = 4 - r/2. \quad (27.9)$$

At the start, $f(0) = 4\mathbf{X}^{-1}$. After four units, it is two. After eight, it is zero. After twelve, it is minus two. The field changes continuously because the Gaussian potential has continuous derivatives.

The field is a directional diagnostic. We introduce no mobility coefficient or automatic flux law here. Knowing that one infinitesimal direction lowers the potential does not mean the model selects that direction, or that actors must choose the quantity where the field becomes zero.

27.3 Calculate the exact endpoint difference

At quantity q , the endpoint potential is

$$V(z + S_e q) = \frac{(8 - q)^2}{8} + \frac{(-8 + q)^2}{8} \quad (27.10)$$

$$= \frac{(8 - q)^2}{4}. \quad (27.11)$$

The starting value is sixteen, and $C = 0$. Therefore

$$E(q) = 16 - \frac{(8 - q)^2}{4} \quad (27.12)$$

$$= 16 - \frac{64 - 16q + q^2}{4} \quad (27.13)$$

$$= 4q - \frac{q^2}{4}. \quad (27.14)$$

Here and below, numerical quantities are expressed in the declared stock unit. The first term is the initial field multiplied by quantity. The second is the exact curvature correction. It is not a numerical safety allowance or a guessed penalty. It follows from expanding the square.

For $q = 4$, the starting-field product is sixteen and the correction is four, giving twelve. For $q = 8$, the product is thirty-two and the correction is sixteen, giving sixteen. Extending the starting slope indefinitely would count gains that are no longer present along the finite movement.

27.4 Recover the same answer by integration

The field along the path is $4 - r/2$. Integrating it from zero to the performed quantity gives

$$\int_0^q f(r) dr = \int_0^q (4 - r/2) dr \quad (27.15)$$

$$= \left[4r - r^2/4 \right]_0^q \quad (27.16)$$

$$= 4q - q^2/4. \quad (27.17)$$

This agrees with the endpoint difference. The agreement is an instance of the fundamental theorem of calculus under the stated differentiability and path assumptions. We are not numerically approximating the integral in this example; both evaluations are exact algebra.

Another parameterization runs from $s = 0$ to $s = 1$, with state $z + s\delta$ and fixed increment $\delta = S_e q$. In that notation the field contribution is

$$- \int_0^1 \nabla V(z + s\delta)^\top \delta ds. \quad (27.18)$$

The factor δ contains the full quantity. Omitting it would integrate a marginal but fail to obtain the finite action value. The negative sign appears because positive EBU means a reduction in potential.

If a model declares constant-rate simultaneous execution, the straight path can be that model's realized path. When only endpoints are supplied, it is a chosen interpolation used to decompose the difference. Arithmetic alone does not prove which physical time path occurred.

27.5 Use the Hessian to see the general pattern

For fixed positive scales, the Hessian matrix is

$$H = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/4 \end{pmatrix} X^{-2}. \quad (27.19)$$

It records how marginals change with stock. The exact quadratic identity is

$$V(z + \delta) - V(z) = \mu(z)^\top \delta + \frac{1}{2} \delta^\top H \delta. \quad (27.20)$$

For $\delta = (-4, 4)^\top X$, the first term is minus sixteen. The matrix product gives

$$\delta^\top H \delta = \frac{(-4)^2}{4} + \frac{4^2}{4} = 8. \quad (27.21)$$

Half of that is four. Thus the potential change is $-16 + 4 = -12$, and the EBU field reduction is twelve.

Because every diagonal entry of H is positive, $\delta^\top H \delta > 0$ for every nonzero increment. A tangent estimate of the field reduction therefore overstates the exact finite reduction by this positive quadratic term. That conclusion relies on this fixed quadratic model. It should not be stretched into a claim about every conceivable potential or hidden action-dependent parameter.

27.6 Read positive, zero, and negative values on one curve

Factor the exact expression:

$$E(q) = \frac{q(16 - q)}{4}. \quad (27.22)$$

For positive quantities below sixteen, this expression is positive. At zero it is zero because no action occurred. At sixteen it is also zero, but for a very different reason: the state becomes $(2, 18)$, whose potential equals the original sixteen. A large movement took place even though its net field value is zero.

At seventeen, the endpoint is $(1, 19)$, with potential $81/4 = 20.25$. The field value is $16 - 20.25 = -4.25$. This endpoint still has nonnegative stocks. Thus physical feasibility under the simple stock constraint and positive EBU are different questions.

The zero of the field at quantity eight and the second zero of the total value at sixteen are also different. At eight, an additional tiny forward movement begins to increase potential, but the completed action has already reduced potential by sixteen. Confusing a marginal zero with a total zero would erase the distinction between the next unit and the whole action.

27.7 Recalculate a short execution

Suppose a proposal and accepted plan both specify four units, but the recorded execution is only three. The endpoint is $(15, 5)$, potential 6.25, and

exact field value 9.75. The unperformed fourth unit must not receive the planned value. Its incremental contribution, had it occurred next, would have been $12 - 9.75 = 2.25$.

That incremental number can be checked from the live state. At $(15, 5)$, the edge field is 2.5. A further one-unit transfer has exact value $2.5(1) - 1/4 = 2.25$. The two live values, 9.75 and 2.25, add to twelve. Two independent three- or one-unit values taken from the old state would answer a different question.

The example shows why a receipt needs the actual quantity and the correct predecessor. For a short execution within the original declared domain, evaluate the precommitted schedule at the measured quantity. Do not substitute the planned four-unit value or change the schedule after the event. Baseline, field and physical-map assumptions must still match. A numerical result should never outlive the assumptions that gave it meaning.

27.8 What changes when process burden is declared

Keep the physical four-unit transfer and suppose a separate, non-overlapping action burden is explicitly declared as $C = 0.5$. The field reduction remains twelve. Net value becomes $E = 11.5$. The identity reads

$$11.5 + (4 - 16) + 0.5 = 0. \quad (27.23)$$

The process line changes the action account, not the already computed end-point potential. This example does not establish a real calibration for any vehicle or medical delivery. Its purpose is to show where a justified burden would enter. A consequence already charged through the represented state change and evaluated potential must not be subtracted again. Any additional process term must cover a distinct, declared physical contribution not already counted there. Merely recording a loss in a conservation ledger does not establish whether the chosen potential has charged it; that valuation boundary must also be explicit.

27.9 Guided practice

1. Calculate the initial field, exact value, and tangent overstatement for a two-unit transfer from the base state.
2. At quantity six along the path, calculate both marginals and the edge field. Then find the value of the completed six-unit transfer.
3. Starting from the reference $(10, 10)$, calculate the value of a two-unit forward transfer. Explain why an initial field of zero does not imply that its finite value is zero.

4. Divide a four-unit transfer into two consecutive two-unit actions. Calculate each live value and their sum.
5. Repeat the fourth problem but quote both actions independently from the original state. What sum do you obtain, and why is it different?
6. Use scales four at Ridge and two at Vale. Derive the initial field and the curvature coefficient for this changed model.
7. Explain whether a negative field forbids a movement by mathematics alone.

27.10 Worked answers and diagnostic questions

The initial field in the first problem is four. Exact value at two is $8 - 1 = 7$. The tangent predicts eight and overstates by one. In the second problem the intermediate stocks are twelve and eight, so marginals are 0.5 and -0.5 , field one. Completed value is $24 - 9 = 15$. The current field one is not the total value fifteen.

At the reference, the initial marginal vector is zero. The two-unit movement has curvature term $q^2/4 = 1$, so its value is minus one. It leaves the minimum of the potential. The absence of a first-order change is compatible with a nonzero second-order change.

For two consecutive two-unit actions from $(18, 2)$, the first value is seven and the live state is $(16, 4)$, with potential nine. The second reaches $(14, 6)$, potential four, so its value is five. Total twelve matches the one-step endpoint comparison. The independent same-baseline calculation instead gives seven plus seven, or fourteen. It repeats the more favourable starting conditions and overlooks the shared curvature correction.

With scales four and two, the marginals at $(18, 2)$ are 0.5 and -2 , so the initial field is 2.5. The finite expression is

$$E(q) = 2.5q - \frac{1}{2} \left(\frac{1}{16} + \frac{1}{4} \right) q^2 = 2.5q - \frac{5}{32} q^2. \quad (27.24)$$

At four it gives ten minus two and a half, or seven and a half, agreeing with the previous studio. This is a useful independent check because it reaches the same number by slopes and curvature instead of direct local squares.

For the final question, a field sign evaluates a direction under the declared potential. Permission and selection require separately stated rules. An actor could perform a physically allowed negative-value movement under a policy that admits it, possibly using an explicitly defined capacity account. This studio supplies neither that policy nor its authorization. It keeps the evaluation transparent so that a later policy can be examined honestly.

27.11 An exact midpoint check

Because the marginal of a quadratic is affine, the average marginal along a straight path equals the marginal at the path midpoint. This gives a third way to check the finite field difference. For a four-unit transfer, the midpoint is $(16, 4)$, where the edge field is three. Multiplying that midpoint field by the full quantity four gives twelve, exactly.

To derive the claim rather than memorize a shortcut, write

$$\mu(z + s\delta) = \mu(z) + sH\delta. \quad (27.25)$$

Integrating over s yields $\mu(z) + \frac{1}{2}H\delta$, which is precisely $\mu(z + \delta/2)$. Therefore

$$V(z) - V(z + \delta) = -\mu(z + \delta/2)^T \delta. \quad (27.26)$$

This is exact for the fixed quadratic and straight interpolation. It is not a general claim that every nonlinear potential can be integrated from one midpoint evaluation. With another potential, midpoint evaluation may only approximate the integral, and its error would need analysis appropriate to that function.

For the eight-unit transfer, the midpoint is $(14, 6)$, where field two times quantity eight gives sixteen. For sixteen units, the midpoint is the reference and its field is zero, giving zero total value even though the path first lowers and then raises the potential. The positive and negative portions cancel in the exact integral.

27.12 Inspect a transfer from an unequal-scale state

Use the changed scales from the last exercise: four at Ridge and two at Vale. The initial field is 2.5 and its slope along the transfer is $-5/16$, so

$$f(r) = 2.5 - \frac{5}{16}r. \quad (27.27)$$

For a four-unit action, the midpoint quantity is two and its field is $2.5 - 10/16 = 1.875$. Multiplying by four gives 7.5, exactly the endpoint difference. The same shortcut works because the field is affine, not because the scales are equal.

The field becomes zero at quantity eight in this particular changed-scale example because both cells still share the compatible reference ten and the transfer path passes through it. With different reference totals or a higher-dimensional constrained movement, a zero along one edge need not mean that every coordinate equals its reference. It means the two endpoint marginals balance along that permitted direction.

For example, take two local stocks fourteen and eleven, both with reference ten, but scales four and two respectively. Their marginals are both one quarter, despite unequal stocks and nonzero deviations. The lossless field between them is zero at that instant. A small finite transfer on that edge still adds a positive curvature term and therefore has negative field value. This local example need not be the whole conserved system: additional cells can hold the compensating deficit required for a compatible global reference. Equal local marginals express a balance of sensitivity along one edge, not a declaration that the entire represented world is at its reference.

27.13 The difference between a direction and a destination

A positive starting field says that sufficiently small forward movement lowers this differentiable potential. It does not identify a unique finite action. Several quantities in the menu can lower it, and their other consequences can differ. Even the continuous maximum of the finite expression is a feature of the evaluation curve, not evidence that an actor should or can choose it.

This distinction protects the intended research question. If every example quietly uses the optimum as the performed quantity, readers may come away believing that EBU itself contains an optimizing controller. Our examples instead calculate declared quantities and preserve adverse or zero outcomes. The field can then be used as an account of action without smuggling a preferred dynamics into its definition.

27.14 The useful habit

Whenever you see a product of starting field and finite quantity, ask where the curvature went. For the Gaussian landscape, there is an exact answer: half the Hessian quadratic form. Whenever you see an integral, ask which path and parameterization it describes. Whenever you see a signed value, ask which actual predecessor and quantity produced it. Those questions turn a compact equation into a reconstructible account.

Chapter 28 will place several such accounts beside one another. Comparing them is arithmetic. Choosing among them is a further rule, and we will keep that rule visible.

Studio: compare actions without hiding a policy

Physical picture: a menu is not a command

Four, eight, and twelve units can all move from Ridge toward Vale in a suitably permissive route. They produce different endpoints and different EBU values. Writing those values in a table helps us understand the alternatives. It does not establish that the largest value must be selected. An account describes consequences; a policy specifies how a choice is made.

This studio develops a patient comparison of finite alternatives while keeping the purpose of each comparison visible. We will not build an EBU-maximizing controller into the definition of value. The current research question leaves actor choice and repeated behaviour open. The exact quote can be used within several policies, including a declared random choice among admitted candidates. Which policy is scientifically useful must be separately specified and assessed.

Our initial state, reference, and scales remain the familiar Ridge–Vale data. The ideal process burden is zero. For the main table, declare the finite candidate menu

$$\mathcal{Q} = \{0, 2, 4, 8, 12, 16, 17\}X. \quad (28.1)$$

Assume a route cap of eighteen for this illustration, so every listed quantity passes the simple route and nonnegative-source checks. A finite menu is a declared restriction on candidates. The underlying stock is divisible; the table does not claim that these are the only physically conceivable quantities.

28.1 Complete every column before comparing

The exact formula from Chapter 27 is

$$E(q) = 4q - q^2/4. \quad (28.2)$$

For each candidate, record its quantity, endpoint, endpoint potential, total value, and value per unit when that ratio is defined:

q	Endpoint	V_{after}	$E(q)$	$E(q)/q$
0	(18, 2)	16	0	undefined
2	(16, 4)	9	7	3.5
4	(14, 6)	4	12	3
8	(10, 10)	0	16	2
12	(6, 14)	4	12	1
16	(2, 18)	16	0	0
17	(1, 19)	20.25	-4.25	-0.25

Quantities and endpoint entries are in X; ratio entries have inverse-stock units. The table's total values are dimensionless signed EBU numbers under the declared convention.

Take time to reconstruct the row for twelve. The source falls from eighteen to six, the destination rises from two to fourteen, and each endpoint is four units from its reference. Each local contribution is two, total four. The difference from sixteen is twelve. The transfer crossed the reference after eight units and continued beyond it, but its endpoint still has less potential than the original state.

The four-unit and twelve-unit rows share total value twelve while producing different states. A scalar tie does not imply equivalent future opportunities. Ridge has fourteen in one case and six in the other. A later demand at Ridge may therefore distinguish the two states even though the current potential does not.

28.2 Totals and ratios ask different questions

Among the positive rows, the two-unit action has the largest value per unit, while the eight-unit action has the largest total field reduction. This is not a contradiction. The ratio asks how much average field reduction accompanies each transferred unit. The total asks how much the whole candidate changes the field.

For positive q , division gives

$$\frac{E(q)}{q} = 4 - q/4. \quad (28.3)$$

The average decreases with quantity in this example. The first units traverse the steepest part of the landscape. Later units approach and may pass the reference. Total value can continue increasing while the average falls, just as a larger collection can have a larger sum but a smaller average.

At $q = 0$, total value is exactly zero, but the ratio is $0/0$ and is undefined. The limiting ratio as positive quantity tends to zero is four; that derivative-like limit must not be entered as the ratio of a performed zero action. A software table that silently substitutes zero or four needs an explicit display convention so that readers do not mistake it for division.

Neither column is automatically the selection rule. A researcher who substitutes the ratio for a registered total-based policy changes the experiment. A researcher who selects the total maximum when the programme specifies unweighted random choice also changes the experiment. Correct arithmetic is compatible with an incorrect implementation of the intended question.

28.3 Three possible rules on the same physical menu

To see the separation clearly, consider three hypothetical policies. The first selects the greatest total EBU. The second selects the greatest positive EBU per transferred unit. The third selects uniformly among the declared physically admissible candidates, including rest. On the table above, the first selects eight, the second selects two, and the third has seven possible outcomes.

These are definitions of three different choice mechanisms. The first two are useful comparisons for understanding what columns mean, but this studio adopts neither as EBU law. The third is also incomplete until its menu, duplicate handling, eligibility, and random mechanism are specified. Calling a policy “random” does not supply those details.

Suppose the same physical four-unit candidate appears twice in the menu under two labels. Uniform choice over records would then give that physical transformation twice the weight. Uniform choice over distinct transformations would not. If group membership also matters, deciding what counts as a duplicate becomes more subtle. The menu is therefore part of the scientific specification rather than a clerical prelude.

28.4 Permission can remove the apparent best row

Change only the route cap to 6X. Among the listed candidates, zero, two, and four remain admissible. The reference-reaching eight-unit row is now a diagnostic outside the available menu. Its attractive value does not open the route.

Do not replace the excluded eight-unit candidate by six without saying so. Six is physically possible under the new cap, but it was not a member of the declared finite menu. Adding it changes the menu. Its value would be fifteen, which can be calculated analytically, but an experiment that freezes a menu

must preserve that distinction between possible quantities and registered candidates.

This may feel fussy in a small example. It becomes important when researchers inspect outcomes and begin adding convenient alternatives. A menu that is revised after seeing which choices improve the desired metric is no longer the same prospective comparison. The book's arithmetic can explore alternatives freely; a scientific study needs the exploration and the frozen test to be clearly distinguished.

28.5 Keep capacity accounting visible

The prospective programme also considers persistent signed capacity accounts. Such an account is different from source stock and from route capacity. Its update rule, ownership map, group netting, and initial conditions are institutional model choices. They do not follow from the equation for E .

For a narrow illustration only, suppose one actor owns one action record, the rule updates its account by the full signed value, and its balance must remain nonnegative. If its balance is three, the candidate with value -4.25 would fail that particular affordability rule because $3 - 4.25 < 0$. If the balance were five, the same candidate would pass it. The physical endpoint would be identical in both cases.

This illustrates what an affordability filter means; it does not register the programme's general settlement policy. Simultaneous groups require declared attribution and ownership, and same-event netting raises additional questions. Zero-value candidates also require a clear rule: a positive physical movement with zero net value can pass a simple nonnegative-balance check, so the account does not automatically prevent cycles.

The value law remains the same across the two hypothetical balances. What changes is which candidates the policy admits. If a later study aims to compare random physical choice with random affordable choice, the common physical menu and the additional accounting filter must be separately recorded. Otherwise an observed difference might be caused by unequal physical opportunities.

28.6 Add a declared process burden

For this section only, replace $C = 0$ with the illustrative schedule $C(q) = 0.25q$, where the coefficient has the units needed to return a dimensionless burden. It must represent a separately justified action burden, not a second charge for

the same endpoint effects. Then

$$E_C(q) = 3.75q - q^2/4. \quad (28.4)$$

At two, four, and eight, the values become 6.5, 11, and 14. At sixteen the value becomes minus four, although the endpoint field potential still equals its starting value. A movement with zero field difference can have negative net value when real process burden is included.

The physical states in the table do not change merely because this extra accounting term is introduced. If the burden corresponds to a represented energy use, however, that physical coordinate must already be included in the broader account. A coefficient in a value formula cannot substitute for a missing energy balance. The exercise isolates the algebraic role of a burden after its meaning has been declared.

28.7 A changed baseline creates a new schedule

Return to $C = 0$, but suppose a separately recorded conservative disturbance has moved the state to (16, 4) before the actor's epoch. Its potential is nine and its edge field is three. The new exact schedule is

$$E_{\text{new}}(q) = 3q - q^2/4. \quad (28.5)$$

The four-unit value is now eight, not twelve. The reference is reached at six. A stored quote from (18, 2) therefore cannot be treated as timeless pricing for the route.

A rest candidate still gives zero because it compares the new baseline with itself. The external mechanism's reduction from sixteen to nine is not assigned to the resting actor. If an audit adds it to every candidate's value, it creates a fictitious reward for doing nothing and duplicates the same external change across alternatives.

28.8 A worksheet for an honest comparison

A useful comparison sheet keeps six matters visible: the shared baseline; the exact potential definition; the physical candidate and accepted quantity; the permission verdict; the signed finite value; and the declared selection rule. It also retains rejected candidates with their reasons when that is part of the audit design. Rejection is information, not a value of zero.

There is a further distinction between the selected proposal and the performed transformation. If four was selected but three was actually delivered in the lossless account, the successor is (15, 5) and the value is 9.75. The difference

from the planned twelve should be explained through the recorded execution, not hidden by retaining the old quote.

The comparison table can support learning: a reader can see which quantities overshoot, which caps bind, and which process burdens matter. Learning from it does not automatically authorize an optimizer, a policy change, or an experimental run. The policy must remain visible precisely because it shapes what the accounting will later observe.

28.9 Practice cases

1. Use the base menu $\{0, 1, 3, 5\}X$. Calculate endpoint potential, total EBU, and defined per-unit ratios. Which rows would total and ratio maximizers choose?
2. Put a physical cap of $4X$ on that menu. List admitted candidates. Should the five-unit row be reported as zero EBU?
3. An actor with illustrative balance one faces value -0.5 . Under the simple signed update described above, calculate its successor balance. Does this calculate physical permission?
4. At base state, compare a four-unit transfer with a twelve-unit transfer on potential and final Ridge stock. Explain why neither number alone specifies an all-purpose winner.
5. A menu contains 0, 2, 4, 4, 8, and selection is uniform over entries. What probability is assigned to the four-unit transformation? Compare with uniform choice over distinct quantities.
6. From $(16, 4)$, calculate the exact value of a two-unit action and identify the external audit contribution that must not be included.

28.10 Solutions

For the first menu, the endpoint potentials at one, three, and five are 12.25, 6.25, and 2.25. Their total values are 3.75, 9.75, and 13.75. Their ratios are 3.75, 3.25, and 2.75. A total maximizer would choose five; a positive-ratio maximizer would choose one. These answers identify hypothetical rules, not a recommendation.

With cap four, the admitted listed quantities are zero, one, and three. Five is physically excluded. Its mathematical value under a hypothetical unconstrained movement is still 13.75, but that value does not make it selectable.

Reporting exclusion as zero would erase the difference between an unavailable action and a genuine zero-value action.

The balance update in the third problem is $1 - 0.5 = 0.5$, which is nonnegative. Physical permission remains a separate check. A balance can permit an account update without establishing stock availability, a usable route, or destination capacity.

Four and twelve both reduce potential by twelve, but they leave Ridge stocks fourteen and six respectively. They also deliver different quantities to Vale. A policy concerned with immediate delivery, future Ridge supply, route use, or another registered metric may distinguish them. The field total should be reported faithfully without being promoted into every possible objective.

Uniform choice over the five entries assigns probability $2/5$ to quantity four. Uniform choice over the four distinct quantities assigns $1/4$. This difference exists before any physical event. The sampling unit is part of the model.

Finally, from $(16, 4)$, a two-unit action gives $3(2) - 4/4 = 5$. The earlier forcing reduction from sixteen to nine was seven. It belongs to the external record. The complete reduction from $(18, 2)$ to the final $(14, 6)$ is twelve, decomposed as seven external and five actor-associated under this chronology.

28.11 Service records need their own definitions

Suppose a separate service task requests four units at Vale during the action. For this paragraph only, define immediate fulfilled service as $\min(q, 4)$, and unfilled requested service as $4 - \min(q, 4)$. These are declared service metrics in stock units. At quantities two, four, and eight, fulfilled service is two, four, and four respectively, while field value is seven, twelve, and sixteen.

The eight-unit action has a larger field reduction than the four-unit action, but it does not fulfill more of this particular four-unit request. Its additional delivery may become inventory, may serve another declared need, or may be unnecessary. The service definition must say which. This example shows why service and EBU should be reported separately even when both improve over some range.

Conversely, a positive service number does not prove that a delivery lowers the field. From the reference, a two-unit transfer could satisfy a newly declared request while increasing the chosen stock potential. That tension may reveal that the state representation needs a demand or consumption mechanism, or that a distinct institutional decision is required. It should not be resolved by replacing the negative value with zero or changing the reference after observing the result.

A future study about sustained provision must represent how demand affects stocks or other state variables. A request that merely caps proposals does not by itself remove resources from a clinic, create a deficit, or generate regeneration. The dynamics must include the mechanisms whose effects the study intends to examine. This book's finite examples teach an accounting foundation; they do not claim that a closed stock transfer alone models continuing life support.

28.12 An honest negative result

Imagine a registered comparison later finds that the proposed random-affordable policy refuses many necessary intermediate actions while the physical-random comparator performs more service. If the world, metrics, and information boundaries were properly specified, that would be an informative adverse finding. The account should make it possible to see why refusal occurred: physical impossibility, lack of declared action capacity, or another explicit condition.

Changing the menu after seeing this result to remove the inconvenient comparator choices would answer another question. A new design may be worth studying, but it should be recorded as new. Scientific usefulness comes from learning which assumptions support which behaviour, not from arranging the reporting vocabulary so that every outcome sounds successful.

28.13 Carry the separation forward

Exact value permits comparison; it does not choose an action. A study must remain open to refusal, cycling, recovery or poor performance. The next studio follows a medicine route through successive baselines, including a useful journey with a negative intermediate value.

Studio: a medicine route in local epochs

Physical picture: a distant need is reached through local acts

Medicine is stored at Ridge, passes through a regional hub, and reaches Vale clinic. The journey has an order: the hub must receive stock before it can dispatch it. Each transfer changes a live local state. Keeping those changes visible lets the route close without inventing a special force for distance or crediting the same clinic improvement twice.

This cold-chain teaching scene uses a fully declared lossless Gaussian account. It is a stock-routing illustration, not a model of clinical treatment. Medicine is treated as one homogeneous divisible stock, measured in X . This is appropriate for arithmetic, while an operational medical system would need product identity, dose, expiry, temperature, sterility, and other non-interchangeable properties. Arrival at a clinic is also different from administration to a patient.

The route is deliberately simple enough to reconstruct line by line. Its importance lies in the accounting pattern: atomic transformations, live predecessors, resource-specific ledgers, and a final comparison that cancels intermediate terms only when the assumptions allow it.

29.1 Declare the route's changed data

This is a three-cell example, so do not import the two-cell reference without adjustment. The coordinate order is warehouse W , hub H , clinic C . Define

$$x^0 = (10, 0, 2)^T X, \quad x^* = (6, 0, 6)^T X, \quad \sigma = (2, 2, 2)^T X. \quad (29.1)$$

Both actual and reference totals are twelve. The reference assigns zero persistent stock to the hub in this simplified model. That does not mean a real hub should hold no safety inventory. It is a declared teaching assumption that makes temporary inventory visible as a deviation.

The two lossless edges are $W \rightarrow H$ and $H \rightarrow C$. Their columns are

$$S_1 = (-1, 1, 0)^T, \quad S_2 = (0, -1, 1)^T. \quad (29.2)$$

We inspect the specified quantities $q_1 = 5X$ and $q_2 = 4X$. Assume the corresponding route capacities permit these quantities. No external change occurs between the two action epochs, and the same potential remains fixed. The ideal process burdens are $C_1 = C_2 = 0$.

The potential is

$$V(x) = \frac{(x_W - 6)^2 + x_H^2 + (x_C - 6)^2}{8}. \quad (29.3)$$

At the start it is $(16 + 0 + 16)/8 = 4$. This equation and its data define the scope of every value below. They are not clinical valuations of a real warehouse or patient.

29.2 First epoch: move five units to the hub

The first atomic transfer has increment $(-5, 5, 0)^T X$. Its successor is

$$x^1 = (5, 5, 2)^T X. \quad (29.4)$$

Warehouse stock falls by five, hub stock rises by five, and clinic stock remains two. The total is still twelve. The resulting local contributions are

$$V_W(5) = (-1)^2/8 = 0.125, \quad (29.5)$$

$$V_H(5) = 5^2/8 = 3.125, \quad (29.6)$$

$$V_C(2) = (-4)^2/8 = 2. \quad (29.7)$$

Together they give $V(x^1) = 5.25$, so the first field receipt is

$$E_1 = 4 - 5.25 = -1.25. \quad (29.8)$$

The first step increases the represented potential. It creates temporary hub inventory and takes the warehouse slightly below its reference before improving the clinic stock. This result is not concealed because the route has a desirable final purpose. The finite equation evaluates the state change specified for this epoch.

The negative line is also not a judgement that the carrier is bad or that the patient's need is illegitimate. It is a scoped result under the chosen intermediate-state potential. A policy that requires every individual step to have nonnegative value might block this route. Whether to permit grouped commitments, finance an intermediate step, choose another path, or change the institutional arrangement is a separate question. The arithmetic does not select the answer.

29.3 Second epoch: dispatch four units to Vale

The second transfer begins from x^1 , where the hub contains five. It withdraws four and delivers four, leaving one at the hub:

$$x^2 = (5, 1, 6)^T X. \quad (29.9)$$

The clinic now equals its declared reference. The final local contributions are 0.125, 0.125, and zero, total 0.25. The second receipt is therefore

$$E_2 = 5.25 - 0.25 = 5. \quad (29.10)$$

Notice that the second action reduces potential at both the hub and clinic. It does not change warehouse stock. The warehouse contribution is present in a global evaluation but cancels between the two sides. A local evaluation can omit that unchanged contribution while retaining the full field difference.

If the second action had been quoted independently against x^0 , it would attempt to withdraw four from an empty hub. The vector formula would produce a negative coordinate, but the physical action would be inadmissible. Correct live ordering is thus needed for feasibility as well as for value.

29.4 Close the physical route

Track the five units removed from the warehouse:

$$5X = 4X \text{ at the clinic} + 1X \text{ remaining at the hub.} \quad (29.11)$$

Both transfer efficiencies are one and both declared losses are zero. The warehouse's final five plus the hub's one plus the clinic's six reconstruct the original twelve units.

The retained hub unit may be useful later, but this receipt does not invent its future delivery. It is present inventory. Likewise, the clinic's receipt of four does not prove that four units were administered effectively. Consumption, treatment, wastage, and disposal would require further physical transformations and any additional state needed to describe them.

This is why a route record is more informative than one net sentence saying "five units were sent to help Vale." That sentence does not reveal where the fifth unit remains, what the hub did, or whether the clinic actually received the planned amount. The local records preserve those distinctions.

29.5 Close the value route by cancellation

Add the two signed results:

$$E_1 + E_2 = -1.25 + 5 = 3.75. \quad (29.12)$$

The endpoint calculation gives the same answer:

$$V(x^0) - V(x^2) = 4 - 0.25 = 3.75. \quad (29.13)$$

Write the unsimplified sum to see why:

$$E_1 + E_2 = [V(x^0) - V(x^1)] + [V(x^1) - V(x^2)] \quad (29.14)$$

$$= V(x^0) - V(x^2). \quad (29.15)$$

The same intermediate potential occurs once with a minus sign and once with a plus sign. Its cancellation requires the second receipt to use the actual state left by the first, under the same potential definition.

For a longer sequence with process burdens,

$$\sum_{k=1}^m E_k = V(x^0) - V(x^m) - \sum_{k=1}^m C_k \quad (29.16)$$

under the action-only, fixed-potential assumptions. The identity does not promise a useful endpoint. A route that returns to its starting state has zero total field difference and, with nonnegative process burden, nonpositive net value. A route that increases the potential has a negative field total even if every record is perfectly consistent.

29.6 Preserve actor records without inventing ownership

Suppose Carrier A performed the first transfer and Carrier B performed the second. The physical action register can attach each signed receipt to the responsible record. Summing those records gives $-1.25 + 5 = 3.75$. Carrier B's five is not a transfer of five from Carrier A. Closure includes the decrease of the field potential.

Attaching a record to a performer also does not decide who owns a spendable balance, how an employer and employee share an entitlement, or how a public service is funded. Those are institutional definitions. A ledger can preserve who performed which transformation while leaving such questions open. The book uses actor names to maintain accountability, not to deduce a complete legal economy from telescoping.

If one organization performed both transfers, its recorded route total would still be 3.75. Renaming performers changes neither the states nor their potential. Splitting one physical act into several labels must not duplicate the same state change. Each performed transformation belongs once in the chain of endpoints.

29.7 A separately declared process account

For an algebraic extension, suppose justified non-overlapping burdens are $C_1 = 0.2$ and $C_2 = 0.3$. The physical medicine endpoints remain the same. The net results become -1.45 and 4.7 , summing to 3.25 . Direct closure gives

$$4 - 0.25 - (0.2 + 0.3) = 3.25. \quad (29.17)$$

These illustrative burdens should not be described as measured refrigeration costs. In a real plant, refrigeration would involve electricity, heat, equipment, and timing. If their consequences are represented and evaluated elsewhere, the separate burden must exclude duplication. The stock-only idealization can be useful without pretending to provide that complete calibration.

Different route lengths matter through declared physical mechanisms such as travel time, energy use, loss, storage, and capacity. There is no universal inverse-square EBU price of geographical distance in this account. Two routes of equal length can have different physical effects; two routes of different length can have similar effects. Their actual transformations must decide the comparison.

29.8 Do not hide a physical chain inside a route name

The medicine route is one resource account. To see how another resource chain should be represented, consider a separate illustrative pumping chain with explicit energy carriers. One megajoule of fuel energy is withdrawn. A declared conversion produces 0.3 megajoules of electrical energy and 0.7 megajoules of heat. The energy balance is $1 = 0.3 + 0.7$.

A following delivery stage transfers the 0.3 megajoules to the pump boundary, delivering 0.24 and recording 0.06 as heat. The pump stage then uses the delivered 0.24 megajoules to perform a specified water movement of $2X$, with its complete mechanical and thermal endpoint accounted for; in this deliberately simplified endpoint example all 0.24 ultimately appears as heat and no energy remains stored mechanically. The water ledger separately records $-2X$ at its source and $+2X$ at its destination.

The completed energy account is one megajoule of initial fuel energy represented by one megajoule of final heat across the declared stages. The water account conserves its own quantity. We do not add two water units to one megajoule. Nor does conservation alone establish that the stated conversion or pump performance describes real equipment. Those rates and mechanisms are illustrative declarations that a real engineering model would need to justify.

This linked description prevents two opposite mistakes. The upstream fuel use does not disappear into a single “water delivery” label, and an extra burden is not invented merely because several named stages exist. Each physical consequence enters its proper account once. Any cross-resource scalar requires a separate justified conversion rule; it does not follow from the existence of the chain.

29.9 What if something happens between route epochs?

Return to the medicine route with zero process burden. After $x^1 = (5, 5, 2)$, suppose an external conservative mechanism moves one unit from the hub back to the warehouse, giving the next baseline $z^2 = (6, 4, 2)$. Its potential is four. The external potential change is $4 - 5.25 = -1.25$.

The second carrier now moves four units from this new baseline, reaching $(6, 0, 6)$, potential zero. Its receipt is four. Actor-associated receipts sum to $-1.25 + 4 = 2.75$. The total endpoint reduction from the original state is four, so it would be wrong to assign all four to the actors. The missing 1.25 of reduction belongs to the intervening external event.

Equivalently, with signed external potential increment $D_{\text{ext}} = -1.25$,

$$\sum E_k = V(x^0) - V(x^2) + D_{\text{ext}} = 4 - 1.25 = 2.75. \quad (29.18)$$

The sign is worth checking: an external improvement lowers potential, so its signed increment is negative and it reduces the part attributable to the actions. A disturbance that raises potential would have the opposite sign. External does not mean harmful by definition.

29.10 Practice route variants

1. In the original route, keep the first quantity five but change the second to three. Calculate both endpoint states, final potential, second receipt, and route total.
2. Instead use quantities four and four. Does the first receipt remain negative? Find both values and the final state.
3. Keep the original five-and-four route, but set the separately justified burdens to 0.4 and 0.6. Calculate the net route total.
4. Explain why the hub-to-clinic action cannot simply be swapped to the first position in the original route.
5. If all route medicine is ultimately administered, what new physical record is missing from this studio?

6. The first carrier calls its negative value an error because the patient needs medicine. What part of the calculation, if any, follows from that need alone?

29.11 Solutions

With quantities five and three, the first successor stays $(5, 5, 2)$ and the final state is $(5, 2, 5)$. Its deviations are $(-1, 2, -1)$, squared sum six, potential 0.75. The second receipt is $5.25 - 0.75 = 4.5$, and the route total is $-1.25 + 4.5 = 3.25$. Two units remain at the hub, three were delivered to the clinic, and the source withdrawal five closes as $2 + 3$.

With quantities four and four, the intermediate state is $(6, 4, 2)$, potential four, giving first receipt zero. The final state is the reference $(6, 0, 6)$, potential zero, giving second receipt four. The route total is four. These numbers compare declared alternatives; they do not establish a universal rule that every route must use the best endpoint under this potential.

The burdens in the third problem sum to one, so net route value is $3.75 - 1 = 2.75$. In the fourth problem, the original hub has no medicine. Dispatching four first violates nonnegative stock. A schedule with borrowing, advance reservations, or another supply would be a changed physical or institutional arrangement and would need declaration.

Administration requires a further transformation describing what leaves usable clinic inventory, what physical state or carrier receives it, and which outcomes are evaluated over which horizon. Arrival is not treatment success. Finally, the patient's need motivates the route and may motivate support for it, but does not change the arithmetic under the declared potential. If the potential omits a relevant function, that is a model criticism to make openly. Silently changing an inconvenient intermediate result is not a correction.

29.12 Planning a route is not performing its future

Before departure, a planner can calculate the specified five-and-four route from estimated states. The prospective sum 3.75 is then a forecast under those assumptions. It must not be recorded as a completed improvement merely because the two arithmetic steps telescope. The hub may receive less, a vehicle may be delayed, or another event may change clinic stock before delivery.

A useful route plan therefore retains its planned local states and quantities, while performed receipts retain observed or otherwise verified local states and quantities. The difference is not a reason to discard the plan. It is evidence

about prediction quality and operational conditions. Over many properly recorded journeys, such differences may motivate a better route model or infrastructure changes.

Those later improvements also require their own physical accounts. Building a regional storage facility might reduce future transport burdens, but construction itself uses resources and does not automatically create a present stock-field credit equal to every forecast saving. A forecast and a performed transformation belong to different evidence categories. Any scheme for recognizing persistent future effects needs a declared horizon, state representation, and validation method.

29.13 Boundary changes during a journey

Suppose a shipment leaves the represented region at one checkpoint and re-enters at another. A local observer may see an outward boundary flux followed later by an inward flux. The wider route account may represent the in-transit stock continuously. Both descriptions can be coherent if their boundaries and identifiers are explicit.

Trouble arises when a report uses the narrow boundary to claim that stock disappeared and the wide boundary to claim that it was available, without recording the change of scope. The same shipment must remain traceable through the chosen carriers. A transit coordinate can be useful because it prevents an in-between quantity from being counted simultaneously at the warehouse and clinic or being omitted from both without explanation.

In the present three-cell example, the hub supplies that visible intermediate location. It is not a universal model of transport time. An actual cold chain may need several transit and condition coordinates. The principle is to add enough physical state to answer the stated question, then evaluate the affected factors consistently. More named coordinates are useful only when they correspond to declared quantities or conditions that the account can support.

29.14 A route is an explanation, not just a total

The route total is compact because the local record is complete. Its meaning can be unfolded into quantities, places, ordering, external events, and separate resources. That is the value of patient accounting. A long journey need not become a mysterious distant interaction, and a useful final purpose need not erase difficult intermediate states. The next studio turns from ordered links to several actors sharing one simultaneous field, where a correct group total still leaves a distinct attribution question.

Studio: several actors at one source

Physical picture: two known movements still share one curve

Two carriers can report their quantities perfectly and still share the same changing source stock. Knowing each movement does not make a squared potential linear. We must first calculate the joint physical endpoint and its exact field value. A declared path can then divide that value arithmetically, while questions of ownership and fair settlement remain distinct.

The examples in this studio are deliberately small. Their arithmetic can be inspected without a simulation, and their limitations can be stated precisely. No actor is allowed to claim the entire benefit of the group's common change. At the same time, we do not solve a general institutional allocation problem by calling an exact sum a fair share.

Start with the base Ridge–Vale state $(18, 2)^T X$, reference $(10, 10)^T X$, scales two, and zero process burden. Suppose two actors, A and B, each move two units on the same lossless edge during a declared joint event. Physical permission admits their total quantity four. Their names identify child action records.

30.1 Compute the tempting independent sum

If A acted alone from the frozen state, the endpoint would be $(16, 4)$, whose potential is nine. Its standalone field value would be $16 - 9 = 7$. B's separate same-baseline hypothetical calculation is identical. Adding the two hypothetical values gives fourteen.

But the actual joint increment is $(-4, 4)^T X$, and the joint endpoint is $(14, 6)$, whose potential is four. The group value is twelve. The independent sum therefore exceeds the group value by two.

Both standalone calculations are mathematically correct as counterfactual one-action comparisons. The error appears when they are described as additive records of the same joint event. Each grants its child the complete original source and destination conditions. The second copy fails to include how the other child changes the shared field.

This is a useful example because no uncertainty, missing actor identity, or physical loss is needed to create the discrepancy. Even exact quantities and a very simple separable potential do not authorize naive independent addition.

30.2 Identify the cross term

Let δ_A and δ_B be the two increments. The Gaussian group field reduction is

$$F_G = -\mu(z)^\top (\delta_A + \delta_B) - \frac{1}{2}(\delta_A + \delta_B)^\top H(\delta_A + \delta_B). \quad (30.1)$$

Expanding the quadratic and subtracting the standalone sum gives

$$F_G - (F_A^{(0)} + F_B^{(0)}) = -\delta_A^\top H \delta_B. \quad (30.2)$$

For $\delta_A = \delta_B = (-2, 2)^\top X$ and $H = \text{diag}(1/4, 1/4)$, the product is

$$\delta_A^\top H \delta_B = \frac{4}{4} + \frac{4}{4} = 2. \quad (30.3)$$

The diagnostic correction is minus two, exactly the difference between twelve and fourteen.

The potential's Hessian is diagonal, yet the actions interact in this comparison because they change the same coordinates. A diagonal state potential does not imply that every pair of action values is additive. The action supports and the potential structure must both be examined.

30.3 Compare a genuine live sequence

If A physically goes first, its value is seven and the live state becomes (16, 4). B then begins from that state and reaches (14, 6), giving value $9 - 4 = 5$. The live sum is twelve. If B goes first, the named actors exchange seven and five, while the total remains twelve.

This sequence has the same endpoint as the simultaneous group, under our fixed quantities and permission assumptions. Its total field value is therefore equal to the group's. The same-baseline correction of minus two is not evidence of an extra physical benefit or harm caused uniquely by simultaneity. It compares the joint endpoint with a naive sum of counterfactual standalone values.

In the terminology of the bridge developed in the main chapters, endpoint-equivalent serial and parallel accounts can have zero comparator-relative difference while still having nonzero same-baseline non-additivity. The two diagnostics answer different questions. Calling both simply “interaction” would make the numerical example hard to interpret.

30.4 Declare one common path attribution

Let the group increment be $\delta_G = \delta_A + \delta_B$. Along the declared straight path $z + s\delta_G$, define the field attribution

$$R_a = - \int_0^1 \nabla V(z + s\delta_G)^\top \delta_a ds. \quad (30.4)$$

For the fixed Gaussian model this is exactly

$$R_a = -\mu(z)^\top \delta_a - \frac{1}{2} \delta_a^\top H \delta_G. \quad (30.5)$$

In the symmetric example, the first term for A is eight. The cross product with the full group increment is four, so half is two. Thus $R_A = 8 - 2 = 6$, and likewise $R_B = 6$. They sum to twelve.

Closure follows by summing the increments inside the integral:

$$\sum_a R_a = - \int_0^1 \nabla V(z + s\delta_G)^\top \delta_G ds \quad (30.6)$$

$$= V(z) - V(z + \delta_G). \quad (30.7)$$

This is an exact mathematical decomposition under the declared path. It does not prove that six is an actor's property, wage, causal entitlement, or fair reward. Those meanings need additional rules and arguments. A sum can be exact without its terms acquiring every interpretation one might want.

If the physical model explicitly realizes constant-rate child movements, the path can describe that model's actual trajectory through stock space. If only endpoints are known, the same arithmetic is an interpolation convention. We must state which interpretation is in force. Neither can be generalized to an unobserved real physical path by changing the name of the formula.

30.5 Use unequal children and different destinations

Add a third cell, a second destination called Hill. This changes the coordinate list and the reference. Set

$$z = (18, 2, 10)^\top X, \quad x^* = (10, 10, 10)^\top X, \quad \sigma = (2, 2, 2)^\top X. \quad (30.8)$$

The total is thirty, and initial potential is sixteen. A sends four to Vale, so $\delta_A = (-4, 4, 0)^\top X$. B sends two to Hill, so $\delta_B = (-2, 0, 2)^\top X$. Assume the joint six-unit source withdrawal and both destination receipts are permitted.

The endpoint is $(12, 6, 12)$, with squared deviations $4 + 16 + 4 = 24$, potential three. Hence the group value is thirteen. A alone has value twelve. B alone

reaches (16, 2, 12), potential thirteen, and has value three. The independent sum is fifteen, again two above the group result.

The initial marginal vector is $(2, -2, 0)^T X^{-1}$. For A, the linear field term is sixteen. Its Hessian product with $\delta_G = (-6, 4, 2)^T X$ is

$$\frac{(-4)(-6) + 4(4) + 0(2)}{4} = 10. \quad (30.9)$$

Half of this is five, so $R_A = 11$. For B, the linear field term is four and the Hessian product is

$$\frac{(-2)(-6) + 0(4) + 2(2)}{4} = 4, \quad (30.10)$$

giving $R_B = 4 - 2 = 2$. The path attributions close as $11 + 2 = 13$.

Equal division would instead give 6.5 to each child. Division proportional to quantity would give $26/3$ and $13/3$. Each pair sums to thirteen, but the pairs are different. Closure alone therefore cannot identify a unique split. The declared path adds mathematical structure; a settlement institution would need to justify why that structure should govern its balances.

30.6 Check both sequential orders in the unequal case

When A goes first, it contributes twelve. The intermediate state is (14, 6, 10), potential four. B then reaches the final potential three, contributing one. The pair is twelve and one.

When B goes first, it contributes three. The intermediate state is (16, 2, 12), potential thirteen. A then reduces that to three, contributing ten. The pair assigned to A and B is ten and three. Both orders sum to thirteen, and the common-path pair eleven and two lies between these two order-specific assignments in this quadratic example.

That numerical relation is instructive, but it should not be casually promoted into a universal fairness theorem. It depends on the particular path and model. More general mechanisms can change endpoints with order, forbid some serializations, or change accepted quantities. A comparison must name the sequential schedule and say whether it keeps the group quantities fixed or reruns a resolver.

30.7 Opposing movement and the meaning of cancellation

Return to the two-cell base state. Let A move two units from Ridge to Vale and B move two in the reverse direction. Declare that the joint physical mechanism permits these opposing transfers. The net increment is zero, so the group endpoint equals the baseline and the ideal field value is zero.

A alone has value seven. B alone reaches $(20, 0)$, potential twenty-five, and has value minus nine. The standalone sum is minus two, while the group value is zero. The same-baseline correction is therefore plus two.

On the constant-state common path, the group increment is zero. The path formula gives $R_A = 8$ and $R_B = -8$, which sum to zero. These terms are a decomposition using the declared child increments; they are not evidence that eight units of physical stock or money passed from B to A. They also do not prove that real opposing transport has no cost. The ideal account set process burden to zero. Any actual energy use or loss must remain in its proper physical account.

This example corrects another tempting inference: positive non-additivity need not demonstrate a special cooperative mechanism. Here it comes from cancelling increments under a curved potential. To claim a physical synergy of parallel execution, one must compare with an admissible named alternative and establish a difference in complete endpoints or declared burdens.

30.8 Physical permission is still aggregate

Take the unequal three-cell example with total proposed withdrawal six. If a newly declared source handling cap is five, that exact group is inadmissible. The mathematical value thirteen does not remove the violation. We must not execute six and then clip the source to a preferred number while leaving both destinations unchanged.

A resolver could reject the group, select a subset, or produce a new accepted vector under a separately declared rule. Whatever it does, value and attribution must use the accepted physical increments. The old calculations for quantities four and two cannot simply be attached to a resized event.

Group assembly also has an information boundary. A calculation that needs the shared source and two destinations is local to that support, but the procedure that searches many groups and selects one may require broader information. Local valuation does not automatically prove that selection is decentralized. A future implementation must specify both.

30.9 Practice and answers

1. In the two-cell base example, let both actors move one unit forward. Calculate standalone values, group value, the correction, and both path attributions.

2. Let A move one and B move three forward on the same edge. Find the group value and common-path attributions. Compare with A-first live values.
3. In the unequal three-cell example, add separately declared child process burdens 0.5 and 0.25. What is the net group result if these burdens are additive and not already represented?
4. Two actions touch disjoint coordinate pairs in a diagonal Gaussian potential. What does the cross product say about their same-baseline correction, assuming both standalone and joint actions are admissible?
5. Why does giving an exact attribution formula a new name fail to establish ownership?

For the first problem, each standalone value is $4 - 1/4 = 3.75$. Their sum is 7.5, while the two-unit group has value seven, so the correction is -0.5 . Each common-path attribution is 3.5. A sequential order instead yields 3.75 followed by 3.25.

For quantities one and three, the total quantity is four and group value twelve. Along the common path, the average field is three, giving attributions three and nine. A first alone has value 3.75, and the subsequent three-unit action contributes $12 - 3.75 = 8.25$. The difference is an attribution consequence of path or order, not missing stock.

The net group result in problem three is $13 - 0.5 - 0.25 = 12.25$. The field attributions eleven and two still sum to thirteen. Subtracting the separately assigned burdens gives 10.5 and 1.75, sum 12.25, under the expressly declared additive burden allocation. If a process burden belongs to the group mechanism and is not separable by child, its own allocation needs a declaration.

Disjoint supports make $\delta_A^T H \delta_B = 0$ for a diagonal Hessian, so the correction is zero under the stated fixed-increment assumptions. This does not establish independence for a nonseparable potential, shared constraints, or a resolver that changes quantities across counterfactual subsets.

Finally, naming a formula establishes vocabulary. Ownership requires a model or institutional rule identifying who may hold or spend a record, and why. The closure proof supplies neither. Preserving that boundary makes the mathematics more useful because it can support several explicitly debated institutions without claiming to have already proved their legitimacy.

30.10 Splitting a label under a fixed common path

There is one narrow invariance that the path formula makes easy to verify. Suppose an action increment δ_A is divided into two child records δ_{A_1} and δ_{A_2} whose sum is exactly δ_A . Hold the complete group increment, baseline, potential, path, and every other accepted increment fixed. Since the formula is linear in the child increment,

$$R_{A_1} + R_{A_2} = R_A. \quad (30.11)$$

Thus merely dividing one recorded increment into two labels does not change its combined field attribution under these fixed conditions.

For the unequal three-cell example, split A's four-unit movement into two two-unit child records on the same edge. Each receives 5.5 under the unchanged common path, and their sum is eleven. The group value remains thirteen. This is a direct algebraic check, not evidence from actors trying to manipulate a live system.

The qualifications matter. If splitting changes how a resolver allocates capacity, which group is sampled, who owns a record, or which candidate passes affordability, the fixed conditions no longer hold. The simple linearity proof then does not establish incentive compatibility of the whole mechanism. A complete policy must examine those effects separately. This is why a locally neat attribution identity should not be advertised as a universal solution to strategic request splitting.

30.11 When a subset comparison is undefined

The standalone values in this studio exist because each child transformation is individually meaningful and physically admissible from the common state. That need not hold for every group. In the medicine route, the hub-to-clinic child cannot occur first when the hub is empty. Treating it as an independently admissible subset would invent a counterfactual outside the physical model.

More elaborate subset decompositions require an explicit protocol for each subset and every relevant subset must be admissible for a full Boolean-lattice formula. If removing one child changes the quantities allowed for the others, those resolved quantities must be recorded. The fixed-increment quadratic cross-term calculation cannot silently stand in for such a re-resolved family.

It is perfectly acceptable for a diagnostic to be undefined. That is more informative than filling a table with values for impossible actions. The joint event can still have an exact endpoint account even when a proposed collection of counterfactual child comparisons does not exist.

30.12 What the complete group record contains

A trustworthy group record keeps the common baseline, each child increment, aggregate permission, the joint endpoint, and the exact group field reduction. If path attributions are displayed, it also records the declared path semantics. If a sequential comparison is made, it records the order and the accepted quantities in that comparison. These entries let another reader reconstruct the distinctions instead of inferring them from a single headline number.

The final studio will use this same discipline to audit a complete account from input card to closing identity. The aim is not to multiply paperwork. It is to make every compact claim reversible into a physical statement, a definition, a calculation, and a clearly delimited interpretation.

Studio: run the complete physical account

Physical picture: one action, every question in order

A four-unit delivery is simple enough to describe in one sentence. A defensible account must still identify its state, units, permission, reference, exact field change, actual quantity, and closing identities. This last studio begins with a blank sheet and reconstructs that chain. The aim is to make the short EBU equation fully readable, not to turn it into an automatic social decision.

The word “run” in this chapter title means working through an account on paper. The examples are illustrative arithmetic, not trajectories from an executed scientific study. A sequence of specified states can teach telescoping without supplying evidence that a controller will produce such a sequence in a real or simulated world.

Keep the main Ridge–Vale fixture. We will first audit one complete lossless transfer, then a short execution, an external-change variant, and a flawed report. Every variation states which data change. This habit is especially valuable near the end of a long calculation, when familiar numbers can encourage the reader to skip the conditions that made them valid.

31.1 Prepare the frozen input card

Write the card with separate columns for role and value:

Object	Declared value	Role
Coordinate order	Ridge, Vale	Physical identity
Frozen state	$(18, 2)^T X$	Present stock
Reference	$(10, 10)^T X$	Evaluation definition
Scales	$(2, 2)^T X$	Evaluation definition
Edge	Ridge to Vale	Directed transformation
Transfer efficiency	One	Lossless model
Finite route cap	6X	Permission input
Proposed quantity	4X	Candidate action
Process burden	Zero	Ideal account
External change in epoch	None	Baseline scope

The route cap is a physical restriction, the scale is part of evaluation, and the proposed quantity is an actor input. Their units and roles differ. We do not derive four from the potential's maximum or from a flux law. It is the given candidate that this account evaluates.

The card's values are illustrative, not clinical recommendations. A real application would need measurement provenance for stock, engineering support for capacity, a justified reference and scale, and a complete declaration of relevant physical effects. The table makes those dependencies visible without pretending to resolve them by arithmetic.

31.2 Check the action before evaluating it

The physical column is $S_e = (-1, 1)^T$. The proposed withdrawal four is nonnegative, below route cap six, and below source stock eighteen. Under the simple restrictions stated on the card, the candidate is permitted.

Its increment and successor are

$$\delta = S_e q = (-4, 4)^T X, \quad z' = z + \delta = (14, 6)^T X. \quad (31.1)$$

Both successor stocks are nonnegative. The source withdrawal is four, the destination receipt is four, and the declared loss is zero. The original and successor totals both equal twenty.

If another physical restriction existed, this check would be incomplete. For example, a clinic capacity below six would forbid the endpoint even though the source and route checks passed. Do not infer absence of a constraint from its absence in the arithmetic. The account is complete relative to the declared small world.

31.3 Calculate the before and after potentials

Before action, the standardized deviations are four and minus four. The two local contributions are eight and eight, giving $V(z) = 16$. After action, the deviations are four stock units and minus four stock units, or two and minus two scales. The local contributions are two and two, giving $V(z') = 4$.

The exact signed result is

$$E = V(z) - V(z') - C = 16 - 4 - 0 = 12. \quad (31.2)$$

Read it as a sentence: the specified four-unit lossless transfer reduces the declared Gaussian potential by twelve from the frozen state, with no separate process burden in this ideal account. That sentence preserves the baseline and assumptions that the shorter phrase “twelve EBU” leaves implicit.

Now check the result independently through the marginal and curvature. The initial marginal is $(2, -2)^T X^{-1}$. Its negative dot product with δ is sixteen. The curvature correction is four. The finite value is therefore $16 - 4 = 12$, agreeing with the endpoint calculation.

Two algebraic routes reaching twelve are a useful consistency check. They are not two independent experiments, and agreement does not establish that the reference describes a real clinic. The kind of evidence supplied by a check matters as much as the number of checks.

31.4 Record proposal, acceptance, and performance separately

For the complete-match case, the proposal is four, acceptance is four, and illustrative measured performance is four. A receipt can then retain the planned and performed values together. Each label refers to a different stage even though the numbers happen to agree.

Now change only performance to three. The performed increment becomes $(-3, 3)^T X$, successor $(15, 5)^T X$, and potential 6.25. The performed result is

$$E_{\text{perf}} = 16 - 6.25 = 9.75. \quad (31.3)$$

The proposal did not create a claim to the missing 2.25. It remains a planning record. The receipt should explain why acceptance and performance differed if the available evidence can establish that reason. If it cannot, the discrepancy remains unresolved rather than being erased.

The same principle applies when performance is larger than accepted. An exact endpoint value does not excuse an unauthorized quantity. The physical permission record and the value record can each be assessed, and one can fail while the other is arithmetically correct.

31.5 Close the three elementary accounts

For the complete four-unit case, stock closure is

$$18 + 2 = 14 + 6 = 20X. \quad (31.4)$$

Transfer closure is $4 = 4 + 0$: withdrawal equals delivery plus declared loss. Field-and-action closure is

$$E + [V(z') - V(z)] + C = 12 + (4 - 16) + 0 = 0. \quad (31.5)$$

The final zero includes the field change. It does not say that another person must lose twelve EBU. No payer–payee relation has been derived.

Local-to-global closure follows because only Ridge and Vale change under a separable potential. If an untouched third cell contributed nine to the global potential before and after, the global difference would be $(16+9) - (4+9) = 12$. The untouched nine cancels. This does not permit omission of a changed coupling factor in a different potential. Local evaluation must include every affected contribution.

For the short execution, the same identities close using three and 9.75: stock remains twenty, and $9.75 + (6.25 - 16) = 0$. A correct account does not require performance to match the plan. It requires the performed transformation to be represented honestly.

31.6 One return action tests the sign convention

From the complete endpoint $(14, 6)$, suppose a separately permitted reverse action returns four units to Ridge. It restores $(18, 2)$. The reverse field result is $4 - 16 = -12$. Forward and reverse values add to zero under the fixed potential and ideal zero-burden assumptions.

If a justified process burden of 0.2 accompanies each movement, the values become 11.8 and -12.2 , summing to -0.4 . The physical stocks return to their start, but the account retains the two process burdens. Real resource use cannot be cancelled merely because the main stock coordinate completes a cycle.

This example proves a conditional accounting statement for the specified cycle. It does not prove that all actors will avoid cycles or that a proposed balance policy makes cycles impossible. A policy can permit zero-value or affordable negative-value movements. The dynamics of repeated choice remain a distinct research question.

31.7 Reconstruct an external event before action

Consider a variant whose previous recorded state is the reference $x = (10, 10)$, potential zero. An external conservative displacement moves eight from Vale to Ridge, creating the same frozen baseline $z = (18, 2)$, potential sixteen. The external signed potential increment is

$$D_{\text{ext}} = V(z) - V(x) = 16. \quad (31.6)$$

The actor then performs the familiar four-unit forward transfer, leaving $(14, 6)$, potential four, and recording twelve. The actor has reduced part of the displacement; the remaining potential is four. The identity is

$$V(z') + E - D_{\text{ext}} = 4 + 12 - 16 = 0 = V(x). \quad (31.7)$$

The external change did not earn actor credit. It established the baseline against which the later action was evaluated.

One prospective accounting policy would map signed actor attributions into persistent nonnegative capacity balances and record external increments in a separate audit variable J . If the policy validly assigned this single action's twelve to a balance starting at zero, then $V + B - J = 4 + 12 - 16 = 0$. This is a conditional cancellation under declared update rules. It does not establish ownership, permission to spend, fairness, or a recovery theorem. A full policy must supply those missing definitions before an implementation can use the identity operationally.

31.8 A smaller account is sometimes the honest result

Suppose a real receipt has reliable medicine stock observations but no reliable energy record. The medicine field difference may still be computable under its declared model. It should be reported as that resource-specific result with an incomplete physical coverage note. We should not invent an energy burden to make the receipt look comprehensive, and we should not describe the medicine result as a complete ecological evaluation.

Similarly, a measured endpoint does not identify every cause of its change. If external disturbance and actor action overlap without enough information to separate them, the actor-specific interpretation may remain unresolved even when the total endpoint potential is known. The accounting architecture should be able to preserve the measurement without forcing a stronger causal claim.

These are limitations of coverage and identification, not excuses for arbitrary numbers. They tell us what additional observation or modelling would be needed. An account that states an unresolved component clearly can be more useful than a single polished total that conceals it.

31.9 Audit a flawed report line by line

Consider this report: “The starting state was eighteen and two, and the route could carry six. Two carriers each proposed four, so both proposals were allowed. Each used the original state and obtained twelve EBU. The combined receipt is twenty-four. Only six units actually arrived, but the two accepted proposals justify payment for eight. We clipped Ridge to twelve to restore conservation. Since potential fell, this proves the policy maintains homeostasis.”

First, the route cap is shared, so a joint proposal of eight fails the six-unit cap. Second, two same-baseline four-unit values cannot be added as the joint result. A lossless eight-unit transfer would reach (10, 10), with group field reduction sixteen, not twenty-four, if it were physically permitted under a different cap.

Third, actual delivery six does not confirm an eight-unit lossless transfer. The source and destination measurements must be reconciled. If exactly six moved losslessly from the original state, the correct endpoint would be (12, 8), potential one, value fifteen. If eight left but six arrived, the present two-cell lossless model is inadequate and the missing two require an explicit physical account.

Fourth, post hoc clipping replaces a physical observation with a preferred number. Conservation can reveal inconsistency, but it does not tell us which observation to invent. Finally, one positive field difference does not demonstrate the long-run behaviour of a policy. The report leaps from an unverified event to an untested dynamic claim.

An audit should record all these defects without quietly repairing the world to make the final claim pass. Different repairs correspond to different events and therefore different evidence. The first step is to determine which physical account is actually supported.

31.10 Independent reconstruction exercise

Set aside the preceding arithmetic and reconstruct the following case. The state is $(17, 3)^T X$, references ten at each cell, scales two, route cap five, proposed quantity four, accepted quantity four, and performed quantity $3.5X$. There is no external change during the epoch and no process burden. Find the complete performed receipt. Then compare its value with the accepted proposal’s value and explain the difference.

Begin with the state, not with a remembered quote. Initial deviations are seven and minus seven, so initial potential is $98/8 = 12.25$. The accepted proposal would leave (13, 7), potential $18/8 = 2.25$, giving ten. The performed quantity leaves (13.5, 6.5), with deviations 3.5 and -3.5 , potential $24.5/8 = 3.0625$. Its exact value is 9.1875.

The performed transfer closes as $3.5 = 3.5 + 0$, and the endpoint total is twenty. The value closure is $9.1875 + (3.0625 - 12.25) = 0$. The quote-to-performance difference is $10 - 9.1875 = 0.8125$. It is the field value of the unperformed half unit from the live short-execution endpoint, not half of the original average value.

Check that last statement independently. At $(13.5, 6.5)$, the edge field is 1.75. A further half-unit would have exact value $1.75(0.5) - (0.5)^2/4 = 0.875 - 0.0625 = 0.8125$. The calculation links the local marginal, finite curvature, and receipt difference in one small example.

31.11 Final practice questions

1. If the independent reconstruction includes a separately justified burden $C = 0.1875$, what net value and closing identity result?
2. If the actual destination were 6.4 rather than 6.5, while source stock remained 13.5, what fails first in the declared world?
3. If a second actor changes a third, disjoint stock pair, what assumptions let its field result be added to this one?
4. If the reference is revised after performance, should the original receipt be silently recalculated under the new definition?
5. Classify the statement “the two endpoint calculations agree” and the statement “the policy keeps clinics supplied for a year.” What evidence does each require?

With the added burden, net value is nine and closure is $9 + (3.0625 - 12.25) + 0.1875 = 0$. The physical endpoint is unchanged. In the second question the endpoint total is 19.9, so the declared lossless closed-stock account fails before any value interpretation. An explicit loss carrier or corrected observation would be needed.

For disjoint actions, additivity needs complete disjoint affected factors under the declared potential, compatible baselines, and jointly valid physical permission. Merely drawing edges far apart is insufficient if a potential factor or shared constraint couples them. A changed reference should receive its own version and audit treatment; historical receipts retain the definition under which they were made.

Agreement of two calculations is a mathematical or implementation consistency finding, depending on how it is established. Year-long supply is a claim about a specified process, demand, disturbances, and service criteria. It needs

the relevant theorem or registered evidence. The first statement cannot substitute for the second.

31.12 Write a receipt that another reader can challenge

A short final report for the base example might read: “Illustrative lossless stock transfer, Ridge to Vale; baseline eighteen and two; reference ten and ten; scales two and two; quantity proposed, accepted, and performed four; successor fourteen and six; stock total twenty before and after; potential sixteen before and four after; separate process burden zero by idealization; signed field result twelve.”

That sentence is intentionally specific. A reader can challenge the stock observations, the route permission, the choice of reference, the scope of the process idealization, or the arithmetic without confusing them. A claim such as “the action helped by twelve” gives fewer handles for a careful review.

The receipt should also say whether its quantities are illustrative, predicted, or observed. In this book they are illustrative. In a deployed system, the recorded observation method and event identity would matter. Exact arithmetic on an illustrative card is educational evidence about the formula’s use; it is not a certificate for a physical shipment.

There is a practical advantage to keeping the longer record even when users see a compact display. If a question arises later, the account can be unfolded. A compact interface need not imply a vague scientific record. The book has taken many pages to teach that unfolding because trust should rest on reconstructible meaning rather than on the impressive appearance of a single number.

31.13 What the reader can now reconstruct

The full account begins with physical identities and ends with a carefully scoped interpretation. Between them lie quantities, graph support, permission, a fixed potential, exact finite differences, performance records, and closing identities. You can now rebuild each link rather than treating EBU as an unexplained number at the bottom of a table.

The final chapter returns to the research programme with that foundation in place. Definitions and conditional identities can already be taught and checked. Claims about recovery, sustained homeostasis, institutional usefulness, and fairness still need the additional work appropriate to them. Keeping those questions open is part of the foundation’s strength: the account is clear enough for future work to fail honestly as well as to succeed.

What is established and what comes next

32.1 The evidence ladder belongs inside the book

We began with a clinic, a payment record and a physical question. We now have a declared state, a potential, an exact finite action account, a local cancellation proof, a group decomposition and an auditable way to compose transitions. The final task is to say what each of those achievements does and does not establish.

A project becomes misleading when a definition, a successful software check and a hope about society are reported in the same voice. The solution is not to weaken every sentence into uncertainty. It is to make the kind of statement clear. An identity can be exact under explicit assumptions. An empirical conclusion can be limited to its registered observations. An institutional proposal can be stated plainly as a proposal.

This edition is a theoretical and explanatory revision. Its arithmetic examples have been supplied for teaching and checked as such. They are not new scientific runs. Their purpose is to let a reader reconstruct the calculation and inspect the assumptions before any future result is interpreted.

Kind of statement	Example and appropriate interpretation
Definition	$V_i = (x_i - x_i^*)^2 / (2\sigma_i^2)$ declares the model's ruler. It does not discover a law of nature.
Conditional identity	The exact quadratic expansion follows from that definition with fixed parameters.
Theorem under assumptions	Complete affected factors cancel; consistently linked endpoint differences telescope.
Implementation check	A named function matches a specified calculation at checked inputs. This requires its own code and test record.
Scientific observation	A registered execution produced particular outcomes under a frozen protocol.
Research hypothesis	A proposed actor/capacity mechanism may improve recovery under declared disturbances.

Kind of statement	Example and appropriate interpretation
Institutional choice	An organization decides how to authorize action, support access or assign responsibility.

32.2 The declared physical world

The principal worked model contains one homogeneous, conserved stock distributed over finitely many locations. Stocks are nonnegative; an internal lossless transfer removes exactly what it adds elsewhere. Any further capacity or action constraint must be stated explicitly. Physical feasibility does not follow from a favourable potential value.

The reference vector is nonnegative and compatible with the fixed total stock. Every scale is positive and fixed during the account. These conditions make the Gaussian potential well defined and put the chosen reference in the basic fixed-mass domain. They do not prove that every action menu can reach it.

This small model deliberately omits many real features. It contains no automatic regeneration, no biological population threshold law and no restoring flow hidden inside the field definition. A future domain can include additional processes, but their physical carriers, state effects and assumptions have to be declared.

The simplification is useful because it exposes the accounting logic. It becomes misleading only if conclusions are transferred to the omitted features without justification. A clear theoretical model is an instrument for asking questions, not a miniature planet whose every property generalizes.

32.3 The Gaussian potential and its exact consequences

With fixed x^* and positive σ_i ,

$$V(x) = \frac{1}{2} \sum_i \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2, \quad (32.1)$$

$$\mu_i(x) = \frac{x_i - x_i^*}{\sigma_i^2}, \quad H = \text{diag}(1/\sigma_i^2). \quad (32.2)$$

The potential is nonnegative and zero exactly at the declared reference. Its gradient is a vector of local slopes. The Hessian specifies the curvature. All three objects are properties of the declared function.

For any fixed admissible increment δ , expansion gives

$$V(z + \delta) = V(z) + \mu(z)^T \delta + \frac{1}{2} \delta^T H \delta. \quad (32.3)$$

The expression is exact for the quadratic, not a short-step approximation. Positive definiteness of H makes the quadratic term positive for nonzero δ . Thus the initial tangent field reduction overstates the full field reduction by that term.

The general action equation is

$$E_a = V_{\text{loc}}(z) - V_{\text{loc}}(z + \delta_a) - C_a. \quad (32.4)$$

The principal examples set $C_a = 0$. Where a residual process burden is used, it must describe declared physical burden, use compatible units and avoid duplicating represented effects. No price, wage or arbitrary penalty becomes physical burden merely by being placed in that term.

32.4 The conditional laws in plain language

First, a physical transfer must have a complete quantity account. In the main lossless model, the stock removed at one endpoint appears at another. A potential difference is not a substitute for this check.

Second, local valuation can be exact. Every unchanged factor has the same value before and after and therefore cancels. A coupled factor expands the required read neighbourhood; omitting it can even reverse the sign.

Third, a finite value includes curvature. A positive initial field does not guarantee a positive result for every finite quantity. The action must be evaluated at its complete endpoint, subject to the declared physical constraints.

Fourth, a group has a joint value. Fixed increments add in the state update, but their singleton values need not add through a nonlinear potential. The Gaussian cross-term formula explains the discrepancy without introducing extra physical stock or additional reward.

Fifth, common-path contributions sum to the group field difference under their stated assumptions. The sum follows from calculus. The interpretation of those contributions as physical path decompositions requires a model-realized path; their interpretation as ownership or capacity updates requires an additional rule.

Sixth, consecutive endpoint differences telescope under one fixed field. Closed undriven sequences have net value equal to minus the declared nonnegative process burden. External changes and revisions of the field definition require explicit additional terms.

32.5 What the potential does not contain

A potential tells us how a declared state is scored. It does not by itself choose an action, assign an actor, select an amount, specify a timescale or guarantee that a path is physically realizable. Each of those objects needs a model of its own.

In particular, the Gaussian shape is not evidence that actual states are normally distributed. The conserved total couples coordinates even if the Hessian is diagonal. A proposed probability law on the fixed-mass domain would need an explicit construction and empirical interpretation.

Nor does a reference automatically become a dynamic equilibrium. If a future transition rule leaves the reference unchanged, that establishes a particular invariance property. Attraction concerns what happens from other states. Homeostatic function concerns whether relevant systems continue operating under disturbances. These are related research questions, not alternative names for the minimum of a quadratic.

32.6 The historical mathematics keeps its own scope

The preserved Parts II and III contain earlier models with threshold penalties, flow laws, discrete-time bounds and preservation controllers. Their proofs and results remain statements about their specified definitions. This edition removes those mechanisms from the introductory core because they are not required to teach the general finite account.

It would be wrong to say that changing the introductory potential invalidates every historical theorem. It would be equally wrong to relabel old experimental findings as Gaussian findings. A theorem transfers when its assumptions hold; an observation transfers only through an additional justified argument about the relevant systems.

For example, unchanged-factor cancellation applies to the new separable quadratic by the same algebraic reasoning. A result that depended on an explicitly chosen threshold controller does not become a result for random-affordable actors merely because both programmes use the name EBU.

The book therefore retains the historical volumes as sources and explains the revised model directly. Compatibility notes and a chapter-disposition ledger accompany the manuscript. They identify where teaching responsibility moved or examples changed, rather than rewriting the past into a seamless but false narrative.

32.7 A narrow historical evidence example

The repository contains a finalized Gate 1D-C execution manifest recording thirty registered runs and six thousand trace rows. Its protocol, implementation identities and result artifacts define what that study tested. This is actual historical execution, distinct from the illustrations in this book.[15]

The existence of those records matters: the project contains scientific work as well as prospective designs. Their existence does not establish that the new Gaussian mechanism has run, that all future harnesses have passed review, or that a real economy would behave as intended.

Detailed historical outcomes belong with their original comparisons and limitations. A short introductory paragraph cannot replace the frozen protocol or turn a positive finding on one outcome into general superiority. Readers who want to evaluate that evidence should follow the manifest and its identified sources.

32.8 Why a software test is a different achievement

A function that computes $4q - q^2/4$ correctly at chosen rational quantities can be checked without simulating a world. Such a check can catch a sign or coefficient error. It does not observe an actor selecting an action or a population recovering from disturbance.

A trajectory test is another class of work. It advances model states under declared rules. Its interpretation depends on the world, initialization, randomness, horizon, comparisons and outcomes fixed before the run. Calling it a unit test does not remove those scientific requirements.

Performance measurements are different again. Fast evaluation may make a large study practical, but speed is not evidence of model validity. A benchmark can establish a measured software property on a named host. It cannot establish whether the actions it would evaluate preserve homeostasis.

The distinction helps protect both kinds of work. Implementation checks can be celebrated for the errors they detect without being asked to support claims they never tested. Scientific evidence can be evaluated on its registered question without being confused with a build log.

32.9 The current prospective question

The new research direction asks what happens when actors choose by unweighted random selection among affordable, physically feasible actions or groups. Affordability is a separately declared rule based on persistent signed

capacity accounts. The sampling unit and finite opportunity menu still need to be specified. EBU supplies the field calculation. It does not secretly choose the action that maximizes the value.

That separation makes the result scientifically open. A mechanism might recover, cycle, remain trapped, refuse useful actions or perform worse than a comparison rule. These are possible outcomes to distinguish, not defects to tune away after inspection.

The proposed study also needs a physical comparison arm using the same relevant opportunities without the capacity restriction. This would help identify the effect of the mechanism being studied. The comparator must be specified fairly; it should not be handicapped by a private information restriction or an artificially weaker physical menu.

No experimental parameter values, horizon or success threshold are chosen by this book. The examples teach equations. They do not quietly preregister a study because their numbers happen to be convenient.

32.10 Decisions a study must make before execution

A scientific protocol must define its event order. Does forcing occur before the action baseline? How are physically infeasible disturbances handled? How do the actions compose, and what is actually measured? Small changes can define different stochastic processes.

It must define the sampling unit and opportunity menu. Choosing an edge, an actor or a group uniformly need not produce the same distribution of physical actions. Duplicate descriptions of an identical action can change probabilities unless the menu convention addresses them. An empty affordable menu needs an explicit response.

The capacity proposal needs an owner map, a path interpretation, signed netting rules and a numerical policy for boundary comparisons. Group closure alone supplies none of these choices. They affect which actions can occur and therefore belong in the registered model.

Finally, the protocol needs outcome measures capable of distinguishing its claims. It should report physical validity, service where applicable, deviation, refusals and account behaviour without combining them into an unexplained winner label. A mechanism can improve one measure and worsen another.

32.11 The same random seed is not always the same exposure

Imagine two comparison arms receiving identical raw disturbance proposals. After their states diverge, one proposal may be feasible in one arm and in-

feasible in the other. If infeasible proposals are rejected, the applied physical disturbances differ even though the raw random numbers match.

This does not make paired comparison impossible. It means the coupling must be described precisely. The study can report both the shared proposals and the realized inputs, or use a design that makes the intended equality feasible. It should not claim identical physical exposure solely because a seed is shared.

The point is another example of the book's reading habit: identify the object. A random draw, an accepted disturbance and an observed state change are different objects. Scientific conclusions depend on which one was controlled.

32.12 What would count as recovery?

Recovery needs a declared reference condition, disturbance and time criterion. Does it mean exact return, approach within a specified neighbourhood, re-stored service, or a distribution concentrated near functional states? These are different propositions.

On a finite fixed-mass domain with positive fixed scales, deviation is bounded even under many unsuccessful mechanisms. Boundedness alone is therefore a weak discriminator. An idle or trapped system can remain bounded forever. The study needs a stronger outcome if its claim concerns useful restoration.

With continuing forcing, a physical state distribution might settle while cumulative audit or capacity accounts drift. Stationarity of the physical state is not stationarity of the augmented state containing all balances. Any claim about a long-run process must identify which coordinates it covers.

These distinctions should be made before results are observed. Otherwise it becomes too easy to describe whichever quantity looks stable as the intended homeostatic outcome.

32.13 Calibration is a scientific and institutional task

The field uses references and scales. Domain evidence must support their physical interpretation; institutional procedures must determine who may adopt or revise them. A mathematically convenient value is not automatically a defensible public standard.

Calibration can change whose actions receive positive or negative values. The reference determines the centre; the scale determines how strongly departures count relative to one another. A doubling of all scales divides this quadratic potential and its field values by four. That is a change in the numerical ruler, not a physical improvement.

Sensitivity analysis can show how conclusions depend on these choices. Independent measurements can constrain them. Appeals and review can address contested representation. None of these steps is replaced by high arithmetic precision.

32.14 A revised clinic capstone

Return to Vale with three linked needs: water, medicine storage and a pump repair. For the water demonstration, use $z = (18, 2)X$, references $(10, 10)X$, scales $(2, 2)X$ and an accepted lossless four-unit transfer. The endpoint is $(14, 6)X$. Conservation gives twenty units before and after, potential changes from sixteen to four, and field value is twelve.

Now add a separately reported delivery requirement of five units for the stated interval. The action supplies four, leaving one unmet on that requirement. The field improved and the service requirement was not fully met. Reporting both is more informative than assigning an unqualified success label.

The pump repair uses a material part and labour. Neither is silently converted into water stock. Its physical transformations need their own records. If the current water state contains no equipment-capacity coordinate, the present water potential does not issue an invented repair benefit. Later water movements can be observed under the repaired action map.

If medicine refrigeration also changes an energy account, preserve the separate resource coordinate. A water value of twelve does not automatically offset an energy burden of another numerical amount. The clinic's actual functioning depends on linked services, timing and quality, not solely on their scalar accounts.

Finally, a public institution might support the repair or guarantee access to medicine. That decision can be justified by need and evidence while leaving the physical record intact. The patient is neither the debit nor the balancing entry.

32.15 Questions a reader should now be able to answer

Is EBU physical energy? Not by the name alone. In this Gaussian model, the standardized potential is dimensionless. A physical application must declare its quantities and calibration. Energy is one possible represented carrier, not an automatic interpretation of every value.

Is the scale a measurement error? No. Here it is a declared width of the reference geometry. Sensor uncertainty is a separate statement about our

knowledge of stock. It may affect a later uncertainty analysis but should not silently redefine the field.

Does zero potential prove a functioning clinic? It proves that the represented stocks match the declared references. A clinic also needs quality, timing, equipment, staff and access. Unrepresented requirements are not certified by that equality.

Does positive marginal field prove positive finite value? Only a sufficiently small movement may inherit the initial directional sign when the assumptions and process burden permit it. The finite equation includes curvature and the full endpoint.

Must every action be sequential? No. The book represents simultaneous groups. Their joint value is calculated at the combined endpoint. A historical theorem with a restricted execution rule retains that scope; it does not prohibit separate analytical treatment of groups.

Does the path attribution settle ownership? No. It provides a declared mathematical decomposition. Physical-path interpretation, causal identification, ownership and spendability need their own assumptions and authority.

Can a negative action still be necessary? Yes. The represented field and an urgent service requirement can point in different directions. The report should expose the conflict, and the relevant institution should decide how to respond.

Can a positive action be refused? Yes. It may be physically infeasible, unauthorized, unaffordable under a proposed capacity rule, or simply not selected. A value calculation is not a command.

Why preserve a zero or negative example? It tests whether we understand the equation rather than treating it as a reward label. Such examples can reveal missing state, overshoot, incomplete service or a genuine tradeoff.

Can a field be improved later? Yes. Scientific models should respond to evidence. Versioning and explicit model-change records prevent an improved description from being misreported as a physical action in the old history.

What would justify a claim about an economy? Evidence about measurement, behaviour, access, incentives, institutions and physical outcomes at the relevant scale. A finite accounting identity is one component, not the complete argument.

32.16 A reader's final reconstruction exercise

Choose a finite action with one or two physical edges. Write its source-side quantity, complete increment and predecessor state. Check feasibility and

conservation before evaluating the potential. Declare references and positive scales with units and a compatible total.

Calculate the before and after potentials. Obtain the field value by subtraction and independently by the exact quadratic expansion. If there is a group, calculate the joint increment and compare with the same-baseline singleton sum. If you display path attributions, state what path they use and verify that they close.

Write a short receipt containing the equation version, physical inputs, calculation and evidence status. Add a separately stated service question. Then identify one human or institutional question your field does not answer. A complete response should allow another reader to reproduce the mathematics without guessing what the action or the claim was.

If a step remains uncertain, revisit the corresponding practice studio before drawing a scientific conclusion.

32.17 The equation and the work ahead

The core account can now be read without treating its symbols as a black box:

$$\begin{aligned} \text{signed EBU} &= \text{before potential} - \text{after potential} \\ &\quad - \text{declared residual burden.} \end{aligned}$$

The state and physical map say what changes. The potential says how that represented change is measured. Feasibility says which changes the physical model permits. An actor rule says what is chosen. Evidence says what happened. Institutions decide how people may use the account.

The foundation in one connected statement

With a declared physical boundary, fixed field and complete action support, exact finite differences provide consistent local action values. Their group and sequential identities follow under explicit assumptions. Calibration, realized dynamics, useful recovery, fair access and an action-accounted economy remain substantive work to be demonstrated.

Part II continues at Chapter 33: from the action's domain and derivations to physical permission, software records, tests and historical evidence. The question stays the same: what changed, under which assumptions, and what can we conclude? A limitation, counterexample or failed comparison can be a useful next result. Scientific integrity preserves those possibilities as carefully as the equation.

Plain-language dictionary and symbol guide

Action. A declared physical transformation with a specified boundary, quantity or other input, predecessor state and successor. It can be a transfer or another represented change. Naming a useful activity does not by itself define its state effect.

Actor. The person, organization or modeled decision-maker associated with an action. Its identity is distinct from the action's field value. The field equation does not assign moral worth to the actor or recipient.

Affected factors. Terms in a potential that depend on at least one changed coordinate. A complete local calculation includes every such term. In the separable Gaussian model, these are the changed endpoint terms.

Baseline, z . The common state from which an action and its no-action alternative are compared. In the prospective forcing-first order it is the post-forcing, pre-action state. A baseline must identify both data and model version.

Capacity account, B_i . A proposed persistent account controlling later modeled affordability. It is not the physical stock, storage capacity, household income or an automatically legitimate entitlement. Its update and use require a separately declared rule.

Cell or node. An indexed location or subsystem holding a represented state. A cell is a modeling boundary rather than necessarily a household, person or geographic square.

Common path, $\gamma(s)$. A single declared state path used to calculate the group integral and its child contributions. The path parameter is not automatically clock time. A physical-path reading needs a physical model that realizes it.

Conservation. A complete physical quantity balance within a named boundary. Lossless internal transfers preserve the sum of homogeneous stocks. Transformations, sinks and boundary crossings must be named when they are present.

Curvature. The way a slope changes as the state changes. For this quadratic it is represented by the fixed matrix H . Curvature explains why an initial slope times a finite quantity differs from the exact finite value.

EBU, E . The signed before/after potential difference minus declared nonduplicated process burden. In the present model its field component is dimen-

sionless standardized deviation reduction. The name does not make it physical energy or money.

Edge. A declared possible directed physical process. For a lossless transfer, its incidence column has -1 at the source and $+1$ at the destination. A line drawn between places is not sufficient evidence that a real process is available.

Equilibrium. A property of a specified dynamical system, typically a state unchanged by its evolution rule. It is distinct from a declared reference or a minimum of a chosen potential unless the required relationship is established.

External audit, J . The proposed cumulative signed potential change from external forcing. It explains externally introduced or removed deviation. It is not a spendable “nature wallet.”

Factor. One term in a decomposed potential, with a declared set of input coordinates. A factor can be local without depending on only one coordinate. Its dependencies determine the required evaluation neighbourhood.

Feasibility. Satisfaction of the declared physical conditions on an action, such as nonnegative stock, permitted edges and shared capacity. A favourable EBU does not establish feasibility.

Field. The declared state, reference geometry and relevant action relationships used for valuation. Its gradient supplies local directional information. The word does not assert a new fundamental physical force.

Field attribution, R_a . A child’s signed derivative integral along a declared group path. It sums to the group field reduction under the stated conditions. It is not automatically a uniquely observed causal contribution or an ownership share.

Forcing. A declared external physical change preceding the action baseline in the prospective study. Raw random proposals and physically applied forcing can differ if feasibility rules reject some proposals.

Functional balance. A description of conditions under which relevant functions can continue. A single numerical reference is one representation of part of that idea, not a complete definition for every living system.

Gaussian geometry. The quadratic relative negative-log shape used here to measure standardized deviation. It does not establish an observed normal distribution of actual states, nor an autonomous restoring process.

Gradient, $\mu = \nabla V$. The vector of derivatives of the potential. Each entry describes local change in potential per small change of its coordinate. The gradient is not an actor, a physical stock or a selection algorithm.

Group, G . Accepted actions evaluated through a common physical transition because of their declared dependencies. The child actions remain identifiable. A group total does not by itself determine every child allocation.

Hessian, H . The matrix of second derivatives. For the separable Gaussian model, $H_{ii} = 1/\sigma_i^2$ and off-diagonal entries vanish. Shared action coordinates can still produce group cross terms.

Homeostasis. Maintaining relevant function through continuing change and regulation. The term does not mean that every component is static, that resources are unlimited or that any proposed accounting rule has already achieved regulation.

Incidence column, S_e . A signed template turning an edge quantity into its state increment. Multiplying S_e by a finite quantity gives the source and destination changes in stock units.

Locality. A statement about necessary support or information. Mathematical locality, distributed computation and decentralized governance are distinct properties. None should be inferred from the others without a specific argument.

Marginal field, f_e . For a lossless directed transfer, $\mu_i - \mu_j$. It gives the initial rate of field reduction per added action quantity. Finite value also includes curvature and any declared process burden.

No action. A genuine zero physical increment with zero process burden. Under the shared-baseline convention it has zero field value. Missing evidence about an action is not the same as verified no action.

Potential, V . The declared scalar function of state. In this model, $V = \frac{1}{2} \sum_i (x_i - x_i^*)/\sigma_i^2$. It measures represented standardized discrepancy, not total welfare or conserved matter.

Process burden, C . Declared physical action burden not already charged through the evaluated state and potential. Its carrier, units and nonduplication must be explained. The principal ideal examples set it to zero.

Quantity, q . The finite source-side amount in this edition. A rate j has stock-per-time units and becomes quantity after integration over time. For a constant rate, $q = \Delta t j$.

Receipt. A provenance-linked record of a physical event, its accepted comparison, evidence and signed calculation. A record can exist before an institution has decided whether its number changes a transferable balance.

Reference, x_i^* . A declared local comparison value. It may be informed by functional evidence, but it is not automatically an observed mean, ethical optimum or dynamic equilibrium.

Scale, $\sigma_i > 0$. A fixed positive stock-unit width determining how much a given deviation contributes to the potential. It is separate from measurement uncertainty unless an explicit model establishes a relationship.

Service. A separately specified outcome such as delivery of a required quantity before a deadline. A positive field difference can coexist with unmet service; a negative one can coexist with necessary service.

State, x . The ordered collection of represented physical coordinates. Complete state identity includes units, meanings and ordering. Equal potential does not imply equal state.

Support, W . Coordinates directly changed by the action. With coupled factors, the set of coordinates required to evaluate the effect can be larger than this write support.

Telescoping. Cancellation of intermediate values in a sum of consecutive differences. It requires consistent definitions and correctly linked boundaries; it does not reconstruct missing physical history.

Vector account. Separate resource-specific results retained together without an undeclared conversion. Even dimensionless scores can have different calibrations and meanings.

The compact symbol card

Symbol	Object	Units or interpretation
x_i, z_i	Stock at a location	Stock, X
x_i^*	Declared reference	Stock, X
σ_i	Positive reference scale	Stock, X
q	Finite action quantity	Stock, X
j	Action rate	Stock per time
δ	State increment	Stock in each coordinate
V	Gaussian potential	Dimensionless account value
μ_i	Marginal potential	Inverse stock
H	Curvature matrix	Inverse squared stock
E	Signed finite value	Compatible potential units
R_a	Common-path attribution	Compatible potential units
C	Residual process burden	Compatible potential units
B_i	Proposed action balance	Declared capacity-account units
J	External potential audit	Compatible potential units

Notation alone establishes neither physical calibration nor spending rights.

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How to follow a mathematical claim

The social and environmental references support the bounded claims attached to them in the opening chapters, not a claim that EBU succeeds. The project sources identify definitions, conditional mathematics and historical records. Source versions, PDF hashes and the full revision disposition accompany the editable manuscript.

The local-to-global argument in this edition is the cancellation of unchanged factors. Its Gaussian examples use a newly declared quadratic; they do not reinterpret historical threshold fixtures. The exact finite expansion is derived directly from that quadratic. The path sum uses linearity, the chain rule and the fundamental theorem of calculus. The sequence and closed-cycle identities use consecutive endpoint cancellation.

The field definition, the path convention and any proposed capacity rule have different roles. A proof about the sum of path contributions cannot supply a missing ownership map. Likewise, a physical conservation equation cannot by itself choose which consequences belong in a potential. These boundaries are part of the claim, including when a source file is committed in Git.